

Stellar Structure and Evolution

Lecture Notes

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Contents

Preface	vii
Course Overview	1
1 Distance by Parallax	3
1.1 Small-angle approximation	3
1.2 Definition of the parsec	4
1.3 Magnitudes	5
2 Stellar Magnitudes	7
2.1 Hipparchus's scale and Pogson's relation	7
2.2 The Vega system and the reference spectrum	7
2.3 Stellar colors	8
2.4 Interstellar extinction and reddening	8
2.5 Absolute magnitude and distance modulus	9
2.6 Bolometric magnitude and correction	10
2.7 Blackbody radiation	11
2.7.1 Wien's displacement law and the Planck function	12
3 Thermodynamics	15
3.1 HR diagrams and related topics	15
3.2 Thermodynamics review	15
3.3 First law of thermodynamics	15
3.4 Thermodynamic relations	17
3.5 Thermal expansion coefficients	20
3.6 Alternative form of the first law	21
3.7 The ideal gas case	22
3.8 The adiabatic gradient	22
4 Radiative Processes	25
4.1 Definition of a star	25
4.2 Fundamentals of radiative transport	25
4.3 Specific intensity	25
4.4 Mean intensity	27
4.5 Radiative energy density	28

4.6	Radiation flux	29
4.7	Moments of the radiation field	31
4.8	Radiation field in any astrophysical environment	33
4.9	Radiative transfer equation	36
4.10	Optical depth and source function	38
4.11	Solutions to the transfer equation (PP)	39
4.12	Eddington-Barbier approximation	49
4.13	Limb darkening	50
4.14	Source function and thermodynamic equilibrium	51
4.15	Diffusion Approximation	52
4.16	Mean Intensity and Radiation Flux	53
4.17	Rosseland Approximation (Equation)	54
4.18	Eddington Luminosity	55
4.19	Astrophysical Applications	57
5	Stellar Structure Equations	61
5.1	Introduction	61
5.2	Mass equation	62
5.3	Gravitational field	64
5.4	Hydrostatic equilibrium	65
5.5	Estimating central values for pressure and temperature	67
5.6	Timescales	69
5.7	Virial Theorem	71
5.8	Kelvin–Helmholtz timescale	76
5.9	Nuclear timescale	77
5.10	Energy conservation	79
5.11	Energy transport equations	82
5.12	Convective energy transport	89
5.13	Summary: the four stellar structure equations	99
5.14	Convective mixing timescale	99
5.15	The 5 th equation of internal structure	101
6	Polytropic Stars	105
6.1	Introduction	105
6.2	Closing the system of equations	105
6.3	Gravitational potential in spherical symmetry	106
6.4	The gravitational Poisson equation	107
6.5	Derivation of the Lane–Emden equation	108
6.6	Mass distribution and boundary conditions	109
6.7	Important polytropic cases	111
6.8	Solutions of the Lane–Emden equations	112
7	Star and Planet Formation	115

7.1	Introduction	115
7.2	Molecular clouds and prestellar cores	115
7.3	The Jeans instability	116
7.4	Collapse dynamics: inside-out collapse and the accretion rate	120
7.5	The angular momentum problem	122
7.6	From protostar to pre-main sequence	124
7.7	Observational classification: Classes 0, I, II, and III	126
7.8	Structure and evolution of protoplanetary disks	128
7.9	Magnetospheric accretion and the T Tauri phase	134
7.10	Synthesis: from star formation to planet formation	139
8	Stellar Equations of State	147
8.1	Introduction	147
8.2	Ideal gas pressure	147
8.3	Decomposition for a fully ionized gas	148
8.4	Mean molecular weight of a fully ionized gas	148
8.5	Degenerate conditions in the stellar interior	149
8.5.1	Motivation: an electron gas confined in 1D	149
8.5.2	Generalization to 3D	150
8.6	Occupation distribution and electron density	151
8.7	Momentum flux and mechanical pressure	153
8.8	Total non-relativistic degeneracy	154
8.9	Total ultra-relativistic degeneracy	156
8.10	Total relativistic degeneracy	157
8.10.1	Ultra-relativistic limit: cross-check	158
8.11	Partial degeneracy	159
8.11.1	Non-relativistic case	159
8.12	Density–temperature diagram	160
8.13	Total equation of state	161
8.14	Crystallization and neutronization	162
8.14.1	Crystallization	162
8.14.2	Neutronization	163
9	White Dwarfs	165
9.1	Introduction	165
9.2	Chandrasekhar’s theory	165
9.3	Combining the equation of state with hydrostatic equilibrium	166
9.4	Chandrasekhar’s differential equation	166
9.5	Boundary conditions, mass, and radius	168
9.6	Limits, the mass-radius relation, and the Chandrasekhar mass	168
10	Energy Generation Processes (Nucleosynthesis)	173
10.1	Introduction: stellar nucleosynthesis	173

10.2	Formalism of nuclear reactions	173
10.3	Cross section and reaction rate	174
10.4	Derivation of $\langle\sigma v\rangle$ and the three physical factors	175
10.5	Gamow factor and the Gamow peak	176
10.6	Electron screening	178
10.7	Nuclear reaction network formalism	180
10.8	Example: destruction of ^{25}Al	181
10.9	Proton-Proton Chain	184
10.10	Comparative overview of energy generation processes	185
11	Stellar Evolution with MIST Tracks and MESA Simulations: A Didactic Approach	189
11.1	Introduction	189
11.2	Evolutionary Trajectories for Low- and High-Mass Stars	191
11.2.1	Low-Mass Stars	192
11.2.2	High-Mass Stars	192
11.2.3	Key Points	195
11.2.4	Effects of Stellar Rotation on the Evolutionary Trajectories	196
11.3	Time Evolution of the Parameters	201
11.3.1	Radius	202
11.3.2	Effective Temperature	205
11.3.3	Energy-Generation Processes	207
11.3.4	Core Temperature	213
11.3.5	Core Density	215
11.3.6	Convective and Radiative Zones	217
11.3.7	Mass Loss	221
11.4	Isochrones	226
11.4.1	Effect of Metallicity on Fixed-Age Isochrones	228
11.4.2	Age Effects on Fixed-Metallicity Isochrones	230
11.5	Stellar Evolution Model	235
11.5.1	Physical Configuration of the Model	236
11.5.2	Simulation Results	237
11.5.3	HR Diagram	238
11.5.4	Time Evolution of the Luminosity	238
11.5.5	Evolution of the Effective Temperature	239
11.5.6	Evolution of the Radius	239
11.5.7	Evolution of the Mass	239
11.5.8	Evolutionary Timeline	241
11.5.9	Kippenhahn Diagram	241
11.5.10	Conclusion	241
11.6	Nucleosynthesis	243
11.7	Conclusions	248
11.8	Appendix A: Inlist Used	249
11.9	Appendix B: Synthesis Mechanisms and Physical Environments of the Chemical Elements	251

Preface

This material collects the lecture notes for the Stellar Structure and Evolution course, taught in the Graduate Program and in the undergraduate Physics/Astrophysics program at the Federal University of Sergipe.

The content is based on a range of material available in the literature, including the lecture notes of Prof. Wagner Marcolino (UFRJ), the notes of O. R. Pols and Jørgen Christensen-Dalsgaard, and the textbooks by Kippenhahn, Weigert & Weiss and by Kepler de Oliveira Filho. The final part, devoted to stellar evolution, was developed together with the students of the 2025.2 semester and resulted in a paper submitted to the *Revista Brasileira de Ensino de Física*.

The notes were originally handwritten and later digitized with the help of Claude Code (Sonnet 5.0 model, medium effort), for which I wrote prompts that converted the handwritten text into L^AT_EX. The colored boxes that appear throughout the text bring complementary astrophysical context generated with the help of artificial intelligence and reviewed by me.

Suggestions for topics to include, chapters to expand, or any other improvement are welcome by e-mail: diogosouto@academico.ufs.br

I hope this material contributes to your education in astrophysics. =)

Course Overview

Bibliography

- Clayton (1983) — *Principles of Stellar Evolution and Nucleosynthesis*
- Kippenhahn et al. (2012) — *Stellar Structure and Evolution*

Supporting material

- Oliveira Filho and Saraiva (2017) — *Astronomia e Astrofísica*
- Christensen-Dalsgaard (2008) — *Lecture Notes on Stellar Structure and Evolution*

Assessment methods

- Two practical exams *or* two exams plus a final project;
- Average of the problem-set grades;
- Class participation and presentations;
- The final grade is the arithmetic mean of these assessments;
- There is no make-up exam. The make-up grade is composed of the average of the highest grades, with the lowest one excluded.

Further Reading and Current References

Beyond the classic texts listed above, it is worth following the *data releases* of the *Gaia* satellite (Gaia Collaboration, 2023), which today provide the reference observational basis for parallaxes, distances, and fundamental properties of stars — a topic we already introduce in Chapter 1.

Chapter 1

Distance by Parallax

The distance to a celestial body is one of the most important physical quantities in astronomy. The easiest way to obtain the distance to an astronomical object directly is through **parallax**. The method consists of observing an object — a star, for example — from two different points. Since astronomers knew from the start that the Earth orbits the Sun in a nearly circular orbit, it would be possible to observe a star at least twice over a six-month period, so that the Earth would be on the opposite side of its orbit relative to the first measurement.

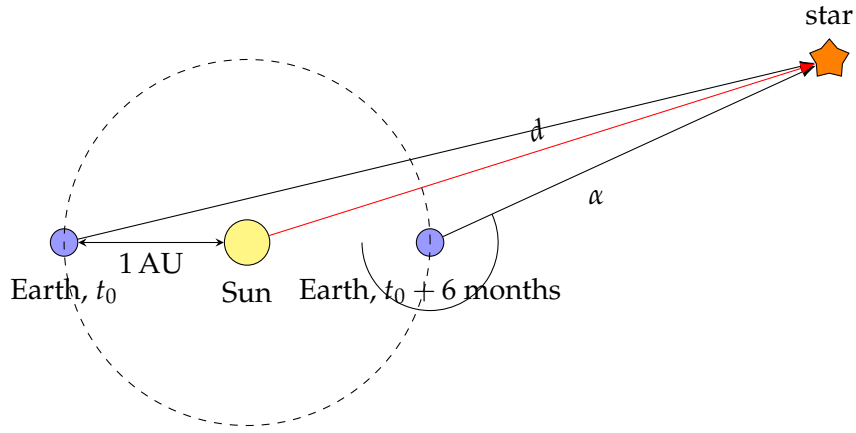


Figure 1.1: Geometry of stellar parallax: the Earth occupies opposite positions of its orbit six months apart, which allows measuring the small angular displacement α of the star relative to the background of more distant stars.

Once the distance from the Sun to the Earth is known, simple trigonometry can be used to determine the distance between the object and the Earth (or, equivalently and more commonly, between the object and the Sun), so that:

$$\tan \alpha = \frac{1 \text{ AU}}{d} \implies d = \frac{1 \text{ AU}}{\tan \alpha}. \quad (1.1)$$

1.1 Small-angle approximation

Since $\tan \alpha = \sin \alpha / \cos \alpha$, and stellar parallax angles are always extremely small (less than $1''$ even for the closest stars), we can approximate $\sin \alpha$ and $\cos \alpha$ by a first-order Taylor series expansion around $\alpha = 0$. Recalling the general form of the Taylor series,

$$f(x) = \sum_{n=0}^{\infty} a_n (x-a)^n, \quad a_n = \frac{f^{(n)}(a)}{n!}, \quad (1.2)$$

the first terms are

$$f(x) = f(a)|_{n=0} + f'(a) \frac{(x-a)^1}{1!} \Big|_{n=1} + f''(a) \frac{(x-a)^2}{2!} + \dots \quad (1.3)$$

Expansion of $\sin \alpha$ and $\cos \alpha$ around $\alpha = 0$

For $f(\alpha) = \sin \alpha$:

$$f(0) = \sin(0) = 0, \quad f'(0) = \cos(0) = 1, \quad f''(0) = -\sin(0) = 0$$

$$\implies f(\alpha) \approx 0 + \frac{1 \cdot \alpha}{1!} + \dots \approx \alpha. \quad (1.4)$$

For $f(\alpha) = \cos \alpha$:

$$f(0) = \cos(0) = 1, \quad f'(0) = -\sin(0) = 0, \quad f''(0) = -\cos(0) = -1$$

$$\implies f(\alpha) \approx 1 + \frac{0 \cdot \alpha}{1!} + \frac{(-1)\alpha^2}{2!} \approx 1. \quad (1.5)$$

Hence, to first-order approximation,

$$\sin \alpha \approx \alpha, \quad \cos \alpha \approx 1, \quad \implies \boxed{\tan \alpha \approx \alpha \quad (\text{in radians}).} \quad (1.6)$$

Validity of the approximation

The approximation $\tan \alpha \approx \alpha$ is only good for small angles *expressed in radians*. The relative error is of order $\alpha^2/3$; for the largest known stellar parallax angle (Proxima Centauri, $\alpha \approx 0.768'' \approx 3.7 \times 10^{-6}$ rad), this error is completely negligible — of order 10^{-12} . The approximation is therefore excellent for any star.

1.2 Definition of the parsec

Combining Equation (1.1) with the small-angle approximation (Equation (1.6)), the distance to an object can be expressed in general as

$$\boxed{d = \frac{1}{\alpha}}, \quad \text{with } d \text{ in parsecs and } \alpha \text{ in arcseconds.} \quad (1.7)$$

The unit of distance obtained by parallax is the **parsec** (*parallax + second of arc*). An astronomical object with $\alpha = 1''$ is, by definition, at a distance of 1 parsec.

Where the value of 1 parsec in AU comes from

The angle of $1''$ (one arcsecond), expressed in radians, is:

$$1'' = \left(\frac{1}{3600} \right) \left(\frac{2\pi}{360} \right) = 4.848 \times 10^{-6} \text{ rad.} \quad (1.8)$$

Hence,

$$1 \text{ pc} = \frac{1 \text{ AU}}{4.848 \times 10^{-6}} \approx 206\,265 \text{ AU.} \quad (1.9)$$

The modern official definition of the parsec

The derivation above — motivating the parsec from the small-angle geometric relation $d = 1/\alpha$ — is the historical and conceptual origin of the unit, and correctly reproduces the numerical value used throughout this book. Since 2015, however, the International Astronomical Union (IAU) has fixed the parsec by a purely numerical

definition, independent of any geometric approximation: the notes of IAU Resolution B2 (2015) define $1 \text{ pc} \equiv 648\,000/\pi \text{ AU}$ as an *exact* value — itself resting on Resolution B2 of the XXVIIIth IAU General Assembly (2012), which fixed the astronomical unit at $1 \text{ AU} = 149\,597\,870\,700 \text{ m}$ exactly. Numerically, this coincides with the value obtained from the small-angle geometric relation ($206\,264.806 \dots \text{ AU}$, rounded to $206\,265 \text{ AU}$ in the Example above), so no result in this book is affected — but it is worth knowing that, formally, today’s parsec no longer depends on the approximation $\tan \alpha \approx \alpha$ that historically motivated it (Mamajek et al., 2015).

Reach of the method and astrometric missions

Trigonometric parallax is the only truly *direct* distance determination method in astronomy — every other distance indicator (spectroscopic parallax, the Cepheid period–luminosity relation, Type Ia supernovae, etc.) is calibrated, directly or indirectly, from it. From Earth, the *Hipparcos* satellite (1989–1993) measured parallaxes with milliarc-second precision for about 118,000 stars (Perryman et al., 1997). The *Gaia* mission, from ESA, revolutionized the method by measuring parallaxes with tens-of-microarcsecond precision for more than 1.5 billion stars (Gaia Collaboration, 2023), reaching direct distances for stars tens of thousands of parsecs away. An important caveat: for very small α (i.e., very distant stars), and even more so in the presence of observational noise, the relation $d = 1/\alpha$ should not be inverted naively — an appropriate (Bayesian) statistical treatment is required, as discussed in Bailer-Jones et al. (2021), along with systematic corrections to the zero-point of the measured parallax itself (Lindgren et al., 2021).

1.3 Magnitudes

Stellar magnitudes — and, likewise, those of all celestial objects — are a very important parameter in characterizing astronomical objects, since they are easy to determine and carry a great deal of information about these bodies. This topic is developed in detail in Chapter 2 (Stellar Magnitudes).

Further Reading and Current References

For the connection between parallax, distance, and absolute magnitude (the so-called *distance modulus*, $m - M = 5 \log_{10} d_{\text{pc}} - 5$), see Chapter 3 of Oliveira Filho and Saraiva (2017) and Chapter 2 of Christensen-Dalsgaard (2008). The Wikipedia discussion is also excellent.

Chapter 2

Stellar Magnitudes

Stellar magnitudes — and, likewise, those of any celestial body — are a very important parameter in characterizing astronomical objects, since they are easy to determine and carry a great deal of information about these bodies.

2.1 Hipparchus's scale and Pogson's relation

Around 130 BCE, Hipparchus catalogued about 10^3 stars, classifying them on a magnitude scale: 1st-magnitude stars were the brightest, and 6th-magnitude stars the faintest. Hipparchus's observations were made with the naked eye alone, as objects became visible at nightfall.

Pogson (1856) proposed that the system described by Hipparchus was based on the perception of the human eye, which follows a logarithmic scale, so that a magnitude-1 star is 100 times brighter than a magnitude-6 star, where

$$m_1 - m_2 = k \log_{10} \left(\frac{F_1}{F_2} \right). \quad (2.1)$$

Hence,

$$1 - 6 = k \log_{10}(100) \implies -5 = k \cdot 2 \implies \boxed{k = -2.5}. \quad (2.2)$$

Thus, we can describe the apparent magnitude of a star as

$$\boxed{m_\lambda = -2.5 \log_{10}(F_\lambda) + \text{const}}, \quad (2.3)$$

where m_λ and F_λ are quantities that depend on the observed wavelength of the object: they take different values for each λ .

Historical origin of Pogson's relation

The choice $k = -2.5$ is not arbitrary: it was calibrated precisely to reproduce, with a round number, the perceptual six-magnitude scale already in use since Hipparchus. The negative sign is responsible for the sometimes counterintuitive convention that *brighter objects have smaller magnitudes* (including negative ones, as for Sirius, Venus, or the Sun itself).

2.2 The Vega system and the reference spectrum

By convention, the star Vega is defined to have magnitude zero at any observed wavelength, so that

$$m_\lambda^{\text{Vega}} = 0. \quad (2.4)$$

If a star has a magnitude greater than that of Vega, $m_{S,\lambda} > 0$; if smaller, $m_{S,\lambda} < 0$; if similar, $m_{S,\lambda} \approx 0$.

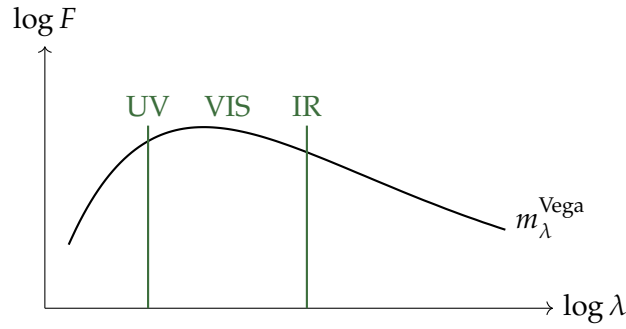


Figure 2.1: Schematic representation of the continuous flux spectrum of the star Vega (illustrative shape, not to physical scale), adopted as the zero-magnitude reference: $m_{\lambda}^{\text{Vega}} = 0$ at every observed wavelength. The ultraviolet (UV), visible (VIS), and infrared (IR) bands are indicated.

2.3 Stellar colors

We define color as the difference between two apparent magnitudes:

$$\boxed{\text{color} = m_{\lambda_1} - m_{\lambda_2}}. \quad (2.5)$$

$B - V$ is a color. If $B - V < 0$, the star is said to be blue, with the B band having the stronger flux. It is also brighter than Vega.

The UBVRI (Johnson-Cousins) photometric system

The $B - V$ color belongs to the most traditional broadband photometric system in astronomy, the *Johnson-Cousins* system (*UBVRI*), whose bands span from the near ultraviolet (U , ≈ 365 nm) to the near infrared (I , ≈ 806 nm), passing through the blue (B , ≈ 445 nm) and the visible (V , ≈ 551 nm). This system was defined from a set of standard stars whose magnitudes are known with high precision, allowing observations made with any telescope to be calibrated (Bessell, 2005).

Precision photometry in modern surveys

The Johnson-Cousins system remains the historical reference, but modern surveys define their own photometric systems, optimized for large data volumes: the *Sloan Digital Sky Survey* (SDSS, filters u, g, r, i, z), *Pan-STARRS1*, and ESA's *Gaia* mission. *Gaia* provides photometry in three bands (G , G_{BP} , G_{RP}) and low-resolution spectra for hundreds of millions of sources, from which it is possible to synthesize magnitudes in essentially any standard photometric system, including *UBVRI* itself (Gaia Collaboration et al., 2023). For a complete review of standard photometric systems and their mutual transformations, see Bessell (2005).

2.4 Interstellar extinction and reddening

We also have that

$$E(B - V) = A_B - A_V, \quad (2.6)$$

a term known as the color excess.

We can also define the total-to-selective extinction ratio R_V as

$$A_V = R_V E(B - V). \quad (2.7)$$

The value of R_V is obtained empirically and varies between 3 and 5 depending on the direction in the Galaxy.

In the presence of an interstellar medium,

$$m_\lambda^i = m_\lambda^{\text{obs}} - A_\lambda, \quad (2.8)$$

where m_λ^i is the intrinsic magnitude, m_λ^{obs} is the observed magnitude, and $A_\lambda > 0$ is the extinction.

The value of R_V and the extinction law

The value $R_V \approx 3.1$ is the one most commonly adopted for the diffuse interstellar medium of the Galaxy, but it can reach ~ 5 in dense molecular cloud regions, where dust grains tend to be larger. The full dependence of extinction on wavelength (the *extinction law*) is more complex than a simple constant ratio, and is tabulated and parametrized as a function of R_V by Fitzpatrick (1999).

2.5 Absolute magnitude and distance modulus

The apparent magnitudes of stars, or of any celestial body, are an estimate or measurement of the object's relative brightness as seen by an observer on Earth. This parameter depends on both the intrinsic brightness and the distance to the source, as we saw from Equation (2.3), where $m_\lambda = -2.5 \log(F_\lambda) + \text{const}$.

However, it is possible to define an absolute expression for stellar magnitude that describes its intrinsic brightness, independent of the distance to the observer on Earth. We define absolute magnitude as the theoretical magnitude the star would have if it were at a distance of 10 pc from us. Hence,

$$M = -2.5 \log_{10}(F(10 \text{ pc})) + \text{const}. \quad (2.9)$$

Thus, the difference between apparent and absolute magnitude is expressed as

$$\begin{aligned} m - M &= [-2.5 \log(F(r)) + \text{const}] - [-2.5 \log(F(10 \text{ pc})) + \text{const}] \\ &= -2.5 \log(F(r)) + 2.5 \log(F(10 \text{ pc})) \\ &= -2.5 \log_{10} \left(\frac{F(r)}{F(10 \text{ pc})} \right). \end{aligned} \quad (2.10)$$

The constant K (the photometric zero-point) is the same in both magnitudes, since both refer to the same photometric system — which is why it cancels out in the subtraction.

The radiative flux is defined as $F = L / (4\pi R^2)$; hence,

$$\begin{aligned} m - M &= -2.5 \log_{10} \left(\frac{L/4\pi r^2}{L/4\pi (10 \text{ pc})^2} \right) = -2.5 \log_{10} \left(\frac{(10 \text{ pc})^2}{r^2} \right) \\ &= -2.5 (2 - 2 \log_{10} r) \\ &= -5 + 5 \log_{10} r, \end{aligned} \quad (2.11)$$

with r measured in parsecs. This is the equation for the **distance modulus**, obtained here from the definition of absolute magnitude.

We can also write the distance as a function of the magnitudes, where

$$m - M + 5 = 5 \log_{10} r \implies \log_{10} r = \frac{m - M + 5}{5}, \quad (2.12)$$

$$r = 10^{(m-M+5)/5}. \quad (2.13)$$

Once we have the apparent magnitude and the distance, we can determine the absolute magnitude.

Sirius, Vega, and the Sun: real magnitudes

The values below illustrate the relation between apparent magnitude, absolute magnitude, and distance for three well-known objects:

Object	Spectral type	m_V	M_V	$B - V$	d
Sun	G2V	-26.74	+4.83	+0.65	1 AU
Sirius A	A1V	-1.46	+1.43	+0.01	2.64 pc
Vega	A0V	+0.03	+0.58	-0.00	7.68 pc

Note that Vega, by definition of the magnitude system (Equation (2.4)), has an apparent magnitude very close to zero — the small offset (+0.03) arises from refinements to the magnitude system since its original definition. Sirius, despite being the brightest star in the night sky ($m_V = -1.46$), is intrinsically less luminous than Vega ($M_V = +1.43$ versus +0.58); it appears brighter to us simply because it is much closer. Using Equation (2.13) with Sirius's values of m_V and M_V , we obtain $r = 10^{(-1.46-1.43+5)/5} \approx 2.64$ pc, exactly the value measured by parallax — a good cross-check between the two distance determination methods discussed in this book.

2.6 Bolometric magnitude and correction

The bolometric magnitude is defined as the flux across all wavelengths, m_{bol} . It is defined as

$$L = 4\pi R^2 \int_0^\infty F_\nu d\nu = 4\pi R^2 F_{\text{bol}}, \quad (2.14)$$

where the integral over ν covers the *entire* electromagnetic spectrum — from radio to gamma rays, passing through infrared, optical, and ultraviolet — unlike a monochromatic magnitude m_λ , which captures only the fraction of the flux within a specific spectral band.

We can write the difference between bolometric magnitudes as

$$m_{1,\text{bol}} - m_{2,\text{bol}} = -2.5 \log_{10} \left(\frac{L_1/4\pi d_1^2}{L_2/4\pi d_2^2} \right). \quad (2.15)$$

Assuming a distance of 10 pc, we have

$$m_{1,\text{bol}} - m_{2,\text{bol}} = -2.5 \log_{10} \left(\frac{L_1}{L_2} \right), \quad d_1 = d_2. \quad (2.16)$$

Assuming star 2 is our Sun, we have

$$M_{S,\text{bol}} - M_{\odot,\text{bol}} = -2.5 \log_{10} \left(\frac{L}{L_{\odot}} \right) \iff \log_{10} \left(\frac{L}{L_{\odot}} \right) = -0.4 (M_{S,\text{bol}} - M_{\odot,\text{bol}}). \quad (2.17)$$

The official value of $M_{\odot,\text{bol}}$ for the Sun, fixed by IAU Resolution B2 (2015) — which defines $M_{\text{Bol}} = 0$ for a source with luminosity $L_0 = 3.0128 \times 10^{28}$ W and is combined with the nominal value of the solar luminosity from Resolution B3 of the same year, $L_{\odot} = 3.828 \times 10^{26}$ W (Prsa et al., 2016) —, is

$$M_{\odot,\text{bol}} \approx 4.74, \quad (2.18)$$

according to Mamajek et al. (2015); hence, the bolometric magnitude of a star can be represented by

$$M_{\text{bol}} = 4.74 - 2.5 \log_{10} \left(\frac{L}{L_{\odot}} \right). \quad (2.19)$$

The traditional value $M_{\odot,\text{bol}} = 4.72$

In texts and papers published before 2015, and still today in some references, it is common to find $M_{\odot,\text{bol}} = 4.72$ — a value that varies between 4.72 and 4.75 depending on the reference, the adopted value of $L_{\odot}$, and the photometric zero-point assumed before official standardization. The ~ 0.02 -mag difference between 4.72 and the official value 4.74 is small and does not affect the qualitative conclusions of this course, but it is important to be aware of the convention adopted when comparing results with work published before 2015.

In practice, it is very difficult to measure the bolometric magnitude of an object because some wavelengths are absorbed by the Earth's atmosphere. An alternative is to determine it from an absolute magnitude in a single band, M_{λ} (typically the V band), where

$$M_{\text{bol}} = M_{\lambda} - CB, \quad (2.20)$$

with CB a bolometric correction factor that varies for each stellar class — since the bolometric magnitude sums the flux over *the entire* spectrum, while M_{λ} captures only one band, CB is, in this convention, always a positive or zero quantity, and grows both for very hot and for very cool stars (whose spectral peak, by Wien's law discussed in the next section, falls farther and farther from the chosen band λ).

Sign convention for the bolometric correction

The modern literature most often defines the **bolometric correction** with the opposite sign to the one used here, $BC \equiv M_{\text{bol}} - M_V = -CB$, so that $BC \leq 0$ *always* (no single band can capture more flux than the entire spectrum) — the Sun, for example, has $BC_V \approx -0.08$. Both conventions carry exactly the same physical information, only with the sign flipped; the difference matters in practice when comparing bolometric correction values with published tables or papers, which almost always report $BC \leq 0$ (Carroll and Ostlie, 2017; Kippenhahn et al., 2012).

2.7 Blackbody radiation

A blackbody is defined as a body that absorbs and emits all of the radiation incident upon it. Ideal blackbodies do not exist in nature; however, stars are the bodies that come closest to this behavior.

In 1893, the German physicist Wilhelm Wien discovered empirically that the radiation peak of a body is proportional to a constant times the temperature of the radiation, where:

2.7.1 Wien's displacement law and the Planck function

Wien's empirical observation, made even before the complete formulation of blackbody radiation, is known today as **Wien's Displacement Law**:

$$\boxed{\lambda_{\max} T = b}, \quad b = 2.898 \times 10^{-3} \text{ m K}, \quad (2.21)$$

where $\lambda_{\max}$ is the peak wavelength of the radiation and T is the temperature of the body. The relation was obtained by Wien in a purely empirical way in 1893 — seven years before Max Planck derived, in 1900, the complete spectral distribution of blackbody radiation, from which Wien's law can be deduced as a special case.

The Planck function, which describes the spectral radiance of a blackbody as a function of wavelength, is given by

$$B_{\lambda}(T) = \frac{2hc^2}{\lambda^5} \frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1}, \quad (2.22)$$

or, as a function of frequency,

$$B_{\nu}(T) = \frac{2h\nu^3}{c^2} \frac{1}{\exp\left(\frac{h\nu}{k_B T}\right) - 1}, \quad (2.23)$$

where $h = 6.626 \times 10^{-34} \text{ J s}$ is the Planck constant, $c = 2.998 \times 10^8 \text{ m/s}$ is the speed of light in vacuum, and $k_B = 1.381 \times 10^{-23} \text{ J/K}$ is the Boltzmann constant. The physical role of each constant is straightforward: h quantizes the energy of the radiation into packets $E = h\nu$, k_B connects that energy to the thermal energy scale $k_B T$ of the emitting body, and c converts between the frequency and wavelength descriptions.

Wien's law is obtained directly from the Planck function, by finding the maximum of $B_{\lambda}(T)$ at fixed temperature, that is, by solving $\frac{dB_{\lambda}}{d\lambda} = 0$. Defining the dimensionless variable $x \equiv hc/(\lambda k_B T)$, so that $B_{\lambda} \propto x^5/(e^x - 1)$, the maximum condition reduces to the transcendental equation

$$(5 - x)e^x = 5, \quad (2.24)$$

whose numerical solution is $x \approx 4.9651$. Since $\lambda_{\max} = hc/(x k_B T)$, it follows that

$$\lambda_{\max} T = \frac{hc}{k_B x} \equiv b = 2.898 \times 10^{-3} \text{ m K}, \quad (2.25)$$

recovering exactly Equation (2.21).

Wien's Displacement Law can also be written as a function of the peak frequency $\nu_{\max}$, a form complementary to Equation (2.21) and equally useful in astrophysics:

$$\boxed{h \nu_{\max} = 2.821 k_B T}, \quad (2.26)$$

where h is the Planck constant, $\nu_{\max}$ is the peak frequency of the radiation, k_B is the Boltzmann constant, and T is the temperature of the body.

This form is obtained in an entirely analogous way to the previous one, but maximizing $B_\nu(T)$ instead of $B_\lambda(T)$. Now defining $x \equiv h\nu/(k_B T)$, so that $B_\nu \propto x^3/(e^x - 1)$, the condition $\frac{dB_\nu}{d\nu} = 0$ reduces to the transcendental equation

$$(3 - x)e^x = 3 \iff \frac{x e^x}{e^x - 1} = 3, \quad (2.27)$$

whose numerical solution is $x \approx 2.8214$ — the value 2.821 from the lecture notes. Since $\nu_{\max} = x k_B T/h$, Equation (2.26) follows directly.

$\lambda_{\max}$ and $\nu_{\max}$ are not the same point of the spectrum

A common mistake is to assume that $\nu_{\max} = c/\lambda_{\max}$, that is, that the peak of B_ν corresponds to the same physical point of the spectrum as the peak of B_λ . This is **false**. B_λ and B_ν are different spectral densities — the relation between them involves the Jacobian factor of the change of variables, $B_\nu |d\nu| = B_\lambda |d\lambda|$, i.e., $B_\nu = B_\lambda (c/\nu^2)$ — and maximizing one is not equivalent to maximizing the other. Numerically, Equation (2.21) gives $\lambda_{\max} T = hc/(4.9651 k_B) = 2.898 \times 10^{-3} \text{ m K}$, while the wavelength corresponding to $\nu_{\max}$ would be $(c/\nu_{\max}) T = hc/(2.8214 k_B) \approx 5.10 \times 10^{-3} \text{ m K}$ — nearly twice as large. In other words, the peak in frequency of the spectrum corresponds to a much longer wavelength than the true peak in wavelength, $\lambda_{\max}$. When citing "the peak wavelength," it is essential to make clear which of the two forms of Wien's law (Equation (2.21) or (2.26)) was used.

Connection with stellar effective temperature

Since stars behave, to a good approximation, as blackbodies (as already discussed in the lecture notes), the peak of the observed stellar continuum spectrum provides a direct estimate of the star's effective temperature T_{eff} , through Wien's Displacement Law. This shift of the spectral peak — from long wavelengths (red/infrared) in cool stars to short wavelengths (blue/ultraviolet) in hot stars — is the physical basis behind the relation between color and temperature discussed in Section 2.3 (the $B - V$ index), and complements more refined spectroscopic methods for determining T_{eff} , based on fitting absorption lines and the stellar continuum.

The spectral peak for different stellar types

Applying Wien's Displacement Law (Equation (2.21)) to three stars with very different effective temperatures:

- **The Sun** ($T_{\text{eff}} \approx 5772 \text{ K}$, the nominal value adopted by the IAU (Prsa et al., 2016)):

$$\lambda_{\max} = \frac{b}{T_{\text{eff}}} = \frac{2.898 \times 10^{-3}}{5772} \approx 502 \text{ nm},$$

right in the middle of the visible range — it is no coincidence that human vision evolved with peak sensitivity near this wavelength.

- **A hot O-type star** ($T_{\text{eff}} \approx 35000 \text{ K}$):

$$\lambda_{\max} \approx \frac{2.898 \times 10^{-3}}{35000} \approx 82.8 \text{ nm},$$

in the extreme ultraviolet, well beyond what the Earth's atmosphere allows to be observed from the ground — hence the importance of space telescopes for characterizing very hot stars.

- **A cool M dwarf** ($T_{\text{eff}} \approx 3200 \text{ K}$):

$$\lambda_{\text{max}} \approx \frac{2.898 \times 10^{-3}}{3200} \approx 906 \text{ nm},$$

already in the near infrared, consistent with the characteristic reddish color of these stars.

Modern determination of stellar T_{eff}

Large-scale spectrophotometric surveys now allow T_{eff} to be estimated for hundreds of millions of stars simultaneously. The *Gaia* mission, through its *Apsis* module, derives effective temperatures, surface gravities, and metallicities from the low-resolution spectra (*BP/RP*) and photometry of more than 470 million sources (Fouesneau et al., 2023). These methods combine fitting the overall shape of the observed spectrum — in a spirit very close to the Wien-law approach discussed here, though using full atmosphere models — with statistical comparisons to large spectral libraries, achieving precisions of order hundreds of kelvin even in regions of high interstellar extinction.

Chapter 3

Thermodynamics

3.1 HR diagrams and related topics

3.2 Thermodynamics review

Thermodynamics describes the dynamic nature of thermal processes and transformations. Since stars are made of gas, thermodynamics describes the main energy transport mechanisms in their interior.

The thermodynamic study is formulated from three main variables: temperature, pressure, and volume.

Zeroth law of thermodynamics

If two bodies, A and B , are separate and in equilibrium with a third body C , then A and B are also in equilibrium with each other.

The zeroth law is important in defining temperature scales, such as Kelvin, Fahrenheit, Celsius, etc.

3.3 First law of thermodynamics

The first law of thermodynamics tells us that the change in the internal energy of a system (ΔU) between any two states is described by the difference between the amount of heat and work exchanged with the surroundings, where

$$dU = dQ - dW, \quad (3.1)$$

recalling that the work done by a force is

$$dW = F dx. \quad (3.2)$$

And that pressure can be expressed as

$$P = \frac{F}{A}, \quad (3.3)$$

we have $dW = P A dx = P dV$, hence

$$\boxed{dU = dQ - P dV}, \quad [dU] = \text{erg}. \quad (3.4)$$

The $P dV$ term in stellar contraction and expansion

The work term $P dV$ is not a formal detail: it is what governs the evolution of a star whenever its radius changes. In the simplest case, a pre-main-sequence protostar has not

yet established nuclear fusion; all the luminosity it radiates comes from the conversion of gravitational energy into heat as the star contracts ($dV < 0$, so $-P dV > 0$ heats the gas). This is the **Kelvin–Helmholtz contraction** mechanism, whose characteristic timescale is

$$t_{\text{KH}} \sim \frac{GM^2}{RL}. \quad (3.5)$$

For the present-day Sun, $t_{\text{KH}} \approx 30$ Myr; at the top of the Hayashi track, when the Sun had a much larger radius ($R \sim 10\text{--}100 R_{\odot}$) and higher luminosity, t_{KH} was much shorter — of order 10^4 to 10^6 years, depending on the evolutionary stage (Kippenhahn et al., 2012). The same term $P dV$, with opposite sign, also governs phases of expansion (for example, when the stellar core contracts but the envelope expands during the transition to a red giant).

We can obtain the first law per unit mass, where

$$du = dq - P dv, \quad [du] = \text{erg/g}, \quad (3.6)$$

with $v = \rho^{-1}$ the **specific volume**, and $\rho = m/V$.

Recalling the empirical definitions of heat capacity and specific heat: heat capacity indicates the amount of heat a body must receive or give up for its temperature to change by 1 degree, and specific heat indicates the amount of heat each unit of mass of the body must receive or give up for the temperature to change by 1 degree, where

$$C = \frac{dQ}{dT}, \quad [C] = \text{erg/K}, \quad \text{and} \quad c = \frac{dq}{dT}, \quad [c] = \text{erg/K/g}. \quad (3.7)$$

Thus, we can define the specific heat at constant pressure and at constant volume by

$$C_P = \left(\frac{dq}{dT} \right)_P, \quad C_V = \left(\frac{dq}{dT} \right)_V. \quad (3.8)$$

$\gamma = C_P/C_V$: from the typical stellar interior to dynamical instability

For a non-relativistic, non-degenerate monatomic ideal gas — the condition holding in much of the stellar interior, including the ionized H/He plasma in the solar core — each particle has only 3 translational degrees of freedom, so that $C_V = \frac{3}{2}R/\mu$ and $C_P = C_V + R/\mu = \frac{5}{2}R/\mu$ (anticipating the result $C_P - C_V = R/\mu$ that we will derive later), hence

$$\gamma = \frac{C_P}{C_V} = \frac{5}{3} \approx 1.667. \quad (3.9)$$

This is the reference value for most main-sequence stars. The limit $\gamma \rightarrow 4/3$, in turn, is the **dynamical stability threshold**: by the virial theorem, a star in hydrostatic equilibrium is stable against radial perturbations only if the mean adiabatic index Γ_1 satisfies $\Gamma_1 > 4/3$ (for $\Gamma_1 < 4/3$, any contraction releases more gravitational energy than pressure can resist, and the collapse accelerates). This limit is approached when a photon gas (radiation), which has $\gamma = 4/3$ exactly, comes to dominate over gas pressure — the case of very massive stars ($M \gtrsim 100 M_{\odot}$), where $P_{\text{rad}}/P_{\text{gas}}$ becomes large, and also of fully degenerate, relativistic matter, as in the Chandrasekhar limit for white dwarfs (Kippenhahn et al., 2012; Clayton, 1983).

3.4 Thermodynamic relations

Thermodynamic relations help us describe physical variables as functions of different parameters. Let us derive a relation for $C_P - C_V$ starting from the first law of thermodynamics, where

$$du = dq - P dv, \quad dq = du + P dv, \quad u = u(v, T). \quad (3.10)$$

Review: partial derivatives and total differentials

Let $f(x, y)$ be a differentiable function of two variables. The **partial derivative** of f with respect to x ,

$$\left(\frac{\partial f}{\partial x}\right)_y \equiv \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x},$$

measures the rate of change of f when only x varies, holding y *constant*. In thermodynamic notation the subscript is indispensable: the same quantity can have numerically distinct partial derivatives depending on which variables are held fixed — for example, $(\partial u / \partial T)_v \neq (\partial u / \partial T)_P$ in general, since holding v constant and holding P constant are physically different conditions.

The **total differential** of $f(x, y)$ combines the contributions from each variable:

$$df = \left(\frac{\partial f}{\partial x}\right)_y dx + \left(\frac{\partial f}{\partial y}\right)_x dy.$$

The infinitesimal change in f is simply the sum of the changes produced by each variable acting in isolation.

Concrete example. Take $f(x, y) = x^2y + y^3$. Computing the partial derivatives (the other variable is treated as constant):

$$\left(\frac{\partial f}{\partial x}\right)_y = 2xy,$$

$$\left(\frac{\partial f}{\partial y}\right)_x = x^2 + 3y^2.$$

The total differential is then

$$df = 2xy dx + (x^2 + 3y^2) dy.$$

Chain rule — change of variables. If $f = f(x, y)$ and the original variables depend on a new pair (u, v) — that is, $x = x(u, v)$ and $y = y(u, v)$ — then differentiating f with respect to u (with v fixed) requires:

$$\left(\frac{\partial f}{\partial u}\right)_v = \left(\frac{\partial f}{\partial x}\right)_y \left(\frac{\partial x}{\partial u}\right)_v + \left(\frac{\partial f}{\partial y}\right)_x \left(\frac{\partial y}{\partial u}\right)_v.$$

This is precisely the tool used next: to rewrite $u(v, T)$ in terms of $u(P, T)$, one treats v as a function of (P, T) and applies the chain rule, expressing $(\partial u / \partial T)_P$ and $(\partial u / \partial P)_T$ in terms of the already known derivatives of u with respect to (v, T) .

The definitions of C_P and C_V and the derivation of $C_P - C_V$ that follow are direct applications of these two tools: the total differential, which decomposes du and dq into the contributions of each independent variable, and the chain rule, which allows switching the set of thermodynamic variables from (v, T) to (P, T) .

Taking the partial derivative, we have

$$\begin{aligned} dq &= \left(\frac{\partial u}{\partial v}\right)_T dv + \left(\frac{\partial u}{\partial T}\right)_v dT + P dv \\ &= \left[\left(\frac{\partial u}{\partial v}\right)_T + P\right] dv + \left(\frac{\partial u}{\partial T}\right)_v dT. \end{aligned} \quad (3.11)$$

We see, from Equation (3.11), that C_V is the term

$$C_V = \left(\frac{\partial u}{\partial T}\right)_v, \quad (3.12)$$

since it is the only one at constant volume.

Since the volume is no longer held constant, the internal energy is now a function of the other two thermodynamic variables, $u = u(P, T)$; hence, using these variables in the first law of thermodynamics, we obtain:

$$\begin{aligned} dq &= du(P, T) + P dv(P, T) \\ &= \left(\frac{\partial u}{\partial P}\right)_T dP + \left(\frac{\partial u}{\partial T}\right)_P dT + P \left[\left(\frac{\partial v}{\partial P}\right)_T dP + \left(\frac{\partial v}{\partial T}\right)_P dT \right] \\ &= \left[\left(\frac{\partial u}{\partial P}\right)_T + P \left(\frac{\partial v}{\partial P}\right)_T \right] dP + \left[\left(\frac{\partial u}{\partial T}\right)_P + P \left(\frac{\partial v}{\partial T}\right)_P \right] dT. \end{aligned} \quad (3.13)$$

In this case, the specific heat at constant pressure is

$$C_P = \left(\frac{\partial u}{\partial T}\right)_P + P \left(\frac{\partial v}{\partial T}\right)_P. \quad (3.14)$$

Thus, we obtain that

$$C_P - C_V = \left(\frac{\partial u}{\partial T}\right)_P - \left(\frac{\partial u}{\partial T}\right)_v + P \left(\frac{\partial v}{\partial T}\right)_P. \quad (3.15)$$

Returning to the partial functions, where $u(v, T) \rightarrow u(P, T)$, $v(P, T)$:

$$du = \left(\frac{\partial u}{\partial v}\right)_T dv + \left(\frac{\partial u}{\partial T}\right)_v dT, \quad \text{but } v = v(P, T), \quad (3.16)$$

$$\begin{aligned} du &= \left(\frac{\partial u}{\partial v}\right)_T \left[\left(\frac{\partial v}{\partial P}\right)_T dP + \left(\frac{\partial v}{\partial T}\right)_P dT \right] + \left(\frac{\partial u}{\partial T}\right)_v dT \\ &= \left[\left(\frac{\partial u}{\partial v}\right)_T \left(\frac{\partial v}{\partial P}\right)_T \right] dP + \left[\left(\frac{\partial u}{\partial v}\right)_T \left(\frac{\partial v}{\partial T}\right)_P + \left(\frac{\partial u}{\partial T}\right)_v \right] dT. \end{aligned} \quad (3.17)$$

Hence, we can identify $(\partial u / \partial T)_P$ as the first term of Equation (3.17) associated with dT , where:

$$\left(\frac{\partial u}{\partial T}\right)_P = \left(\frac{\partial u}{\partial v}\right)_T \left(\frac{\partial v}{\partial T}\right)_P + \left(\frac{\partial u}{\partial T}\right)_v. \quad (3.18)$$

Substituting Equation (3.18) into (3.15), we obtain:

$$\begin{aligned} C_P - C_V &= \left(\frac{\partial u}{\partial v}\right)_T \left(\frac{\partial v}{\partial T}\right)_P + \cancel{\left(\frac{\partial u}{\partial T}\right)_v} - \cancel{\left(\frac{\partial u}{\partial T}\right)_v} + P \left(\frac{\partial v}{\partial T}\right)_P \\ &= \left(\frac{\partial v}{\partial T}\right)_P \left[\left(\frac{\partial u}{\partial v}\right)_T + P \right]. \end{aligned} \quad (3.19)$$

Using the definition of entropy, where $dq = T dS$, we can proceed similarly, with $S = S(v, T)$:

$$dq = T dS = P dv + du. \quad (3.20)$$

Substituting $du = (\partial u / \partial v)_T dv + (\partial u / \partial T)_v dT$, we have:

$$\begin{aligned} dS &= \frac{P}{T} dv + \frac{1}{T} du \\ &= \frac{P}{T} dv + \frac{1}{T} \left[\left(\frac{\partial u}{\partial v} \right)_T dv + \left(\frac{\partial u}{\partial T} \right)_v dT \right] \\ &= \frac{1}{T} \left[P + \left(\frac{\partial u}{\partial v} \right)_T \right] dv + \frac{1}{T} \left(\frac{\partial u}{\partial T} \right)_v dT. \end{aligned} \quad (3.21)$$

Comparing with the total differential

$$dS = \left(\frac{\partial S}{\partial v} \right)_T dv + \left(\frac{\partial S}{\partial T} \right)_v dT, \quad (3.22)$$

we see that

$$\left(\frac{\partial S}{\partial v} \right)_T = \frac{1}{T} \left[P + \left(\frac{\partial u}{\partial v} \right)_T \right], \quad \left(\frac{\partial S}{\partial T} \right)_v = \frac{1}{T} \left(\frac{\partial u}{\partial T} \right)_v. \quad (3.23)$$

Maxwell relation

Using the fact that the mixed partial derivatives of S commute, $\partial^2 S / \partial T \partial v = \partial^2 S / \partial v \partial T$, we can write

$$\frac{\partial}{\partial T} \left\{ \frac{1}{T} \left[P + \left(\frac{\partial u}{\partial v} \right)_T \right] \right\}_v = \frac{\partial}{\partial v} \left\{ \frac{1}{T} \left(\frac{\partial u}{\partial T} \right)_v \right\}_T. \quad (3.24)$$

Applying the derivative to each side of Equation (3.24), we have:

$$-\frac{P}{T^2} + \frac{1}{T} \left(\frac{\partial P}{\partial T} \right)_v - \frac{1}{T^2} \left(\frac{\partial u}{\partial v} \right)_T + \frac{1}{T} \frac{\partial^2 u}{\partial T \partial v} = \frac{1}{T} \frac{\partial^2 u}{\partial v \partial T}. \quad (3.25)$$

Since the mixed derivatives on the right-hand side and in the last term on the left-hand side are equal, they cancel, leaving

$$-\frac{P}{T^2} - \frac{1}{T^2} \left(\frac{\partial u}{\partial v} \right)_T + \frac{1}{T} \left(\frac{\partial P}{\partial T} \right)_v = 0. \quad (3.26)$$

Multiplying by T^2 and isolating $(\partial u / \partial v)_T$:

$$\boxed{\left(\frac{\partial u}{\partial v} \right)_T = -P + T \left(\frac{\partial P}{\partial T} \right)_v}. \quad (3.27)$$

Substituting Equation (3.27) into (3.19), we obtain:

$$\begin{aligned} C_P - C_V &= \left(\frac{\partial v}{\partial T} \right)_P \left[-P + T \left(\frac{\partial P}{\partial T} \right)_v + P \right] \\ &= T \left(\frac{\partial v}{\partial T} \right)_P \left(\frac{\partial P}{\partial T} \right)_v. \end{aligned} \quad (3.28)$$

$$\boxed{C_P - C_V = T \left(\frac{\partial v}{\partial T} \right)_P \left(\frac{\partial P}{\partial T} \right)_v} \quad (3.29)$$

We obtain an expression that is independent of the internal energy u and depends only on the main thermodynamic quantities. The derivatives can be determined from the derivative of the equation of state.

3.5 Thermal expansion coefficients

To obtain a simplified relation for Equation (3.28), we can define the thermal expansion coefficient α :

$$\alpha \equiv \left(\frac{\partial \ln \rho}{\partial \ln P} \right)_T = \left(\frac{\partial \rho / \rho}{\partial P / P} \right)_T = \frac{P}{\rho} \left(\frac{\partial \rho}{\partial P} \right)_T, \quad \text{since } v = \frac{1}{\rho}, \quad (3.30)$$

$$\alpha = Pv \left[\frac{\partial}{\partial P} \left(\frac{1}{v} \right) \right]_T = Pv \left(-\frac{1}{v^2} \right) \left(\frac{\partial v}{\partial P} \right)_T = -\frac{P}{v} \left(\frac{\partial v}{\partial P} \right)_T. \quad (3.31)$$

Similarly, we obtain the coefficient δ :

$$\delta \equiv - \left(\frac{\partial \ln \rho}{\partial \ln T} \right)_P = \frac{T}{v} \left(\frac{\partial v}{\partial T} \right)_P. \quad (3.32)$$

With this, the ratio δ/α reduces to

$$\frac{\delta}{\alpha} = \frac{\frac{T}{v} \left(\frac{\partial v}{\partial T} \right)_P}{-\frac{P}{v} \left(\frac{\partial v}{\partial P} \right)_T} = -\frac{T}{P} \cdot \frac{\left(\frac{\partial v}{\partial T} \right)_P}{\left(\frac{\partial v}{\partial P} \right)_T}. \quad (3.33)$$

By the triple product rule (the cyclic rule for partial derivatives), $(\partial v / \partial T)_P / (\partial v / \partial P)_T = -(\partial P / \partial T)_v$ — the sign flips upon inversion — hence

$$\boxed{\frac{\delta}{\alpha} = \frac{T}{P} \left(\frac{\partial P}{\partial T} \right)_v}. \quad (3.34)$$

Returning to Equation (3.28) and substituting the expressions obtained, with $(\partial v / \partial T)_P = \delta v / T$ and $(\partial P / \partial T)_v = (\delta / \alpha)(P / T)$:

$$C_P - C_V = T \cdot \frac{\delta v}{T} \cdot \frac{\delta P}{\alpha T} = \boxed{\frac{P \delta^2 v}{\alpha T}} = \frac{P \delta^2}{\alpha T \rho}. \quad (3.35)$$

Beyond the ideal gas: degeneracy and partial ionization

The reason we derived $C_P - C_V$ in the general form above — in terms of α and δ , without assuming an ideal gas — is that this is exactly the form needed to handle two regimes where $\alpha \neq 1$ and/or $\delta \neq 1$, and Equation (3.46) (the result R/μ , still to come) no longer holds:

- **Degenerate gas** (white dwarf interiors): in the limit of complete, non-relativistic degeneracy, the electron gas pressure $P \propto \rho^{5/3}$ *does not depend on temperature*, so that $(\partial P / \partial T)_v \rightarrow 0$ and, by Equation (3.34), $\delta \rightarrow 0$. Since $C_P - C_V \propto \delta^2$, this implies $C_P \approx C_V$ in strongly degenerate matter — a classic result of white dwarf physics.
- **Partial ionization zones** (H, He I, He II), in layers with $T \sim 10^4$ – 10^5 K in the stellar envelope: part of the thermal energy supplied to the gas is consumed by ionization rather than raising T , causing α and δ to deviate significantly from 1. This alters $C_P - C_V$ and, as we shall see in Section 3.8, has a dramatic effect on ∇_{ad} — the physical mechanism behind the **instability strip** of Cepheids and RR Lyrae stars in the HR diagram.

For the complete treatment of these corrections via the equation of state, including the Saha equation for partial ionization, see Kippenhahn et al. (2012).

3.6 Alternative form of the first law

From the first law of thermodynamics, where $u = u(v, T)$, we have

$$dq = P dv + \left(\frac{\partial u}{\partial v} \right)_T dv + \left(\frac{\partial u}{\partial T} \right)_v dT, \quad (3.36)$$

hence

$$dq = \left[\left(\frac{\partial u}{\partial v} \right)_T + P \right] dv + \left(\frac{\partial u}{\partial T} \right)_v dT. \quad (3.37)$$

From Equations (3.12) and (3.27), we have

$$dq = \left[T \left(\frac{\partial P}{\partial T} \right)_v + P - P \right] dv + C_V dT = T \left(\frac{\partial P}{\partial T} \right)_v dv + C_V dT. \quad (3.38)$$

Similarly, we can seek differential expressions in terms of α and δ , considering that $\rho = \rho(P, T)$:

$$\frac{d\rho}{\rho} = \frac{1}{\rho} \left[\left(\frac{\partial \rho}{\partial P} \right)_T dP + \left(\frac{\partial \rho}{\partial T} \right)_P dT \right]. \quad (3.39)$$

Substituting into the definitions of α and δ [Equations (3.31) and (3.32), rewritten in terms of ρ : $(\partial \rho / \partial P)_T = \alpha \rho / P$ and $(\partial \rho / \partial T)_P = -\delta \rho / T$], we have

$$\boxed{\frac{d\rho}{\rho} = \alpha \frac{dP}{P} - \delta \frac{dT}{T}}. \quad (3.40)$$

Using the first law again from Equation (3.38), with $dv = -d\rho / \rho^2 = -(1/\rho)(d\rho / \rho)$ and $(\partial P / \partial T)_v = (P/T)(\delta / \alpha)$:

$$\begin{aligned} dq &= T \cdot \frac{P\delta}{T\alpha} dv + C_V dT = \frac{P\delta}{\alpha} \left(-\frac{1}{\rho} \right) \frac{d\rho}{\rho} + C_V dT \\ &= -\frac{P\delta}{\alpha\rho} \left[\alpha \frac{dP}{P} - \delta \frac{dT}{T} \right] + C_V dT \end{aligned} \quad (3.41)$$

$$= -\frac{P\delta}{\rho} \frac{dP}{P} + \left[\frac{P\delta^2}{\alpha\rho T} + C_V \right] dT. \quad (3.42)$$

Recognizing, from Equation (3.35), that the term in brackets is exactly C_P , we arrive at the alternative form of the first law of thermodynamics:

$$\boxed{dq = C_P dT - \delta \frac{dP}{\rho}}. \quad (3.43)$$

The first law of thermodynamics written in this form simplifies the study of stellar evolution, where temperature, density, and pressure change simultaneously.

3.7 The ideal gas case

For an ideal gas, we know that the equation of state is described by the Clapeyron equation, $PV = nRT$. In our case, it can also be written as

$$P = \frac{\rho RT}{\mu}. \quad (3.44)$$

Recalling that

$$\alpha = \left(\frac{\partial \ln \rho}{\partial \ln P} \right)_T = \frac{P}{\rho} \left(\frac{\partial \rho}{\partial P} \right)_T = \frac{P}{\rho} \left[\frac{\partial}{\partial P} \left(\frac{P\mu}{RT} \right) \right]_T = \frac{P}{\rho} \frac{\mu}{RT}, \quad (3.45)$$

but notice that this is exactly the term from the ideal gas equation [Equation (3.44) inverted, $\rho = P\mu/(RT)$], hence $\alpha = 1$. By analogy, we can also verify that $\delta = 1$.

Thus, Equation (3.35) reduces to

$$C_P - C_V = \frac{P}{T\rho} = \frac{\rho RT}{\mu T\rho} = \left[\frac{R}{\mu} \right]. \quad (3.46)$$

Equation (3.46) recovers the difference $C_P - C_V$ for the case of an ideal gas.

$C_P - C_V$ in the solar core and the limits of ideality in the photosphere

In the solar core, the hydrogen and helium plasma is fully ionized, with mean molecular weight $\mu \approx 0.6$ for typical solar composition. Using the gas constant $R = 8.314 \times 10^7 \text{ erg g}^{-1} \text{ K}^{-1}$ (the universal constant per gram, with μ dimensionless), Equation (3.46) gives

$$C_P - C_V = \frac{R}{\mu} \approx \frac{8.314 \times 10^7}{0.6} \text{ erg g}^{-1} \text{ K}^{-1} \approx 1.39 \times 10^8 \text{ erg g}^{-1} \text{ K}^{-1}, \quad (3.47)$$

a well-defined value because the core plasma is well above the ionization temperature and behaves as an ideal gas ($\alpha = \delta = 1$).

In the layers just below the photosphere, however, where T drops to $\sim 10^4 \text{ K}$, hydrogen begins to partially recombine: the effective mean molecular weight increases (from $\mu \approx 0.5$ for fully ionized hydrogen to $\mu \approx 1$ for neutral hydrogen), and — more importantly — α and δ no longer equal 1. In this region, Equation (3.46) *no longer holds*; it is necessary to use the general form $C_P - C_V = P\delta^2/(\alpha T\rho)$ derived earlier, with α and δ computed from an equation of state that includes partial ionization (typically via the Saha equation). Section 3.8 shows the concrete effect of this on ∇_{ad} .

3.8 The adiabatic gradient

We can also define a term that is the **adiabatic gradient**, which recurs frequently in astrophysics, where

$$\nabla_{\text{ad}} \equiv \left(\frac{\partial \ln T}{\partial \ln P} \right)_S = \frac{P}{T} \left(\frac{\partial T}{\partial P} \right)_S. \quad (3.48)$$

This equation is similar to others seen earlier, except that now the entropy of the system is held constant.

From the first law of thermodynamics, we have Equation (3.43), where, for an adiabatic system, there is no heat exchange with the surroundings, so $dq \equiv 0$. Thus,

$$dq = C_P dT - \delta \frac{dP}{\rho} = 0 \implies C_P dT = \frac{\delta}{\rho} dP \implies \left(\frac{\partial T}{\partial P} \right)_s = \frac{\delta}{\rho C_P}. \quad (3.49)$$

Substituting Equation (3.49) into Equation (3.48), we have:

$$\frac{T}{P} \nabla_{\text{ad}} = \frac{\delta}{\rho C_P} \implies \boxed{\nabla_{\text{ad}} = \frac{P\delta}{T\rho C_P}}. \quad (3.50)$$

In the ideal-gas approximation, $\delta = 1$ and $P/(T\rho) = R/\mu$ [Equation (3.44)]; hence, Equation (3.50) reduces to

$$\nabla_{\text{ad}} = \frac{R}{\mu C_P}. \quad (3.51)$$

Combining with $\gamma = 5/3$ and $C_P = \frac{5}{2}R/\mu$, valid for a non-relativistic monatomic ideal gas (as discussed earlier for the typical stellar interior), we obtain the more usual form of this result,

$$\nabla_{\text{ad}} = \frac{\gamma - 1}{\gamma} = \frac{2}{5} = 0.4. \quad (3.52)$$

The Schwarzschild criterion and the Sun's convective zone

The adiabatic gradient is the term of comparison in the **Schwarzschild criterion** for convective stability: a stellar layer is unstable to convection whenever the local radiative gradient exceeds ∇_{ad} , that is, $\nabla_{\text{rad}} > \nabla_{\text{ad}}$. The value $\nabla_{\text{ad}} = 2/5 = 0.4$ [Equation (3.52)] is the ideal-gas reference used to identify, in a stellar model, where energy transport ceases to be purely radiative. In the Sun's case, helioseismic measurements place the base of the convective zone at $r = (0.713 \pm 0.003) R_{\odot}$ (Christensen-Dalsgaard et al., 1991) — that is, the outer $\sim 29\%$ of the solar radius (often rounded to "beyond $0.71 R_{\odot}$ ") is convective, exactly the region where ∇_{rad} exceeds ∇_{ad} .

The hydrogen partial ionization zone and the origin of solar convection

The Schwarzschild criterion alone does not explain why solar convection begins so close to the surface, where the density is low and the composition is predominantly neutral hydrogen. The answer lies in Equation (3.50): in layers with $T \sim 10^4$ K (roughly $\log T \approx 4.1$), hydrogen is being partially ionized. Part of the available thermal energy is consumed by the ionization process instead of raising the gas temperature — which translates, via α and δ , into a sharp drop in ∇_{ad} : detailed solar models show ∇_{ad} falling to values of order 0.1 in this region, well below the ideal-gas value 0.4. Since ∇_{rad} in the outer layers is already typically high (low density, high opacity), the combination of an elevated ∇_{rad} with a reduced ∇_{ad} makes convective instability practically inevitable in this layer — this is the physical, not merely geometric, reason the solar envelope is convective. The same mechanism, acting in the partial ionization zones of H and He II in stars somewhat hotter than the Sun, is responsible for driving the κ mechanism of pulsation that defines the instability strip of Cepheids and RR Lyrae stars in the HR diagram.

From thermodynamics to stellar structure

The thermodynamic relations derived in this chapter are not an abstract exercise: C_P , C_V , α , δ , and ∇_{ad} appear directly in the equations that govern stellar structure and evolution. The coefficients α and δ (often written as $\chi_\rho \equiv 1/\alpha$ and $\chi_T \equiv \delta/\alpha$ in other conventions) describe how pressure responds to small changes in density and temperature — essential information in any equation of state used in numerical models of stellar interiors. For a complete derivation of these relations in the context of stellar structure, including their generalizations to partially degenerate gases and ionization regions, see Kippenhahn et al. (2012).

Modern equations of state in stellar models

Current stellar evolution codes do not use the ideal-gas equation directly, except as a low-density approximation: they combine different equation-of-state tables and formulations, valid in different regimes of density and temperature (ionized gas, partially degenerate matter, Coulomb effects, etc.). The MESA code (*Modules for Experiments in Stellar Astrophysics*), widely used in current research, combines the OPAL, SCVH, FreeEOS, HELM, PC, and Skye equations of state, automatically choosing the most appropriate one at each point of the stellar mesh (Jermyn et al., 2023). The Skye module, in particular, was developed to be analytically differentiable — an important property for the numerical stability of modern stellar evolution models (Jermyn et al., 2021). The depth of the solar convective zone, used throughout this chapter as an observational reference for the Schwarzschild criterion, is one of the most precise constraints available for calibrating these equations of state and the opacity in the partial ionization layers, through comparison between the sound speed inferred from helioseismology and that predicted by the models (Christensen-Dalsgaard et al., 1991).

Wrap-up: the adiabatic gradient for an ideal gas

The lecture notes explicitly return to Equation (3.51),

$$\nabla_{\text{ad}} \Big|_{\text{ideal}} = \frac{R}{\mu C_P}, \quad (3.53)$$

as the expected value for an ideal gas — confirming the derivation already presented in Section 3.8.

For a non-relativistic monatomic ideal gas, $C_V = \frac{3}{2}R/\mu$ and, consequently, $C_P = C_V + R/\mu = \frac{5}{2}R/\mu$. Substituting this result into Equation (3.51),

$$\nabla_{\text{ad}} \Big|_{\text{monatomic}} = \frac{R/\mu}{(5/2)R/\mu} = \frac{2}{5} = 0.4, \quad (3.54)$$

recovering the same value $\nabla_{\text{ad}} = 2/5 = 0.4$ already obtained in Equation (3.52) — a finite, non-zero number, as required by the first law of thermodynamics for a gas with finite heat capacity ($C_P < \infty$): under adiabatic compression or expansion, the temperature of a parcel of monatomic gas must necessarily change.

Chapter 4

Radiative Processes

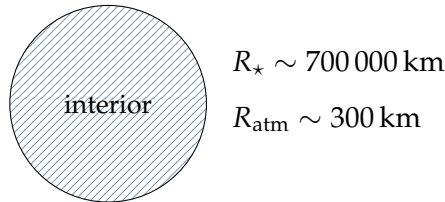
The study of the radiative processes that occur in stars helps us understand some of the most important processes in astrophysics, since stars are among the most abundant astronomical objects in the universe.

4.1 Definition of a star

For an initial definition of stars, we can say that a star is a large ball of gas compressed by the action of gravitational force. Such compression implies an increase in the particle density in the central region of the star, where atomic fusion processes occur, which generate energy and brightness (radiation) for the body. Throughout this course, we will see more details about stellar properties.

4.2 Fundamentals of radiative transport

Initially, let us recall that a star, roughly speaking, can be subdivided into two layers: the stellar interior and the stellar atmosphere, where



Let us first study the radiative processes that occur in the stellar interior. That is, photon gases for the most part.

To describe the radiation field of a star, we can define three necessary variables:

- I_{ν} : specific luminous intensity;
- J_{ν} : mean luminous intensity;
- F_{ν} : luminous radiation flux.

4.3 Specific intensity

The specific intensity is defined by $I_{\nu}(\vec{r}, \hat{n}, t)$. It describes the intensity of the radiation at frequency ν , at any point $\vec{r}$, with direction $\hat{n}$, at an instant of time t .

Consider the case of a solid-angle element in polar coordinates:

Now consider a light beam crossing the area element shown below:

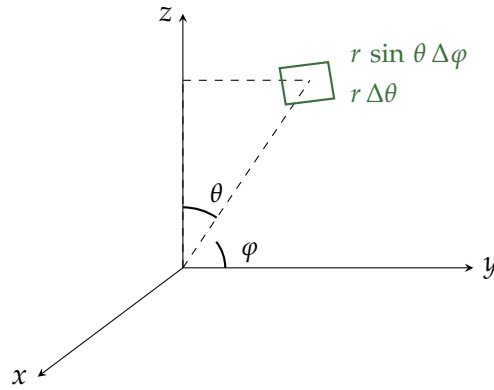


Figure 4.1: Solid-angle element in spherical coordinates. The area of the sphere of radius r bounded by $(\theta, \theta + \Delta\theta)$ and $(\varphi, \varphi + \Delta\varphi)$ is $r^2 \sin \theta \Delta\theta \Delta\varphi$, so that the corresponding solid angle is $\Delta\Omega = \sin \theta \Delta\theta \Delta\varphi$.

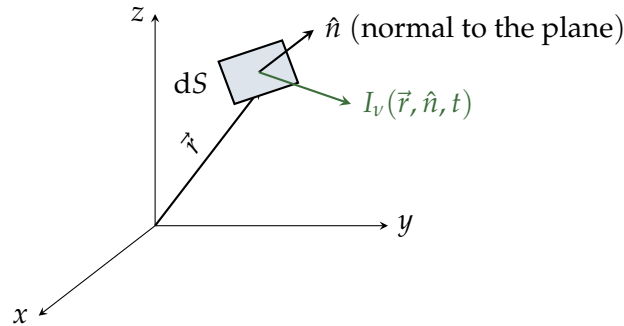


Figure 4.2: Light beam crossing an area element dS , with normal $\hat{n}$.

Thus, the amount of energy that crosses the plane dS can be expressed as

$$dE = \underbrace{I_\nu(\vec{r}, \hat{n}, t)}_{\text{specific intensity}} \cdot \underbrace{(\hat{n} \cdot d\vec{S})}_{\substack{\text{area element,} \\ d\vec{S} = dS \hat{s}}} \cdot \underbrace{dt}_{\text{time}} \cdot \underbrace{d\omega}_{\substack{\text{solid} \\ \text{angle}}} \cdot \underbrace{d\nu}_{\text{freq.}}, \quad (4.1)$$

which is the amount of energy transported.

The specific intensity has units of

$$I_\nu = \left[\frac{\text{erg}}{\text{s m}^2 \text{ ster Hz}} \right]. \quad (4.2)$$

What a telescope actually measures

No instrument measures I_ν along a single mathematical direction $\hat{n}$: every detector has a finite angular aperture (the telescope's field of view, or the spectrograph slit) and a finite collecting area (the mirror or lens). What a telescope of effective area A , observing through a solid angle $\Delta\omega$, actually records is

$$dE_{\text{measured}} = \left[\int_{\Delta\omega} I_\nu(\vec{r}, \hat{n}, t) (\hat{n} \cdot \hat{s}) d\omega \right] A dt d\nu, \quad (4.3)$$

that is, an integral of I_ν over the solid angle of the beam accepted by the instrument. For an unresolved point source (like most stars, except the Sun), this integral reduces essentially to the flux F_ν (Section 4.6); for spatially resolved objects (the solar disk, nebulae,

nearby galaxies), it is possible to recover $I_\nu(\hat{n})$ point by point — this is exactly what an astronomical image records, pixel by pixel.

The specific intensity can be represented in different coordinate systems:

- i) **Cartesian plane:** $I_\nu(x, y, z, \hat{n}, t)$;
- ii) **Spherical plane:** $I_\nu(r, \theta, \varphi, \Theta, \Phi, t)$, where (Θ, Φ) is the angle describing the direction of the normal $\hat{n}$.

The general expression for dE_ν is

$$dE_\nu = I_\nu(x, y, z, \theta, \varphi, t) \underbrace{\cos \theta dA}_{\hat{n} \cdot d\vec{S}} dt d\nu d\omega. \quad (4.4)$$

Plane-parallel geometry

Stars like the Sun can be studied using plane-parallel geometry (stratification of the atmosphere into layers), which simplifies the calculations, since $R_{\text{atm}} \ll R_\star$:

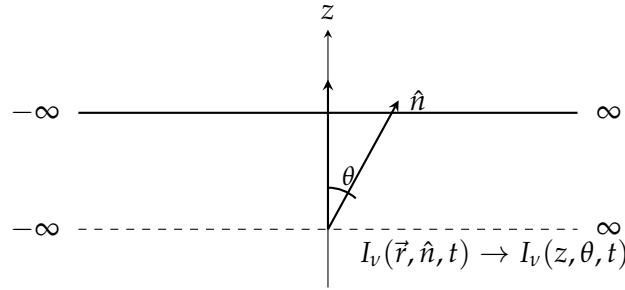


Figure 4.3: Plane-parallel geometry: the atmosphere is treated as semi-infinite horizontal layers, so that I_ν depends only on the depth z and the angle θ relative to the normal.

The luminous intensity, I_ν , completely describes the stellar radiation field.

4.4 Mean intensity

The mean intensity, as the name suggests, describes the average distribution of the luminous intensity per area element. It is defined as

$$J_\nu(\vec{r}, t) = \frac{\oint I_\nu(\vec{r}, \hat{n}, t) d\omega}{\oint d\omega}, \quad (4.5)$$

where, in spherical coordinates, $d\omega = \sin \theta d\theta d\varphi$ (the area of the sphere of radius r bounded by $(\theta, \theta + \Delta\theta)$ and $(\varphi, \varphi + \Delta\varphi)$ is $r^2 \sin \theta \Delta\theta \Delta\varphi$).

Recalling that

$$\begin{aligned} \oint d\omega &= 4\pi = \int_0^{2\pi} \int_0^\pi \sin \theta d\theta d\varphi \\ &= \underbrace{\left[-\cos \theta \right]_0^\pi}_{=2} \cdot \underbrace{\int_0^{2\pi} d\varphi}_{=2\pi} = 2 \cdot 2\pi = 4\pi, \end{aligned} \quad (4.6)$$

we therefore have

$$J_\nu(\vec{r}, t) = \frac{1}{4\pi} \oint I_\nu(\vec{r}, \hat{n}, t) d\omega. \quad (4.7)$$

Note that both J_ν and I_ν are scalar quantities (same dimension). In the case of an ideal blackbody, the specific intensity can be described as $I_\nu = B_\nu$ (Planck), if it is a blackbody. Thus,

$$J_\nu = \frac{1}{4\pi} \oint B_\nu(T) d\omega = B_\nu. \quad (4.8)$$

4.5 Radiative energy density

Let us consider a photon, or light beam, crossing a cylindrical region of length dl and cross-sectional area ds , as represented below.

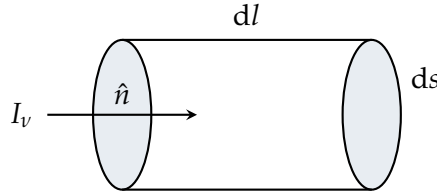


Figure 4.4: Cylinder of photons of length dl and cross-sectional area ds , used to derive the radiative energy density.

The amount of energy that enters the cylinder is equal to

$$dE = I_\nu \cdot ds \cdot dt \cdot d\omega \cdot d\nu. \quad (4.9)$$

In the case of photons, their speed is constant (c), so we can write $c = dl/dt \Rightarrow dt = dl/c$, hence

$$dE = I_\nu \frac{ds dl}{c} d\omega d\nu, \quad \text{but } ds dl \equiv dV, \quad (4.10)$$

so

$$\frac{dE}{dV d\omega d\nu} = \frac{I_\nu}{c} \equiv du_\nu(\omega), \quad (4.11)$$

where du_ν is the radiative energy density (per solid angle and frequency).

We can integrate Equation (4.11) over ω , so

$$u_\nu(\vec{r}, t) = \frac{1}{c} \oint I_\nu d\omega = \frac{4\pi}{c} J_\nu, \quad (4.12)$$

using Equation (4.7). In the case where $I_\nu = B_\nu$, that is, a blackbody,

$$u_\nu = \frac{4\pi}{c} J_\nu = \frac{4\pi}{c} B_\nu, \quad [u_\nu] = \left[\frac{\text{erg}}{\text{cm}^3} \right]. \quad (4.13)$$

Integrating Equation (4.13) over frequency space, and using the result of the Planck integral over frequency, $\int_0^\infty B_\nu d\nu = \sigma T^4/\pi$ (which we will derive further ahead, Equation (4.128)), we obtain

$$u = \frac{4\pi}{c} \int_0^\infty B_\nu d\nu = \frac{4\pi}{c} \cdot \frac{\sigma T^4}{\pi} = a T^4, \quad a = \frac{4\sigma}{c}, \quad (4.14)$$

where a is the radiation constant — note that the two factors of π cancel exactly, so that a carries no π : numerically, $a = 4\sigma/c \approx 7.566 \times 10^{-15} \text{ erg cm}^{-3} \text{ K}^{-4}$.

Astrophysical environment	Energy density
Synchrotron	$3 \times 10^{-18} \text{ erg/cm}^3$
Dust	$6 \times 10^{-13} \text{ erg/cm}^3$
H α emission	$3 \times 10^{-16} \text{ erg/cm}^3$

Where these energy densities occur

The three values above illustrate the enormous dynamic range of radiative energy densities present in the interstellar medium and in high-energy environments:

- **Synchrotron:** emitted by relativistic electrons spiraling in magnetic fields, this is the dominant mechanism in active galactic nucleus (AGN) jets and in supernova remnants, where particles are accelerated to relativistic energies by shocks. It is typically the component with the lowest energy density among the three, since the fraction of relativistic electrons in the medium is small.
- **Dust:** dust grains in the diffuse interstellar medium absorb starlight in the optical/UV and re-emit it thermally in the infrared; the average interstellar radiation field (stellar + dust) has an energy density of order $\sim 10^{-13}$ – $10^{-12} \text{ erg/cm}^3$ (comparable, in order of magnitude, to the energy density of the cosmic microwave background itself, $u_{\text{CMB}} \approx 4.2 \times 10^{-13} \text{ erg/cm}^3$).
- **H α emission:** traces HII regions, clouds of hydrogen ionized by young, hot stars (types O and B), where electronic recombination to the $n = 3 \rightarrow 2$ level of hydrogen produces the H α line at $6563 \text{ \AA}$ — one of the most widely used tracers of recent star formation in galaxies.

4.6 Radiation flux

The radiation flux describes the total energy that crosses an area perpendicular to the plane of the radiation per second, where

$$\vec{F}(\vec{r}, t) = \oint I_\nu(\vec{r}, \hat{n}, t) \hat{n} d\omega. \quad (4.15)$$

Assuming a blackbody, $I_\nu = B_\nu$, so:

$$\vec{F}(\vec{r}, t) = B_\nu(T) \oint \hat{n} d\omega. \quad (4.16)$$

$F \neq 0$: why a real star is not in strict LTE

If the star were a blackbody, $F(\vec{r}, t) \equiv 0$, since all absorbed radiation would have to be re-emitted isotropically, with no preferred direction. Since $F(\vec{r}, t)$ is always $\neq 0$ — it is precisely the flux that escapes the star and that we observe as luminosity —, we have that the equality $I_\nu = B_\nu$ is only an approximation.

The Local Thermodynamic Equilibrium (LTE) approximation

The observation above is the entry point to one of the most important — and most delicate — approximations in stellar physics: **Local Thermodynamic Equilibrium** (LTE). A gas is in complete thermodynamic equilibrium only if the radiation is isotropic and given exactly by the Planck function at all frequencies ($I_\nu = B_\nu$ in all directions), which would require $F_\nu = 0$ — never satisfied in a real star, which must radiate energy into space. The LTE approximation works around this problem by assuming that, although the radiation field is *not* in thermal equilibrium with matter, the *local* properties of matter (atomic level populations, ionization, velocity distribution) are well described by the equilibrium thermal formulas (Boltzmann, Saha, Maxwell) evaluated at the local temperature and density.

- **Where LTE holds well:** in the dense, optically thick stellar interior, where the photon mean free path is much shorter than the scale over which T and ρ vary — matter and radiation collide (via absorption/emission and scattering) frequently enough to keep the atomic populations locally thermalized, even if the radiation field as a whole is not isotropic.
- **Where LTE fails:** in the optically thin outer layers of a stellar atmosphere — the solar chromosphere and corona, stellar winds, and low-density regions in general —, where the photon mean free path becomes long, collisions between particles become rare, and the atomic populations come to be determined by the balance between radiative and collisional processes, requiring a full **non-LTE** (NLTE) treatment.

Analyzing the radiation flux in the direction of an observer, we have:

$$\vec{F} \cdot \hat{z} = \hat{z} \cdot \oint B_\nu(T) \hat{n} d\omega. \quad (4.17)$$

Since $\hat{z} \cdot \hat{n} = \cos \theta$ (unit vectors), we have:

$$\begin{aligned} \vec{F} \cdot \hat{z} &= B_\nu \int_0^{2\pi} \int_0^{\pi/2} \cos \theta \sin \theta d\theta d\varphi \\ &= B_\nu \cdot 2\pi \cdot \underbrace{\left[\frac{\sin^2 \theta}{2} \right]_0^{\pi/2}}_{=1/2} = \pi B_\nu. \end{aligned} \quad (4.18)$$

We again see that the radiation flux is nonzero, hence it is not a blackbody and is not in local thermodynamic equilibrium.

In Cartesian coordinates, we can describe the flux by

$$\vec{F} = (F_x, F_y, F_z), \quad \begin{cases} F_x = \oint I_\nu n_x d\omega, \\ F_y = \oint I_\nu n_y d\omega, \\ F_z = \oint I_\nu n_z d\omega, \end{cases} \quad (4.19)$$

where, in spherical coordinates, the unit vector $\hat{n}$ has components

$$n_x = \sin \theta \cos \varphi, \quad n_y = \sin \theta \sin \varphi, \quad n_z = \cos \theta. \quad (4.20)$$

Thus, for a plane-parallel case (where $I_\nu = I_\nu(z, \theta)$ does not depend on φ):

$$F_x = \int_0^{2\pi} \int_0^\pi I_\nu(z, \theta) \sin^2 \theta \cos \varphi \, d\theta \, d\varphi = 0, \quad (4.21)$$

$$F_y = \int_0^{2\pi} \int_0^\pi I_\nu(z, \theta) \sin^2 \theta \sin \varphi \, d\theta \, d\varphi = 0, \quad (4.22)$$

$$F_z = \int_0^{2\pi} \int_0^\pi I_\nu(z, \theta) \cos \theta \sin \theta \, d\theta \, d\varphi = 2\pi \int_0^\pi I_\nu(z, \theta) \cos \theta \sin \theta \, d\theta. \quad (4.23)$$

The integrals of Equations (4.21) and (4.22) vanish due to the cyclicity of the sine and cosine functions in φ (and also because I_ν does not depend on φ).

We thus have a plane-parallel geometry that depends only on a single component z , so $\vec{F}(\vec{r}) = F(0, 0, F_z)$.

Analyzing the plane-parallel case, we have that the depth of an atmosphere can be characterized by the parameters I_ν , J_ν , F_ν :

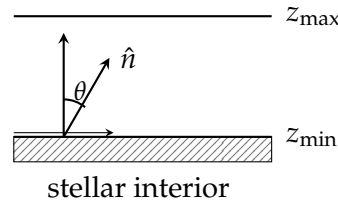


Figure 4.5: Depth of a plane-parallel atmosphere, bounded between $z_{\min}$ (base, adjacent to the stellar interior) and $z_{\max}$ (top).

4.7 Moments of the radiation field

Recalling Equation (4.7), we can work a bit more with the mean intensity. Defining $\cos \theta \equiv \mu$,

$\mu = \cos \theta$ in this chapter — notation collision with Chapter 3

From here on, μ denotes the **direction cosine** $\mu \equiv \cos \theta$, the angle between the photon direction $\hat{n}$ and the reference axis (typically the z axis, normal to the plane-parallel atmosphere). This is the universal convention in radiative transfer, and will be used throughout the remainder of this book whenever the context is radiation.

Warning: in Chapter 3 (Thermodynamics), μ was used with a completely different meaning — the *mean molecular weight* of the equation of state ($P = \rho RT / \mu$). This symbol collision is real and common in the stellar astrophysics literature (both conventions are widely established in their respective subfields), not an error in this text. We choose to keep $\mu = \cos \theta$ here, without introducing an alternative symbol, precisely because this is the universally recognized notation in radiative transfer — departing from it would make this chapter difficult to compare with any reference in the field. Whenever there is real ambiguity between the two chapters, the meaning of μ will be explicitly reinforced in the text.

hence, $d\mu = -\sin \theta \, d\theta$, and

$$J_\nu = \frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi I_\nu(z, \mu) \sin \theta \, d\theta \, d\varphi. \quad (4.24)$$

Since I_ν does not depend on φ in the plane-parallel case,

$$\begin{aligned} J_\nu &= \frac{1}{4\pi} \cdot 2\pi \int_0^\pi I_\nu \sin \theta \, d\theta = \frac{1}{2} \int_{-1}^1 I_\nu (-d\mu) \\ &= \frac{1}{2} \int_{-1}^1 I_\nu \, d\mu, \end{aligned} \quad (4.25)$$

hence

$$J_\nu(z) = \frac{1}{2} \int_{-1}^1 I_\nu(z, \mu) \, d\mu. \quad (4.26)$$

Equation (4.26) is known as the **zeroth-order moment** of the radiation field. The moment describes the momentum/displacement of something; in this particular case, we are studying the evolution of the luminous intensity in the stellar interior.

There are still two other degrees for the moment, with a general expression given by

$$M_n = \frac{1}{2} \int_{-1}^1 I_\nu \mu^n \, d\mu, \quad (4.27)$$

$$n = 0 : J_\nu = \frac{1}{2} \int_{-1}^1 I_\nu \, d\mu, \quad n = 1 : H_\nu = \frac{1}{2} \int_{-1}^1 I_\nu \mu \, d\mu, \quad n = 2 : K_\nu = \frac{1}{2} \int_{-1}^1 I_\nu \mu^2 \, d\mu. \quad (4.28)$$

For the first- and second-order moments, a detailed description is beyond the scope of this review, but their physical meanings can be examined.

The order-1 moment function is directly related to the radiation flux and is known as the **Eddington flux**, where:

$$H_\nu = \frac{F_\nu(z, t)}{4\pi}. \quad (4.29)$$

For $n = 2$, the second-order moment, we obtain a direct relation to the **radiation pressure**. When we consider a photon gas, its classical pressure is

$$P = \frac{F}{A}, \quad (4.30)$$

which in the quantum case results in

$$dP_r = dp_\nu \cdot \frac{1}{dt} \cdot \frac{1}{dA}, \quad (4.31)$$

that is, pressure as the rate of momentum delivery per unit area. With $p_\nu = h\nu/c$ (momentum of a photon), we obtain

$$dP_r = \frac{h}{c} \frac{d\nu}{dt} \cdot \frac{1}{dA}. \quad (4.32)$$

Eddington flux, radiation pressure, and the Eddington Luminosity

H_ν and K_ν are not merely formal exercises: they are the basis of one of the most important limits in stellar astrophysics, the **Eddington Luminosity**, L_{Edd} . It is defined as the maximum luminosity that a star (or any accreting object) can sustain in hydrostatic equilibrium: above it, the radiation pressure — governed by the second-order moment K_ν — overcomes the gravitational force confining the gas, and the excess matter is literally pushed outward by radiation. For a spherical source of mass M with opacity dominated

by Thomson scattering (κ_{es}),

$$L_{\text{Edd}} = \frac{4\pi GMc}{\kappa_{\text{es}}}. \quad (4.33)$$

This limit is central to understanding why very massive stars ($M \gtrsim 100 M_{\odot}$) have such intense winds and are unstable (radiation pressure already competes with gravity in their outer layers), and it is equally crucial in accretion physics: accretion disks around black holes and neutron stars in X-ray binaries, and in active galactic nuclei, typically saturate near L_{Edd} , regulating the rate at which these objects can grow (Kippenhahn et al., 2012).

By substituting and integrating the values in the monochromatic energy differential equation, we can obtain

$$dP = \frac{4\pi}{c} K_{\nu}, \quad (4.34)$$

indicating that K_{ν} is a measure of the radiation pressure.

Radiative transfer in modern stellar atmospheres

The formulation of I_{ν} , J_{ν} , F_{ν} and the moments H_{ν} , K_{ν} presented here is the theoretical basis on which modern stellar spectral synthesis codes are built. The TLUSTY code, one of the most widely used for hot-star (O, B) atmospheres, solves radiative transfer coupled to the statistical equilibrium of atomic levels outside of LTE (non-LTE), using the hybrid complete-linearization/ accelerated-lambda-iteration method (Hubeny and Lanz, 1995). The PHOENIX code, in turn, is widely used for cooler atmospheres (Sun-like stars, M dwarfs, exoplanet atmospheres), including detailed treatment of molecular lines (Husser et al., 2013). For a recent and comprehensive review of the 3D non-LTE radiative transfer problem — increasingly relevant as hydrodynamic simulations of stellar atmospheres replace classical 1D models — see Bergemann and Hoppe (2025).

4.8 Radiation field in any astrophysical environment

We have seen that the radiation field is completely characterized if we know the luminous intensity I_{ν} . However, how can this quantity be calculated and derived?

Radiation-matter interaction

Let us define two essential macroscopic quantities: the opacity of the gas and the emissivity of the gas. Consider a light beam I_{ν} crossing any gas (atoms, ions, molecules, protons, etc.):

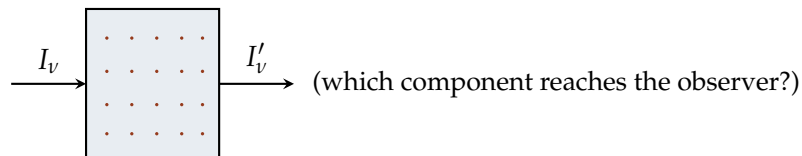


Figure 4.6: Light beam crossing an arbitrary gas. Electromagnetic interactions with the medium modify I_{ν} , resulting in I'_{ν} .

There are three main processes by which radiation can be modified: absorption, emission, and scattering.

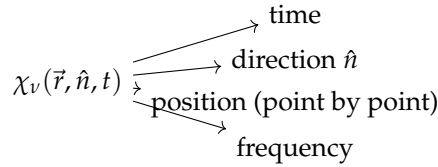
- **Absorption:** the photon is converted into thermal energy. Its energy is incorporated into the medium.
- **Emission:** this is the transition energy that was released by some volume.
- **Scattering:** photons can be emitted or absorbed in any direction.

Gas opacity

We define the term $\chi_\nu(\vec{r}, \hat{n}, t)$ through the amount of energy absorbed in a volume element of the gas:

$$dE = I_\nu(\vec{r}, \hat{n}, t) \cdot \underbrace{\chi_\nu(\vec{r}, \hat{n}, t)}_{\text{opacity}} \cdot dS \cdot ds \cdot dt \cdot d\nu \cdot d\omega. \quad (4.35)$$

$\chi_\nu(\vec{r}, \hat{n}, t)$ is the gas opacity. The opacity depends on:



The opacity also depends on the optical depth of the star (see below).

In the particular case we are analyzing, plane-parallel geometry, we have that $\chi_\nu(z)$ is a function of z alone. It is also isotropic, not depending on $\hat{n}$. Under these circumstances, χ_ν must have units of cm^{-1} .

The unit of χ_ν does not depend on the geometry

The opacity χ_ν is defined as energy absorbed per unit of path length traveled by the beam [Equation (4.35)], and therefore has units of cm^{-1} (inverse length) *regardless* of the coordinate system used to describe the position — switching from plane-parallel to spherical geometry changes the functional form of $\chi_\nu(\vec{r})$, not its physical dimension.

Thus, we can define the quantity $1/\chi_\nu = l$, which has units of distance. This quantity describes the **mean free path** of the photons in the stellar interior:

$$l = \frac{1}{\chi_\nu}. \quad (4.36)$$

Note that the larger the absorption coefficient, the shorter the mean free path of the photons.

The radiative random walk: how long does a photon take to escape the Sun?

The mean free path $l = 1/\chi_\nu$ in the solar core is surprisingly short — on the order of fractions of a millimeter to a few centimeters, many orders of magnitude smaller than the solar radius ($R_\odot \approx 7 \times 10^{10} \text{ cm}$). A photon created by fusion in the core does not travel in a straight line to the surface: it is absorbed and re-emitted repeatedly, in random directions, performing a **random walk** of $N = (R_\odot/l)^2$ steps before escaping. The careful calculation by Mitalas and Sills (1992), using a realistic solar model (where l varies with depth, rather than a single value assumed *a priori*), finds an effective mean free path of $\approx 9 \times 10^{-4} \text{ m}$ ($\approx 0.9 \text{ mm}$) and a diffusion time of

$$t_{\text{diffusion}} \approx 1.7 \times 10^5 \text{ years}, \quad (4.37)$$

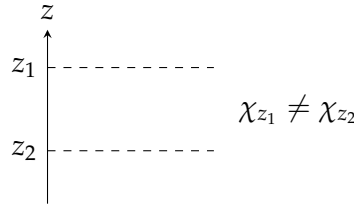
considerably longer than naive estimates using $l \sim 1$ cm (which give $t \sim 10^4$ yr, a value frequently cited but underestimated by about an order of magnitude). In both cases, the qualitative result is striking: the light reaching Earth today carries energy generated in the solar core tens to hundreds of thousands of years ago, even though the final photon, once free in space, takes only ≈ 8 minutes to cross the Earth-Sun distance.

Other important definitions:

$$\chi_\nu = n \sigma_\nu; \quad n = \text{number density (cm}^{-3}\text{)}, \quad \sigma_\nu = \text{cross section at } \nu \text{ (cm}^2\text{)}, \quad (4.38)$$

$$\chi_\nu = \kappa_\nu \rho; \quad \kappa_\nu = \text{cross section per unit mass (cm}^2\text{/g)}, \quad \rho = \text{density (g/cm}^3\text{)}. \quad (4.39)$$

χ_ν is one of the most fundamental quantities in astrophysics. Also recall that it is a function of the optical depth:



Where κ_ν comes from: the main sources of stellar opacity

The specific opacity κ_ν is not a single physical quantity, but the sum of several radiation-matter interaction processes, whose relative importance changes drastically with temperature and composition:

- **Kramers opacity** (free-free and bound-free): dominates in dense, hot stellar interiors, where free electrons scatter or are accelerated in the field of ions; it scales approximately as $\kappa \propto \rho T^{-3.5}$, the basis of the earliest analytical estimates of stellar structure.
- **H⁻ ion opacity**: dominates in the atmospheres of cool stars (Sun-like and cooler), where the abundance of free electrons (donated by low-ionization-potential metals) allows the formation of the H⁻ ion, an efficient absorber in the visible/near-infrared — responsible for much of the opacity of the solar photosphere.
- **Thomson scattering** by free electrons: dominates in hot, massive stars (types O, B), where ionization is nearly complete; it is the opacity that sets the Eddington limit (Section 4.6 and following).

The precise calculation of κ_ν (or of its averages, such as the Rosseland opacity, used in the stellar structure equations) for all relevant combinations of temperature, density, and chemical composition is the task of large coordinated computational projects: the **OPAL** project, at Lawrence Livermore National Laboratory (Iglesias and Rogers, 1996), and the **Opacity Project** (OP), an independent international collaboration that serves as a cross-check of the OPAL results. Both remain, decades later, the basis of the opacity tables used in stellar evolution codes such as MESA (already mentioned in Chapter 3).

Gas emissivity

The emissivity has a physical meaning similar to that of opacity, except that in this case there is an increase in the intensity of the radiation beam, so:

$$dE = \underbrace{\eta_\nu(\vec{r}, \hat{n}, t)}_{\text{emissivity}} \cdot \underbrace{dS \cdot d\mathbf{s}}_{\text{volume}} \cdot dt \cdot d\nu \cdot d\omega. \quad (4.40)$$

In the plane-parallel case, we have $\eta_\nu(z)$. The unit of $\eta_\nu(z)$ is $\text{erg}/(\text{cm}^3 \text{ Hz s ster})$, similar to that of I_ν .

Gas scattering

Gas scattering can be represented by an absorption or emission coefficient, since the energy can change into any direction. Thus, the coefficients η_ν and χ_ν also describe scattering transitions.

4.9 Radiative transfer equation

Once χ_ν and η_ν of the gas have been defined, we can determine the radiative transfer equation, that is, the equation describing the macroscopic radiation-matter interaction.

Consider the following situation:

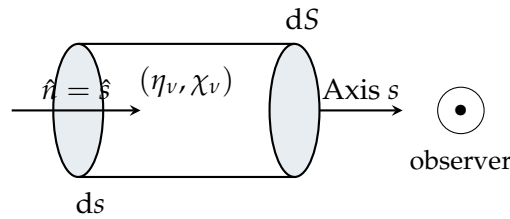


Figure 4.7: Cylinder of gas with opacity coefficient χ_ν and emissivity η_ν , traversed along the axis s .

In this case, $I_\nu(s, t)$ enters the cylinder, and $I_\nu(s + ds, t + dt)$ exits the cylinder. The energy absorbed and the energy emitted inside the cylinder can be described as

$$E_{\text{abs}} = \chi_\nu I_\nu dS ds dt d\nu d\omega, \quad (4.41)$$

$$E_{\text{emi}} = \eta_\nu dS ds dt d\nu d\omega. \quad (4.42)$$

On the other hand, the energy entering the cylinder is

$$E_1 = I_\nu(s, t) dS d\omega dt d\nu, \quad (4.43)$$

and the energy leaving the cylinder is

$$E_2 = I_\nu(s + ds, t + dt) dS d\omega dt d\nu. \quad (4.44)$$

Thus, by conservation of energy of the system, we can study the variation of energy, where

$$E_2 - E_1 = \Delta E = [I_\nu(s + ds, t + dt) - I_\nu(s, t)] dS d\omega dt d\nu. \quad (4.45)$$

But this energy variation is a consequence of the absorption and emission processes of the gas, so we can also represent ΔE by

$$\Delta E = E_{\text{emi}} - E_{\text{abs}} = (\eta_\nu - \chi_\nu I_\nu) dS ds dt d\nu d\omega. \quad (4.46)$$

Equating the two expressions for ΔE [Equations (4.45) and (4.46)] and dividing both sides by $dS dt d\nu d\omega$ (note that Equation (4.45) does *not* have the factor ds , while Equation (4.46) does — the factor ds survives the division only on the right-hand side):

$$I_\nu(s + ds, t + dt) - I_\nu(s, t) = (\eta_\nu - \chi_\nu I_\nu) ds. \quad (4.47)$$

In the infinitesimal limit, the left-hand side is, by definition, the total differential dI_ν , hence

$$dI_\nu = (\eta_\nu - \chi_\nu I_\nu) ds. \quad (4.48)$$

Knowing that I_ν is a function of s and t , we can write, using partial derivatives, that

$$\left(\frac{\partial I_\nu}{\partial s}\right) ds + \left(\frac{\partial I_\nu}{\partial t}\right) dt = (\eta_\nu - \chi_\nu I_\nu) ds. \quad (4.49)$$

Dividing Equation (4.49) by ds and using that, for a photon, $dt/ds = 1/c$:

$$\frac{\partial I_\nu}{\partial s} + \frac{1}{c} \frac{\partial I_\nu}{\partial t} = \eta_\nu - \chi_\nu I_\nu. \quad (4.50)$$

This is the radiative transfer equation along the photon beam.

For a displacement in an arbitrary direction, $dI_\nu/ds \Rightarrow (\hat{n} \cdot \nabla) I_\nu$, so that Equation (4.50) generalizes to the **general vector form** of the radiative transfer equation:

$$\left[\hat{n} \cdot \nabla + \frac{1}{c} \frac{\partial}{\partial t} \right] I_\nu(\vec{r}, \hat{n}, t) = \eta_\nu(\vec{r}, \hat{n}, t) - \chi_\nu(\vec{r}, \hat{n}, t) I_\nu(\vec{r}, \hat{n}, t). \quad (4.51)$$

Plane-parallel, stationary case

In a typical star, for the outermost layers of the atmosphere, we have a stationary plane-parallel geometry (with no explicit time dependence):

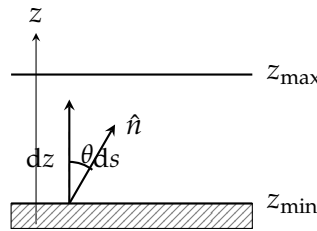


Figure 4.8: Displacement ds along the direction $\hat{n}$, with vertical component $dz = ds \cos \theta$.

From Equation (4.48), we have that $dI_\nu/ds = \eta_\nu - \chi_\nu I_\nu$. But, by the plane-parallel geometry, $\vec{r} = (0, 0, z)$, so

$$\frac{dI_\nu}{ds} = \frac{dI_\nu}{dz} \cdot \frac{dz}{ds}. \quad (4.52)$$

From the analysis of components in Figure 4.8, we have that $\cos \theta = dz/ds$, so we can once again define $\cos \theta \equiv \mu$, such that

$$\frac{dI_\nu}{ds} = \mu \frac{dI_\nu}{dz}. \quad (4.53)$$

Thus, the radiative transfer equation in plane-parallel geometry takes the form

$$\mu \frac{dI_\nu}{dz}(z, \mu) = \eta_\nu(z) - \chi_\nu(z) I_\nu(z, \mu). \quad (4.54)$$

4.10 Optical depth and source function

Optical depth is a measure of χ_ν from the top of the star's atmosphere to a point z . It indicates how much of a light beam will be absorbed as it travels through the atmosphere. We define τ by

$$\tau_\nu(z) = - \int_z^{z_{\max}} \chi_\nu(z') dz', \quad d\tau_\nu = -\chi_\nu dz. \quad (4.55)$$

Note that τ is a dimensionless quantity; it varies with frequency and with the height in the atmosphere.

Why the photosphere is defined at $\tau \approx 2/3$

The optical depth gives a precise operational meaning to the notion of a "surface" for a star — something that, for a ball of gas without a solid edge, is not obvious *a priori*. A typical photon, produced at a layer of optical depth τ (measured from the top of the atmosphere), has a probability of escaping without being reabsorbed of order $e^{-\tau}$. The layer conventionally identified as the **photosphere** — from which most of the light we observe originates — is defined as the one at which $\tau_\nu \approx 2/3$: it can be shown that, for an atmosphere in radiative equilibrium with $S_\nu \approx B_\nu(T)$ varying smoothly with depth, the emergent intensity at the top is approximately equal to the source function evaluated exactly at $\tau = 2/3$. This is why we define the effective temperature T_{eff} (Chapter 2, via $L = 4\pi R^2 \sigma T_{\text{eff}}^4$) as the temperature of the layer at $\tau \approx 2/3$: it is this layer that dominates the emergent spectrum and, therefore, the total observed flux.

Substituting τ into the radiative transfer equation in the plane-parallel case [Equation (4.54)], with $dz = -d\tau_\nu/\chi_\nu$:

$$\mu \frac{dI_\nu}{-d\tau_\nu/\chi_\nu} = \eta_\nu - \chi_\nu I_\nu \implies \mu \frac{dI_\nu}{d\tau_\nu}(\tau_\nu, \mu) = I_\nu - \frac{\eta_\nu(\tau)}{\chi_\nu(\tau)}. \quad (4.56)$$

We can define the coefficient η_ν/χ_ν as the **source function**, $S_\nu(\tau)$ or $S_\nu(z)$. This quantity describes the increase or decrease that a given point z contributes to the radiative beam:

$$S_\nu(\tau) = \frac{\eta_\nu}{\chi_\nu}, \quad [S_\nu] = \left[\frac{\text{erg}}{\text{s cm}^2 \text{ ster Hz}} \right]. \quad (4.57)$$

When thermodynamic equilibrium holds, we have $S_\nu = B_\nu$.

In this way, we can write the radiative transfer equation as

$$\mu \frac{dI_\nu}{d\tau} = I_\nu - S_\nu, \quad \text{where } S_\nu(\tau_\nu), \quad I_\nu(\tau_\nu, \mu). \quad (4.58)$$

This is the final form of the radiative transfer equation in a plane-parallel geometry.

4.11 Solutions to the transfer equation (PP)

So far, we have seen that the radiation field can be fully characterized by the parameters I_ν , J_ν , F_ν . The gas and matter can be studied through the coefficients χ_ν and η_ν , and the radiation-matter interaction is described by the radiative transfer equation,

$$\mu \frac{dI_\nu}{dz}(z, \mu) = \eta_\nu - \chi_\nu I_\nu, \quad \text{or} \quad \mu \frac{dI_\nu}{d\tau}(\tau, \mu) = I_\nu - S_\nu(\tau). \quad (4.59)$$

In this topic, we will determine the solutions to the radiative transfer equation for four cases. Let us see:

Case 1: vacuum ($\chi_\nu = \eta_\nu = 0$)

The case where there is neither absorption nor emission, so: $\chi_\nu = 0$ and $\eta_\nu = 0$. In this case, Equation (4.59) reduces to

$$\mu \frac{dI_\nu}{dz}(z, \mu) = 0 \implies I_\nu(z, \mu) = \text{constant}. \quad (4.60)$$

This case describes an environment with no gas present. Thus, the radiation beam does not change: $I_\nu \equiv \text{constant}$.

Why surface brightness does not change with distance — but flux does

Case 1 seems trivial, but it encapsulates a subtle and important conceptual point: in vacuum (interstellar or intergalactic space, to a good approximation), the *specific intensity* I_ν along a ray of light does not change with distance — this is exactly what Equation (4.60) states. This differs from everyday intuition ("light sources get fainter the farther away they are"): what falls off as $1/r^2$ is the *flux* F_ν , not I_ν . The reason is geometric: as a beam moves away from a source, the solid angle subtended by the source decreases as $1/r^2$, but this is compensated because the specific intensity is measured *per unit solid angle*, and this combination is exactly what keeps it constant along the ray. In practice, this means that the surface brightness of a resolved source (such as the surface of the Sun, or a nearby galaxy) does not depend on the observer's distance — only how large, angularly, that source appears (and, therefore, how much total flux it delivers) decreases.

Case 2: emission only ($\chi_\nu = 0$)

The case where the opacity is zero ($\chi_\nu = 0$), so we have only emission:

$$\mu \frac{dI_\nu}{dz}(z, \mu) = \eta_\nu \implies dI_\nu(z, \mu) = \frac{\eta_\nu}{\mu} dz. \quad (4.61)$$

Solving the integral, we have

$$\int_{I_\nu(z_1, \mu)}^{I_\nu(z_2, \mu)} dI_\nu = \int_{z_1}^{z_2} \frac{\eta_\nu(z)}{\mu} dz, \quad (4.62)$$

where $I_\nu(z_1, \mu)$ is the initial intensity of the beam and $I_\nu(z_2, \mu)$ is the final intensity. Hence,

$$I_\nu(z_2, \mu) - I_\nu(z_1, \mu) = \frac{1}{\mu} \int_{z_1}^{z_2} \eta_\nu(z) dz, \quad (4.63)$$

that is,

$$I_\nu(z_2, \mu) = I_\nu(z_1, \mu) + \frac{1}{\mu} \int_{z_1}^{z_2} \eta_\nu(z) dz. \quad (4.64)$$

From Equation (4.64), we can already see that the beam *gains* energy, and this gain is of order $\frac{1}{\mu} \int_{z_1}^{z_2} \eta_\nu dz$: as it travels the path from z_1 to z_2 (receiving energy at each volume element crossed), the beam emerges more intense than it entered.

Example: planetary nebulae

An important example of this case are **planetary nebulae**: a hot, compact central star (the exposed remnant core of a low-/ intermediate-mass star at the end of its evolution), surrounded by a nebula of ionized gas that the star itself ejected. In this geometry, the gas surrounding the star emits light (but does not significantly reabsorb it), such that the notes show that the nebulae have an observed spectrum of the form

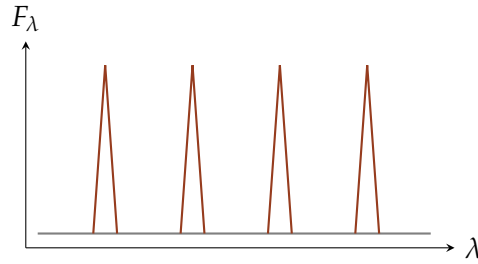


Figure 4.9: Observed spectrum of a planetary nebula: a weak continuum (the central starlight, attenuated by distance) added to strong, narrow emission lines, produced by photon recombination in the surrounding nebula's ionized gas.

and the star+nebula structure can be represented as

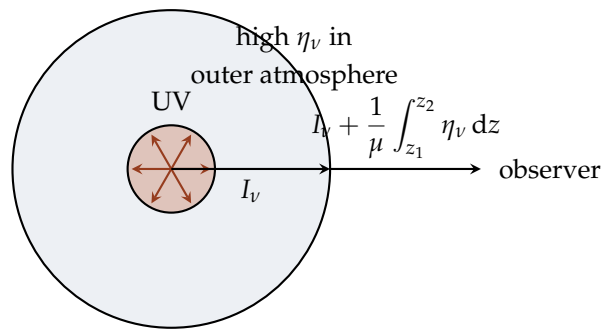


Figure 4.10: Star+nebula structure: the hot central star emits UV radiation that ionizes the surrounding gas, producing high emissivity η_ν in the outer layer (the nebula). The beam I_ν coming from the central stellar disk crosses the nebula and reaches the observer added to the emission accumulated along the path, in accordance with Equation (4.64).

What the observer sees is, therefore, the central spectrum of the star added to the emission spectrum (from photon recombination) of the surrounding nebula — exactly the sum described by Equation (4.64) and illustrated in Figure 4.10.

Optically thin nebulae: HII regions

Case 2 — pure emission, without absorption — is also the reference approximation for **HII regions**, clouds of hydrogen ionized by young, hot stars (types O and B), such as those that produce the $H\alpha$ emission already mentioned in the table of energy densities. Just like the planetary nebulae discussed above, these nebulae tend to be optically thin ($\tau_v \ll 1$) over most of the visible spectrum: the line of sight crosses the ionized gas *accumulating* emission (each volume element along the path contributes $\eta_v dz/\mu$) without significant absorption back. This is exactly why images of HII regions (the Orion Nebula being the most observed example) can be modeled, to a first approximation, by simply summing the emissivity along the line of sight — without solving the full radiative transfer problem, which would also require treating absorption.

Case 3: absorption only ($\eta_v = 0$)

The case where the emissivity is zero ($\eta_v = 0$), that is, there is only absorption. In this case, the source function is zero, since $S_v(\tau) = \eta_v/\chi_v = 0$. Hence, we can work with the radiative transfer equation as a function of optical depth, $I_v(\tau, \mu)$:

$$\mu \frac{dI_v}{d\tau} = I_v - S_v \implies \frac{dI_v}{I_v} = \frac{d\tau}{\mu}. \quad (4.65)$$

Solving the differential equation (by separation of variables), we obtain

$$\int_{I_v(\tau_1, \mu)}^{I_v(\tau_2, \mu)} \frac{dI_v}{I_v} = \frac{1}{\mu} \int_{\tau_1}^{\tau_2} d\tau \implies \ln I_v \Big|_{I_v(\tau_1, \mu)}^{I_v(\tau_2, \mu)} = \frac{1}{\mu} (\tau_2 - \tau_1), \quad (4.66)$$

that is,

$$\ln \left(\frac{I_v(\tau_2, \mu)}{I_v(\tau_1, \mu)} \right) = \frac{1}{\mu} (\tau_2 - \tau_1). \quad (4.67)$$

Applying the exponential to both sides,

$$\boxed{I_v(\tau_2, \mu) = I_v(\tau_1, \mu) e^{(\tau_2 - \tau_1)/\mu}}. \quad (4.68)$$

If we want to determine the emergent I_v from the star, we must consider the beam at the top of the atmosphere, that is, $\tau_2 = 0$ and $\tau_1 = \tau$. Thus, Equation (4.68) takes the form

$$I_v(0, \mu) = I_v(\tau, \mu) e^{-\tau/\mu}. \quad (4.69)$$

In the case where $\tau_2 = \tau(z_{\max})$ and $\tau_1 = \tau$ is any point, with $\mu = 1$ (beam emerging perpendicular to the surface), we have the geometry of Figure 4.11: Assuming $\mu = 1$, Equation (4.69) reduces to

$$I_v(0, 1) = I_v(\tau, 1) e^{-\tau}. \quad (4.70)$$

Furthermore, if $\tau = 1$, we have

$$I_v(0, 1) = \frac{1}{e} I_v(1, 1). \quad (4.71)$$

That is, in this case, the light beam was reduced by a factor of $1/e$. This value, in turn, is proportional to the absorption of the medium.

In general terms, we can say that:

- if $\tau \gg 1$, we have a **thick** medium, with more possible absorptions;
- if $\tau \ll 1$, we have a **thin** medium, where the radiation travels more freely.

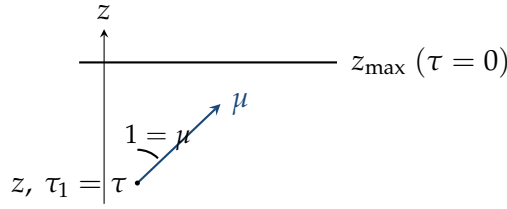


Figure 4.11: Emergent beam with $\mu = 1$: path from an arbitrary point z (depth $\tau_1 = \tau$) to the top of the atmosphere $z_{\max}$ ($\tau = 0$).

Typical optical depths in different astrophysical environments

Equation (4.68) makes the notion of "thick" and "thin" in radiative transfer concrete, but it is worth comparing the scale of τ in real astrophysical environments — the differences span many orders of magnitude:

- **Stellar interior (core to solar surface):** using the mean free path $l \approx 0.9$ mm estimated by Mitalas and Sills (1992) (already discussed in the box on the radiative random walk) and $R_{\odot} \approx 7 \times 10^{10}$ cm, the typical optical depth between the center and the surface is $\tau \sim R_{\odot}/l \sim 10^{11}$ — an extremely thick medium, in which radiation moves only by diffusion.
- **Solar corona:** by contrast, the corona is optically thin to Thomson scattering ($\tau \ll 1$) over most of its volume — this is exactly why we can observe the corona in white light during eclipses, with the background photospheric light passing through it without much attenuation, and why the technique of coronagraphy (blocking the solar disk) is necessary to isolate its much fainter brightness.
- **Diffuse interstellar medium:** the average interstellar extinction in the V band is of order ~ 1 – 2 mag/kpc in typical directions of the Galactic disk (Fitzpatrick, 1999), which corresponds to $\tau_V = A_V/1.086 \sim 10^{-3}$ per parsec — a very thin medium on the scale of individual clouds, but which accumulates to $\tau_V \sim 1$ – 2 over the entire Galactic disk.

Case 4: general case ($\chi_{\nu} \neq 0$ and $\eta_{\nu} \neq 0$)

This is the most important and most general case: neither the opacity nor the emissivity is zero. We start, once again, from the radiative transfer equation as a function of optical depth,

$$\mu \frac{dI_{\nu}}{d\tau} = I_{\nu}(\tau, \mu) - S_{\nu}(\tau), \quad (4.72)$$

whose solution we will derive step by step using the **integrating factor method**, the standard technique for solving first-order linear differential equations.

Step (a): standard form. We divide Equation (4.72) by μ to place it in the standard form of a first-order linear ODE:

$$\mu \frac{dI_{\nu}}{d\tau} = I_{\nu} - S_{\nu} \quad (4.73)$$

$$\frac{dI_{\nu}}{d\tau} - \frac{1}{\mu} I_{\nu} = -\frac{S_{\nu}}{\mu}. \quad (4.74)$$

Comparing with the standard form $\frac{dy}{dx} + P(x)y = Q(x)$, we identify $y \equiv I_\nu$, $x \equiv \tau$, $P(\tau) = -1/\mu$ (a constant, since μ is fixed along a given ray) and $Q(\tau) = -S_\nu(\tau)/\mu$.

Step (b): the integrating factor method. Before applying it, it is worth recalling the general idea behind the method, useful for any first-order linear ODE $dy/dx + P(x)y = Q(x)$. The strategy is to multiply the whole equation by an auxiliary function $\mathcal{F}(x)$ — the *integrating factor* — chosen specifically so that the left-hand side of the equation becomes the derivative of a product, $d[\mathcal{F}(x)y]/dx$. This happens exactly when

$$\mathcal{F}(x) = \exp\left(\int P(x) dx\right), \quad (4.75)$$

since then $d\mathcal{F}/dx = P(x)\mathcal{F}(x)$, and

$$\frac{d}{dx}[\mathcal{F}(x)y] = \mathcal{F} \frac{dy}{dx} + y \frac{d\mathcal{F}}{dx} = \mathcal{F} \frac{dy}{dx} + \mathcal{F} P y = \mathcal{F} \left(\frac{dy}{dx} + P y \right) = \mathcal{F} Q. \quad (4.76)$$

Once the left-hand side becomes an exact derivative, it suffices to integrate both sides and isolate y . (Note that we use the symbol $\mathcal{F}$ here for the integrating factor, rather than μ , precisely so as not to collide with the direction cosine $\mu = \cos \theta$ already in use in this chapter — this is merely a naming coincidence in the ODE literature.)

Step (c): the integrating factor of our equation. Since $P(\tau) = -1/\mu$ is constant,

$$\mathcal{F}(\tau) = \exp\left(\int \left(-\frac{1}{\mu}\right) d\tau\right) = e^{-\tau/\mu}. \quad (4.77)$$

Watch the sign of the integrating factor

It is essential not to confuse this result with an attempt to use $e^{+\tau}$ as the integrating factor — which is precisely what appears, partially crossed out, in the middle of the original derivation in the handwritten notes (page 44). The standard form of our equation [Equation (4.74)] has $P(\tau) = -1/\mu$, not $P(\tau) = +1$: the correct integrating factor *must* carry exactly this coefficient in the exponent, that is, $e^{-\tau/\mu}$, not e^τ . Substituting $e^{+\tau}$ in place of $e^{-\tau/\mu}$ does not make the left-hand side of the equation become the derivative of a product [the condition of Equation (4.76) no longer holds], and this is precisely why that path did not close algebraically and was abandoned partway through the calculation. We will verify below that $e^{-\tau/\mu}$ exactly reproduces the final result that the professor had already recorded — and this is also the convention adopted in standard reference texts in the field, such as *Radiative Processes in Astrophysics* (Rybicki & Lightman) and Kippenhahn et al. (2012).

Step (d): recognizing the product rule "backwards". Applying Equation (4.76) to our specific case ($\mathcal{F} = e^{-\tau/\mu}$, $y = I_\nu$), we explicitly verify that

$$\begin{aligned} \frac{d}{d\tau} [I_\nu(\tau, \mu) e^{-\tau/\mu}] &= \frac{dI_\nu}{d\tau} e^{-\tau/\mu} + I_\nu \cdot \left(-\frac{1}{\mu}\right) e^{-\tau/\mu} \\ &= e^{-\tau/\mu} \left(\frac{dI_\nu}{d\tau} - \frac{I_\nu}{\mu} \right), \end{aligned} \quad (4.78)$$

which is exactly the product rule used "backwards": instead of differentiating a product to obtain two terms, we recognize that the two terms on the right-hand side of Equation (4.74), when multiplied by $e^{-\tau/\mu}$, recombine into the derivative of a single product.

Step (e): integrating. Multiplying Equation (4.74) by $e^{-\tau/\mu}$ and using Equation (4.78),

$$\begin{aligned} e^{-\tau/\mu} \left(\frac{dI_v}{d\tau} - \frac{I_v}{\mu} \right) &= -\frac{S_v}{\mu} e^{-\tau/\mu} \\ \frac{d}{d\tau} \left[I_v(\tau, \mu) e^{-\tau/\mu} \right] &= -\frac{S_v(\tau)}{\mu} e^{-\tau/\mu}. \end{aligned} \quad (4.79)$$

Integrating both sides from τ_1 to τ_2 (with $\tau_2 > \tau_1$),

$$\begin{aligned} \int_{\tau_1}^{\tau_2} \frac{d}{dt} \left[I_v(t, \mu) e^{-t/\mu} \right] dt &= -\frac{1}{\mu} \int_{\tau_1}^{\tau_2} S_v(t) e^{-t/\mu} dt \\ I_v(\tau_2, \mu) e^{-\tau_2/\mu} - I_v(\tau_1, \mu) e^{-\tau_1/\mu} &= -\frac{1}{\mu} \int_{\tau_1}^{\tau_2} S_v(t) e^{-t/\mu} dt, \end{aligned} \quad (4.80)$$

where we used the Fundamental Theorem of Calculus on the left-hand side. Isolating $I_v(\tau_1, \mu)$,

$$\begin{aligned} I_v(\tau_1, \mu) e^{-\tau_1/\mu} &= I_v(\tau_2, \mu) e^{-\tau_2/\mu} + \frac{1}{\mu} \int_{\tau_1}^{\tau_2} S_v(t) e^{-t/\mu} dt \\ I_v(\tau_1, \mu) &= I_v(\tau_2, \mu) e^{-(\tau_2-\tau_1)/\mu} + \frac{1}{\mu} \int_{\tau_1}^{\tau_2} S_v(t) e^{-(t-\tau_1)/\mu} dt, \end{aligned} \quad (4.81)$$

where, in the last step, we multiplied both sides by $e^{\tau_1/\mu}$ and absorbed the exponential into the integral (since τ_1 does not depend on t).

Step (f): final result. Writing $\tau \equiv \tau_1$ (the point at which we evaluate the intensity), we arrive at the **formal solution** for the radiation intensity in a stellar atmosphere in plane-parallel geometry:

$$I_v(\tau, \mu) = I_v(\tau_2, \mu) e^{-(\tau_2-\tau)/\mu} + \frac{1}{\mu} \int_{\tau_1}^{\tau_2} S_v(t) e^{-(t-\tau)/\mu} dt. \quad (4.82)$$

As an additional consistency check: if $S_v = 0$ (Case 3, absorption only), the integral vanishes and Equation (4.82) reduces to $I_v(\tau_1, \mu) = I_v(\tau_2, \mu) e^{-(\tau_2-\tau_1)/\mu}$ — exactly Equation (4.68), as expected.

The result of Equation (4.82) has a direct physical interpretation: the intensity at τ_1 is the sum of (i) the intensity that entered at τ_2 , attenuated exponentially by its passage through the medium up to τ_1 , with (ii) the contribution of each intermediate layer $t \in [\tau_1, \tau_2]$, emitting according to its local source function $S_v(t)$ and also attenuated exponentially by the stretch of medium between t and τ_1 .

The central equation of every stellar spectral synthesis code

Equation (4.82) is not merely an ODE exercise: it is, literally, the equation that every modern spectral synthesis code solves numerically to predict the emergent spectrum of a star. The real complexity of these codes — TLUSTY (Hubeny and Lanz, 1995), PHOENIX (Husser et al., 2013), and the 3D non-LTE methods reviewed by Bergemann and Hoppe (2025), already mentioned in this chapter — lies almost entirely in how $S_v(t)$ is calculated at each depth: in local thermodynamic equilibrium (LTE), $S_v = B_v[T(t)]$

suffices (see the next section); outside of LTE, the source function depends on the simultaneous solution of statistical equilibrium equations for the atomic populations, which in turn depend on the radiation field I_ν itself — a coupled, nonlinear problem, solved iteratively. But once $S_\nu(t)$ is known at each point, Equation (4.82) (or its generalization to spherical/3D geometries) is always the final step: the formal integral along the line of sight that converts the physical structure of the atmosphere model into an observable spectrum.

Boundary conditions

The formal solution, Equation (4.82), depends on two boundary values, $I_\nu(\tau_2, \mu)$ and the source function $S_\nu(t)$ over the whole interval. We now need to specify these boundary conditions for the cases of physical interest.

Stellar interior ($\tau \rightarrow \infty$). In PP geometry, we consider the boundary condition of an atmosphere bounded between $z_{\max}$ (top, $\tau = 0$) and z_{base} (base, in the stellar interior), as illustrated in Figure 4.12. In studying the interior of the star, we can assume that $\tau \rightarrow \infty$, and that the

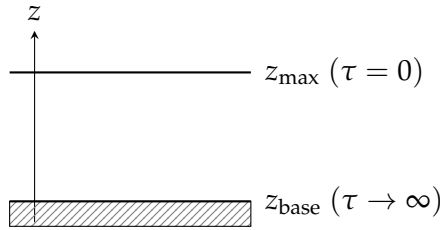


Figure 4.12: General boundary condition in PP geometry: $z_{\max}$ (top of the atmosphere, $\tau = 0$) and z_{base} (base, taken as $\tau \rightarrow \infty$).

intensity I_ν remains finite at these depths (it cannot physically diverge). Hence, we also have:

$$\lim_{\tau \rightarrow \infty} I_\nu(\tau, \mu) e^{-\tau/\mu} \rightarrow 0. \quad (4.83)$$

The first term of the radiative transfer solution [the term $I_\nu(\tau_2, \mu) e^{-(\tau_2 - \tau_1)/\mu}$ of Equation (4.82)] vanishes under these circumstances.

Case where $\mu > 0$ (leaving the star). The geometry of this case is that of Figure 4.13. Thus,

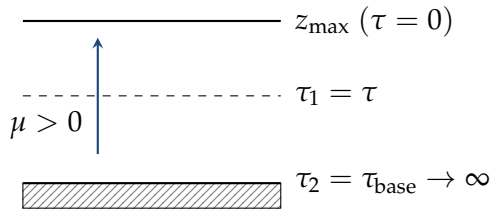


Figure 4.13: Case $\mu > 0$: the beam propagates from $\tau_2 \rightarrow \infty$ (interior) to $\tau_1 = \tau$, toward the surface.

the specific intensity can be represented by (using Equation (4.83) to cancel the first term):

$$I_\nu(\tau, \mu) = 0 + \frac{1}{\mu} \int_\tau^\infty S_\nu(t) e^{-(t-\tau)/\mu} dt, \quad \text{for } \mu > 0. \quad (4.84)$$

Note that, at the top of the atmosphere ($\tau = 0$),

$$I_\nu(0, \mu) = \frac{1}{\mu} \int_0^\infty S_\nu(t) e^{-t/\mu} dt, \quad (4.85)$$

and, if $\mu = 1$,

$$I_\nu(0, 1) = \int_0^\infty S_\nu(t) e^{-t} dt. \quad (4.86)$$

Case where $\mu < 0$ (entering the star). The geometry of this case is that of Figure 4.14. In

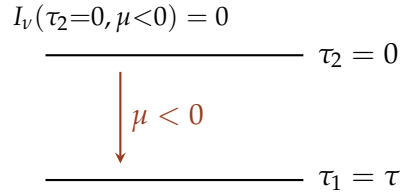


Figure 4.14: Case $\mu < 0$: with no external radiation incident on the star, $I_\nu(\tau_2 = 0, \mu < 0) = 0$ is the natural boundary condition at the top of the atmosphere.

this case, we use $I_\nu(\tau_2 = 0, \mu < 0) = 0$ (there is no radiation incident from outside the star). Hence, the luminous intensity takes the form:

$$I_\nu(\tau, \mu) = \frac{1}{\mu} \int_\tau^0 S_\nu(t) e^{-(t-\tau)/\mu} dt, \quad \text{for } \mu < 0. \quad (4.87)$$

Note that this treatment applies only to the interior of the star. Hence, the radiation field in an atmosphere can be expressed in general, in any direction, as

$$I_\nu(\tau, \mu) = \begin{cases} \frac{1}{\mu} \int_\tau^\infty S_\nu(t) e^{-(t-\tau)/\mu} dt, & \mu > 0, \\ \frac{1}{\mu} \int_\tau^0 S_\nu(t) e^{-(t-\tau)/\mu} dt, & \mu < 0, \end{cases} \quad (4.88)$$

recalling the definition of optical depth given in Equation (4.55). The term τ is what defines the separation between the stellar interior and the atmosphere.

Moments of the radiation field as a function of τ and the Schwarzschild–Milne equations

Recalling the zeroth- and first-order moments we defined earlier [Equation (4.28)], now written explicitly as functions of τ :

$$J_\nu(\tau) = \frac{1}{2} \int_{-1}^1 I_\nu(\tau, \mu) d\mu, \quad H_\nu(\tau) = \frac{1}{2} \int_{-1}^1 I_\nu(\tau, \mu) \mu d\mu. \quad (4.89)$$

Substituting the formal solution $I_\nu(\tau, \mu)$ [Equation (4.88)] into these integrals over μ , we obtain a set of integral equations relating the moments of the radiation field, $J_\nu(\tau)$, $H_\nu(\tau)$ and $K_\nu(\tau)$, to the source function $S_\nu(\tau)$ throughout the atmosphere — the so-called **Schwarzschild–Milne equations**. They are the formal link between the radiation field (via its moments) and the physical structure of the atmosphere (via S_ν , which in turn depends on T , ρ , and the chemical composition) — it is precisely this link that makes it possible, in principle, to solve the structure of a stellar atmosphere self-consistently. Let us now formally derive the Schwarzschild–Milne equations, mentioned above.

Derivation of the Schwarzschild–Milne Equations

Starting point. We have already established the formal solution of the transfer equation for a semi-infinite plane-parallel medium, separated into the two cases of beam direction [Equations (4.84) and (4.87)]:

$$I_\nu(\tau, \mu) = \int_\tau^\infty S_\nu(t) e^{-(t-\tau)/\mu} \frac{dt}{\mu}, \quad \mu > 0, \quad (4.90)$$

$$I_\nu(\tau, \mu) = - \int_0^\tau S_\nu(t) e^{-(t-\tau)/\mu} \frac{dt}{\mu}, \quad \mu < 0. \quad (4.91)$$

Physically, Equation (4.90) describes a ray traveling toward the surface ($\mu > 0$): it carries the sum (weighted by $e^{-(t-\tau)/\mu}$) of everything emitted at depths $t > \tau$, that is, *below* the point of observation. Equation (4.91), in turn, describes a ray traveling into the star ($\mu < 0$): using the boundary condition that there is no external radiation incident at $\tau = 0$, it carries only the contribution of the layers $t < \tau$, that is, *above* the point of observation.

The exponential integral function. To compactify the final result, it is convenient to introduce the family of functions

$$E_n(x) \equiv \int_1^\infty \frac{e^{-xt}}{t^n} dt = \int_0^1 \mu^{n-2} e^{-x/\mu} d\mu, \quad (4.92)$$

known as the **exponential integral function**. Two properties will be useful: the recurrence relation $dE_n/dx = -E_{n-1}(x)$, and the fact that $E_n(x)$ decays monotonically with x — physically, it is this property that controls how "local" the influence of the source function is on the radiation field at distant depths.

Assembling the moments. Starting from the definition of the moments already established [Equation (4.28)], written explicitly as a function of τ ,

$$J_\nu(\tau) = \frac{1}{2} \int_{-1}^1 I_\nu(\tau, \mu) d\mu, \quad H_\nu(\tau) = \frac{1}{2} \int_{-1}^1 \mu I_\nu(\tau, \mu) d\mu, \quad K_\nu(\tau) = \frac{1}{2} \int_{-1}^1 \mu^2 I_\nu(\tau, \mu) d\mu, \quad (4.93)$$

we separate each integral over μ into the two halves $\mu \in (0, 1]$ and $\mu \in [-1, 0)$, and substitute the formal solutions [Equations (4.90) and (4.91)] into each half. Taking $J_\nu(\tau)$ as an example,

$$J_\nu(\tau) = \frac{1}{2} \int_0^1 \left[\int_\tau^\infty S_\nu(t) e^{-(t-\tau)/\mu} \frac{dt}{\mu} \right] d\mu - \frac{1}{2} \int_{-1}^0 \left[\int_0^\tau S_\nu(t) e^{-(t-\tau)/\mu} \frac{dt}{\mu} \right] d\mu. \quad (4.94)$$

The central step of the derivation is to **swap the order of integration** between μ and t : instead of integrating first over t (for fixed μ) and then over μ , we integrate first over μ (for fixed t) and then over t . In the first term, t ranges over (τ, ∞) and, for each t , μ ranges over $(0, 1]$; in the second term, t ranges over $(0, \tau)$ and, for each t , μ ranges over $[-1, 0)$ — the limits of t and μ simply swap roles, without changing the region covered by the double integral:

$$J_\nu(\tau) = \frac{1}{2} \int_\tau^\infty S_\nu(t) \left[\int_0^1 e^{-(t-\tau)/\mu} \frac{d\mu}{\mu} \right] dt - \frac{1}{2} \int_0^\tau S_\nu(t) \left[\int_{-1}^0 e^{-(t-\tau)/\mu} \frac{d\mu}{\mu} \right] dt. \quad (4.95)$$

Solving the integral over μ . In the first term, $t > \tau$, so $(t - \tau) > 0$; with the substitution $\mu \rightarrow \mu$ directly in definition (4.92) (with $n = 1$), $\int_0^1 e^{-(t-\tau)/\mu} d\mu/\mu = E_1(t - \tau)$. In the second term, $t < \tau$, so $(t - \tau) < 0$; making $\mu \rightarrow -\mu$ in the integral (which reverses the limits to $\mu \in (0, 1]$ and flips the sign of $d\mu/\mu$ twice, canceling out), we likewise obtain $\int_{-1}^0 e^{-(t-\tau)/\mu} d\mu/\mu = E_1(\tau - t)$. In both cases the argument of E_1 is the distance $|t - \tau|$, so that

$$J_\nu(\tau) = \frac{1}{2} \int_0^\infty S_\nu(t) E_1(|t - \tau|) dt. \quad (4.96)$$

The same procedure, repeated for $H_\nu(\tau)$ and $K_\nu(\tau)$ — with an additional factor of μ or μ^2 inside the integral, which shifts the index of the resulting exponential integral function from E_1 to E_2 and E_3 , respectively, and introduces a sign flip between the two halves in the case of H_ν (since μ changes sign, while μ^2 does not) —, yields the other two Schwarzschild–Milne equations:

$$H_\nu(\tau) = \frac{1}{2} \int_0^\infty S_\nu(t) \operatorname{sgn}(t - \tau) E_2(|t - \tau|) dt, \quad (4.97)$$

$$K_\nu(\tau) = \frac{1}{2} \int_0^\infty S_\nu(t) E_3(|t - \tau|) dt. \quad (4.98)$$

What the Schwarzschild–Milne equations say physically

Equation (4.96) shows that $J_\nu(\tau)$ is a **weighted average** of the source function S_ν over all optical depths t in the atmosphere, with weight $E_1(|t - \tau|)$. Since E_1 decays rapidly with its argument, the mean radiation field at a given depth is determined mainly by the source function in the nearby vicinity — typically within one optical mean free path ($|t - \tau| \sim 1$) —, and only weakly by what happens at distant depths. This is exactly the physical intuition behind the Diffusion Approximation (Section 4.15), where $S_\nu(t)$ is expanded in a Taylor series around τ : the expansion only makes sense because the influence of S_ν is predominantly local, a property that here appears explicitly through the weight $E_1(|t - \tau|)$.

Exponential integral functions in stellar atmospheres

The formalism of the Schwarzschild–Milne equations, with the exponential integral functions $E_n(x)$ as the weighting kernel, is standard treatment in stellar atmosphere textbooks — see Mihalas (1978), Chapter 2, for the complete derivation and its applications to the numerical solution of radiative transfer, and Hubeny and Mihalas (2014) for a modern and more extensive treatment, including the generalization to atmospheres out of local thermodynamic equilibrium.

Problems and difficulties

Before moving on to a practical approximation, it is worth making explicit why the formal solution, Equation (4.82), is not yet a *usable* solution without further work:

- The source function $S_\nu = \eta_\nu/\chi_\nu$ varies from point to point and with frequency.
- The gas can change with (ρ, T) at each point, including its predominant chemical composition.

That is, $S_\nu(t)$ inside the integral of Equation (4.82) is not, in general, a function known *a priori* — it is part of what we want to determine. This motivates the search for simplifying approximations, such as the one we will see next.

4.12 Eddington-Barbier approximation

To avoid all the complexity of the source function, $S_\nu = \eta_\nu / \chi_\nu$, we can assume a first-degree approximation, where:

$$S_\nu = a + b t, \quad (4.99)$$

with a and b constants. Thus, we have a linear form of the source function with optical depth. Substituting into Equation (4.84) (case $\mu > 0$, at the top of the atmosphere, $\tau = 0$):

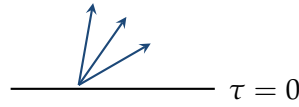


Figure 4.15: Beam emerging from the top of the atmosphere ($\tau = 0$) in different directions μ .

$$I_\nu(0, \mu > 0) = \frac{1}{\mu} \int_0^\infty S_\nu(t) e^{-t/\mu} dt = \frac{1}{\mu} \int_0^\infty a e^{-t/\mu} dt + \frac{1}{\mu} \int_0^\infty b t e^{-t/\mu} dt. \quad (4.100)$$

Calculating the integrals separately. For the first, with the substitution $x = e^{-t/\mu}$ (hence $dx = -\frac{1}{\mu} e^{-t/\mu} dt$):

$$\frac{1}{\mu} \int_0^\infty a e^{-t/\mu} dt = a \int_0^\infty \frac{1}{\mu} e^{-t/\mu} dt = a \left[-e^{-t/\mu} \right]_0^\infty = a (0 - (-1)) = a. \quad (4.101)$$

For the second integral, we use integration by parts, $\int u dv = uv - \int v du$, with

$$u = t, \quad dv = e^{-t/\mu} dt \quad \implies \quad du = dt, \quad v = -\mu e^{-t/\mu}. \quad (4.102)$$

Hence,

$$\begin{aligned} \frac{1}{\mu} \int_0^\infty b t e^{-t/\mu} dt &= \frac{b}{\mu} \left\{ \left[t \cdot (-\mu e^{-t/\mu}) \right]_0^\infty - \int_0^\infty (-\mu e^{-t/\mu}) dt \right\} \\ &= \frac{b}{\mu} \left\{ \underbrace{\left[-\mu t e^{-t/\mu} \right]_0^\infty}_{=0} + \mu \int_0^\infty e^{-t/\mu} dt \right\} \\ &= \frac{b}{\mu} \cdot \mu \cdot \left[-\mu e^{-t/\mu} \right]_0^\infty = \frac{b}{\mu} \cdot \mu \cdot \mu = b \mu, \end{aligned} \quad (4.103)$$

where the boundary term $[-\mu t e^{-t/\mu}]_0^\infty$ vanishes at both limits ($t = 0$ gives zero directly; $t \rightarrow \infty$ gives zero because the exponential decays faster than t grows).

Substituting Equations (4.101) and (4.103) back into Equation (4.100), we obtain

$$\boxed{I_\nu(0, \mu > 0) = a + b \mu}, \quad (4.104)$$

where a and b are the constants of the linear approximation of the source function, Equation (4.99). For $\mu = 0$, we have directly that $I_\nu(0, 0) = a$; for $\mu = 1$, $I_\nu(0, 1) = a + b$.

Eddington–Barbier approximation: when it works

Equation (4.104) is known as the **Eddington–Barbier relation**: it states that the emergent intensity in direction μ is approximately equal to the source function evaluated at the optical depth $\tau = \mu$ (compare $S_\nu(\tau) = a + b\tau$ with $I_\nu(0, \mu) = a + b\mu$ — they are the same expression, with $\tau \rightarrow \mu$). This approximation is good when S_ν varies smoothly (approximately linearly) on a scale of $\Delta\tau \sim 1$, which is reasonable for the stellar continuum in the photosphere, but fails in regions where S_ν varies rapidly with τ (for example, cores of strong spectral lines, or opacity discontinuities).

4.13 Limb darkening

Notice that $\mu = \cos \theta$; with $\theta = 0$, we have $\hat{n} \equiv \hat{s}$ (the line of sight coincides with the normal to the surface) and I_ν is maximum. As μ decreases (that is, θ increases), I_ν also decreases, by Equation (4.104) (assuming $b > 0$, the usual case in which the atmosphere heats inward). Thus, I_ν is maximum at the center of the stellar disk, and decreases as one moves away from the center. This produces **limb darkening**, since the star appears darker toward the edges (near the apparent border of the disk).

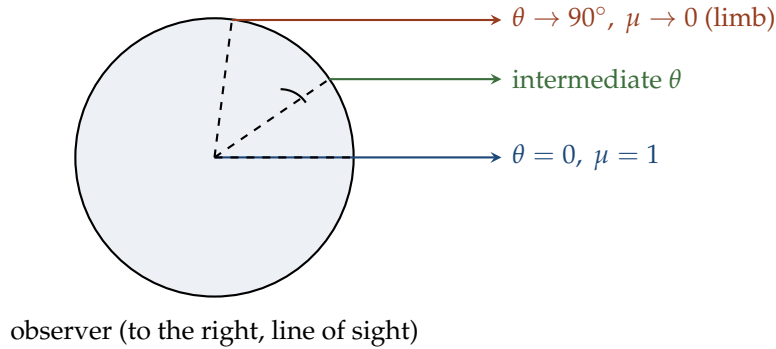


Figure 4.16: Limb darkening: in plane-parallel geometry, outer, cooler layers of the atmosphere dominate the emergent intensity as the angle θ between the line of sight and the local normal increases ($\mu = \cos \theta$ decreases) — that is, as one observes closer and closer to the edge of the stellar disk.

The physical explanation is as follows: by the Eddington–Barbier relation, observing in direction μ is equivalent to “seeing” the source function at optical depth $\tau \approx \mu$. At the center of the disk ($\mu = 1$), we look along the normal and see down to $\tau \approx 1$, a relatively deep and hot layer. Near the limb ($\mu \rightarrow 0$), the same physical depth z corresponds to a much longer oblique path through the gas [since $ds = dz/\mu$, Equation (4.53)], so that the optical depth $\tau = 1$ is reached in a layer much closer to the surface — more external and, as the temperature increases toward the interior of the star, cooler. Since $S_\nu \approx B_\nu(T)$ increases with T (Chapter 2), a cooler layer emits less, which is why the limb appears darker than the center of the disk.

Solar limb darkening and the VAL/FAL semi-empirical models

The Sun is the only case in which limb darkening is observed **directly and spatially resolved** — for any other star, the disk is too small to be resolved, and the effect can only be inferred indirectly (for example, in exoplanet transit light curves or in eclipsing binaries). Historically, it is precisely the careful measurement of $I_\nu(0, \mu)$ as a function of μ across the solar disk, combined with the Eddington–Barbier relation [Equation (4.104)],

that allowed the reconstruction of the temperature profile $T(\tau)$ of the solar photosphere: each observed μ gives $S_\nu \approx B_\nu[T(\tau \approx \mu)]$, and by varying μ (from center to limb) and the observed frequency ν , $T(\tau)$ is mapped layer by layer. This is the logic behind the **semi-empirical models** of the solar atmosphere, of which the most influential are the VAL model, by Vernazza et al. (1981), and its update, the FAL model, by Fontenla et al. (1993) — both still widely used as a reference for the temperature structure of the photosphere and low chromosphere in modern solar physics work (Carlsson et al., 2019).

Limb darkening in the era of exoplanet transit spectroscopy

For any star with a transiting planet, the precise shape of the light curve depends directly on the limb darkening profile of the host star: as the planetary disk crosses regions of the stellar disk with different brightness (brighter at the center, darker near the edge), a poorly characterized limb darkening translates into a systematic bias in the inferred planetary radius. This matters increasingly because **transmission spectroscopy** — measuring the apparent radius of the planet across multiple spectral bands to detect molecular signatures in its atmosphere — requires precision of tens of parts per million, comparable to the magnitude of the errors introduced by imprecise limb darkening itself, especially in narrow optical/near-infrared bands observed by HST/WFC3 and the JWST instruments (NIRSpec, NIRISS, NIRCams, MIRI) (Morello et al., 2017). The functional form chosen to parametrize limb darkening (linear, quadratic, etc.) also introduces wavelength-dependent systematic biases in the extracted transmission spectrum (Coulombe et al., 2024). More surprisingly still: standard theoretical limb darkening models (based on atmospheres such as PHOENIX, ATLAS, and MARCS) predict darkening toward the limb that is systematically stronger than observed in high-precision light curves from Kepler, TESS, and JWST — a discrepancy attributed to the absence of small-scale magnetic fields in the models, and which remains an active area of research (Kostogryz et al., 2024). In observational practice, limb darkening coefficients for planning JWST proposals are obtained with tools such as the *ExoCTK Limb Darkening Calculator*, maintained by STScI (Space Telescope Science Institute, 2023). In all cases, the underlying physics is exactly the same as derived in this chapter — source function, optical depth, solution of the transfer equation —, but instead of a simple analytical approximation like Eddington–Barbier, modern exoplanetary science requires high-precision numerical atmosphere models.

4.14 Source function and thermodynamic equilibrium

Stating that a system is in thermodynamic equilibrium implies, as one of its consequences, that the rate of absorption of a gas must equal the rate of emission. Thus, recalling that we defined

$$dE = \chi_\nu I_\nu dt dV d\omega d\nu \quad \text{and} \quad dE = \eta_\nu dt dV d\omega d\nu, \quad (4.105)$$

we have that the energy variation must be conserved, where:

$$\left(\frac{\Delta E}{\Delta t \Delta V \Delta \omega \Delta \nu} \right)_{\text{abs}} \equiv \left(\frac{\Delta E}{\Delta t \Delta V \Delta \omega \Delta \nu} \right)_{\text{emi}}. \quad (4.106)$$

Also, if the absorption and emission rates are similar, the radiation flux must be zero, $\vec{F} = 0$, that is, emission and absorption are balanced.

As we saw earlier, in thermodynamic equilibrium, $I_\nu = B_\nu$, but $I_\nu \chi_\nu = \eta_\nu$, hence $I_\nu = \eta_\nu / \chi_\nu$. Thus:

$$B_\nu = \frac{\eta_\nu}{\chi_\nu} = S_\nu \implies \boxed{B_\nu = S_\nu}, \quad \text{valid in thermodynamic equilibrium.} \quad (4.107)$$

4.15 Diffusion Approximation

As we move toward the center (interior) of the star, we know intuitively that the density, pressure, and temperature increase, where ($\rho \uparrow$, $T \uparrow$, $P \uparrow$). Thus, the opacity of the medium will be larger, which implies that the photon mean free path is much shorter than in the atmosphere:

$$l_\gamma = \frac{1}{\chi_\nu} = \frac{1}{\kappa \rho}. \quad (4.108)$$

For comparison with solar values, we have

$$\rho = \rho_\odot = \frac{M_\odot}{V_\odot} = \frac{M_\odot}{\frac{4}{3}\pi R_\odot^3} = 1.4 \text{ g/cm}^3. \quad (4.109)$$

With κ the scattering cross section, described by Thomson electron scattering, we have $\kappa = 0.4 \text{ cm}^2 \text{ g}^{-1}$, so

$$l_\gamma = \frac{1}{0.4 \cdot 1.4} \sim 2 \text{ cm}. \quad (4.110)$$

That is, the photon has a mean free path of order 2 cm. Every 2 cm it is absorbed or re-emitted in an arbitrary direction.

We therefore have the following considerations:

- (i) the radiation field is in local thermodynamic equilibrium;
- (ii) $S_\nu = B_\nu = I_\nu$.

Recalling that, from Equation (4.84), we can describe the radiation field in the stellar interior as

$$I_\nu(\tau, \mu) = \frac{1}{\mu} \int_\tau^\infty S_\nu(t) e^{-(t-\tau)/\mu} dt, \quad \text{for } \mu > 0. \quad (4.111)$$

To obtain the diffusion approximation, we can use the Taylor series, recalling that, by definition,

$$f(x) = \sum_{n=0}^{\infty} \frac{d^n f(x)}{dx^n} \bigg|_{x_0} \frac{(x - x_0)^n}{n!}. \quad (4.112)$$

Considering that our approximation is made for $S_\nu(t)$ in an opaque medium, where $\tau \gg 1$, we have

$$S_\nu(t) = \sum_{n=0}^{\infty} \frac{d^n B_\nu}{dt^n} \bigg|_{\tau \gg 1} \frac{(t - \tau)^n}{n!}, \quad (4.113)$$

where we have already used $S_\nu \rightarrow B_\nu$ [item (ii) above]. Substituting Equation (4.113) into Equation (4.111), we have:

$$I_\nu(\tau, \mu) = \frac{1}{\mu} \int_\tau^\infty \sum_{n=0}^{\infty} \frac{d^n B_\nu}{dt^n} \bigg|_{\tau} \frac{(t - \tau)^n}{n!} e^{-(t-\tau)/\mu} dt. \quad (4.114)$$

Solving the integral. Analyzing the integral over t : let $x = (t - \tau)/\mu$, so $dx = dt/\mu$. Then

$$\int_0^\infty x^n \mu^n e^{-x} \mu dx = \mu^{n+1} \int_0^\infty x^n e^{-x} dx. \quad (4.115)$$

But the integral $\int_0^\infty x^n e^{-x} dx$ can be obtained from tables.

$$\int_0^\infty e^{-x} x^n dx = \Gamma(n+1) = n!, \quad n \in \mathbb{Z}_0^+. \quad (4.116)$$

Substituting into Equation (4.114), we have:

$$I_\nu(\tau, \mu) = \frac{1}{\mu} \sum_{n=0}^\infty \frac{d^n B_\nu}{dt^n} \Big|_\tau \mu^{n+1} \cdot \frac{1}{n!} \cdot n!, \quad (4.117)$$

that is,

$$I_\nu(\tau, \mu) = \sum_{n=0}^\infty \mu^n \frac{d^n B_\nu}{dt^n} \Big|_\tau. \quad (4.118)$$

First-order approximation

For an approximation with the first two terms of the series, we have:

$$I_\nu(\tau, \mu) \cong \underbrace{B_\nu(T)}_{n=0} + \underbrace{\mu \frac{dB_\nu}{dt} \Big|_\tau}_{n=1}. \quad (4.119)$$

The second term is the **anisotropy term**. It is essential: if $I_\nu = B_\nu$ exactly (isotropic, with no dependence on μ), the flux $F_\nu = \oint I_\nu \hat{n} d\omega$ would necessarily be zero — the contribution from each direction $\hat{n}$ would be exactly canceled by the opposite direction, $-\hat{n}$. This cannot be true for a real star: we know that $F \neq 0$ (it is, after all, the very energy flux that the star radiates!). The term $\mu dB_\nu/dt$ is, therefore, the small but essential correction that breaks the perfect isotropy — and it is this term, not the isotropic term $B_\nu(T)$, that carries all the information about the effective transport of radiative energy out of the star.

4.16 Mean Intensity and Radiation Flux

As defined in previous lectures, the mean intensity and the radiation flux are defined as the zeroth- and first-order moments of the radiation field [Equation (4.28)]. For J_ν , we have:

$$\begin{aligned} J_\nu(\tau) &= \frac{1}{2} \int_{-1}^1 I_\nu(\tau, \mu) d\mu = \frac{1}{2} \int_{-1}^1 \sum_{n=0}^\infty \mu^n \frac{d^n B_\nu}{dt^n} \Big|_\tau d\mu \\ &= \frac{1}{2} \sum_{n=0}^\infty \frac{d^n B_\nu}{dt^n} \Big|_\tau \int_{-1}^1 \mu^n d\mu. \end{aligned} \quad (4.120)$$

Solving the integral over μ ,

$$J_\nu(\tau) = \frac{1}{2} \sum_{n=0}^\infty \frac{d^n B_\nu}{dt^n} \Big|_\tau \frac{\mu^{n+1}}{n+1} \Big|_{-1}^1 = \frac{1}{2} \sum_{n=0}^\infty \frac{d^n B_\nu}{dt^n} \Big|_\tau \frac{1}{n+1} (1^{n+1} - (-1)^{n+1}). \quad (4.121)$$

Note that, when n is odd, $n + 1$ is even, and $1^{n+1} - (-1)^{n+1} = 1 - 1 = 0$: the term vanishes. Renaming $n \rightarrow 2n$ to keep only the surviving even terms, we obtain the mean intensity as a function of the Planck function:

$$J_\nu(\tau) = \sum_{n=0}^{\infty} \frac{d^{2n} B_\nu}{dt^{2n}} \Big|_\tau \frac{1}{2n+1}. \quad (4.122)$$

Truncating Equation (4.122) at the first nonzero correction, we obtain the diffusion approximation for J_ν :

$$J_\nu(\tau) \cong B_\nu(T) + \frac{1}{3} \frac{d^2 B_\nu}{dt^2} \Big|_{\tau \gg 1}. \quad (4.123)$$

Similarly, we can obtain a relation for the first-order moment, that is, the radiative energy flux. Hence:

$$\begin{aligned} H_\nu(\tau) &= \frac{1}{2} \int_{-1}^1 I_\nu(\tau, \mu) \mu d\mu = \frac{1}{2} \sum_{n=0}^{\infty} \frac{d^n B_\nu}{dt^n} \Big|_\tau \int_{-1}^1 \mu^{n+1} d\mu \\ &= \frac{1}{2} \sum_{n=0}^{\infty} \frac{d^n B_\nu}{dt^n} \Big|_\tau \frac{\mu^{n+2}}{n+2} \Big|_{-1}^1 = \frac{1}{2} \sum_{n=0}^{\infty} \frac{d^n B_\nu}{dt^n} \Big|_\tau \frac{1}{n+2} (1^{n+2} - (-1)^{n+2}). \end{aligned} \quad (4.124)$$

Now, it is when n is *even* that $n + 2$ is even and the term vanishes; the odd-order terms survive. Renaming $n \rightarrow 2n + 1$,

$$H_\nu(\tau) = \sum_{n=0}^{\infty} \frac{1}{2n+3} \frac{d^{2n+1} B_\nu}{dt^{2n+1}} \Big|_\tau. \quad (4.125)$$

In this way, the order-0 approximation (the first term of this sum, $n = 0$) is already sufficient for most practical cases, where:

$$H_\nu(\tau) \cong \frac{1}{3} \frac{dB_\nu}{dt} \Big|_{\tau \gg 1}. \quad (4.126)$$

This is the radiation flux in the stellar interior, that is, the net energy leaving the star. We thus see that I_ν , J_ν and H_ν describe the stellar interior well.

By choosing $\tau \gg 1$, we are studying regions of greater optical depth, that is, deeper inside the star. Thus, studying H_ν means studying the transport of energy from the innermost layers to the outermost layers of stars.

4.17 Rosseland Approximation (Equation)

Integrating Equation (4.126) over frequency, we obtain:

$$H = \int_0^\infty \frac{1}{3} \frac{dB_\nu}{dt} \Big|_\tau d\nu = \frac{1}{3} \frac{d}{dt} \int_0^\infty B_\nu d\nu. \quad (4.127)$$

$$\int_0^\infty B_\nu(T) d\nu = \frac{\sigma T^4}{\pi}, \quad (4.128)$$

so

$$H = \frac{1}{3} \frac{d}{dt} \left(\frac{\sigma T^4}{\pi} \right) \Big|_\tau = \frac{4\sigma T^3}{3\pi} \frac{dT}{dt} \Big|_\tau. \quad (4.129)$$

Since H is the Eddington flux, defined by $H = F/(4\pi)$ [Equation (4.29), now integrated over frequency], and recalling that the optical depth is $d\tau = -\chi_\nu dr$, which in turn is $\chi = \kappa\rho$ [Equation (4.39)], we have, by the chain rule,

$$\left. \frac{d}{dt} \right|_\tau \equiv \frac{d}{d\tau} = -\frac{1}{\chi} \frac{d}{dr} = -\frac{1}{\kappa\rho} \frac{d}{dr}. \quad (4.130)$$

Substituting into Equation (4.129) and using $H = F/(4\pi)$:

$$\frac{F}{4\pi} = -\frac{4\sigma T^3}{3\pi\kappa\rho} \frac{dT}{dr} \implies \boxed{F = -\frac{16\sigma T^3}{3\kappa\rho} \frac{dT}{dr}}. \quad (4.131)$$

This result exactly matches the final value recorded in the handwritten notes, and is also the standard form found in the literature (e.g. Kippenhahn & Weigert, radiative transport chapter) — the two factors of π (one from the Planck integration, the other from the relation $H = F/4\pi$) cancel exactly in the correct calculation.

This relation is known as the **Rosseland Approximation** for the energy flux. It is commonly called the **radiative diffusion equation**. Using $\chi = \kappa\rho$, an equivalent form of Equation (4.131) is:

$$\boxed{F = -\frac{16\sigma T^3}{3\chi} \frac{dT}{dr}}. \quad (4.132)$$

Thus, the flux equation can be described, in a physical sense, as a diffusion flux analogous to **Fourier's law** of heat conduction, $F \propto \nabla T$ — with the role of the "thermal conductivity" played by the coefficient $16\sigma T^3/(3\kappa\rho)$.

4.18 Eddington Luminosity

As we have seen, a star cannot be considered a perfect blackbody for several reasons, but mainly because its radiation flux is nonzero. The **Eddington Luminosity** describes a maximum luminosity limit that a body (e.g., a star) can attain while equilibrium exists between the radiation force (acting from inside the star outward) and the gravitational force (acting from outside inward).

We saw that the absorption coefficient, or the opacity of any gas, is defined by [Equation (4.35)]:

$$dE = \chi_\nu I_\nu dA ds dt d\nu d\omega. \quad (4.133)$$

In absorption processes, there is momentum and energy transport from the radiation field (photons) to matter (gases). Thus, we can use the relativistic energy-momentum relation,

$$E^2 = (m_0 c^2)^2 + (pc)^2, \quad (4.134)$$

which, for the photon ($m_\gamma = 0$), reduces to

$$E = pc, \quad (4.135)$$

which, for an infinitesimal element, gives $dE = c dp$. Substituting into Equation (4.133),

$$c dp = \chi_\nu I_\nu dA ds dt d\nu d\omega. \quad (4.136)$$

Since $\vec{F} = d\vec{p}/dt$ (force),

$$\frac{d\vec{p}}{dt} = \vec{F} = \frac{1}{c} \chi_\nu I_\nu dV d\nu d\omega, \quad (4.137)$$

where we used $dA ds = dV$ (volume element). This is the **radiative force**.

We can define the **radiative force per unit volume**,

$$\frac{d\vec{p}}{dt dV} \equiv \vec{f}. \quad (4.138)$$

Hence, the volumetric force element that depends on ν and ω is

$$\frac{d\vec{f}}{d\nu d\omega} = \frac{1}{c} \chi_\nu I_\nu \hat{n}. \quad (4.139)$$

The term $\hat{n}$ must appear in this step because of the premise that $d\vec{p} = \hat{n} dE/c$ — momentum is a vector quantity.

From Equation (4.139), we can express the volumetric radiative force by:

$$\vec{F}_{\text{rad}}(\vec{r}, t) = \frac{1}{c} \int_0^\infty \chi_\nu \left(\oint I_\nu \hat{n} d\omega \right) d\nu = \frac{1}{c} \int_0^\infty \chi_\nu \vec{F}_\nu d\nu, \quad (4.140)$$

where we recognize $\oint I_\nu \hat{n} d\omega \equiv \vec{F}_\nu$ — the very definition of flux F_ν already established in this chapter — and thus obtain the volumetric radiative force for any gas.

Question: can the radiative force balance, or even exceed, the gravitational force? Yes! At the Eddington limit.

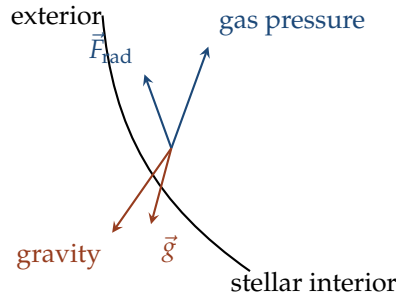


Figure 4.17: Force balance on a gas element in the star: the gas pressure and the radiative force $\vec{F}_{\text{rad}}$ (outward) balance the gravitational force $\vec{g}$ (inward). At the Eddington limit, the radiative force alone equals the gravitational force.

We can assume that we have a pure hydrogen gas and that the opacity coefficient does not depend on frequency (we have only Thomson scattering), so $\chi_\nu \equiv \chi$. Equation (4.140) takes the form

$$\vec{F}_{\text{rad}} = \frac{\chi}{c} \int_0^\infty \vec{F}_\nu d\nu, \quad (4.141)$$

which is the definition of bolometric flux, over all frequencies. Hence:

$$\boxed{\vec{F}_{\text{rad}} = \frac{\chi}{c} \vec{F}}. \quad (4.142)$$

For there to be equilibrium with the gravitational force, we have $F_{\text{rad}} \equiv F_g$, where F_g is the volumetric gravitational force ($F_g = \rho g = GM\rho/r^2$, with $\rho = m/V$). Hence:

$$\frac{\chi F}{c} = \frac{GM\rho}{r^2}. \quad (4.143)$$

Knowing that $F = L/(4\pi r^2)$ (bolometric flux at a distance r from a source of luminosity L), we have:

$$\frac{\chi}{c} \cdot \frac{L}{4\pi r^2} = \frac{GM\rho}{r^2} \implies L = \frac{4\pi GM\rho c}{\chi}. \quad (4.144)$$

As defined earlier [Equation (4.38)], $\chi = n_H\sigma_T$ (for pure, fully ionized hydrogen, with σ_T the Thomson scattering cross section). The density can also be expressed as $\rho = n_H m_H$. Substituting these values into Equation (4.144), we have:

$$L_{\text{Edd}} = \frac{4\pi GM n_H m_H c}{n_H \sigma_T} \implies L_{\text{Edd}} = \frac{4\pi GM m_H c}{\sigma_T}. \quad (4.145)$$

Under these assumptions, Equation (4.145) describes the luminosity limit at which radiation and gravitational force are in equilibrium. Note that it is exactly equivalent to the general form already presented earlier in this chapter (context box right after the definition of the Eddington flux H_ν), $L_{\text{Edd}} = 4\pi GMc/\kappa_{\text{es}}$: one simply needs to identify the specific Thomson-scattering opacity per ionized hydrogen, $\kappa_{\text{es}} = \sigma_T/m_H$ (cross section per unit mass).

We can calculate, for example, the Eddington luminosity as a function of the solar luminosity, where:

$$\frac{L_{\text{Edd}}}{L_\odot} = \frac{1}{L_\odot} \cdot \frac{4\pi GM m_H c}{\sigma_T} \cdot \frac{M_\odot}{M_\odot} = \left(\frac{M_\odot}{L_\odot}\right) \frac{4\pi G m_H c}{\sigma_T} \left(\frac{M}{M_\odot}\right), \quad (4.146)$$

that is,

$$\left[\frac{L_{\text{Edd}}}{L_\odot} = \alpha \left(\frac{M}{M_\odot}\right)\right], \quad \alpha \equiv \left(\frac{M_\odot}{L_\odot}\right) \frac{4\pi G m_H c}{\sigma_T}. \quad (4.147)$$

The numerical value of α

Substituting the values in CGS units ($G = 6.674 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}$, $m_H = 1.673 \times 10^{-24} \text{ g}$, $c = 2.998 \times 10^{10} \text{ cm/s}$, $\sigma_T = 6.652 \times 10^{-25} \text{ cm}^2$, $M_\odot = 1.989 \times 10^{33} \text{ g}$, $L_\odot = 3.828 \times 10^{33} \text{ erg/s}$), we obtain

$$\alpha = \left(\frac{M_\odot}{L_\odot}\right) \frac{4\pi G m_H c}{\sigma_T} \approx 3.3 \times 10^4, \quad (4.148)$$

in excellent agreement with the standard value cited in the literature, $L_{\text{Edd}} \approx 3.2\text{--}3.3 \times 10^4 (M/M_\odot) L_\odot$ (Kippenhahn et al., 2012).

This radiative force is one of the components that give stars their stability.

4.19 Astrophysical Applications

We close this chapter by connecting its two central results — the diffusion/Rosseland approximation and the Eddington luminosity — to real, observable astrophysical phenomena, from the solar interior to supermassive black holes.

From the Rosseland equation to the opacity tables of modern codes

Equation (4.131) is not a result of merely historical interest: it is, literally, the equation that **every** modern stellar evolution code (MESA, GARSTEC, among others) uses to calculate the transport of radiative energy in the interior of a star, layer by layer, throughout its entire evolution. In practice, however, κ is not a single simple function of (ρ, T) : it is the result of summing the absorption and scattering of *all* relevant frequencies,

weighted in a specific way (the so-called **Rosseland mean**, which gives more weight to the frequencies where the gas is more transparent — exactly the ones that dominate the energy transport). Calculating this average for all relevant combinations of temperature, density, and chemical composition is the task of large computational projects: the **OPAL** tables (Iglesias and Rogers, 1996) and those of the **Opacity Project**, already mentioned in this chapter, are today read directly as input tables by codes such as MESA (Jermyn et al., 2023), making Equation (4.131) operational at every step of a stellar evolution simulation.

Near-Eddington winds: Wolf-Rayet stars and luminous blue variables

Very massive stars ($M \gtrsim 50 M_{\odot}$) can have luminosities that approach a substantial fraction of L_{Edd} [Equation (4.145)], which drastically reduces the effective gravity in their outer layers and drives very intense, dense stellar winds — the observational signature of **Wolf-Rayet** stars. An even more extreme case is that of **luminous blue variables** (LBVs), such as η Carinae: these stars undergo eruptive **super-Eddington** episodes, in which the luminosity momentarily exceeds L_{Edd} and the excess matter is physically ejected from the star (the historical Great Eruption of η Carinae, in the 19th century, is the most studied example of this phenomenon). The exact role of the Eddington limit — and whether episodes like this are better explained by instantaneous proximity to L_{Edd} or by other mechanisms, such as mergers in binary systems — remains an active area of research (Smith, 2025).

Super-Eddington accretion: X-ray binaries and ultraluminous sources

The Eddington limit also regulates how fast compact objects — neutron stars, stellar-mass black holes, and supermassive black holes in active galactic nuclei — can grow by mass accretion: accretion rates that would produce $L > L_{\text{Edd}}$ tend to be self-limiting, since the excess radiation pushes away the very matter feeding the object. However, it has been found that this limit can be exceeded (or apparently exceeded) under certain conditions: **ultraluminous X-ray sources** (ULXs) exhibit X-ray luminosities that, if emitted isotropically, would exceed L_{Edd} for a stellar-mass black hole by factors of 10–100. The currently most accepted explanation is genuine **super-Eddington accretion**: very high accretion rates inflated in a thick disk, which collimates the radiation into a narrow cone (*beaming*) toward the observer — a scenario reinforced by the discovery of coherent pulsations in some ULXs, revealing that the accreting object can be a strongly magnetized neutron star, rather than a black hole (Kaaret et al., 2017).

Back to the solar photon diffusion time

At the beginning of this chapter, when discussing the photon mean free path in the stellar interior, we saw that a photon produced in the solar core takes $\sim 1.7 \times 10^5$ years to escape by diffusion, despite the final straight-line path lasting only minutes (Mitalas and Sills, 1992). We now finally have the tool that makes this number quantitatively calculable: Equation (4.131) is exactly the equation that, integrated layer by layer along the profile $T(r)$, $\rho(r)$, and $\kappa(r)$ of a complete solar model, allows the calculation of the radiative diffusion time from center to surface. It is this same equation — the Rosseland approximation, built in this chapter from the Taylor series expansion of the source function in an optically thick medium — that connects the small mean free path of $l_{\gamma} \sim 2$ cm [Equation (4.110)] to the timescale, of hundreds of thousands of years, that governs the

thermal evolution of the entire solar interior.

Radiative processes: references for further study

This chapter covered the theory of radiative transfer from the definition of specific intensity to the diffusion approximation and the Eddington luminosity — the theoretical basis on which all the physics of stellar interiors and atmospheres is built. For a complete and systematic treatment of radiative transfer, radiative equilibrium, and the Rosseland approximation in the context of stellar structure, the standard reference remains Kippenhahn et al. (2012). The Rosseland opacity tables that make Equation (4.131) usable in practice are calculated and maintained by the **OPAL** (Iglesias and Rogers, 1996) and **Opacity Project** projects, both still in active use in modern stellar evolution codes such as MESA (Jermyn et al., 2023). For the physics of super-Eddington accretion and its observational signatures in ultraluminous X-ray sources, see the review by Kaaret et al. (2017); for the phenomenology of stars near the Eddington limit at the other extreme of stellar mass — the winds and eruptions of luminous blue variables such as η Carinae — see Smith (2025).

Chapter 5

Stellar Structure Equations

5.1 Introduction

The simplest discussion of a star relies on a few approximations:

- **Spherical:** no flattening or shape distortions.
- **Isolated:** no mass exchange, tidal effects, etc.
- **Non-rotating:** no deviations from spherical symmetry (1D), no rotation-related instabilities.
- **No magnetic fields.**
- **No mass loss:** no stellar winds.

When these approximations break down

Each of these five approximations is violated by entire classes of real stars — and it is precisely where they break down that much of the interesting physics emerges:

- **Isolated:** close binary systems, especially interacting or contact binaries, exchange mass through the Roche lobe and experience significant tidal effects, making the evolution of each star dependent on its companion.
- **No magnetic fields:** Ap/Bp stars have strong, organized surface magnetic fields (tens to thousands of gauss), which alter the diffusion of chemical elements and the structure of the atmosphere; at the opposite extreme, magnetars — compact remnants of massive stars — have fields of up to 10^{15} G, many orders of magnitude beyond any main-sequence star.
- **Non-rotating:** Be stars and other rapid rotators spin close to the critical (centrifugal breakup) velocity, which flattens the star, induces meridional circulation, and can form circumstellar disks — none of these effects is captured by a spherical 1D model.
- **No mass loss:** stars with very intense winds, such as Wolf-Rayet stars, lose mass at rates that significantly alter their evolution; not by coincidence, these same stars have luminosities close to the Eddington limit L_{Edd} (Section 4.18 of Chapter 4), which is exactly what drives such strong radiative winds.

Models that relax each of these approximations (rotation, magnetic fields, mass loss, binary effects) are necessary extensions of the 1D formalism presented here, and remain active areas of research in stellar physics.

We will show that the internal structure of stars can be studied through 5 coupled nonlinear differential equations (4 + 1), where the main parameters are:

ρ	density
P	pressure
T	temperature
l	luminosity
X_i	mass fraction of each chemical element

The first four equations describe, respectively, mass, hydrostatic equilibrium, energy transport, and energy generation; the fifth (the “+1”) describes the evolution of the chemical composition of the gas over time, via nuclear reactions. In this chapter, we begin with the **mass equation** and **hydrostatic equilibrium**; the others will be developed in future parts.

5.2 Mass equation

Initially, let us assume that only two forces act on stars: the gravitational force and the pressure force.

Considering that a star is spherical, we have the geometry shown in Figure 5.1. The mass

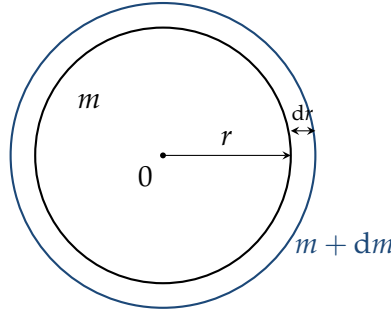


Figure 5.1: Spherical shell of thickness dr at a distance r from the center of the star: the mass contained up to r is m , and the mass contained up to $r + dr$ is $m + dm$.

contained up to a radius r is, in general, also a function of time, $m(r, t)$. Hence, by partial derivatives, we can describe the term dm as:

$$dm = \left(\frac{\partial m}{\partial r} \right)_t dr + \left(\frac{\partial m}{\partial t} \right)_r dt, \quad (5.1)$$

where we have one term constant in time (depending only on r) and another constant in r (depending only on t).

Recalling that $\rho = m/V$, for an infinitesimal element we have

$$dm = \rho(r) dV, \quad \text{but knowing that } dV = 4\pi r^2 dr, \quad (5.2)$$

we get:

$$dm = 4\pi r^2 \rho(r) dr, \quad \text{or} \quad m = \int_0^r 4\pi r'^2 \rho(r') dr'. \quad (5.3)$$

Comparing Equation (5.3) with Equation (5.1), we obtain:

$$\boxed{\left(\frac{\partial m}{\partial r} \right)_t = 4\pi r^2 \rho(r)}, \quad \text{with } t \text{ constant.} \quad (5.4)$$

This is the **mass equation** proper — the first of the 5 stellar structure equations.

Since (Equation (5.3)) $dm = 4\pi r^2 \rho(r) dr$, and $v = dr/dt$ is the radial velocity of the gas, we have $dm = 4\pi r^2 \rho(r) v dt$, hence:

$$\left(\frac{\partial m}{\partial t}\right)_r = -4\pi r^2 \rho v. \quad (5.5)$$

A negative sign appears because the mass flux has a direction pointing radially outward from the star: if $v > 0$ (outward flow), the mass contained within a *fixed* radius r decreases with time — the material is, literally, crossing the boundary of the shell under consideration and leaving it.

Thus, we can write Equation (5.1) as:

$$dm = 4\pi r^2 \rho dr - 4\pi r^2 \rho v dt. \quad (5.6)$$

Recalling the property of partial derivatives whereby $\partial^2 m / \partial r \partial t = \partial^2 m / \partial t \partial r$ (mixed partial derivatives commute), we have:

$$\frac{\partial}{\partial r} \left(-4\pi r^2 \rho v \right)_t = \frac{\partial}{\partial t} \left(4\pi r^2 \rho \right)_r. \quad (5.7)$$

Since the radius does not depend on time (it is merely the radial coordinate), we can assume that r has no temporal dependence, hence:

$$r^2 \frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial r} (r^2 \rho v), \quad (5.8)$$

that is,

$$\boxed{\frac{\partial \rho}{\partial t} = -\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \rho v)}, \quad (5.9)$$

which is the **continuity equation**.

Assuming spherical symmetry, we obtain the well-known hydrodynamic continuity equation, in its general vector form:

$$\boxed{\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \vec{v})}. \quad (5.10)$$

Continuity beyond “no mass loss”

In this chapter, we adopt the approximation of a star with no mass loss ($v = 0$ at the outer boundary), which makes the structure stationary in time. But Equation (5.10), in its complete form, already contains everything needed to describe phases of stellar evolution in which mass loss is *not* negligible: stellar winds throughout the main sequence, and — far more dramatically — the final phases of red giants and the asymptotic giant branch (AGB), where mass loss rates of 10^{-7} to $10^{-4} M_\odot/\text{yr}$ remove a large fraction of the star’s envelope within a few million years, eventually exposing the degenerate core that will become a white dwarf. The same formalism used here for a static star generalizes directly to these cases — one simply need not impose $v = 0$.

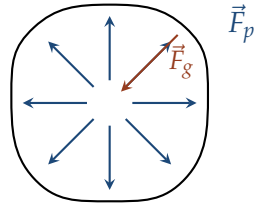


Figure 5.2: Self-gravitating gas cloud: when the gravitational force $\vec{F}_g$ overcomes the gas pressure force $\vec{F}_p$, the cloud collapses — the onset of Jeans instability.

5.3 Gravitational field

Let us recall what a star is. A star begins to be born when there is enough gas in a region where the gravitational force is able to aggregate matter. In physical terms, the gravitational force exceeds the gas pressure force, as in Figure 5.2.

This instability is known as **Jeans instability**. Once the mass of the system equals the Jeans mass, the instability begins, also triggering the star formation process.

Once the gas compresses, accreting more matter through the increase in density and the action of the gravitational force, hydrogen fusion into helium may begin, generating energy in the core (locally) and causing the pressure to rise again.

The Jeans mass and star formation

The Jeans instability condition can be expressed, in order of magnitude, by comparing the gravitational energy with the thermal energy of a spherical gas cloud. The result — the **Jeans mass** — is

$$M_J \sim \left(\frac{5kT}{G\mu m_H} \right)^{3/2} \left(\frac{3}{4\pi\rho} \right)^{1/2}, \quad (5.11)$$

where μ is the mean molecular weight of the gas, m_H the mass of the hydrogen atom, T the temperature, and ρ the density of the cloud. Clouds with $M > M_J$ are gravitationally unstable and collapse.

For a typical cold, dense molecular cloud, with $T \sim 10$ K, composition dominated by molecular H_2 ($\mu \approx 2.3$), and number density $n \sim 10^2$ – 10^4 cm^{-3} ($\rho = n\mu m_H$), we obtain:

- for $n = 10^2$ cm^{-3} ($\rho \approx 3.8 \times 10^{-22}$ g/cm^3): $M_J \sim$ a few hundred $M_\odot$;
- for $n = 10^4$ cm^{-3} (dense, prestellar cores, $\rho \approx 3.8 \times 10^{-20}$ g/cm^3): $M_J \sim$ a few tens of $M_\odot$.

In both cases, the result is of order **solar masses to a few tens/hundreds of solar masses** — consistent with the fact that giant molecular clouds (10^4 – 10^6 $M_\odot$) do not collapse all at once, but fragment hierarchically into smaller cores, each of order M_J , which give rise to individual stars or small clusters. This fragmentation is one of the central physical ingredients behind the shape of the **initial mass function** (IMF) — the statistical distribution of masses with which stars are born — as discussed in the recent review by Hennebelle and Grudić (2024).

At the moment when the pressure force is balanced by the gravitational force, we define that **hydrostatic equilibrium** exists, and from this moment the star enters the main sequence of the HR diagram:

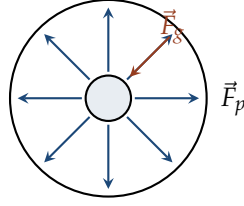


Figure 5.3: Star in hydrostatic equilibrium: the pressure force $\vec{F}_p$ (outward, in every direction) exactly balances the gravitational force $\vec{F}_g$ (inward) in each layer — $F_g = F_p$.

$$F_g = F_p \implies \text{Hydrostatic equilibrium.} \quad (5.12)$$

We know that we can obtain the gravitational force from Newton's law, where:

$$d\vec{F}_g = dm \vec{g} = -\frac{Gm dm}{r^2} \hat{r}, \quad (5.13)$$

or, the volumetric force,

$$\vec{f}_g = \frac{d\vec{F}_g}{dV} = -\frac{Gm\rho}{r^2} \hat{r}, \quad (5.14)$$

where $m = m(r)$ is the mass contained up to radius r [Equation (5.3)].

Hydrostatic equilibrium and the Virial Theorem

The hydrostatic equilibrium condition, $F_g = F_p$, is the local (layer-by-layer) version of a broader, global result: the **Virial Theorem**, which for a self-gravitating system in equilibrium establishes $2K + U = 0$, where K is the total kinetic energy (thermal, in the case of a gas) and U is the total gravitational potential energy. We already mentioned this theorem in Chapter 3 when discussing the $\Gamma_1 = 4/3$ threshold of dynamical stability. The violation of hydrostatic equilibrium — when $F_g \neq F_p$ in some layer — is precisely what characterizes the *dynamic* (non-static) phases of stellar evolution: the collapse of the iron core in a core-collapse supernova (where $F_g \gg F_p$ suddenly), the radial pulsations of Cepheid and RR Lyrae variable stars (oscillations around equilibrium), and the Kelvin–Helmholtz contraction itself during star formation, before nuclear fusion is established. In all these cases, it is the *imbalance* between F_g and F_p that governs the dynamics, and the characteristic timescale of this response is precisely the subject of Section 5.6, next.

5.4 Hydrostatic equilibrium

We can obtain an expression for the pressure force acting on a star by studying a small element of area and mass. Figure 5.4 shows the decomposition of the forces acting on the gas shell under consideration. (Note that we are considering only two forces, the simplest case.)

We can, initially and directly, conclude that the laterally applied pressures cancel out, since pressure depends only on height (**Stevin's law**). Hence, we assume that the resultant of the forces being applied to the system is

$$\vec{F}_r = m \vec{a}, \quad (5.15)$$

$$\vec{F}_g + \cancel{\vec{F}_p} + \vec{F}_{p,\text{top}} + \vec{F}_{p,\text{bot}} = dm \frac{d^2\vec{r}}{dt^2}, \quad (5.16)$$

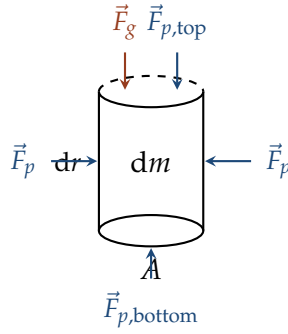


Figure 5.4: Cylindrical gas element used to derive hydrostatic equilibrium: gravitational force $\vec{F}_g$ and pressure forces at the top, bottom, and sides — the latter cancel by Stevin's law.

$$F_g - F_{p,\text{top}} + F_{p,\text{bot}} = dm \frac{d^2 r}{dt^2}. \quad (5.17)$$

From Figure 5.4, we can define the pressure variation as

$$dF_p = dF_{p,\text{top}} - dF_{p,\text{bot}}, \quad (5.18)$$

substituting into Equation (5.17),

$$F_g - dF_p = dm \frac{d^2 r}{dt^2}. \quad (5.19)$$

But we know that $F_g = -Gm \, dm \, \hat{r} / r^2$ [Equation (5.14), integrated over the element], and that, by definition, $P = F / A$, hence $dF = dP \cdot A$. Thus:

$$-\frac{Gm \, dm}{r^2} - dP \cdot A = dm \frac{d^2 r}{dt^2}. \quad (5.20)$$

Using the volumetric expressions, as usual ($dm = \rho \, dV = \rho A \, dr$):

$$-\frac{Gm\rho}{r^2} - \frac{dP \cdot A}{dr \cdot A} = \frac{dm}{dV} \frac{d^2 r}{dt^2} \implies -\frac{Gm\rho}{r^2} - \frac{dP}{dr} = \rho \frac{d^2 r}{dt^2}. \quad (5.21)$$

Assuming the star is in **hydrostatic equilibrium**, there is no acceleration, hence $d^2 r / dt^2 = 0$, which yields:

$$\frac{dP}{dr} + \frac{Gm\rho}{r^2} = 0 \implies \boxed{\frac{dP}{dr} = -\frac{Gm(r)\rho(r)}{r^2}}, \quad (5.22)$$

the **hydrostatic equilibrium equation** — the second of the 5 stellar structure equations.

Checking the sign of Equation (5.22)

It is worth stopping to explicitly check the sign of this central result, step by step: G , $m(r)$, $\rho(r)$, and r^2 are all positive, so the right-hand side of Equation (5.22) is always *negative*. This means $dP/dr < 0$: pressure **decreases** with r — physically correct, since pressure must be maximal at the center and fall to (approximately) zero at the surface. The negative sign arises from two places in the derivation: (i) the gravitational force [Equation (5.14)] is already defined as attractive (pointing inward, $-\hat{r}$); (ii) when isolating dP/dr in Equation (5.21), the gravitational term moves to the other side without changing sign, and the acceleration term vanishes. The final result is exactly consistent with the

standard literature (e.g., Kippenhahn & Weigert).

We see that pressure itself is not important — what matters is the pressure **gradient**. And it is this term that counterbalances the gravitational force.

Alternative form, in mass

We can obtain an alternative form for the hydrostatic equilibrium equation by multiplying Equation (5.22) by $\partial r / \partial m$:

$$\frac{\partial P}{\partial r} \cdot \frac{\partial r}{\partial m} + \frac{Gm\rho}{r^2} \cdot \frac{\partial r}{\partial m} = 0. \quad (5.23)$$

Using $\partial r / \partial m = 1 / (\partial m / \partial r) = 1 / (4\pi r^2 \rho)$ [Equation (5.4)], the first term is already $\partial P / \partial m$, and in the second term the factor ρ cancels:

$$\frac{\partial P}{\partial m} + \frac{Gm\rho}{r^2} \cdot \frac{1}{4\pi r^2 \rho} = 0, \quad (5.24)$$

that is,

$$\boxed{\frac{\partial P}{\partial m} + \frac{Gm(r)}{4\pi r^4} = 0}. \quad (5.25)$$

Here we have the pressure gradient as a function of the mass contained in the stellar region — a form that is often more convenient when integrating stellar structure using m , rather than r , as the independent variable.

5.5 Estimating central values for pressure and temperature

From Equations (5.22) and (5.25), we can obtain estimates for the central values of pressure and temperature in the interior of the Sun.

We can express $\partial P / \partial m$ as $(P_0 - P_c) / (M/2)$, where $P_0 = 0$ at the surface of the star and P_c is the pressure value at the center of the star. As an approximation, we use $\Delta m = M/2$ (the star's average mass) as a representative scale for the finite difference — and, consistently, we evaluate the gravitational term at a characteristic radius $r = R/2$.

Hence:

$$\frac{P_c}{M/2} \simeq \frac{GM}{4\pi r^4} \Big|_{r=R/2} \implies P_c = \frac{GM^2}{8\pi (R/2)^4} = \frac{GM^2}{8\pi} \cdot \frac{16}{R^4}, \quad (5.26)$$

that is,

$$\boxed{P_c \approx \frac{2GM^2}{\pi R^4}}. \quad (5.27)$$

Substituting reference values for the constants and for the Sun ($G = 6.674 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}$, $M_\odot = 1.989 \times 10^{33} \text{ g}$, $R_\odot \approx 7 \times 10^{10} \text{ cm}$), we obtain

$$P_{c,\odot} \approx 7 \times 10^{15} \text{ dyn/cm}^2. \quad (5.28)$$

To determine the central temperature, we can consider that the gas making up the star is ideal, hence:

$$P_c = \frac{\rho_c R T_c}{\mu} \implies T_c = \frac{\mu P_c}{\rho_c R}, \quad (5.29)$$

where R is the universal gas constant [same convention as Equation (3.44), Chapter 3] and μ is the mean molecular weight of the gas.

Notation note: μ once again denotes mean molecular weight

From here on, μ again denotes the **mean molecular weight** of the gas's chemical composition, exactly as in Chapter 3 (Thermodynamics) — and *not* the direction cosine $\mu = \cos \theta$ used throughout Chapter 4 (Radiative Processes). This reversal of convention accompanies the change of subject: radiative transport uses $\mu = \cos \theta$ universally in the literature, while equations of state and stellar structure use μ for mean molecular weight — both are standard conventions in their respective subfields, and here we resume this chapter's original notation.

Knowing that the mean density is defined by $\bar{\rho} = 3M/(4\pi R^3)$, we can rewrite $1/\rho_c$ in terms of the dimensionless ratio $\bar{\rho}/\rho_c$:

$$T_c = \frac{\mu}{\rho_c R} P_c = \frac{\mu}{R} P_c \cdot \frac{\bar{\rho}}{\rho_c} \cdot \frac{1}{\bar{\rho}} = \frac{\mu}{R} P_c \cdot \frac{\bar{\rho}}{\rho_c} \cdot \frac{4\pi R^3}{3M}, \quad (5.30)$$

substituting $P_c = 2GM^2/(\pi R^4)$ [Equation (5.27)]:

$$T_c = \frac{\mu}{R} \cdot \frac{2GM^2}{\pi R^4} \cdot \frac{4\pi R^3}{3M} \cdot \frac{\bar{\rho}}{\rho_c} = \frac{8}{3} \frac{G\mu}{R} \frac{M}{R} \cdot \frac{\bar{\rho}}{\rho_c}. \quad (5.31)$$

For most stars, density grows monotonically from the surface to the center, hence $\bar{\rho}/\rho_c < 1$, and thus:

$$T_c \lesssim \frac{8}{3} \frac{G\mu}{R} \frac{M}{R}. \quad (5.32)$$

Note that μ is a parameter that depends on the chemical composition of the gas and is very difficult to determine *a priori*. Suppose $\mu = 0.50$ for the Sun (fully ionized hydrogen). Substituting the constants and the values for the Sun into Equation (5.32), we obtain:

$$T_{c,\odot} \lesssim 3 \times 10^7 \text{ K}. \quad (5.33)$$

Note that the temperature and pressure at the center of the Sun (and of any star) are much larger than their surface values.

Comparison with the Standard Solar Model

The rough estimates above, $P_{c,\odot} \approx 7 \times 10^{15} \text{ dyn/cm}^2$ and $T_{c,\odot} \lesssim 3 \times 10^7 \text{ K}$, can be compared to the values obtained by detailed solar models, calibrated by helioseismology (which measures acoustic oscillations at the solar surface to infer the internal structure with great precision). The current Standard Solar Model gives:

$$T_{c,\odot}^{\text{real}} \approx 1.57 \times 10^7 \text{ K}, \quad P_{c,\odot}^{\text{real}} \approx 2.4 \times 10^{17} \text{ dyn/cm}^2, \quad (5.34)$$

according to Vinyoles et al. (2017). The T_c estimate is of the same order of magnitude as the real value (a factor ~ 2 above, consistent with being an upper bound, $\lesssim$). The P_c estimate, however, underestimates the real value by more than an order of magnitude (7×10^{15} versus 2.4×10^{17}). The reason is the approximation $\Delta m = M/2$: it treats the pressure gradient as approximately *linear* throughout the entire star, when, in fact, the real profile of $P(m)$ is extremely nonlinear — pressure falls off much faster near the surface than near the center, because density and temperature also fall off nonlinearly. A simple finite difference, with a single interval $\Delta m = M/2$, cannot capture this concentration of the pressure drop near the center, and so it systematically underestimates P_c . This

is exactly the kind of limitation that full numerical integration of the stellar structure equations (as used in codes like MESA) resolves, propagating the differential equations layer by layer instead of using a single crude finite difference. For a detailed discussion of how the solar chemical composition (the GS98 vs. AGSS09 abundance problem) affects these central values via opacity and the equation of state, see the review by Basu and Antia (2008).

5.6 Timescales

We can ask what happens if a star's hydrostatic equilibrium is lost, or how quickly changes in the star's structure will occur. These answers are obtained by analyzing the hydrostatic equilibrium equation.

For example, suppose the pressure gradient that is supporting gravity suddenly drops, that is, $\nabla P = \partial P / \partial r = 0$.

Free-fall time

In this situation, the pressure force disappears and Equation (5.21) — the dynamic, out-of-equilibrium version of the force balance — loses the pressure term and reduces to

$$-\frac{Gm\rho}{r^2} = \rho \frac{d^2r}{dt^2} \implies \frac{d^2r}{dt^2} = -\frac{Gm}{r^2}, \quad (5.35)$$

that is, each layer of the star begins to fall freely under the exclusive action of gravity — a free fall occurs. The negative sign indicates that the acceleration points inward, into the star (in the direction of the collapse motion).

We can estimate the timescale of this free fall, t_{ff} (*free-fall*), with an order-of-magnitude approximation, $d^2r/dt^2 \sim r/t_{\text{ff}}^2$. Substituting into Equation (5.35):

$$\frac{r}{t_{\text{ff}}^2} \sim \frac{Gm}{r^2} \implies t_{\text{ff}} = \left(\frac{r^3}{Gm} \right)^{1/2}. \quad (5.36)$$

Recalling the definition of the local gravitational acceleration, $g \equiv Gm/r^2$, we can equivalently rewrite this as

$$t_{\text{ff}} = \left(\frac{r}{g} \right)^{1/2}. \quad (5.37)$$

This is the time it would take the star to collapse upon itself due to the exclusive action of the gravitational force.

Confirming $t_{\text{ff}} \approx 27$ minutes for the Sun

Substituting $r = R_{\odot} \approx 6.957 \times 10^{10}$ cm, $m = M_{\odot} = 1.989 \times 10^{33}$ g, and $G = 6.674 \times 10^{-8}$ cm³ g⁻¹ s⁻² into Equation (5.36):

$$t_{\text{ff},\odot} = \left(\frac{(6.957 \times 10^{10})^3}{6.674 \times 10^{-8} \times 1.989 \times 10^{33}} \right)^{1/2} \text{ s} = \left(\frac{3.37 \times 10^{32}}{1.327 \times 10^{26}} \right)^{1/2} \text{ s} \approx 1.59 \times 10^3 \text{ s}, \quad (5.38)$$

that is, $t_{\text{ff},\odot} \approx 1590 \text{ s} \approx 27 \text{ min}$. This is the time the Sun would take to collapse upon itself if the pressure sustaining it suddenly disappeared.

Explosion time (hypothetical case, without gravity)

Analogously, we can consider the opposite case: what if gravity suddenly disappeared, but the pressure gradient remained? In this case the star would expand (explode) instead of collapsing. Starting from the mass version of the equation of motion — combining the gravitational term from Equation (5.25) with an inertial term, analogously to Equation (5.21) —

$$\frac{\partial P}{\partial m} - \frac{Gm}{4\pi r^4} = \frac{1}{4\pi r^2} \frac{d^2 r}{dt^2}, \quad (5.39)$$

and imposing $F_g = 0$ (neglecting the gravitational term compared to the others, since gravity is now absent), we obtain

$$\frac{\partial P}{\partial m} = \frac{1}{4\pi r^2} \frac{d^2 r}{dt^2}. \quad (5.40)$$

Using again the order-of-magnitude approximation $d^2 r / dt^2 \sim r / t^2$, and also $\partial P / \partial r \sim P / r$ — so that, by the chain rule and the continuity equation [$\partial r / \partial m = 1 / (4\pi r^2 \rho)$, Equation (5.4)], $\partial P / \partial m = (\partial r / \partial m)(\partial P / \partial r) \sim P / (4\pi r^3 \rho)$ —, Equation (5.40) becomes

$$\frac{1}{4\pi r^2 \rho} \cdot \frac{P}{r} \sim \frac{1}{4\pi r^2} \cdot \frac{r}{t^2} \implies \frac{P}{\rho r} \sim \frac{r}{t^2} \implies t^2 \sim \frac{r^2 \rho}{P}, \quad (5.41)$$

that is,

$$t \sim r \left(\frac{\rho}{P} \right)^{1/2}, \quad (5.42)$$

the **explosion time**. The larger the pressure compared to the density, the shorter the time needed for the star to fly apart.

Dynamical timescale

The **dynamical timescale** (also called the hydrodynamical time, or quasi-equilibrium time, t_{hid}) describes the time the star takes to return to hydrostatic equilibrium after some perturbation. It is obtained analogously to the free-fall time — also assuming $\nabla P = 0$ —, such that $t_{\text{ff}} \simeq t_{\text{hid}}$; the difference is that we now express the result in terms of the star's mean density, $\bar{\rho} \equiv 3m / (4\pi r^3)$. Substituting $r^3 = 3m / (4\pi \bar{\rho})$ into Equation (5.36):

$$t_{\text{ff}} = \left(\frac{r^3}{Gm} \right)^{1/2} = \left(\frac{3m / (4\pi \bar{\rho})}{Gm} \right)^{1/2} = \left(\frac{3}{4\pi} \right)^{1/2} \frac{1}{\sqrt{G\bar{\rho}}}. \quad (5.43)$$

Since $\sqrt{3/(4\pi)} \approx 0.489 \approx 1/2$ — a reasonable order-of-magnitude approximation, in the same spirit as the other approximations in this section —, we obtain the final form, widely used in the literature:

$$t_{\text{ff}} \approx \frac{1}{2\sqrt{G\bar{\rho}}}. \quad (5.44)$$

For the Sun, $\bar{\rho}_{\odot} = 3M_{\odot} / (4\pi R_{\odot}^3) \approx 1.41 \text{ g/cm}^3$, which gives $t_{\text{ff},\odot} \approx 1630 \text{ s} \approx 27 \text{ min}$ — reproducing the same result obtained directly from Equation (5.36), as expected.

Note that this time is much shorter than the age of the Sun ($\approx 4.5 \times 10^9 \text{ yr}$); this means that, were they to occur, changes in the Sun's hydrostatic structure on this timescale would be easily observed.

The dynamical timescale and the pulsation of variable stars

The dynamical timescale we have just derived, $t_{\text{din}} \sim 1/\sqrt{G\bar{\rho}}$, is not merely a theoretical limit for collapse or explosion — it also governs the physics of **pulsating** stars, such as classical Cepheids and RR Lyrae stars. In these stars, a zone of partial helium ionization (the κ mechanism, an opacity “valve”) acts as a heat engine that sustains periodic radial oscillations around hydrostatic equilibrium, with a pulsation period Π of the same order of magnitude as t_{din} . This leads directly to the famous **period–mean density relation**,

$$\Pi\sqrt{\bar{\rho}} \approx \text{constant}, \quad (5.45)$$

valid (with different numerical constants) for an entire class of radial pulsators. Combined with the empirical period–luminosity relation for Cepheids (discovered by Henrietta Leavitt in 1908 from stars in the Small Magellanic Cloud), this physics is the basis of the first rung of the extragalactic distance ladder. Recent observations from the *James Webb Space Telescope* (JWST) confirmed the photometric robustness of the SH0ES project’s Cepheids in host galaxies of Type Ia supernovae, ruling out the hypothesis that unresolved stellar crowding in Hubble Space Telescope images biased the calibration of the period–luminosity relation — thereby reinforcing the current tension in the value of the Hubble constant measured via the local distance ladder (Riess et al., 2024).

5.7 Virial Theorem

An important consequence of hydrostatic equilibrium is the **Virial Theorem**, of vital importance for understanding stars. The theorem connects two fundamental energy reservoirs in a star — the internal (thermal) energy and the gravitational potential energy —, allowing predictions and interpretations about the internal structure of stars.

Gravitational potential energy

The gravitational force acting on a shell of mass dm [Equation (5.14), integrated over the shell] is

$$d\vec{F}_g = -\frac{Gm dm}{r^2} \hat{r}. \quad (5.46)$$

The gravitational potential energy of this shell, dM_g , is, by definition, the work required to move it from radius r to infinity:

$$dM_g = W(r \rightarrow \infty) = \int_r^\infty d\vec{F}_g \cdot d\vec{r} = -Gm dm \int_r^\infty \frac{dr'}{r'^2} = Gm dm \left(\frac{1}{r'} \right) \Big|_r^\infty = -\frac{Gm dm}{r}. \quad (5.47)$$

We can write this result as $dM_g = \phi dm$, where

$$\phi(r) \equiv -\frac{Gm}{r} \quad (5.48)$$

is the **gravitational potential** (negative, as expected for a gravitationally bound system).

The total gravitational potential energy of the star is the integral of Equation (5.47) over all spherical shells of mass dm , from 0 to M :

$$E_g \equiv - \int_0^M \frac{Gm}{r} dm. \quad (5.49)$$

Since this energy depends on the radius, contractions and expansions of the star modify its gravitational potential energy.

Derivation via the hydrostatic equilibrium equation

There is a second, more fundamental route to relate E_g to the local thermodynamic properties of the gas: integrating the hydrostatic equilibrium equation itself. Starting from the mass form [Equation (5.25)], we multiply both sides by $4\pi r^3$ and integrate over dm , from 0 to M :

$$\int_0^M 4\pi r^3 \frac{\partial P}{\partial m} dm = - \int_0^M \frac{Gm \cdot 4\pi r^3}{4\pi r^4} dm = - \int_0^M \frac{Gm}{r} dm. \quad (5.50)$$

The right-hand side is, by definition (5.49), exactly E_g , hence

$$\int_0^M 4\pi r^3 \frac{\partial P}{\partial m} dm = E_g. \quad (5.51)$$

We solve this integral by parts, with $u = 4\pi r^3$ ($du = 12\pi r^2 \partial r / \partial m dm$) and $dv = (\partial P / \partial m) dm$ ($v = P$):

$$\int_0^M 4\pi r^3 \frac{\partial P}{\partial m} dm = \underbrace{4\pi r^3 P}_{\text{boundary term}} \Big|_0^M - \int_0^M P \cdot 12\pi r^2 \frac{\partial r}{\partial m} dm. \quad (5.52)$$

The boundary term vanishes at *both* limits of integration: at $m = M$ (the star's surface), the pressure is zero, $P = 0$; at $m = 0$ (the star's center), the radius is zero, $r = 0$, so $r^3 = 0$ as well. Thus, the boundary term vanishes completely, and

$$\int_0^M 4\pi r^3 \frac{\partial P}{\partial m} dm = - \int_0^M 12\pi r^2 P \frac{\partial r}{\partial m} dm. \quad (5.53)$$

Combining Equations (5.51) and (5.53):

$$E_g = - \int_0^M 12\pi r^2 P \frac{\partial r}{\partial m} dm. \quad (5.54)$$

Using the continuity equation, $\partial r / \partial m = 1 / (4\pi r^2 \rho)$ [Equation (5.4)], the factor r^2 cancels:

$$E_g = - \int_0^M 12\pi r^2 P \cdot \frac{1}{4\pi r^2 \rho} dm = -3 \int_0^M \frac{P}{\rho} dm, \quad (5.55)$$

that is,

$$\boxed{\int_0^M \frac{3P}{\rho} dm = -E_g}, \quad (5.56)$$

valid for *any* gas — not only an ideal gas —, and therefore this is the most general form of the Virial Theorem applied to a star in hydrostatic equilibrium.

Specialization to a monatomic ideal gas

If we assume that the stellar gas behaves as an ideal gas, its equation of state is $PV = nRT$ or, per unit volume,

$$P = \frac{\rho RT}{\mu} \implies \frac{P}{\rho} = \frac{RT}{\mu}, \quad (5.57)$$

where R is the universal gas constant and μ the mean molecular weight. We already saw, in Chapter 3, that for an ideal gas $C_P - C_V = R/\mu$, hence

$$\frac{P}{\rho} = (C_P - C_V) T = C_V \left(\frac{C_P}{C_V} - 1 \right) T = C_V (\gamma - 1) T, \quad (5.58)$$

where $\gamma \equiv C_P/C_V$. For a monatomic ideal gas, $\gamma = 5/3$, hence $\gamma - 1 = 2/3$ and

$$\frac{P}{\rho} = \frac{2}{3} C_V T. \quad (5.59)$$

By definition, $C_V \equiv (\partial u / \partial T)_V$, where u is the specific internal energy of the gas; integrating (at constant C_V , starting from $u = 0$ at $T = 0$),

$$\int_0^T C_V dT' = \int_0^u du' \implies u = C_V T, \quad (5.60)$$

and therefore

$$\boxed{\frac{P}{\rho} = \frac{2}{3} u}. \quad (5.61)$$

Substituting into Equation (5.56):

$$\int_0^M \frac{3P}{\rho} dm = \int_0^M 3 \cdot \frac{2}{3} u dm = \int_0^M 2u dm = -E_g. \quad (5.62)$$

The integral of the specific internal energy over the entire mass of the star is, by definition, the **total internal energy**, $E_i \equiv \int_0^M u dm$, such that

$$\boxed{2E_i = -E_g}, \quad (5.63)$$

the best-known form of the Virial Theorem, valid for a monatomic ideal gas. Since the energy variation must be conserved, the same holds for variations between any two states of the star:

$$2 \Delta E_i = -\Delta E_g. \quad (5.64)$$

Alternative derivation: moment of inertia and the Lagrange–Jacobi identity

The same result, $2E_i = -E_g$, can be obtained by a route completely different from the one used above — not via the hydrostatic equilibrium equation, but via the mechanics of self-gravitating systems. Define the moment of inertia of the system about the origin,

$$I = \int_V \rho r^2 dV. \quad (5.65)$$

Its first time derivative is

$$\frac{dI}{dt} = 2 \int_V \rho \vec{r} \cdot \vec{v} dV, \quad (5.66)$$

and differentiating once more, using $d(\vec{r} \cdot \vec{v})/dt = \vec{v} \cdot \vec{v} + \vec{r} \cdot \vec{a} = v^2 + \vec{r} \cdot \vec{a}$,

$$\frac{d^2 I}{dt^2} = 2 \int_V \rho v^2 dV + 2 \int_V \rho \vec{r} \cdot \vec{a} dV = 4T + 2 \int_V \rho \vec{r} \cdot \vec{a} dV, \quad (5.67)$$

where $T \equiv \frac{1}{2} \int_V \rho v^2 dV$ is the total kinetic energy of the system. Substituting the acceleration using the fluid's Euler equation, $\rho \vec{a} = -\nabla P - \rho \nabla \Phi$ — equivalent, in local form, to

the hydrostatic equilibrium equation already used in this chapter —,

$$\int_V \rho \vec{r} \cdot \vec{a} dV = - \int_V \vec{r} \cdot \nabla P dV - \int_V \rho \vec{r} \cdot \nabla \Phi dV. \quad (5.68)$$

The first term is resolved by integration by parts (the Divergence Theorem, with $P = 0$ at the surface of the system) and $\nabla \cdot \vec{r} = 3$ in three dimensions, yielding $3 \int_V P dV$; the second term is, by definition, the gravitational potential energy E_g of the system, with the same sign already established above. We thus arrive at the **Lagrange–Jacobi identity**, the most general form of the Virial Theorem:

$$\boxed{\frac{1}{2} \frac{d^2 I}{dt^2} = 2T + 3 \int_V P dV + E_g}. \quad (5.69)$$

In static equilibrium ($d^2 I / dt^2 = 0$), with no large-scale bulk kinetic energy ($T = 0$, thermal support only), and using $3 \int_V P dV = 2E_i$ for the monatomic ideal gas [as already obtained above, Equation (5.56) combined with (5.61)], Equation (5.69) reduces exactly to $2E_i + E_g = 0$ — the same result as Equation (5.63), now obtained without any reference to spherical geometry or to the hydrostatic equilibrium equation in radial form. This is the virtue of the moment-of-inertia derivation: it holds for *any* self-gravitating system in equilibrium — a continuous fluid (like a star) or a discrete set of particles (simply replace the integral by a sum, $I = \sum_i m_i r_i^2$) —, without any assumption about symmetry or shape.

The Virial Theorem beyond an isolated star

The generality of the Lagrange–Jacobi identity [Equation (5.69)] allows the same theorem to be applied on scales very different from that of an individual star.

For a system of N gravitationally bound bodies — stars within a star cluster, or galaxies within a cluster of galaxies — there is no gas pressure term (the system is not a continuous fluid), and the identity reduces to $2T + E_g = 0$, with T the total kinetic energy of the particles' motion. It was exactly this form of the theorem that Zwicky (1933) applied to the observed velocities of galaxies in the Coma cluster, finding a dynamical mass almost an order of magnitude larger than the mass inferred from the luminosity — the first observational evidence of dark matter in an extragalactic system.

In star formation, the theorem is used to assess whether a molecular cloud is gravitationally bound and prone to collapse, through the **virial parameter**, $\alpha_{\text{vir}} \equiv 2T / |E_g|$: values $\alpha_{\text{vir}} \lesssim 1$ –2 generally indicate a gravity-dominated cloud, favorable to collapse and star formation (McKee and Ostriker, 2007).

There is also a more complete generalization, the **tensor virial theorem**, which includes rotational and magnetic energy terms in addition to those seen here — relevant for modeling rapidly rotating stars, white dwarfs, and magnetized neutron stars (Chandrasekhar, 1969). In its various forms, the Virial Theorem is one of the most universal results in astrophysics, valid from the interior of a single star up to the scale of entire clusters of galaxies.

Negative heat capacity and the pre-main-sequence phase

Equation (5.64) describes a profoundly counterintuitive behavior: if the star *loses* total energy ($W = E_i + E_g$ decreases, as we will see next), it becomes *hotter*, not colder. This

happens because a self-gravitating system has a **negative effective heat capacity** — unlike a gas in a rigid box, whose temperature drops when it loses energy, a star contracts when it loses energy, and the contraction releases enough gravitational potential energy not only to replace the lost energy but also to increase the internal energy (and therefore the temperature). This mechanism governs the entire pre-main-sequence contraction phase: a newly formed protostar, still unable to sustain nuclear fusion, radiates energy into space and slowly contracts in response, heating its interior — first along the **Hayashi track** (nearly vertical, at approximately constant T_{eff} , with the star fully convective) and then along the **Henyey track** (nearly horizontal, once a radiative core develops) in the HR diagram, until the central temperature reaches $\sim 10^7$ K and hydrogen fusion begins, halting the contraction by supplying an energy source that no longer depends on gravity (Kippenhahn et al., 2012). It is exactly this same negative heat capacity, acting in reverse, that later drives the heating of the stellar core after the exhaustion of central hydrogen, enabling the onset of helium fusion.

Example: stellar contraction

Let us consider the case of a contracting star, that is, with the radius decreasing between an initial instant (i) and a final instant (f), as illustrated in Figure 5.5. Generalizing Equation (5.64) to the difference between the two states,

$$E_g^f - E_g^i = -2(E_i^f - E_i^i). \quad (5.70)$$

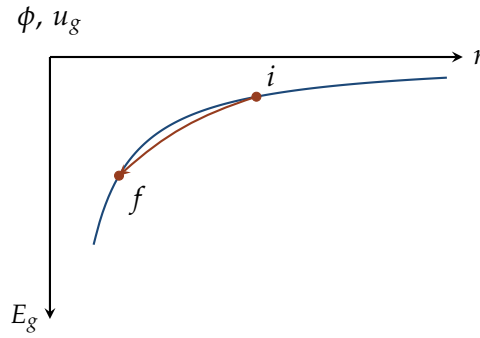


Figure 5.5: Contraction trajectory in the (ϕ, u_g) plane (or, equivalently, E_g) versus r : as the radius decreases from the initial state i to the final state f , the gravitational potential energy becomes more negative (the curve follows $\propto -1/r$).

If the star's radius decreases, the gravitational potential energy E_g becomes more negative (increases in magnitude), since $E_g \sim -1/r$ [Equation (5.49)], that is, $\Delta E_g < 0$. By Equation (5.64),

$$\Delta E_i = -\frac{\Delta E_g}{2} > 0, \quad (5.71)$$

hence if E_g decreases, E_i must *increase*. Since internal energy is proportional to temperature ($\Delta E_i \propto \frac{3}{2}nR\Delta T$, for an ideal gas), the increase in E_i implies an increase in the star's temperature — contraction heats the star. Symmetrically, if there is an expansion, the temperature decreases. This is exactly what happens when stars evolve off the main sequence.

Total energy and luminosity

The total energy of the system is defined by the sum of internal and gravitational energies,

$$W = E_i + E_g. \quad (5.72)$$

If the star has a luminosity, it radiates power into space; hence

$$L = -\frac{dW}{dt}. \quad (5.73)$$

Using the Virial Theorem [Equation (5.63), $E_i = -E_g/2$] to eliminate E_i in favor of E_g ,

$$L = -\frac{d}{dt}(E_i + E_g) = -\frac{d}{dt}\left(-\frac{E_g}{2} + E_g\right) = -\frac{d}{dt}\left(\frac{E_g}{2}\right), \quad (5.74)$$

that is,

$$L \propto -\dot{E}_g \quad \text{and, equivalently,} \quad L \propto \dot{E}_i. \quad (5.75)$$

The star's luminosity is, therefore, proportional to the rate of temporal change of its energy (gravitational or internal).

5.8 Kelvin–Helmholtz timescale

We saw in the previous section that the luminosity can be expressed as $L = \frac{\partial}{\partial t}(-\frac{1}{2}E_g)$ [Equation (5.74)]. We can integrate this expression to obtain a characteristic timescale,

$$\tau_{\text{KH}} = \frac{1}{2} \frac{E_g}{L}. \quad (5.76)$$

As seen previously [Equation (5.49)], $E_g = -\int_0^M (Gm/r) dm$; for an order-of-magnitude estimate, we approximate $|E_g| \approx GM^2/R$ (dropping the exact numerical factor from the integration, of order unity), hence

$$\tau_{\text{KH}} \approx \frac{1}{2} \frac{GM^2}{LR}. \quad (5.77)$$

Computing τ_{KH} for the Sun

Substituting the solar values directly in cgs units ($G = 6.674 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}$, $M_\odot = 1.989 \times 10^{33} \text{ g}$, $L_\odot = 3.828 \times 10^{33} \text{ erg/s}$, $R_\odot = 6.957 \times 10^{10} \text{ cm}$) into Equation (5.77):

$$\tau_{\text{KH},\odot} = \frac{GM_\odot^2}{2L_\odot R_\odot} = \frac{6.674 \times 10^{-8} \times (1.989 \times 10^{33})^2}{2 \times 3.828 \times 10^{33} \times 6.957 \times 10^{10}} \text{ s} \approx 4.96 \times 10^{14} \text{ s} \approx 1.57 \times 10^7 \text{ yr}. \quad (5.78)$$

Generalizing to any star, and recalling that $\tau_{\text{KH}} \propto M^2/(LR)$ — so that the mass, radius, and luminosity ratios appear relative to solar values as $(M/M_\odot)^2$, $(R_\odot/R)$, and $(L_\odot/L)$, respectively —, we obtain the final form:

$$\tau_{\text{KH}} \approx 1.5 \times 10^7 \left(\frac{M}{M_\odot}\right)^2 \left(\frac{R_\odot}{R}\right) \left(\frac{L_\odot}{L}\right) \text{ yr}, \quad (5.79)$$

which exactly reproduces $\tau_{\text{KH},\odot} \approx 1.6 \times 10^7$ yr when $M = M_{\odot}$, $R = R_{\odot}$, $L = L_{\odot}$ — tens of millions of years, not fractions of a second.

Note that, for the Sun, this time is of order 1.6×10^7 years, orders of magnitude larger than the dynamical timescale (27 min). Thus, it is not possible to observe, on human timescales, direct evidence that stars are in thermal equilibrium.

This timescale led 19th-century astronomers, most notably Lord Kelvin, to infer an upper age limit for the Sun of a few tens of millions of years — assuming that all of the Sun’s energy came from gravitational contraction. However, geological studies indicated that the age of the Earth should be of order billions of years (10^9), which raised a problem, since it made no sense for the Earth to be older than the Sun.

The resolution of Kelvin’s paradox

The impasse between Kelvin’s estimated solar age [Equation (5.79)] and the age of the Earth estimated by geology (and, later, with great precision, by radiometric dating) was resolved in two historical steps. First, the discovery of **radioactivity** by Henri Becquerel (1896) and the subsequent work of Marie and Pierre Curie and of Ernest Rutherford showed, already in the first decades of the twentieth century, that radioactive decay within the Earth provides a heat source independent of the Sun, and also a precise geological clock (radiometric dating), confirming an Earth age of billions of years — much larger than any plausible τ_{KH} . Second, and more fundamentally, Arthur Eddington proposed in 1920 that the nuclear fusion of hydrogen into helium, not gravitational contraction, is the dominant energy source of stars (an idea detailed in his 1926 book *The Internal Constitution of the Stars*); but only in 1939 did Hans Bethe manage to describe, in quantitative detail, the responsible nuclear reactions — the proton–proton chain and the CNO cycle —, work for which he received the Nobel Prize in Physics in 1967 (Bethe, 1939). With an energy source able to sustain the Sun for billions of years [see Section 5.9, next], the paradox disappeared: the Sun is, in fact, older than any naively calculated τ_{KH} would suggest, precisely because gravitational contraction is not its dominant energy source over most of its life.

5.9 Nuclear timescale

The Kelvin–Helmholtz timescale does not describe the Sun’s lifetime well. Moreover, a star can remain in thermal equilibrium as long as it has “fuel” to generate thermal energy on its own. We can suppose that it is not gravitational energy, but rather nuclear energy, that provides the largest energy contribution to the star. Thus, in thermal equilibrium ($L = L_{\text{nuc}} = \dot{E}_{\text{nuc}}$),

$$\tau_{\text{nuc}} = \frac{E_{\text{nuc}}}{L}. \quad (5.80)$$

Assuming that the star’s energy comes from the fusion (burning) of H into He, the energy released in this process is approximately $\phi = 0.007$ of the rest mass of the converted hydrogen (the mass–energy conversion efficiency of the proton–proton chain), hence

$$E_{\text{nuc}} = \phi M_{\text{nuc}} c^2, \quad (5.81)$$

where $M_{\text{nuc}} = f_{\text{nuc}} M$, with f_{nuc} the fraction of the star’s mass that actually participates in nuclear fusion (only the stellar core reaches a temperature sufficient for this — the rest of the

mass is never processed). Substituting,

$$\tau_{\text{nuc}} = \frac{\phi f_{\text{nuc}} M c^2}{L}. \quad (5.82)$$

Estimating the Sun's nuclear lifetime

The numerical coefficient in Equation (5.86) can be obtained by directly computing $\phi M_{\odot} c^2 / L_{\odot}$ in cgs units ($c = 2.998 \times 10^{10}$ cm/s, $M_{\odot} = 1.989 \times 10^{33}$ g, $L_{\odot} = 3.828 \times 10^{33}$ erg/s):

$$\frac{M_{\odot} c^2}{L_{\odot}} = \frac{1.989 \times 10^{33} \times (2.998 \times 10^{10})^2}{3.828 \times 10^{33}} \text{ s} \approx 4.67 \times 10^{20} \text{ s} \approx 1.48 \times 10^{13} \text{ yr}. \quad (5.83)$$

Multiplying by the efficiency $\phi = 0.007$:

$$\phi \frac{M_{\odot} c^2}{L_{\odot}} \approx 1.04 \times 10^{11} \text{ yr}, \quad (5.84)$$

the time the Sun would last *if its entire* mass were converted to helium — an unrealistic scenario, since only the core reaches sufficient temperature and density for fusion. Adopting $f_{\text{nuc}} \approx 0.1$ (approximately 10% of the stellar mass participates in hydrogen fusion over the lifetime of a star like the Sun on the main sequence — a standard estimate for the extent of the convective/radiative core where the temperature is sufficient for the p–p chain; Kippenhahn et al., 2012):

$$\tau_{\text{nuc},\odot} \approx 0.1 \times 1.04 \times 10^{11} \text{ yr} \approx 1.0 \times 10^{10} \text{ yr}, \quad (5.85)$$

in excellent agreement with the known literature value for the Sun's main-sequence lifetime, $\sim 10^{10}$ years. Thus, the correct numerical coefficient is

$$\tau_{\text{nuc}} \approx 1 \times 10^{10} \left(\frac{M}{M_{\odot}} \right) \left(\frac{L_{\odot}}{L} \right) \text{ yr}, \quad (5.86)$$

with the ratio $L_{\odot}/L$ (and not $L/L_{\odot}$) for the same dimensional-consistency reason discussed for τ_{KH} [Equation (5.79)]: $\tau_{\text{nuc}} \propto M/L$, hence more luminous — generally more massive — stars have *shorter*, not longer, lifetimes.

The hierarchy of timescales and the quasi-static approximation

Bringing together the three results of this section for the Sun,

$$\tau_{\text{din}} \sim 27 \text{ min} \ll \tau_{\text{KH}} \sim 1.6 \times 10^7 \text{ yr} \ll \tau_{\text{nuc}} \sim 10^{10} \text{ yr}, \quad (5.87)$$

a hierarchy of approximately 8 and 3 orders of magnitude between successive scales. This enormous separation of timescales is what justifies, in practice, all standard stellar evolution modeling: at any instant of its main-sequence life (and during most other phases), a star can be treated as being simultaneously in **hydrostatic equilibrium** ($\tau_{\text{din}} \rightarrow 0$ compared to the other scales — the mechanical structure adjusts “instantaneously”) and in **local thermal equilibrium** ($\tau_{\text{KH}} \ll \tau_{\text{nuc}}$ — the thermal structure also adjusts much faster than the nuclear fuel is depleted). This is the **quasi-static approximation**: instead of solving the full, time-dependent hydrodynamics, stellar evolution codes (such as MESA) solve, at each time step $\Delta t \sim \tau_{\text{nuc}}/1000$, a sequence of models in exact hydrostatic and

thermal equilibrium, with the chemical composition evolving slowly through nuclear burning — without this separation of scales, the problem would be computationally infeasible.

The hierarchy $\tau_{\text{nuc}} \propto M/L$ [Equation (5.86)] also directly explains why more massive stars live much *shorter* lives. Combining with the mass–luminosity relation for main-sequence stars ($L \propto M^{3.5}$, approximately, for $M \sim M_{\odot}$):

$$\tau_{\text{nuc}} \propto \frac{M}{L} \propto \frac{M}{M^{3.5}} = M^{-2.5}. \quad (5.88)$$

A $10 M_{\odot}$ star, for example, has a τ_{nuc} about $10^{2.5} \approx 300$ times shorter than the Sun’s — a few tens of millions of years, instead of tens of billions. This is precisely the physical principle behind the **main-sequence turnoff method** for dating star clusters: in a cluster (where all stars were born, to a good approximation, at the same epoch), the more massive stars have already exhausted their central hydrogen and left the main sequence, while the less massive ones are still on it. The mass (or luminosity/temperature) of the exact point where the observed main sequence bends toward the subgiant branch — the *turnoff* — corresponds to the mass whose τ_{nuc} equals the cluster’s age, providing a direct estimate of that age when compared against theoretical isochrones (Soderblom, 2010). The *Gaia* satellite, with unprecedented-precision photometry and parallaxes for millions of stars in open clusters of the Milky Way, has recently revolutionized the application of this method, both by identifying thousands of new clusters and by refining ages derived through isochrone fitting (Cantat-Gaudin and Casamiquela, 2024).

It is worth noting, to close this section, that the Sun’s energy source gives a timescale of order 10 billion years, which makes a great deal of sense given the Sun’s observed age. For the burning of He and even heavier elements, however, the value of ϕ decreases by a factor of 10 or more relative to H burning — each successive fusion stage releases less energy per unit mass —, which proportionally decreases the nuclear time available in those phases. Even so, the hierarchy of timescales still holds:

$$\tau_{\text{nuc}} \gg \tau_{\text{KH}} \gg \tau_{\text{din}}. \quad (5.89)$$

5.10 Energy conservation

Let us define the term $l(r)$ as the **local luminosity**, that is, the energy crossing a sphere of radius r per unit time. The function $l(r)$ is zero at the origin (the star’s center), $l(0) = 0$, and reaches the value L at the star’s surface, $l(R) = L$. Between these limits, l can be a complicated function, depending on the energy distribution and chemical composition of the region.

Considering a spherical shell of mass m and radius r , with infinitesimal elements dm and dr , we can analyze the energy entering the shell with luminosity l and leaving it with luminosity $l + dl$.

Derivation via spherical shell

Assuming that the star’s only energy source is nuclear fusion processes, with energy generation rate per unit mass and time ε (which depends, in general, on the local temperature, density, and chemical composition), we can determine the amount of energy crossing the

spherical shell:

$$dl = \underbrace{4\pi r^2 \rho dr}_{\text{volume element}} \cdot \underbrace{\varepsilon}_{\text{energy released per unit mass}} . \quad (5.90)$$

Using the mass equation, $dm = 4\pi r^2 \rho dr$ [Equation (5.4)], dl can be expressed more compactly:

$$\boxed{dl = dm \varepsilon} \implies \boxed{\frac{dl}{dm} = \varepsilon} , \quad (5.91)$$

the **energy conservation equation** (or energy generation equation). In general, ε depends on the temperature, density, and chemical composition of the star.

Alternative form: Gauss's Divergence Theorem

An alternative way of obtaining the energy conservation equation is to use **Gauss's Divergence Theorem**. The theorem states that

$$\iiint_V \vec{\nabla} \cdot \vec{F} dV = \oint_S \vec{F} \cdot \hat{n} dS , \quad (5.92)$$

where $\vec{F}$ is a flux (or radiation field), and V and S are the volume and surface elements, respectively.

The radiative flux crossing the surface dS is proportional to the density and the amount of energy generated in the region that surface encloses, hence:

$$\oint_S \vec{F} \cdot \hat{n} dS = \underbrace{\varepsilon \cdot \rho}_{\text{amount of energy per unit volume}} . \quad (5.93)$$

Thus, the differential form of Gauss's Theorem reduces to

$$\vec{\nabla} \cdot \vec{F} = \varepsilon \rho , \quad (5.94)$$

which closely resembles Gauss's law of electromagnetism. In spherical coordinates, we can express

$$\vec{\nabla} \cdot \vec{F} = \frac{1}{r^2} \frac{d}{dr} (r^2 F) = \frac{1}{4\pi r^2} \frac{d}{dr} (4\pi r^2 F) , \quad (5.95)$$

but $F = l/(4\pi r^2)$ (the total flux through the sphere of radius r), hence

$$\vec{\nabla} \cdot \vec{F} = \frac{1}{4\pi r^2} \frac{dl}{dr} , \quad (5.96)$$

and, therefore,

$$\boxed{\frac{dl}{dr} = 4\pi r^2 \rho \varepsilon} , \quad (5.97)$$

which is exactly the same equation obtained previously [differentiating Equation (5.91) with respect to r via the mass equation].

Non-stationary case

If we drop the time-independence condition, $\varepsilon - \partial l / \partial m$ may be nonzero even in the absence of nuclear reactions, since $l = l(r, t)$. A non-stationary shell can change its internal energy and exchange this energy with its surroundings through mechanical work. Thus, in a non-stationary case, the amount of heat in the medium (energy) is expressed by

$$\frac{dq}{dt} = \varepsilon - \frac{\partial l}{\partial m}, \quad [dq] = \text{erg/g}. \quad (5.98)$$

Recalling the first law of thermodynamics, $dq = -P dv + du$ (with $v \equiv 1/\rho$ the specific volume and u the gas's specific internal energy):

$$-P \frac{\partial v}{\partial t} + \frac{\partial u}{\partial t} = \varepsilon - \frac{\partial l}{\partial m} \implies \boxed{\frac{\partial l}{\partial m} = \varepsilon + P \frac{\partial v}{\partial t} - \frac{\partial u}{\partial t}}. \quad (5.99)$$

Knowing that dq can also be described in terms of C_P and δ [Chapter 3, alternative form of the first law], $dq = C_P dT - (\delta/\rho) dP$:

$$\frac{\partial q}{\partial t} = C_P \frac{\partial T}{\partial t} - \frac{\delta}{\rho} \frac{\partial P}{\partial t}. \quad (5.100)$$

Combining with Equation (5.98):

$$\frac{\partial l}{\partial m} = \varepsilon - \frac{\partial q}{\partial t} = \varepsilon - C_P \frac{\partial T}{\partial t} + \frac{\delta}{\rho} \frac{\partial P}{\partial t}, \quad (5.101)$$

that is,

$$\boxed{\frac{\partial l}{\partial m} = \varepsilon - C_P \frac{\partial T}{\partial t} + \frac{\delta}{\rho} \frac{\partial P}{\partial t}}, \quad (5.102)$$

where $\delta \equiv -(\partial \ln \rho / \partial \ln T)_P$ is the same thermodynamic coefficient already defined in Chapter 3.

Neutrino energy loss

We can also include the energy crossing the spherical shell in the form of neutrinos. Since neutrinos practically do not interact with stellar matter (their mean free path is many orders of magnitude larger than the star's radius), the energy they carry escapes immediately, without contributing to local energy transport. When including this neutrino-borne energy, we must subtract this amount from the available nuclear energy, since neutrinos are produced by the fusion reactions themselves. We define

$$L_\nu = \int_0^M \varepsilon_\nu dm, \quad (5.103)$$

where ε_ν is the neutrino energy loss rate, per unit mass. Thus, the final form of the energy conservation equation is

$$\boxed{\frac{\partial l}{\partial m} = \varepsilon_{\text{nuc}} - \varepsilon_\nu - C_P \frac{\partial T}{\partial t} + \frac{\delta}{\rho} \frac{\partial P}{\partial t}}, \quad (5.104)$$

the complete **energy conservation equation**. Note that the dependence of the variables is on (m, t) . The term $-C_P \partial T / \partial t + (\delta/\rho) \partial P / \partial t$ is often called, in the literature, $\varepsilon_{\text{grav}}$ — the rate of

energy generation (or absorption) associated with the contraction or expansion of the star [see the discussion of the Virial Theorem, Section 5.7: it is essentially the same “negative heat capacity” mechanism discussed there, now expressed locally, layer by layer, rather than for the star as a whole].

The actual detection of solar neutrinos

The prediction that the Sun emits an enormous flux of neutrinos [Equation (5.103)] was tested observationally over more than three decades, revealing one of the most instructive episodes in twentieth-century physics: the **solar neutrino problem**. The Homestake experiment (South Dakota, operated by Raymond Davis Jr. starting in 1970, using a tank of perchloroethylene sensitive to electron neutrinos) systematically detected only about a third of the flux predicted by John Bahcall’s Standard Solar Model. For decades, it was unclear whether the problem lay in the nuclear physics/stellar structure of the solar model or in the particle physics of neutrinos. The resolution came from later experiments, sensitive to different neutrino flavors: Super-Kamiokande (Japan) and, especially, the Sudbury Neutrino Observatory (SNO, Canada), which in 2001-2002 separately measured the electron-neutrino flux and the total flux of all flavors, demonstrating that solar neutrinos change flavor (*neutrino oscillation*) on their way to Earth — the “missing” neutrinos were not actually missing, they had simply changed type and become undetectable by Davis’s original method. This result simultaneously confirmed the Standard Solar Model (the *total* neutrino flux matched the prediction exactly) and demonstrated that neutrinos have mass (required by quantum mechanics for oscillation to occur), earning the 2002 Nobel Prize in Physics (Davis) and the 2015 Nobel Prize (Kajita and McDonald, for the discovery of oscillation) (Haxton et al., 2013).

Thus, we now have the **3 stellar structure equations** derived so far:

$$\frac{dm}{dr} = 4\pi r^2 \rho, \quad \frac{dP}{dr} = -\frac{Gm\rho}{r^2}, \quad \frac{dl}{dr} = 4\pi r^2 \rho \varepsilon. \quad (5.105)$$

They successfully describe the process of stellar evolution off the main sequence of the HR diagram. However, they are still insufficient to describe how the energy generated in the core is transported to the star’s outer regions — an equation (or set of equations) for **energy transport** is still missing, the subject of the next section.

The processes that transform H into He have energy generation rates with very different temperature dependencies: the proton–proton chain follows approximately $\varepsilon_{pp} \propto T^4$ at the typical temperatures of the solar core ($T \sim 1.5 \times 10^7$ K), while the CNO cycle, depending on reactions with much higher Coulomb barriers (carbon and nitrogen nuclei, with Z much larger than that of hydrogen), is far more sensitive to temperature, $\varepsilon_{CNO} \propto T^{15-20}$. This difference in thermal sensitivity is, physically, the reason why the pp chain dominates in low-mass stars (cooler cores) while the CNO cycle dominates in stars more massive than the Sun (hotter cores, where the strong T -dependence makes the CNO cycle quickly overtake the pp chain) — with direct consequences for the energy transport regime in the cores of such stars, as we will see in Section 5.12.

5.11 Energy transport equations

In general, stars can be composed of an inner radiative region and a convective one. The main difference between these zones lies in how energy transport is carried out, which is described by the processes of **radiation**, **conduction**, and **convection**.

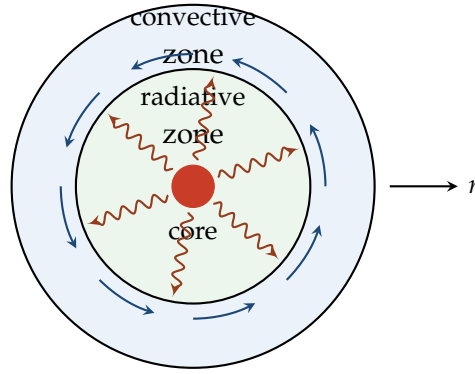


Figure 5.6: Structure of energy transport zones in a solar-type star (schematic, not to scale): core, radiative zone (wavy arrows indicate photons slowly diffusing outward), and convective zone (curved arrows indicate convection cells). In solar-type stars, the radiative zone dominates the inner $\approx 70\%$ of the stellar *radius*, and the convective zone the outer $\approx 30\%$ — but, due to the sharp density drop toward the surface, the convective zone contains only $\approx 2\%$ of the star’s *mass*.

In solar-type stars, $\approx 70\%$ (in radius) is dominated by the radiative region and $\approx 30\%$ (in radius) by the convective region — the Sun has the base of its convective zone at $r \approx 0.71 R_{\odot}$, according to helioseismology-calibrated solar models. It is worth emphasizing that this $\approx 30\%$ fraction is in *radius*, not in *mass*: since density increases very rapidly toward the center, the thin outer convective layer contains only $\approx 2\%$ of the Sun’s total mass, even though it occupies nearly a third of its radius.

Conductive transport is carried out mostly in the radiative region, due to the high collision rate of particles (small mean free path of photons).

Radiative transport

Previously [Chapter 4], we saw that the mean free path of a photon in the solar core ($\ell_{\gamma} \approx 2$ cm) was much smaller than the size of the star ($R_{\odot} \sim 10^{10}$ cm), which allowed us to assume that radiative transport can be approximated by a **diffusion** equation — entirely analogous to **Fick’s law** of particle diffusion,

$$\vec{J} = -D \vec{\nabla} n, \quad (5.106)$$

where $\vec{J}$ is the diffusive flux per unit area and time, D is the diffusion coefficient, and n is the density (of particles, or, in our case, of energy). The negative sign plays the role of a resistance to the flow, in the direction of motion.

Transforming Equation (5.106) to describe our problem, the diffusion coefficient is defined by

$$D = \frac{1}{3} \ell_{\gamma} v_p, \quad (5.107)$$

where ℓ_{γ} is the photon mean free path and v_p is the particle velocity (here, photons, $v_p = c$), with the motion normalized in three radial dimensions (the factor $1/3$). Since $\ell_{\gamma} = 1/(\rho \kappa_{\text{rad}})$, where ρ and κ_{rad} are the density and radiative opacity of the gas [the same opacity κ from Chapter 4, now with the subscript “rad” to distinguish it from the conductive opacity, introduced ahead], the coefficient $\vec{J}$ describes the star’s radiation flux, hence $\vec{J} \equiv \vec{F}$.

We can express the divergence of the energy density as $\vec{\nabla} n = \vec{\nabla} u = (du/dr) \hat{r}$. Substituting the values, we have:

$$\vec{F} = -\frac{1}{3} \ell_{\gamma} v_p \frac{du}{dr} \hat{r}. \quad (5.108)$$

Working with du/dr : since the energy density u can be expressed as $u = aT^4$ [blackbody thermodynamic equation, already established in Chapter 4], the equation takes the form

$$\vec{F} = -\frac{1}{3} \ell_\gamma v_p \frac{d}{dr}(aT^4) = -\frac{1}{3} \ell_\gamma v_p \cdot a \cdot 4T^3 \frac{dT}{dr}, \quad (5.109)$$

and, substituting $\ell_\gamma = 1/(\rho\kappa_{\text{rad}})$ and $v_p = c$:

$$\boxed{\vec{F} = -\frac{1}{3} \frac{4acT^3}{\kappa_{\text{rad}}\rho} \frac{dT}{dr}}. \quad (5.110)$$

Note that this expression can be written as $\vec{F} = \alpha \vec{\nabla} T$, with $\alpha = -\frac{1}{3}(4ac/\kappa_{\text{rad}}\rho)T^3$ — that is, nothing more than a heat conduction equation, with α playing the role of an effective thermal conductivity.

Consistency with the Rosseland Approximation

Equation (5.110), obtained here from a photon diffusion argument (mean free path $\times$ velocity), is *exactly* the same equation already derived in Chapter 4 by a completely different route — the Taylor series expansion of the source function in optically thick media, followed by integration over frequency (the **Rosseland Approximation**):

$$F = -\frac{16\sigma T^3}{3\kappa\rho} \frac{dT}{dr} \quad [\text{Equation (4.131)}]. \quad (5.111)$$

Using the relation between the radiation constant and the Stefan–Boltzmann constant, $\sigma = ac/4$ [already used in Chapter 4]:

$$F = -\frac{16(ac/4)T^3}{3\kappa\rho} \frac{dT}{dr} = -\frac{4acT^3}{3\kappa\rho} \frac{dT}{dr} = -\frac{1}{3} \frac{4acT^3}{\kappa\rho} \frac{dT}{dr}, \quad (5.112)$$

which exactly reproduces Equation (5.110) (with $\kappa \equiv \kappa_{\text{rad}}$). This coincidence is no coincidence: the two arguments — photon diffusion with mean free path $\ell_\gamma = 1/(\rho\kappa)$, and the Rosseland expansion in optically thick media — describe exactly the same radiative transport physics in diffusive regimes, and it is reassuring that the two derivations, made in different chapters of this book with distinct mathematical methods, arrive at the same final result. This is an important internal consistency check between Chapters 4 and 5.

Knowing that the total radiative flux through a sphere of radius r is $F = l/(4\pi r^2)$:

$$\frac{l}{4\pi r^2} = -\frac{1}{3} \frac{4acT^3}{\kappa_{\text{rad}}\rho} \frac{dT}{dr} \implies \boxed{\frac{dT}{dr} = -\frac{3\kappa_{\text{rad}}\rho l}{16\pi ac r^2 T^3}}, \quad (5.113)$$

the radiative transport equation as a function of radius. Note that: (i) radiative energy transport is proportional to dT/dr ; (ii) the flux is larger for more transparent gases (smaller κ_{rad} , the absorption coefficient).

To describe Equation (5.113) as a function of the star's total mass, we use the mass equation, $dr/dm = 1/(4\pi r^2\rho)$ [Equation (5.4)]:

$$\frac{dT}{dm} = \frac{dT}{dr} \cdot \frac{dr}{dm} = \left(-\frac{3\kappa_{\text{rad}}\rho l}{16\pi ac r^2 T^3} \right) \cdot \frac{1}{4\pi r^2\rho} = -\frac{3\kappa_{\text{rad}} l}{64\pi^2 ac r^4 T^3}, \quad (5.114)$$

that is,

$$\frac{dT}{dm} = -\frac{3}{64\pi^2 ac} \cdot \frac{\kappa_{\text{rad}} l}{r^4 T^3}. \quad (5.115)$$

Dividing Equation (5.115) by the hydrostatic equilibrium equation in mass, $dP/dm = -Gm/(4\pi r^4)$ [Equation (5.25)]:

$$\frac{dT}{dP} = \frac{dT/dm}{dP/dm} = \frac{-\frac{3}{64\pi^2 ac} \cdot \frac{\kappa_{\text{rad}} l}{r^4 T^3}}{-\frac{Gm}{4\pi r^4}} = \frac{3}{64\pi^2 ac} \cdot \frac{4\pi}{G} \cdot \frac{\kappa_{\text{rad}} l}{m T^3}, \quad (5.116)$$

that is,

$$\frac{dT}{dP} = \frac{3}{16\pi ac G} \cdot \frac{\kappa_{\text{rad}} l}{m T^3}. \quad (5.117)$$

Physical consistency of the signs

As a physical check, we confirm that the signs of Equations (5.113)–(5.117) are consistent with the expected behavior of temperature in the stellar interior: T must *decrease* with r ($dT/dr < 0$, since temperature falls from the hotter central region toward the surface) — which is already correct in Equation (5.113), since all factors on the right-hand side are positive except the explicit negative sign. Since $dm/dr > 0$ always, the sign of dT/dm [Equation (5.115)] must also be negative, and indeed it is. Meanwhile, T must *increase* with P ($dT/dP > 0$), since both temperature and pressure increase toward the star's center — and since dT/dP is the ratio of two negative terms [$dT/dm < 0$ and $dP/dm < 0$, Equation (5.25)], the result is indeed *positive*, confirming the “+” sign in Equation (5.117). All three signs, therefore, make physical sense.

For high values of κ_{rad} , the radiative flux tends to zero ($F \propto 1/\kappa_{\text{rad}}$), where there will be great resistance to motion, $dT/dr \propto -\kappa_{\text{rad}}$. In this scenario, energy movement in the stellar interior is dominated by conductive transport — or, as we will see in a future section of this chapter, it may become more efficient for the star to transport this energy by **convection** instead of radiation: it is precisely this kind of situation (a very steep radiative gradient) that lies behind the **Schwarzschild criterion for convective stability**, which we will discuss shortly.

From the radiative transport equation to the real $T(r)$ profile

Equation (5.113) — or, equivalently, Equations (5.115) and (5.117) — cannot be solved analytically for a real star, because $\kappa_{\text{rad}} = \kappa_{\text{rad}}(\rho, T, X_i)$ depends, in an extremely nonlinear way, on the local density, temperature, and chemical composition. In practice, stellar evolution codes (such as MESA, already mentioned in previous chapters) solve this problem numerically: at each layer of the star and at each time step, the code consults a pre-computed **opacity table** — typically the **OPAL** (Iglesias and Rogers, 1996) or **Opacity Project** (OP) tables, already mentioned in Chapter 4 —, interpolating $\kappa_{\text{rad}}(\rho, T, X_i)$ for the local values of ρ , T , and chemical composition X_i of that layer. Combining this interpolated opacity with Equation (5.113) (or its mass form) and the other structure equations, the code numerically integrates the complete system of differential equations, layer by layer, building the $T(r)$ profile (and $\rho(r)$, $P(r)$, $l(r)$) of the entire star. It is this procedure — repeated over thousands of time steps — that produces the detailed stellar evolution models used to compare with real observations (HR diagrams, asteroseismology, etc.).

Conductive transport

Conductive transport occurs more efficiently in stars (or stellar regions) with high opacity, with **white dwarfs** being an excellent example. Energy transport in this case is achieved through the exchange of energy from macroscopic collisions of gas particles.

In the radiative transport case, we started from $F_{\text{rad}} = -D_{\text{rad}} dT/dr$. Analogously, we can assume that conductive transport is described by a similar diffusion function,

$$F_{\text{cond}} = -D_{\text{cond}} \frac{dT}{dr}, \quad D_{\text{cond}} = \frac{1}{3} \ell_{\text{cond}} v_p, \quad (5.118)$$

where now ℓ_{cond} and v_p are the mean free path and velocity of the particles responsible for conduction (for example, electrons, in the case of a degenerate gas) — completely analogous to the radiative case, but with $\ell_\gamma \rightarrow \ell_{\text{cond}}$.

Thus, the total energy flux is the sum of the two mechanisms,

$$F = F_{\text{rad}} + F_{\text{cond}} = -(D_{\text{rad}} + D_{\text{cond}}) \frac{dT}{dr}. \quad (5.119)$$

Developing Equation (5.119) leads to the same response as the radiative transport equation as a function of dT/dr , where the only difference will be in the absorption coefficient, now described by a **conductive opacity** κ_{cond} analogous to κ_{rad} :

$$F = -\frac{1}{3} \frac{4ac}{\rho} T^3 \left(\frac{1}{\kappa_{\text{rad}}} + \frac{1}{\kappa_{\text{cond}}} \right) \vec{\nabla} T. \quad (5.120)$$

Total opacity combines as a harmonic mean, not a sum

The combined total opacity is *not* the direct sum $\kappa_T = \kappa_{\text{rad}} + \kappa_{\text{cond}}$. This becomes clear by examining Equation (5.120) itself: the term in parentheses is $1/\kappa_{\text{rad}} + 1/\kappa_{\text{cond}}$, and for this to equal $1/\kappa_T$ (allowing the final result to be written as a single effective opacity) it is necessary that

$$\frac{1}{\kappa_T} = \frac{1}{\kappa_{\text{rad}}} + \frac{1}{\kappa_{\text{cond}}}, \quad (5.121)$$

that is, $\kappa_T = \kappa_{\text{rad}} \kappa_{\text{cond}} / (\kappa_{\text{rad}} + \kappa_{\text{cond}})$ — a **harmonic mean**, not a direct sum. This is exactly the standard form found in the literature (Kippenhahn et al., 2012), and it makes physical sense: as with electrical resistors in parallel, the *more efficient* transport mechanism (smaller κ) dominates the total flux, and κ_T always turns out smaller than the smaller of the two individual κ 's — never larger, as a direct sum would erroneously suggest.

Using Equation (5.121), the final combined form of radiative and conductive energy transport is

$$\vec{F} = -\frac{4acT^3}{3\rho} \cdot \frac{1}{\kappa_T} \vec{\nabla} T, \quad (5.122)$$

valid in any region of the star, radiative, conductive, or a combination of both regimes.

Conduction by degenerate electrons in white dwarfs

Why is conduction so efficient in white dwarfs, to the point of completely dominating over radiation in their interiors? The answer lies in the nature of the electron gas present there: at densities $\rho \sim 10^6\text{--}10^9 \text{ g/cm}^3$, the electrons form a **degenerate** gas (the Pauli exclusion

principle, not temperature, determines its pressure and typical kinetic energy — a topic to be revisited in the White Dwarfs chapter). Degenerate electrons occupy very well-defined momentum states and travel at typically relativistic or quasi-relativistic velocities; combined with the fact that the phase space available for scattering is extremely restricted by the exclusion principle itself (an electron can only be scattered into an unoccupied state, and most low-energy states are already occupied), the result is a very long effective mean free path for energy transport — exactly as in an ordinary metal, where conduction electrons (also a nearly degenerate gas) are responsible for the extremely high thermal and electrical conductivity. This is why ℓ_{cond} [Equation (5.118)] can be orders of magnitude larger than ℓ_{γ} under these conditions, making $\kappa_{\text{cond}} \ll \kappa_{\text{rad}}$ and causing conduction to completely dominate in Equation (5.121). This extremely high thermal conductivity has an important structural consequence: the degenerate core of a white dwarf is, to an excellent approximation, **isothermal** — conduction is so efficient that any temperature gradient there is quickly smoothed out —, with almost the entire temperature gradient of the star concentrated in the thin, non-degenerate outer envelope, where radiative opacity again dominates. It is this nearly isothermal core, acting as a heat reservoir that slowly cools by diffusion through the envelope, that governs the so-called **cooling tracks** of white dwarfs — the basis of methods for dating old stellar populations via the cooling sequence observed in clusters. Recent results show that this cooling can be further significantly delayed by phase-separation processes during core crystallization (lighter ^{22}Ne crystals floating up and releasing additional gravitational energy), which can extend the cooling time by billions of years for massive white dwarfs (Bédard et al., 2024).

If one of the terms, κ_{rad} or κ_{cond} , is much larger than the other in the sum $(1/\kappa_{\text{rad}} + 1/\kappa_{\text{cond}})^{-1}$ [Equation (5.121)], we can neglect the smaller term (the inverse of the larger one dominates the sum). In practice, this means that, in any region of the star, it suffices to consider the *more efficient* transport mechanism — radiative or conductive — and it is not necessary to solve both simultaneously.

Rosseland mean opacity

In determining the radiative transport equations, we did not take into account that opacity is a function of the gas's frequency. Assuming this dependence, we obtain:

$$F_{\nu} = -\frac{1}{3} \ell_{\gamma} v_p \frac{\partial}{\partial r} \left(\frac{4\pi}{c} B_{\nu} \right) \hat{r}, \quad (5.123)$$

where κ_{ν} is the opacity at a specific frequency ν (instead of the bolometric opacity κ used so far), and the term in parentheses, $u_{\nu} \equiv (4\pi/c) B_{\nu}$, is the spectral analog of the energy density $u = aT^4$ used previously [Equation (5.108)] — the same relation $u_{\nu} = 4\pi J_{\nu}/c$, with $J_{\nu} = B_{\nu}$ in LTE, that underlies $u = aT^4 = (4\pi/c) \int_0^{\infty} B_{\nu} d\nu$. Using $\ell_{\gamma} = 1/(\kappa_{\nu}\rho)$ and $v_p = c$ — so that the factor v_p/c cancels exactly —, and applying the chain rule $[\partial B_{\nu}/\partial r = (\partial B_{\nu}/\partial T)(\partial T/\partial r)]$, since $B_{\nu} = B_{\nu}(T)$:

$$F_{\nu} = -\frac{4\pi\ell_{\gamma}}{3} \frac{\partial B_{\nu}(T)}{\partial r} = -\frac{4\pi}{3\kappa_{\nu}\rho} \frac{\partial B_{\nu}}{\partial T} \frac{\partial T}{\partial r}. \quad (5.124)$$

Integrating both sides over the entire frequency space, we have:

$$\int_0^{\infty} F_{\nu} d\nu = - \left(\frac{4\pi}{3\rho} \int_0^{\infty} \frac{1}{\kappa_{\nu}} \frac{\partial B_{\nu}}{\partial T} d\nu \right) \frac{\partial T}{\partial r}, \quad (5.125)$$

that is,

$$F_{\text{bol}} = -\frac{4\pi}{3\rho} \int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B_\nu}{\partial T} d\nu \frac{\partial T}{\partial r}, \quad (5.126)$$

which is the *exact* radiative flux transporting energy (without assuming, a priori, that opacity is independent of frequency).

Notation: F_ν is a radial vector

Equation (5.123) ends with $\hat{r}$: F_ν is a vector in the radial direction, exactly as $\vec{F}$ in the previous equations of this section — the same convention already used, for example, in the volumetric gravitational force [Equation (5.14)].

Recalling that, in local thermodynamic equilibrium (LTE), we had already determined that [Equation (5.110)]

$$F = -\frac{1}{3} \frac{4ac}{\bar{\kappa}\rho} T^3 \frac{\partial T}{\partial r}, \quad (5.127)$$

where $\bar{\kappa}$ now denotes the **mean opacity** (the bolometric analog of κ_ν). Comparing Equation (5.127) with Equation (5.126) — both describe the same physical flux, $F = F_{\text{bol}}$ —, and canceling $\partial T / \partial r$ on both sides:

$$\frac{1}{3} \frac{4ac}{\bar{\kappa}\rho} T^3 = \frac{4\pi}{3\rho} \int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B_\nu}{\partial T} d\nu, \quad (5.128)$$

and also canceling the common factors $4/(3\rho)$:

$$\frac{1}{\bar{\kappa}} = \frac{\int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B_\nu}{\partial T} d\nu}{\frac{acT^3}{\pi}}. \quad (5.129)$$

It can be shown that

$$\int_0^\infty \frac{\partial B_\nu}{\partial T} d\nu = \frac{acT^3}{\pi}, \quad (5.130)$$

an auxiliary identity that follows directly from integrating Planck's law [analogously to the identity $\int_0^\infty B_\nu d\nu = \sigma T^4 / \pi$, already used in Chapter 4, simply differentiating both sides with respect to T and using $\sigma = ac/4$]. Substituting into Equation (5.129):

$$\boxed{\frac{1}{\bar{\kappa}} = \frac{\int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B_\nu}{\partial T} d\nu}{\int_0^\infty \frac{\partial B_\nu}{\partial T} d\nu}}, \quad (5.131)$$

which is defined as the **Rosseland mean opacity**. We can express the opacity of any gas as a harmonic mean, weighted by the derivative of the Planck function with respect to temperature, $\partial B_\nu / \partial T$ — which gives more weight to frequencies where the gas is more transparent (smaller κ_ν) and where the Planck function varies more rapidly with T , exactly as already anticipated in the discussion of the Rosseland Approximation in Chapter 4.

5.12 Convective energy transport

The study of energy transport by convection tends to be quite complex, since it depends on all thermodynamic variables. In this case, we will consider the existence of instability regions, or gas **bubbles**, and analyze whether they remain present given a few constraints. The intuitive idea is simple: inner regions with higher density tend to have higher temperature, expand, and rise toward the outer regions — and the converse is also true (cooler, denser regions tend to sink).

Thermodynamic assumptions for the bubble



Figure 5.7: Schematic of the gas bubble: at the initial instant, it is in equilibrium with the medium (surroundings); at the final instant, it has already risen relative to its original position. A denotes the surrounding medium and B the bubble.

Let us consider the thermodynamic states (P_B, T_B, ρ_B) of the bubble and (P_A, T_A, ρ_A) of the surrounding medium ($A \rightarrow$ medium, $B \rightarrow$ bubble), subject to the following initial conditions:

- $\rho_B > \rho_A, T_B > T_A, P_B > P_A$;
- the bubble is expanding, $P_B \rightarrow P_A$;
- we assume $T_B > T_A$ even after the expansion;
- the expansion is adiabatic, $dq = 0$.

At the end of the thermodynamic transformation, we will have $P_B \equiv P_A$ and $T_B > T_A$. Considering an ideal gas,

$$P = nRT\rho, \quad (5.132)$$

On the notation $n \equiv 1/\mu$ in Equation (5.132)

At first glance, $P = nRT\rho$ appears to have an extra factor ρ compared to the usual ideal gas form. If n denoted molar concentration (mol/volume), the equation of state would already be $P = nRT$, without additional density. The reading consistent with the rest of the book — where the ideal gas equation of state is always written as $P = \rho RT/\mu$ [Equation (3.44), Chapter 3] — is that here $n \equiv 1/\mu$, the **number of moles per unit mass** (the inverse of the mean molecular weight), not the molar concentration. With this identification, $P = nRT\rho = (1/\mu)RT\rho = \rho RT/\mu$, exactly the standard form — there is no inconsistency, only a choice of symbol (n instead of $1/\mu$) that is worth making explicit, since n (thus defined) reappears in the Ledoux Criterion, ahead, as a way of encoding chemical composition differences between the bubble and the medium.

Since at the end of the process $P_B = P_A$, we have

$$n_B R T_B \rho_B = n_A R T_A \rho_A, \quad (5.133)$$

and, since $T_B > T_A$, for this equality to hold, we must have

$$n_B \rho_B < n_A \rho_A. \quad (5.134)$$

If the chemical composition is homogeneous ($n_B = n_A$, the same assumption behind the Schwarzschild Criterion, ahead), it follows directly that $\rho_B < \rho_A$.

Forces on the bubble and the medium: buoyancy

Considering that both gravitational and pressure forces act on the system, we have, for both the bubble and the medium,

$$\frac{\partial P_B}{\partial z} + \frac{G m \rho_B}{r^2} \hat{z} = \rho_B \frac{\partial^2 z}{\partial t^2} \hat{z}, \quad \frac{\partial P_A}{\partial z} + \frac{G m \rho_A}{r^2} \hat{z} = \rho_A \frac{\partial^2 z}{\partial t^2} \hat{z}, \quad (5.135)$$

which will be exactly equal to the hydrostatic equilibrium equation [Equation (5.21)] should there be no acceleration.

But note that the same magnitude of forces is being applied to both the bubble and the medium. This indicates that the resultant force should be zero, and the system would remain at rest. However, there is motion — the bubble rises! This motion can only be explained by the action of an additional external force, acting upward on the bubble: this is the **buoyancy force**.

Schwarzschild instability criterion

The action of buoyancy makes the motion continue. We thus have an instability criterion.

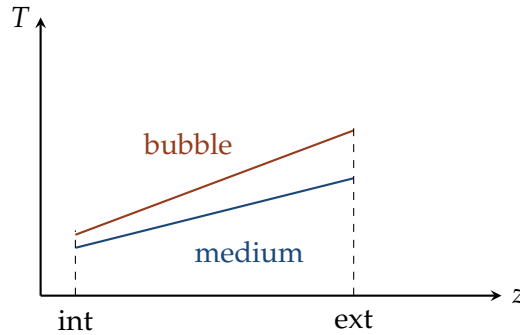


Figure 5.8: Temperature profile T versus z for the bubble (less steep) and the surrounding medium (steeper), from the interior (int) to the exterior (ext) of the region considered. Instability occurs when the bubble cools *more slowly* than the medium as it rises.

The instability can be described by the difference

$$\left(\frac{dT}{dz} \right)_B < \left(\frac{dT}{dz} \right)_A. \quad (5.136)$$

If a buoyancy force is acting on the bubble, it must cool *less* in order to reach the upper layers, compared to the gas of the surrounding medium — this produces the inequality of

Equation (5.136). Knowing that $dT/dz = (dP/dz)(dT/dP)$, we have:

$$\left(\frac{dP}{dz} \frac{dT}{dP}\right)_B < \left(\frac{dP}{dz} \frac{dT}{dP}\right)_A. \quad (5.137)$$

Using the hydrostatic equilibrium equation, $dP/dz = -Gm\rho/r^2$ [Equation (5.22)] — the same on both sides, since the bubble and the medium are, at every instant, at the same position z (same m , same r) —, the common factor $-Gm/r^2$ is negative; canceling it from both sides of the inequality *reverses the direction of the inequality*:

$$\left(\frac{dT}{dP} \rho\right)_B > \left(\frac{dT}{dP} \rho\right)_A. \quad (5.138)$$

Using $\rho = P/(nRT)$ [Equation (5.132)] and assuming homogeneous composition ($n_B = n_A$, canceling nR from both sides):

$$\left(\frac{dT}{dP} \frac{P}{T}\right)_B < \left(\frac{dT}{dP} \frac{P}{T}\right)_A, \quad (5.139)$$

where the inequality returns to its original direction: the *two* sign cancellations — that of $-Gm/r^2$ (negative) and that of the reversal when isolating dT/dP by multiplying by $\rho > 0$ — cancel each other out, and the final result is consistent with Equation (5.136), as it should be. Defining

$$\nabla \equiv \frac{d \ln T}{d \ln P}, \quad (5.140)$$

Equation (5.139) becomes, compactly,

$$(\nabla)_B < (\nabla)_A, \quad (5.141)$$

which can be interpreted as

$$\boxed{\nabla_{\text{ad}} < \nabla_{\text{rad}}} \longrightarrow \text{Schwarzschild criterion}, \quad (5.142)$$

since the bubble, in adiabatic expansion, follows the adiabatic gradient ∇_{ad} , while the medium follows the gradient that would result from purely radiative transport, ∇_{rad} [Equation (5.117), rewritten in logarithmic form].

Physical interpretation of the Schwarzschild Criterion

Physically, the criterion $\nabla_{\text{ad}} < \nabla_{\text{rad}}$ [Equation (5.142)] defines the condition for convective instability: it occurs when the gradient that would result from purely radiative transport, ∇_{rad} , is larger than the adiabatic gradient, ∇_{ad} . This means that the star “would like” to establish a temperature gradient so steep that a bubble displaced adiabatically upward would arrive hotter — and therefore less dense — than the surrounding medium, experiencing positive buoyancy and continuing to rise.

Observational evidence of stellar convection

Convection is not merely a theoretical deduction — it is directly observed on the surface of the Sun in the form of **solar granulation**: a pattern of bright polygonal cells (the tops of ascending hot gas columns, with a typical diameter of ~ 1000 km) surrounded by narrow dark channels (cooler gas descending), the entire structure renewing itself within a few minutes. Very high spatial resolution observations, such as those obtained by the *Hinode*

satellite and, more recently, by the ground-based solar telescope *Daniel K. Inouye Solar Telescope* (DKIST, with a 4 m mirror, the largest in the world dedicated to solar observation), allow direct measurement of the temperature and (line-of-sight) velocity of the gas in individual granules and intergranular lanes, quantitatively testing the predictions of convection theory (Crowley et al., 2025). The radial extent of this convective zone — whose depth helioseismology fixes at $r \approx 0.71 R_\odot$ — has already been discussed earlier in this chapter and in Chapter 3.

Adiabatic process and the calculation of ∇_{ad}

In adiabatic processes, there is no heat exchange, that is, $dq = 0$. With this, we have, from the first law of thermodynamics, that $du + dw = dq \Rightarrow du + dw = 0$. As seen previously, $dw = +P dV$, hence

$$du = -P dV. \quad (5.143)$$

Recalling that $C_V = (\partial u / \partial T)_V$, we can integrate and obtain $u = C_V T + u_0$ (u_0 constant), that is, $du = C_V dT$; for n moles, $du = nC_V dT$. Substituting into Equation (5.143):

$$-P dV = nC_V dT, \quad (5.144)$$

which describes the energy variation in terms of P , V , T . Using the ideal gas equation, $PV = nRT$, and differentiating both sides:

$$d(PV) = nR dT \implies P dV + V dP = nR dT, \quad (5.145)$$

$$V dP = -P dV + nR dT, \quad (5.146)$$

substituting Equation (5.144):

$$V dP = nC_V dT + nR dT = n(C_V + R) dT. \quad (5.147)$$

But, as we saw [Chapter 3], $C_P = C_V + R$, hence

$$V dP = nC_P dT. \quad (5.148)$$

Substituting dT from Equation (5.144) [$dT = -P dV / (nC_V)$] into Equation (5.148):

$$V dP = nC_P \left(\frac{-P dV}{nC_V} \right) \implies \frac{dP}{P} = -\frac{C_P}{C_V} \frac{dV}{V} = -\gamma \frac{dV}{V}, \quad (5.149)$$

where $\gamma \equiv C_P / C_V$. Integrating Equation (5.149):

$$\int_{P_0}^P \frac{dP'}{P'} = -\gamma \int_{V_0}^V \frac{dV'}{V'} \implies \ln \left(\frac{P}{P_0} \right) = -\gamma \ln \left(\frac{V}{V_0} \right), \quad (5.150)$$

that is, $P/P_0 = (V_0/V)^\gamma$, and therefore

$$\boxed{PV^\gamma = \text{const} = P_0 V_0^\gamma}. \quad (5.151)$$

Thus, $P \propto V^{-\gamma}$, or, in terms of density ($\rho \propto 1/V$),

$$P \propto \rho^\gamma. \quad (5.152)$$

Using the equation of state, $PV \propto T$:

$$T \propto \rho^\gamma \cdot \rho^{-1} \implies T \propto \rho^{\gamma-1}. \quad (5.153)$$

We can obtain the adiabatic gradient assuming that the gas is ideal. Hence, since $\nabla_{\text{ad}} = d \ln T / d \ln P$, and from Equations (5.152) and (5.153), $d \ln T \propto (\gamma - 1) d \ln \rho$ and $d \ln P = \gamma d \ln \rho$, we have:

$$\boxed{\nabla_{\text{ad}} = \frac{\gamma - 1}{\gamma}}, \quad (5.154)$$

for a monatomic ideal gas, $\gamma = 5/3$, hence

$$\nabla_{\text{ad}} = \frac{5/3 - 1}{5/3} = \frac{2/3}{5/3} = \frac{2}{5} = 0.4. \quad (5.155)$$

This is exactly the same value already obtained in Chapter 3 [Equation (3.52)], there via an independent route — through entropy and the alternative form of the first law, $dq = C_P dT - \delta dP/\rho = 0$ — instead of the $PV^\gamma = \text{const}$ argument used here. The two derivations agreeing is a welcome consistency check between the two chapters.

Then, the criterion [Equation (5.142)] can be rewritten, in this case, as

$$\boxed{\nabla_{\text{rad}} > \frac{\gamma - 1}{\gamma} \equiv \nabla_{\text{ad}}}. \quad (5.156)$$

If this inequality holds, energy (and matter) transport is carried out by convection.

Convective motion processes occur in weakly or partially ionized regions, and also in regions with very high C_V .

Where convection dominates in the HR diagram

We already saw, in Chapter 3, why the zone of partial hydrogen ionization pushes ∇_{ad} down and triggers convection in the solar envelope. It is worth generalizing this idea to the entire main sequence. In **low-mass** stars (types K and M, $M \lesssim 0.8 M_\odot$), this same mechanism is even more effective — the envelopes are convective throughout nearly their entire extent, and stars with $M \lesssim 0.35 M_\odot$ are, in practice, fully convective. In **massive** stars (types O and B, $M \gtrsim 1.3 M_\odot$), on the other hand, it is the *core* that is convective, not the envelope: the strong temperature dependence of the CNO cycle ($\varepsilon_{\text{CNO}} \propto T^{15-20}$, Section 5.10) concentrates energy generation in a very small central volume, demanding an enormous ∇_{rad} there to transport all that luminosity radiatively — large enough to violate the Schwarzschild criterion and trigger convection in the core, while the envelope, more rarefied and with lower opacity, remains radiative. The Sun, with intermediate mass, illustrates the transition: radiative core and convective envelope, as already discussed in this chapter.

∇_{ad} for other compositions

Equation (5.154) depends only on γ , which allows us to compute ∇_{ad} for any ideal gas. For a **diatomic** ideal gas (e.g., molecular H_2 , relevant in cool atmospheres and planets), $\gamma = 7/5$, hence

$$\nabla_{\text{ad}} = \frac{7/5 - 1}{7/5} = \frac{2/5}{7/5} = \frac{2}{7} \approx 0.286, \quad (5.157)$$

a value smaller than 0.4 — diatomic gases have a shallower adiabatic gradient, which

makes them more susceptible to convective instability (it is easier to satisfy $\nabla_{\text{rad}} > \nabla_{\text{ad}}$) than an equivalent monatomic gas. In the zones of partial H and He ionization in the solar envelope, the situation is even more extreme: part of the energy from the bubble's compression/expansion goes into ionizing (or recombining) atoms instead of changing the temperature, which reduces the gas's *effective* γ to well below 5/3 — in general, γ_{eff} can drop to close to 1 at the peak of an ionization zone. Since $\nabla_{\text{ad}} = (\gamma - 1)/\gamma \rightarrow 0$ as $\gamma \rightarrow 1$, this makes convective instability even more likely precisely in these zones — reinforcing, by a different thermodynamic route, the same physical conclusion as the previous context box.

Ledoux instability criterion

Another way of writing the instability criterion is

$$\left(\frac{\partial \rho}{\partial r}\right)_B < \left(\frac{\partial \rho}{\partial r}\right)_A. \quad (5.158)$$

But we previously defined that [Equation (3.40), Chapter 3]

$$\frac{d\rho}{\rho} = \alpha \frac{dP}{P} - \delta \frac{dT}{T}. \quad (5.159)$$

If the gas's chemical composition can vary, we obtain $F(\rho, P, T, \mu) = 0$ (an extra dimension). In this case, Equation (5.159) takes the form

$$\frac{d\rho}{\rho} = \alpha \frac{dP}{P} - \delta \frac{dT}{T} + \phi \frac{d\mu}{\mu}, \quad \phi \equiv \frac{d \ln \rho}{d \ln \mu}. \quad (5.160)$$

Ledoux vs. Schwarzschild, and semiconvection

Equation (5.160) generalizes the Schwarzschild Criterion [Equation (5.142)] to the case in which the chemical composition is *not* homogeneous — precisely the assumption we made when canceling $n_B = n_A$ in the Schwarzschild derivation. When $\nabla_\mu \equiv d \ln \mu / d \ln P = 0$ (homogeneous composition), the extra term vanishes and the **Ledoux Criterion** reduces exactly to the Schwarzschild one. When there is a *stabilizing* mean-molecular-weight gradient (typically μ increasing toward the center, as occurs above a convective core that converts H into He — heavier), a new possibility arises: a region may be formally unstable by the Schwarzschild criterion (which ignores composition) but stable by the Ledoux criterion (which includes it) — the so-called **semiconvection**. Physically, the μ gradient acts as an additional brake on the bubble's buoyancy, allowing only slow, partial mixing, rather than fully developed convection. Semiconvection is especially relevant in the evolution of massive stars, at the edge of the convective H core, where the choice between the two criteria (and between different prescriptions for semiconvective diffusion) significantly affects the final helium core mass, the tracks in the HR diagram, and even the star's fate as a supernova (Sibony et al., 2023).

Convective energy flux (Mixing Length Theory)

We can describe the energy flux carried by the bubbles as

$$F_{\text{conv}} = \rho_B v_B u_B, \quad (5.161)$$

where v_B is the particle velocity and u_B is the specific energy (per unit mass) carried by the bubble, in excess relative to the medium. Since $u_B = C_p \Delta T$, and $\Delta T \equiv T_B - T_A$, we need a closed expression for ΔT in terms of already known quantities. This leads us to **Mixing Length Theory (MLT)**: we assume that the bubble travels a characteristic distance ℓ_m (the **mixing length**) before dissolving into the medium, accumulating along this path a temperature difference proportional to $(\nabla - \nabla_{\text{ad}})$ — the excess of the actual gradient over the adiabatic one — and to the pressure scale height,

$$H_P \equiv -\frac{dr}{d \ln P} = -P \frac{dr}{dP}. \quad (5.162)$$

The convective flux is then

$$F_{\text{conv}} = \rho_B v_B C_p \left[-\ell_m \frac{T}{H_P} (\nabla - \nabla_{\text{ad}}) \right], \quad (5.163)$$

where we define

$$\Delta T = -\ell_m \frac{T}{H_P} (\nabla - \nabla_{\text{ad}}). \quad (5.164)$$

Dimensional check of Equation (5.164)

It is worth checking this expression term by term. ∇ and ∇_{ad} are dimensionless [Equation (5.140)]; T/H_P has dimension of temperature over length; multiplying by ℓ_m (a length), we obtain a dimension of temperature — as it should be for ΔT . Equation (5.164) is, therefore, dimensionally correct, and algebraically identical to $(\ell_m/H_P)(\nabla - \nabla_{\text{ad}}) T$, the most common way of writing it in the literature. The only discrepancy relative to the full MLT derivation (e.g., Kippenhahn & Weigert) is the **absence of the factor 1/2**: the standard form in the literature is $\Delta T \approx (\ell_m/2H_P)(\nabla - \nabla_{\text{ad}}) T$, with the 1/2 arising from an average over the bubble's displacement along its entire path (which runs from 0 to ℓ_m , with mean displacement $\ell_m/2$), instead of directly using the total displacement ℓ_m . Since the derivation presented here is an order-of-magnitude estimate — not a formal average over the trajectory —, we keep the result without inserting the factor 1/2, but it is worth flagging this difference relative to the more rigorous literature.

Estimating convective velocity and mixing length

We can obtain an estimate for the convective velocity of the particles. If the difference between the density of the gas elements and their surroundings is $\Delta\rho$, we can express the acceleration by

$$a = -g \frac{\Delta\rho}{\rho} \approx g \frac{\Delta T}{T}, \quad (5.165)$$

assuming that P is constant and that $\rho \propto 1/T$ (ideal gas at fixed pressure, $P \propto \rho T \Rightarrow \rho \propto 1/T$).

We can also describe the distance ℓ_m using the fundamental law of vertical displacement, where

$$\ell_m = v_0 t + \frac{1}{2} a t^2. \quad (5.166)$$

Assuming $v_0 \equiv 0$ (the bubble starts from rest), we have

$$\ell_m = \frac{1}{2} a t^2. \quad (5.167)$$

Since the average velocity is described by

$$v_B = \frac{\ell_m}{t} = \frac{\frac{1}{2}at^2}{t} = \frac{1}{2}at, \quad (5.168)$$

we can write v_B as a function of distance rather than time. Isolating t in Equation (5.167), $t = (2\ell_m/a)^{1/2}$, and substituting into Equation (5.168):

$$v_B = \frac{1}{2}a \left(\frac{2\ell_m}{a} \right)^{1/2} = \left(\frac{a^2}{2^2} \cdot \frac{2\ell_m}{a} \right)^{1/2} \approx \left(\frac{1}{2} \ell_m a \right)^{1/2}. \quad (5.169)$$

Substituting $a = g\Delta T/T$ [Equation (5.165)] and, next, ΔT in the MLT form [Equation (5.164)]:

$$v_B \approx \left(\frac{1}{2} \ell_m g \frac{\Delta T}{T} \right)^{1/2} = \left[\frac{1}{2} \ell_m g \frac{1}{T} \cdot \frac{T}{H_p} (\nabla - \nabla_{\text{ad}}) \ell_m \right]^{1/2}, \quad (5.170)$$

that is,

$$v_B \approx \left[\frac{\ell_m^2 g}{2H_p} (\nabla - \nabla_{\text{ad}}) \right]^{1/2}. \quad (5.171)$$

Substituting into the expression for F_{conv} [Equation (5.163)]:

$$F_{\text{conv}} = \rho_B C_P \left[\frac{\ell_m^2 g}{2H_p} (\nabla - \nabla_{\text{ad}}) \right]^{1/2} \cdot \left[-\ell_m \frac{T}{H_p} (\nabla - \nabla_{\text{ad}}) \right], \quad (5.172)$$

which, rearranging the factors, arrives at the final form:

$$F_{\text{conv}} = \rho_B C_P T \left(\frac{\ell_m}{H_p} \right)^2 \left(\frac{1}{2} g H_p \right)^{1/2} (\nabla - \nabla_{\text{ad}})^{3/2}. \quad (5.173)$$

The term $(\nabla - \nabla_{\text{ad}})$ is called the **superadiabatic term**: it measures how much the actual temperature gradient exceeds the adiabatic gradient, and acts as a resistance term to convective motion — the larger the excess, the more vigorous the convection [Equation (5.173), where $F_{\text{conv}} \propto (\nabla - \nabla_{\text{ad}})^{3/2}$].

Estimating the convective temperature gradient

What value of $\nabla - \nabla_{\text{ad}}$ is required to transport *all* of a star's energy flux by pure convection, $F = L/(4\pi r^2)$? We can obtain this estimate by making the following order-of-magnitude approximations:

- $\rho \approx \bar{\rho} = 3M/(4\pi R^3)$ (mean density);
- an order-of-magnitude estimate for $\bar{T}$, obtained from hydrostatic equilibrium: $dP/dr \sim -P/R \sim -GM\rho/R^2$, that is, $P/\rho \sim GM/R$; using the ideal gas equation, $P/\rho = \mathcal{R}T/\mu$, we have $\bar{T} \sim (\mu/\mathcal{R})(GM/R)$;
- $C_P = \frac{5}{2}(\mathcal{R}/\mu)$, valid for a monatomic ideal gas [Chapter 3];
- $\sqrt{gH_p} = (P/\rho)^{1/2} \sim (GM/R)^{1/2}$, again using $P/\rho \sim GM/R$.

Order-of-magnitude estimates for $\bar{T}$ and $\sqrt{gH_P}$

From hydrostatic equilibrium, in order of magnitude, $dP/dr \sim P/R \sim Gm\rho/r^2 \sim GM\rho/R^2$ (substituting $m \rightarrow M, r \rightarrow R$, the crudest possible estimate), that is, $P/\rho \sim GM/R$. Combining with the ideal gas equation, $P = \rho\mathcal{R}T/\mu$ [Equation (3.44), Chapter 3], that is, $P/\rho = \mathcal{R}T/\mu$, we equate the two expressions: $\mathcal{R}\bar{T}/\mu \sim GM/R$, hence $\bar{T} \sim (\mu/\mathcal{R})(GM/R)$. For $\sqrt{gH_P}$, starting from $gH_P = (GM/R^2) \cdot H_P$ and using the definition of H_P [Equation (5.162)] in order of magnitude ($H_P \sim P/(\rho g)$, the very definition of the pressure scale height), we obtain directly $gH_P \sim P/\rho \sim GM/R$, and therefore $\sqrt{gH_P} \sim (GM/R)^{1/2}$.

Substituting these approximations into Equation (5.173) and neglecting numerical terms of order unity (such as $5/2$, $\sqrt{2}$, and the ratio ℓ_m/H_P , all treated as ~ 1 in this estimate), we obtain

$$F_{\text{conv}} \sim \frac{M}{R^3} \left(\frac{GM}{R} \right)^{3/2} (\nabla - \nabla_{\text{ad}})^{3/2}. \quad (5.174)$$

Considering that $F_{\text{conv}} = L/(4\pi r^2) \sim L/R^2$ (using $r \sim R$, and dropping the factor 4π as being of order unity):

$$\frac{L}{R^2} \sim \frac{M}{R^3} \left(\frac{GM}{R} \right)^{3/2} (\nabla - \nabla_{\text{ad}})^{3/2}, \quad (5.175)$$

isolating $(\nabla - \nabla_{\text{ad}})$:

$$(\nabla - \nabla_{\text{ad}})^{3/2} \sim \frac{LR}{M} \left(\frac{R}{GM} \right)^{3/2} \implies \boxed{\nabla - \nabla_{\text{ad}} \sim \left(\frac{LR}{M} \right)^{2/3} \frac{R}{GM}}. \quad (5.176)$$

Substituting reference values for the Sun ($L_{\odot} = 3.828 \times 10^{33}$ erg/s, $R_{\odot} = 6.957 \times 10^{10}$ cm, $M_{\odot} = 1.989 \times 10^{33}$ g, $G = 6.674 \times 10^{-8}$ cm³ g⁻¹ s⁻²), we numerically confirm:

$$\nabla - \nabla_{\text{ad}} \sim 10^{-8}. \quad (5.177)$$

Thus, convection is so efficient that only a tiny fraction of deviation from the adiabatic process is needed to transport the entire stellar flux. That is, in the convective regions of the stellar interior,

$$\nabla \approx \nabla_{\text{ad}} \quad (5.178)$$

is an excellent approximation. This allows us to simplify the mass form of the radiative transport equation [Equation (5.115), generalized to include both regimes] using directly $\nabla \equiv \nabla_{\text{ad}}$ whenever the layer is convectively unstable:

$$\frac{dT}{dm} = -\frac{Gm}{4\pi r^4} \frac{T}{P} \nabla, \quad \nabla = \nabla_{\text{ad}} \text{ (convective regions)}. \quad (5.179)$$

$\nabla \approx \nabla_{\text{ad}}$ holds for the interior, not the surface

The approximation $\nabla - \nabla_{\text{ad}} \sim 10^{-8}$ [Equation (5.177)] holds for the **deep interior** of the convective zone, where density is high and convection transports the flux with enormous efficiency. It does *not* apply to the outermost layers (near the photosphere), where density drops sharply, the bubble's dynamical time becomes comparable to the radiative cooling time, and convection ceases to be efficient — it is precisely there that the deviation

$\nabla - \nabla_{\text{ad}}$ can become large (non-negligible), and the full form of MLT [Equation (5.173)], with the free parameter $\alpha_{\text{MLT}} = \ell_m / H_P$, remains necessary. In other words: in the deep interior, $\nabla \approx \nabla_{\text{ad}}$ makes MLT almost dispensable (the gradient is already known *a priori*); near the surface, it is exactly the opposite, and the choice of α_{MLT} directly affects observables such as the stellar radius.

Limitations of Mixing Length Theory

Despite its practical usefulness — MLT is, even today, the standard method for treating convection in the vast majority of 1D stellar evolution codes —, it is worth highlighting its limitations. The parameter $\alpha_{\text{MLT}} \equiv \ell_m / H_P$ is a **free parameter**, without rigorous physical justification from first principles; it is typically calibrated by fitting a solar model to reproduce the Sun's observed radius and luminosity at the solar age (typically $\alpha_{\text{MLT}} \approx 1.8\text{--}2.0$), and then applied, with the same value, to stars of very different mass, composition, and evolutionary stage — an extrapolation not always justified. Modern 3D hydrodynamic simulations of stellar convection (*radiative hydrodynamics*, RHD — e.g., the works of Trampedach, Magic, and Freytag, among others) show that the simple physical picture of MLT (discrete, isolated bubbles rising a fixed distance ℓ_m) does not correspond to the actual structure of convection, which is strongly asymmetric (narrow ascending columns surrounded by a broader, more diffuse downflow — exactly the pattern observed in solar granulation); these simulations also show that the effective value of α_{MLT} that best reproduces 3D behavior varies systematically with the star's effective temperature, surface gravity, and metallicity (Magic et al., 2015). A recent, accessible review of the state of the art of MLT — including its limitations and attempts to overcome them — can be found in Joyce and Tayar (2023).

Further Reading and Current References

For a complete and rigorous derivation of Mixing Length Theory, including the calculation of the actual gradient ∇ (which, unlike ∇_{ad} and ∇_{rad} , depends on the efficiency of convection and cannot be obtained without simultaneously solving the convective and radiative transport equations), see the convection chapter of Kippenhahn et al. (2012). For a modern, didactic, and up-to-date review specifically on MLT in 1D stellar modeling — including its history, complete formulation, and the main attempts at calibration and replacement by 3D simulations — see Joyce and Tayar (2023), and, for the quantitative calibration of α_{MLT} against 3D RHD simulations, Magic et al. (2015). For the distinction between the Schwarzschild and Ledoux criteria and its consequences for the evolution of massive stars (including semiconvection), see Sibony et al. (2023).

And we can then simply write the energy transport by convection as

$$\frac{dT}{dm} = -\frac{Gm}{4\pi r^4} \frac{T}{P} \nabla, \quad \text{where now } \nabla \equiv \nabla_{\text{ad}}. \quad (5.180)$$

5.13 Summary: the four stellar structure equations

We thus have that the 4 stellar structure equations are:

$$\frac{\partial r}{\partial m} = \frac{1}{4\pi r^2 \rho} \quad (\text{mass conservation}) \quad (5.181)$$

$$\frac{\partial P}{\partial m} = -\frac{Gm}{4\pi r^4} - \frac{1}{4\pi r^2} \frac{\partial^2 r}{\partial t^2} \quad (\text{hydrostatic equilibrium, non-static case}) \quad (5.182)$$

$$\frac{\partial l}{\partial m} = \epsilon_{\text{nuc}} - \epsilon_{\nu} - C_P \frac{\partial T}{\partial t} + \frac{\delta}{\rho} \frac{\partial P}{\partial t} \quad (\text{energy generation/loss}) \quad (5.183)$$

$$\frac{\partial T}{\partial m} = -\frac{GmT}{4\pi r^4 \rho} \nabla \quad (\text{energy transport}) \quad (5.184)$$

where, in the radiative case,

$$\nabla = \nabla_{\text{rad}} = \left(\frac{d \ln T}{d \ln P} \right), \quad (5.185)$$

and, in the convective case,

$$\nabla = \nabla_{\text{ad}} = \left(\frac{P}{T} \frac{dT}{dP} \right)_s = \frac{P\delta}{T\rho C_P} \quad [\text{Equation (3.50), Chapter 3}], \quad (5.186)$$

with the Schwarzschild Criterion [Equation (5.142)] deciding which of the two to use in each layer of the star. Equation (5.182) reduces to the usual hydrostatic equilibrium [Equation (5.25)] when $\partial^2 r / \partial t^2 = 0$ (static case).

5.14 Convective mixing timescale

Besides transporting energy, convective processes can transport chemical elements from inner to outer layers of stars, and vice versa. We can define the **mixing time** (τ_{mix}) as the time taken by a bubble to traverse a convective region.

Let us consider the case of a massive star whose energy is generated by the CNO cycle. In this case, the convective zone is more internal in the star — the opposite of the Sun.

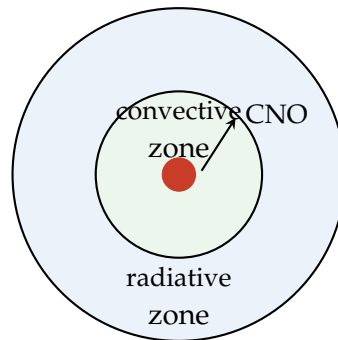


Figure 5.9: Structure of energy transport zones in a massive star (schematic, not to scale): core with energy generation via the CNO cycle, inner convective zone (surrounding the core), and outer radiative zone — the configuration opposite to that of the Sun (Figure 5.6).

We can express the mixing time as

$$\tau_{\text{mix}} = \frac{d}{v_B}, \quad (5.187)$$

where d is the distance defined by $d = q R_*$, with q the fraction of the star's radius included in the convective zone.

Notation collision in “ q ”

The symbol q is used with three different meanings in nearby parts of this material. Here, q is the **fraction of the stellar radius** contained in the convective zone [Equation (5.187)]. Later in this section (in the derivation of the 5th equation of internal structure), the subscripts p, q come to label the **atomic nuclei** involved in a nuclear transition $p \rightarrow q$, and $q_{p,q} \equiv e_{p,q}/m_p$ denotes the **energy generated per unit mass** in that transition. These are entirely independent uses of the same symbol — kept as in the original material, but flagged here to avoid confusion.

v_B , as we defined previously, is $v_B \approx [(\ell_{m8}^2 g)/(2H_P)(\nabla - \nabla_{\text{ad}})]^{1/2}$ [Equation (5.171)]. Substituting reference values for the Sun, we obtain that the mixing time is of order

$$\tau_{\text{mix}} \approx 3 \times 10^7 \text{ s}, \quad (5.188)$$

and, thus, we have that

$$\tau_{\text{mix}} \ll \tau_{\text{KH}} \ll \tau_{\text{nuc}}. \quad (5.189)$$

Why convective mixing can be treated as instantaneous

The hierarchy $\tau_{\text{mix}} \ll \tau_{\text{KH}} \ll \tau_{\text{nuc}}$ has an important practical consequence for stellar evolution modeling: since the mixing time ($\sim 10^7$ s, comparable in order of magnitude to the dynamical timescale itself) is many orders of magnitude smaller than the nuclear timescale ($\sim 10^{10}$ years, the scale on which the chemical composition actually *changes*), a convective zone can be treated, at any instant of evolution, as having **perfectly homogeneous** (well-mixed) chemical composition throughout its extent. This justifies a central numerical simplification of stellar evolution codes: instead of solving the transport of each chemical element layer by layer within a convective zone (which would require coupling the much faster turbulent mixing to the much slower nuclear evolution), it suffices to instantaneously impose the same chemical composition throughout the region identified as convectively unstable, and evolve this mean composition using the nuclear burning rate integrated over the entire zone.

Convective cores and the extent of the main sequence

In massive stars like the one in the example above (Figure 5.9), the convective core has an important evolutionary effect: convective mixing continuously homogenizes the composition of the *entire* core, bringing “fresh” hydrogen from the edges of the core to the center (where burning is fastest) and redistributing the helium already produced throughout the convective volume, rather than letting it accumulate only in the center. The net effect is that a larger fraction of the star's total mass participates in H burning than would if the core were purely radiative (no mixing) — the available fuel “reservoir” is effectively extended to the entire convective volume, prolonging the main-sequence lifetime. This effect is further amplified, in practice, by the phenomenon of **convective overshooting**: convective currents that, by inertia, overshoot the formal boundary of the Schwarzschild Criterion, mixing an additional layer of nominally stable material. The actual extent of the convective core (including overshooting) remains one of the most uncertain parameters of stellar evolution models, and has been increasingly constrained by asteroseismology data — the *Kepler* and *TESS* missions measured pulsation frequencies

(including g-modes, sensitive to the deep structure) of thousands of stars with convective cores, allowing the extent of mixing around the core to be inferred far more directly than through main-sequence tracks alone (Lindsay et al., 2024).

5.15 The 5th equation of internal structure

The chemical composition of stars is very important: it directly influences basic properties, such as the absorption and emission coefficients, and the star's own energy generation source.

Stellar composition is much simpler than that of bodies on Earth. Because of the high temperatures and pressures, there are no complex molecules in the cores of stars, and the vast majority of atoms are fully ionized.

We can define the mass fraction of a given nucleus such that

$$\sum_i X_i = 1, \quad (5.190)$$

the sum of the mass fractions of all chemical elements of a star is 1. The star's chemical composition at a time t and mass m is defined by:

$$X_i = X(m, t), \quad X_i \equiv \frac{m_i n_i}{\rho}, \quad (5.191)$$

where n_i is the number of particles per unit volume and m_i is the mass of the atomic nucleus. In general, one defines

$$X \equiv X_H, \quad Y \equiv X_{He}, \quad Z \equiv 1 - X - Y, \quad (5.192)$$

where Z is the sum of all elements beyond H and He (the **metals**, in the usual convention of stellar astrophysics).

For the structure equation, let us consider the fusion of element “ i ” into other elements, and of other elements into “ i ”. Thus, we can express the variation of chemical composition with time as

$$\frac{\partial X_i}{\partial t} = \frac{m_i}{\rho} \left(\sum_j r_{ji} - \sum_k r_{ik} \right), \quad (5.193)$$

where r is the number of reactions per second (it can convert $j \rightarrow i$, and then i into k).

In some cases, it is more convenient to write the equation as a function of the energy exchanged in each reaction. Considering that a transition $p \rightarrow q$ releases an energy defined by $e_{p,q}$, we can write an energy conservation expression where

$$E = \sum_{p,q} E_{p,q} = \frac{1}{\rho} \sum_{p,q} r_{p,q} e_{p,q}. \quad (5.194)$$

Introducing a term for the energy generated when a unit mass of a nucleus p transforms into q , given by

$$q_{p,q} \equiv \frac{e_{p,q}}{m_p}, \quad (5.195)$$

Notation collision in “ $q_{p,q}$ ”

As already flagged in Section 5.14, the symbol q reappears here with a meaning completely different from the one used for the radius fraction in the convective zone: now, p and q label the **atomic nuclei** involved in a transition $p \rightarrow q$, and $q_{p,q}$ [Equation (5.195)] is the energy released per unit mass in that transition — analogous to ϕ (the mass–energy conversion efficiency) already used in Section 5.9 for the p-p chain, but now defined more generally, transition by transition.

we can rewrite Equation (5.195) as $e_{p,q} = m_p q_{p,q}$; since $E = (1/\rho) \sum_{p,q} r_{p,q} e_{p,q}$ [Equation (5.194)], we have

$$\sum_{p,q} E_{p,q} = \frac{m_p}{\rho} \sum_{p,q} r_{p,q} q_{p,q}, \quad (5.196)$$

that is, term by term, $E_{p,q} = (m_p/\rho) r_{p,q} q_{p,q}$. Isolating $r_{p,q}$ and summing over all transitions:

$$\sum_{p,q} \frac{E_{p,q}}{q_{p,q}} = \frac{m_p}{\rho} \sum_{p,q} r_{p,q}, \quad (5.197)$$

hence, substituting into Equation (5.193), we have

$$\frac{\partial X_i}{\partial t} = \sum_j \frac{E_{j,i}}{q_{j,i}} - \sum_k \frac{E_{i,k}}{q_{i,k}}. \quad (5.198)$$

Intermediate step: isolating $r_{p,q}$

The passage from Equation (5.196) to Equation (5.197) follows from the term-by-term identification, $E_{p,q} = (m_p/\rho) r_{p,q} q_{p,q}$, isolating

$$r_{p,q} = \frac{\rho}{m_p} \frac{E_{p,q}}{q_{p,q}}$$

and summing over all pairs (p, q) .

The energy of H→He fusion and the solar luminosity

For the simplest possible transformation, the fusion of H into He, Equation (5.198) reduces to a single loss term (hydrogen, being primordial, is not produced by any earlier reaction in this simplified chain):

$$\frac{\partial X}{\partial t} = -\frac{E_H}{q_H}, \quad (5.199)$$

which describes the loss of hydrogen (and the associated energy) from the star to its surroundings. The value of q_H has already appeared in this book, albeit under a different symbol: it is exactly ϕc^2 [Equation (5.81), Section 5.9], with $\phi = 0.007$ the fraction of rest mass converted into energy. Numerically,

$$q_H = \phi c^2 = 0.007 \times (2.998 \times 10^{10} \text{ cm/s})^2 \approx 6.3 \times 10^{18} \text{ erg/g}, \quad (5.200)$$

an important reference number: it is the energy released per gram of hydrogen converted into helium. As an order-of-magnitude check, multiplying this value by the H consumption rate in the solar core required to sustain $L_\odot = 3.828 \times 10^{33} \text{ erg/s}$ — about

6×10^{14} g/s, that is, about 600 million tons per second — exactly reproduces the Sun's observed luminosity, closing the loop between the microscopic nuclear physics of this section and the macroscopic nuclear timescale already discussed in Section 5.9.

We will return to analyzing chemical processes in the stellar interior in more detail later (likely in the chapter dedicated to Energy Generation).

Further Reading and Current References

The nuclear reaction network formulation presented here, in terms of rates $r_{p,q}$ and energies per transition $q_{p,q}$, is only the skeleton of the problem — in a real star, dozens of nuclear species and hundreds of reactions compete simultaneously. For the complete formulation of stellar nuclear reaction networks, including the pp chains, the CNO cycle, and the nucleosynthesis of heavier elements, see Clayton (1983), the classic reference on the subject. For the connection between these reaction networks and the complete stellar structure equations (including the coupled evolution of chemical composition and structure), see the corresponding chapter of Kippenhahn et al. (2012).

Chapter 6

Polytropic Stars

6.1 Introduction

A **polytropic process** describes a thermodynamic transformation in which pressure is a function of density alone, $P = P(\rho)$, more precisely:

$$PV^\gamma = K, \quad \text{or} \quad P = K\rho^\gamma, \quad (6.1)$$

where K and γ are constants.

Polytropic stellar models were historically important for describing the theory of stellar structure, although more robust models are currently used for a more realistic description of stellar structure.

6.2 Closing the system of equations

We saw in previous lectures that the dynamics of a star is described by

$$\frac{\partial P}{\partial r} = -\frac{Gm\rho}{r^2} \quad \text{and} \quad \frac{\partial m}{\partial r} = 4\pi r^2 \rho, \quad (6.2)$$

valid for the static case.

In this case, the variables of the system are (P, ρ, m) , that is, 3 unknowns for only 2 equations. But we also know that thermodynamic processes depend on 3 variables, $F(P, \rho, T) = 0$, which implies one more unknown to be obtained. One way to reduce the degree of freedom in this treatment is to assume the polytropic case, where

$$P = K\rho^\gamma = K\rho^{1+1/n}, \quad (6.3)$$

$K \rightarrow$ polytropic constant,

$\gamma \rightarrow$ polytropic exponent,

$n \rightarrow$ polytropic index,

with

$$\gamma = 1 + \frac{1}{n}. \quad (6.4)$$

For an isothermal ideal gas, we have

$$P = \frac{\rho \mathcal{R}T}{\mu} \implies P = K\rho, \quad (6.5)$$

which corresponds to the case where $K = \mathcal{R}T/\mu$ — in an isothermal process, with $n \rightarrow \infty$, $\rho^{1+1/n} \rightarrow \rho$.

If we take the polytropic case, we are left with two equations and two unknowns when we add the hydrostatic equilibrium equation, where:

$$\frac{dP}{dr} = -\frac{Gm\rho}{r^2}, \quad (6.6)$$

we can define the **gravitational potential** as

$$\Phi(r) = \int_0^r \frac{Gm}{r'^2} dr', \quad \text{or} \quad \frac{d\Phi}{dr} = \frac{Gm}{r^2}. \quad (6.7)$$

Substituting into the hydrostatic equilibrium equation, we get:

$$\boxed{\frac{dP}{dr} = -\rho \frac{d\Phi}{dr}}. \quad (6.8)$$

Historical overview: from polytropes to Lane-Emden

Polytropic models have a long history in stellar astrophysics, going back to the work of Jonathan Homer Lane and Robert Emden in the nineteenth century, who first studied self-gravitating systems with the relation $P = K\rho^{1+1/n}$ — the resulting differential equation for the structure of these systems (which we will see in the next part of this chapter) bears the name **Lane–Emden equation**. Arthur Eddington popularized the use of polytropes in the 1920s as a way of obtaining tractable analytical (or semi-analytical) solutions for stellar structure before the computer era — the so-called **Eddington standard model** specifically uses the index $n = 3$ ($\gamma = 4/3$), appropriate for a star in radiative equilibrium dominated by radiation pressure. The index $n = 3$ also notably describes the relativistic case of the degenerate electron gas, directly linked to the **Chandrasekhar mass limit** for the maximum mass of a white dwarf (a topic of the future chapter on White Dwarfs). The index $n = 3/2$ ($\gamma = 5/3$), in turn, describes the non-relativistic case of the same degenerate gas, relevant for low-mass white dwarfs and for the cores of red giants. Polytropic models with these specific indices are still used today as fast and pedagogically important approximations, even when complete numerical models are needed for observational precision (Chandrasekhar, 1939).

6.3 Gravitational potential in spherical symmetry

Let us consider two spherical shells, with mass elements dm (at position $\vec{r}$) and dm' (at position $\vec{r}'$).

Since $\vec{F} = m\vec{a}$,

$$\vec{F} = m\vec{a}, \quad (6.9)$$

it is true that

$$d\vec{F} = m \delta\vec{g}, \quad (6.10)$$

and, by definition,

$$\delta\vec{g} = -G dm' \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3}. \quad (6.11)$$

Knowing that $\rho = dm'/d^3r'$, that is, $dm' = \rho d^3r'$, substituting into Equation (6.11) and integrating, we get

$$\vec{g}(\vec{r}) = \int \delta\vec{g} = -G \int \rho(\vec{r}') \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} d^3r', \quad (6.12)$$

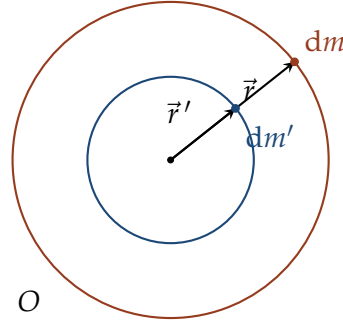


Figure 6.1: Two concentric spherical shells, with mass elements dm (radius r) and dm' (radius r'), used in the derivation of the gravitational potential in spherical symmetry.

where we can assume that the density depends only on r' (not on direction), by the spherical symmetry approximation.

Applying the divergence to both sides of the equation, we get:

$$\vec{\nabla} \cdot \vec{g}(\vec{r}) = -G \int \rho(\vec{r}') \vec{\nabla} \cdot \left[\frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \right] d^3 r'. \quad (6.13)$$

This step points directly to a fundamental identity of the Newtonian potential,

$$\vec{\nabla} \cdot \left[\frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \right] = 4\pi \delta(\vec{r} - \vec{r}'). \quad (6.14)$$

6.4 The gravitational Poisson equation

Substituting Equation (6.14) into Equation (6.13) and using the filtering property of the Dirac delta function in the integration over $d^3 r'$, we obtain

$$\vec{\nabla} \cdot \vec{g}(\vec{r}) = -4\pi G \int \rho(\vec{r}') \delta(\vec{r} - \vec{r}') d^3 r' = -4\pi G \rho(\vec{r}). \quad (6.15)$$

The right-hand side of Equation (6.15) is exactly the expression of the **Poisson equation in the gravitational case**,

$$\vec{\nabla} \cdot \vec{g}(r) = -4\pi G \rho. \quad (6.16)$$

We can verify that this acceleration g has exactly the same magnitude as the gravitational potential already defined in Equation (6.7): for a test particle in orbital equilibrium, $F_g = F \Rightarrow mg = \frac{GMm}{r^2} \Rightarrow g = \frac{Gm}{r^2}$, which is exactly $d\Phi/dr$. Since the gravitational force is central, the acceleration points in the direction opposite to the radial growth of the potential, that is,

$$\vec{g} = -\vec{\nabla} \Phi. \quad (6.17)$$

Substituting Equation (6.17) into Equation (6.16), we arrive at the **Poisson equation of gravitation**,

$$\boxed{\nabla^2 \Phi = 4\pi G \rho}, \quad (6.18)$$

which, assuming spherical symmetry, is written as

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right) = 4\pi G \rho. \quad (6.19)$$

With Equation (6.19), we finally have a closed system of three equations for the three unknowns (P, ρ, Φ) :

$$\begin{cases} \frac{dP}{dr} = -\rho \frac{d\Phi}{dr}, \\ \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right) = 4\pi G \rho, \\ P = K\rho^\gamma. \end{cases} \quad (6.20)$$

It is this system that we will solve next to arrive at the Lane–Emden equation.

6.5 Derivation of the Lane–Emden equation

Substituting the polytropic relation $P = K\rho^\gamma$ into the first equation of system (6.20),

$$K\gamma\rho^{\gamma-1} \frac{d\rho}{dr} = -\rho \frac{d\Phi}{dr} \implies d\Phi = -K\gamma\rho^{\gamma-2} d\rho. \quad (6.21)$$

Integrating both sides, with the boundary condition $\Phi = 0$ where $\rho = 0$ (the surface of the star),

$$\int_0^\Phi d\Phi' = -K\gamma \int_0^\rho \rho'^{\gamma-2} d\rho' \implies \Phi = -\frac{K\gamma}{\gamma-1} \rho^{\gamma-1}. \quad (6.22)$$

Using $\gamma = 1 + 1/n$ (hence $\gamma - 1 = 1/n$) and $\gamma n = n + 1$,

$$\Phi = -K\gamma \cdot n \rho^{1/n} = -(n+1)K\rho^{1/n}. \quad (6.23)$$

Inverting Equation (6.23), we obtain the density as a function of the potential,

$$\rho = \left[-\frac{\Phi}{(n+1)K} \right]^n. \quad (6.24)$$

Substituting Equation (6.24) back into the Poisson equation in spherical symmetry (6.19),

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right) = 4\pi G \left[-\frac{\Phi}{(n+1)K} \right]^n, \quad (6.25)$$

or, expanding the derivative on the left-hand side,

$$\frac{d^2\Phi}{dr^2} + \frac{2}{r} \frac{d\Phi}{dr} = 4\pi G \left[-\frac{\Phi}{(n+1)K} \right]^n. \quad (6.26)$$

We now define the dimensionless variables that simplify Equation (6.26):

$$z = Ar, \quad \omega = \frac{\Phi}{\Phi_c}, \quad A^2 = \frac{4\pi G}{(n+1)^n K^n} (-\Phi_c)^{n-1}, \quad (6.27)$$

where Φ_c is the potential at the center of the star. Thus,

$$\frac{d\Phi}{dr} = A\Phi_c \frac{d\omega}{dz}, \quad \frac{d^2\Phi}{dr^2} = A^2\Phi_c \frac{d^2\omega}{dz^2}. \quad (6.28)$$

Substituting into Equation (6.26),

$$A^2\Phi_c \omega'' + \frac{2A^2\Phi_c}{z} \omega' = 4\pi G \left[\frac{-\Phi_c \omega}{(n+1)K} \right]^n, \quad (6.29)$$

and dividing by $A^2\Phi_c$ and using the definition of A^2 from Equation (6.27) to cancel the common factors, we arrive at the final form:

$$\omega'' + \frac{2}{z}\omega' = -\omega^n, \quad \text{or, equivalently,} \quad \frac{1}{z^2} \frac{d}{dz} \left(z^2 \frac{d\omega}{dz} \right) = -\omega^n, \quad (6.30)$$

the **Lane–Emden equations**. The second form is obtained directly by expanding the derivative of the product in the first:

$$\frac{1}{z^2} \frac{d}{dz} \left(z^2 \frac{d\omega}{dz} \right) = \frac{1}{z^2} (2z\omega' + z^2\omega'') = \omega'' + \frac{2}{z}\omega', \quad (6.31)$$

which is identical to the left-hand side of the first form — confirming the equivalence of the two.

From the definitions of ω and the central density $\rho_c = [-\Phi_c / ((n+1)K)]^n$ [Equation (6.24) evaluated at $r = 0$], and comparing with Equation (6.24),

$$\frac{\rho}{\rho_c} = \left(\frac{\Phi}{\Phi_c} \right)^n \implies \rho = \rho_c \omega^n. \quad (6.32)$$

Using $P = K\rho^\gamma = K\rho^{(n+1)/n}$, the value at the center is

$$P_c = K\rho_c^{(n+1)/n}, \quad (6.33)$$

and dividing P by P_c ,

$$\frac{P}{P_c} = \left(\frac{\rho}{\rho_c} \right)^{1+1/n}. \quad (6.34)$$

Using Equation (6.32) in Equation (6.34),

$$\frac{P}{P_c} = (\omega^n)^{1+1/n} = \omega^n \cdot \omega = \boxed{\omega^{n+1}}, \quad (6.35)$$

which is known as **another solution of the Lane–Emden equation**: knowing $\omega(z)$, we obtain both the density profile [Equation (6.32)] and the pressure profile [Equation (6.35)] of the polytropic star.

6.6 Mass distribution and boundary conditions

We can use the Lane–Emden solution to directly obtain the mass distribution of the polytrope. Starting from the definition of mass in spherical shells and Equation (6.32),

$$m(r) = \int_0^r 4\pi r'^2 \rho \, dr' = 4\pi\rho_c \int_0^r r'^2 \omega^n \, dr'. \quad (6.36)$$

Changing the integration variable to $z = Ar$ (hence $r' = z/A$, $dr' = dz/A$),

$$m(r) = \frac{4\pi\rho_c}{A^3} \int_0^z z'^2 \omega^n \, dz', \quad z = Ar. \quad (6.37)$$

From the Lane–Emden equation itself (6.30), $d(z^2\omega')/dz = -z^2\omega^n$, so that the integrand is an exact derivative:

$$\int_0^z z'^2 \omega^n \, dz' = - \left[z'^2 \omega' \right]_0^z = -z^2 \frac{d\omega}{dz}. \quad (6.38)$$

Substituting into Equation (6.37) and using $1/A^3 = r^3/z^3$,

$$m(r) = -4\pi\rho_c r^3 \left(\frac{1}{z} \frac{d\omega}{dz} \right)_{r=z/A}, \quad (6.39)$$

the mass solution for the polytropic case.

Boundary conditions. Imposing zero density at the surface, $\rho(r = R) = 0$, and using Equation (6.32),

$$\rho_c \omega^n(z_R) = 0 \implies \omega(z_R) = 0, \quad (6.40)$$

where $z_R \equiv AR$ is the value of z at the surface of the star — the so-called **first zero** of the Lane–Emden function, denoted ξ_1 in the standard notation of the literature.

Notation: z_R vs. R

z and r are different variables, related by $z = Ar$ [Equation (6.27)]. The boundary condition at the surface of the star corresponds to ω vanishing at $z = z_R \equiv AR$, the value of z corresponding to the stellar radius R — not at “ $z = R$ ” directly, since that would mix a dimensionless variable with a length dimension. We use z_R (equivalent to the ξ_1 of the standard literature, see Chandrasekhar, 1939; Horedt, 2004) consistently from here on.

Figure 6.2 illustrates the behavior of $\omega(z)$: decreasing from $\omega(0) = 1$ at the center of the star until it vanishes at $z = z_R$, the **polytropic radius**.

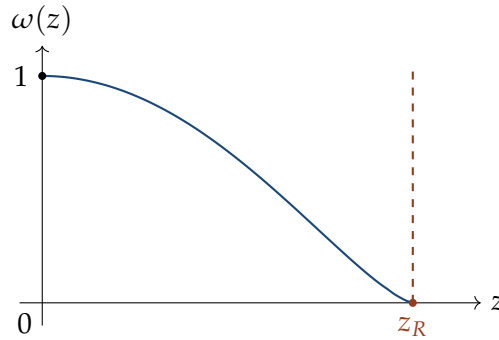


Figure 6.2: Schematic behavior of the Lane–Emden function $\omega(z)$: it decreases monotonically from $\omega(0) = 1$ at the center of the star ($z = 0$) until it vanishes at the polytropic radius $z = z_R$ (dashed line), which marks the stellar surface.

Evaluating Equation (6.39) at $r = R$ (the total mass, M) and comparing with the mean density $\bar{\rho} = 3M/(4\pi R^3)$,

$$\frac{M}{m(r)} = 1 = \frac{4\pi R^3 \bar{\rho}/3}{-4\pi R^3 \rho_c \left(\frac{1}{z} \frac{d\omega}{dz} \right)_{z_R}}, \quad (6.41)$$

that is,

$$\frac{\bar{\rho}}{\rho_c} = \left(-\frac{3}{z} \frac{d\omega}{dz} \right)_{z_R}. \quad (6.42)$$

Sign of $d\omega/dz$ in the mass equation

The mass equation evaluated at the surface, $m(r) = -4\pi r^3 \rho_c \left(\frac{1}{z} \frac{d\omega}{dz} \right)$ [Equation (6.39)], has the explicit minus sign by construction. Since ω decreases monotonically from 1 to 0 between $z = 0$ and $z = z_R$ (Figure 6.2), we always have $d\omega/dz < 0$ for physically acceptable solutions — hence $-\rho_c(1/z)(d\omega/dz)$ is always a *positive* quantity, as a physical mass requires. Equation (6.42), which consistently uses this same form, confirms this reading.

Thus, knowing the observed M and R of a star, Equation (6.42) allows us to determine ρ_c (provided the numerical solution of $\omega(z)$ for the chosen index n is known) and, subsequently, the radial scale

$$A = \frac{z}{r} = \frac{z_R}{R}. \quad (6.43)$$

6.7 Important polytropic cases

Some polytropic indices appear frequently in stellar astrophysics:

	Physical case	γ	n
(i)	Non-relativistic degenerate gas	5/3	3/2
(ii)	Relativistic degenerate gas	4/3	3
(iii)	Isothermal ideal gas	1	$n \rightarrow \infty$
(iv)	Fully convective star	5/3	3/2
(v)	Eddington standard model ($P_T = P_{\text{gas}} + P_{\text{rad}}$)	4/3	3

On the physical origin of $\gamma = 4/3$

A monatomic ideal (non-degenerate) gas always has $\gamma = 5/3$, regardless of chemical composition — the value $\gamma = 4/3$ never arises from the ideal nature of the gas itself. It appears in only two distinct physical contexts: dominant radiation pressure (case (v), Eddington standard model) or a relativistic degenerate electron gas (case (ii)).

The indices $n = 3/2$ and $n = 3$ in modern stellar astrophysics

The two most commonly used polytropic indices remain relevant today. The index $n = 3/2$ ($\gamma = 5/3$) describes well fully convective low-mass stars (including sub-stellar objects) and non-relativistic degenerate white dwarfs. The index $n = 3$ ($\gamma = 4/3$), in turn, describes both the Eddington standard model and relativistic degenerate white dwarfs — and is the basis of the classical derivation of the **Chandrasekhar mass**: using the numerical solution of the Lane–Emden equation for $n = 3$, whose first zero is $\xi_1 \approx 6.897$ and whose derivative at that point is $-\xi_1^2 \omega'(\xi_1) \approx 2.018$, one obtains $M_{\text{Ch}} \approx 1.4 M_\odot$ — the mass limit above which relativistic electron degeneracy pressure can no longer support a white dwarf against gravitational collapse (Chandrasekhar, 1939). This result will be revisited in more detail in the future chapter on White Dwarfs.

6.8 Solutions of the Lane–Emden equations

The Lane–Emden equation has a singularity at $z = 0$ (the term $2\omega'/z$ diverges there, unless $\omega'(0) = 0$). Furthermore, the equation has a closed analytical solution for only three values of n :

$$n = 0 : \quad \omega(z) = 1 - \frac{z^2}{6}, \quad (6.44)$$

$$n = 1 : \quad \omega(z) = \frac{\sin z}{z}, \quad (6.45)$$

$$n = 5 : \quad \omega(z) = \left(1 + \frac{z^2}{3}\right)^{-1/2}. \quad (6.46)$$

For the remaining values of n — the physically most relevant ones, such as $n = 3/2$ and $n = 3$ — the equation must be solved numerically. A simple way to do this is via the finite difference method, using

$$\omega'' \approx \frac{\omega'(z + \Delta z) - \omega'(z)}{\Delta z}, \quad (6.47)$$

combined with the Lane–Emden equation itself [Equation (6.30)] to obtain ω'' at each step, and integrated from the initial conditions at $z = 0$:

$$\omega(z=0) = \frac{\Phi(z=0)}{\Phi_c} = 1, \quad \omega'(z=0) = 0, \quad (6.48)$$

corresponding to the central potential ($\Phi(0) = \Phi_c$, by definition) and to the absence of a potential gradient at the center of the star, by symmetry.

The initial condition $\omega''(0)$

In addition to $\omega(0) = 1$ and $\omega'(0) = 0$, the numerical integration of the Lane–Emden equation requires the value of $\omega''(0)$, obtained directly from the equation itself: since the term $2\omega'/z$ is a $0/0$ indeterminate form at $z = 0$ (given that $\omega'(0) = 0$), we expand $\omega(z)$ in a Taylor series around $z = 0$,

$$\omega(z) \approx \omega(0) + \frac{1}{2}\omega''(0)z^2 + \cdots = 1 + \frac{1}{2}\omega''(0)z^2 + \cdots, \quad (6.49)$$

and require that the Lane–Emden equation be satisfied in this limit. Substituting this expansion into the form $\omega'' + \frac{2}{z}\omega' = -\omega^n$ and using $\omega'(z) \approx \omega''(0)z$, the term $\frac{2}{z}\omega' \rightarrow 2\omega''(0)$, leading to $3\omega''(0) = -\omega(0)^n = -1$, that is, $\omega''(0) = -1/3$ — a standard result used as a starting point for any numerical integration of the Lane–Emden equation (Chandrasekhar, 1939; Horedt, 2004).

Polytropes as an idealization

Polytropic models are mathematically elegant constructions, but they represent an idealization: they assume a simple relation $P = K\rho^\gamma$ that ignores the detailed coupling between energy transport, chemical composition, and nuclear reaction rates that govern a real star. In practice, building realistic stellar models requires full numerical integration of the four stellar structure equations (mass, hydrostatic equilibrium, energy transport, and energy generation) coupled to an equation of state and tabulated opacities. Even so, polytropes remain extremely useful today: as good-quality analytical approximations for

specific objects (white dwarfs, fully convective low-mass stars) and, in computational practice, as an *initial guess* to initialize the integration of modern stellar evolution codes — for example, the MESA code (*Modules for Experiments in Stellar Astrophysics*) uses homogeneous polytropic models to build the initial configuration of a pre-main-sequence star, before relaxing it into a fully consistent model (Paxton et al., 2011).

Further Reading and Current References

For the classical and complete treatment of polytropic models and the detailed derivation of the Lane–Emden equation (including its analytical solutions for $n = 0, 1, 5$ and numerical solutions for the remaining indices), see Chandrasekhar (1939) — still today the starting reference on the subject, despite its nearly nine decades — and the corresponding chapter of Kippenhahn et al. (2012), with notation closer to that used in this book. For a modern and comprehensive reference, covering applications of polytropes far beyond the classical stellar context (including relativistic systems and plasma physics problems), see Horedt (2004). On the use of polytropes as an initial condition in modern stellar evolution codes, see the documentation of the MESA code in Paxton et al. (2011).

Chapter 7

Star and Planet Formation

7.1 Introduction

In the previous chapters we built the formalism of a star in **equilibrium**: the stellar structure equations (Chapter 5) showed how gravity and pressure balance hydrostatically, and the Virial Theorem (Section 5.7), $2E_i = -E_g$ [Equation (5.63)], revealed the relation between internal and gravitational energy that sustains that equilibrium. But no star is born in equilibrium: every star forms through the gravitational collapse of a region of interstellar gas and dust, and it is the same Virial Theorem — now applied to the condition under which equilibrium *ceases* to exist — that determines when that collapse begins. In this chapter we study the earliest stages of star formation, from the molecular cloud to the collapsing prestellar core, with the Jeans instability connecting them, up to the formation of a circumstellar disk by conservation of angular momentum — the nursery of planet formation, the topic with which we close the chapter.

7.2 Molecular clouds and prestellar cores

Stars form out of the cold, dense interstellar medium, out of **molecular clouds**, where the density is high enough for hydrogen to exist predominantly as H_2 rather than atomic or ionized. **Giant molecular clouds** (GMCs) are the largest self-gravitating structures in the Galaxy: masses of 10^4 – $10^6 M_\odot$, tens of parsecs, $T \sim 10$ – 30 K, $n \sim 10^2$ – 10^3 cm^{-3} (Heyer and Dame, 2015). Because H_2 is homonuclear, with no permanent dipole, its rotational transitions are dipole-forbidden and it does not radiate efficiently in the cold — which is why GMCs are mapped via **chemical tracers**, above all CO ($J=1 \rightarrow 0$, $E/k_B \approx 5.5$ K, easily populated at $T \sim 10$ K). The abundance of CO is small ($\text{CO}/H_2 \sim 10^{-4}$), but sufficient as a standard tracer via the factor $X_{\text{CO}} \equiv N(H_2)/W_{\text{CO}}$ (Heyer and Dame, 2015).

Within GMCs, the gas organizes itself into filaments and, at the densest points, into **dense cores**. A gravitationally bound dense core with no detectable protostar is a **prestellar core** — the object of this chapter. The *Herschel* Gould Belt Survey characterized a large sample of these cores: sizes ~ 0.1 pc, $T \sim 10$ K, $n \sim 10^4$ – 10^5 cm^{-3} , median mass $\sim 0.5 M_\odot$ (André et al., 2010) (up to $n \sim 10^6 \text{ cm}^{-3}$ in more concentrated cores, before dynamical contraction). It is this range — $T \sim 10$ K, $n \sim 10^4 \text{ cm}^{-3}$ — that we will use as a reference in Section 7.3.

Where the cold of molecular clouds comes from

A prestellar core at $T \sim 10$ K and $n \sim 10^4 \text{ cm}^{-3}$ is among the coldest environments in the observable universe outside the cosmic microwave background — cooling essential for collapse, since the Jeans mass decreases as $T^{3/2}$ (Section 7.3): warm gas ($T \sim 10^4$ K, like the diffuse ionized medium) would require masses orders of magnitude larger. Cooling occurs by molecular rotational emission in the outer regions and, in denser cores optically thick to that emission, by continuum thermal emission from dust in the far infrared

(André et al., 2010).

7.3 The Jeans instability

A self-gravitating gas cloud remains in hydrostatic equilibrium as long as thermal pressure is able to support its own weight. The precise criterion for when this equilibrium ceases to be possible — the **Jeans instability** — can be obtained by two independent routes: an energy argument, based directly on the Virial Theorem already established in Chapter 5, and a complete hydrodynamic treatment, based on the dispersion relation of linear perturbations in a self-gravitating fluid. The two must agree to within numerical factors of order unity, since they describe the same physics by distinct routes.

Energy method: the Virial Theorem

We have already seen, in the most general form of the Virial Theorem applied to a sphere of self-gravitating gas in hydrostatic equilibrium [Equation (5.56)], that

$$\int_0^M \frac{3P}{\rho} dm = -E_g, \quad (7.1)$$

valid for any gas in hydrostatic equilibrium, regardless of its composition — the derivation uses only the hydrostatic equilibrium equation and the definition of gravitational energy, never the equation of state of the gas. This is convenient here: a cold molecular cloud is composed mainly of *molecular* H_2 , with mean molecular weight $\mu \approx 2.3$ (instead of the $\mu \approx 0.6$ of ionized stellar plasma), but Equation (7.1) remains valid because the pressure of an ideal gas, $P = nk_B T$, depends only on the translational degrees of freedom — the rotational and vibrational degrees of freedom of H_2 contribute to the internal energy, not to the pressure.

For a spherical, uniform, isothermal cloud composed of ideal gas, $P/\rho = k_B T/(\mu m_H)$ is constant throughout the cloud, so that the integral on the left-hand side of Equation (7.1) resolves trivially:

$$\int_0^M \frac{3P}{\rho} dm = \frac{3k_B T}{\mu m_H} \int_0^M dm = \frac{3Mk_B T}{\mu m_H}. \quad (7.2)$$

For the right-hand side, we need the gravitational energy of a uniform sphere of mass M , radius R , and density $\rho = 3M/(4\pi R^3)$:

$$E_g = - \int_0^M \frac{Gm}{r} dm = - \int_0^R \frac{G \left(\frac{4}{3}\pi r^3 \rho \right)}{r} 4\pi r^2 \rho dr = - \frac{16\pi^2 G \rho^2}{3} \int_0^R r^4 dr = - \frac{16\pi^2 G \rho^2 R^5}{15}. \quad (7.3)$$

Substituting $\rho = 3M/(4\pi R^3)$,

$$E_g = - \frac{16\pi^2 G R^5}{15} \cdot \frac{9M^2}{16\pi^2 R^6} = \boxed{- \frac{3}{5} \frac{GM^2}{R}}, \quad (7.4)$$

the standard result for the gravitational energy of a homogeneous sphere — the factor $3/5$ (instead of the 1 of the crude approximation $E_g \sim -GM^2/R$ used in Chapter 5) comes from integrating over the uniform-density profile.

Hydrostatic equilibrium ceases to be possible when $|E_g|$ exceeds the left-hand side of Equation (7.1) at the marginal limit. Equating Equations (7.2) and (7.4):

$$\frac{3Mk_B T}{\mu m_H} = \frac{3}{5} \frac{GM^2}{R} \implies \frac{GM}{R} = \frac{5k_B T}{\mu m_H}. \quad (7.5)$$

Using $R = (3M/4\pi\rho)^{1/3}$ to eliminate R in favor of M and ρ , and solving for M , we obtain the **Jeans mass**:

$$M_J = \left(\frac{5k_B T}{G\mu m_H} \right)^{3/2} \left(\frac{3}{4\pi\rho} \right)^{1/2}. \quad (7.6)$$

Clouds with $M > M_J$ have a gravitational energy larger, in magnitude, than the thermal pressure can support, and collapse; with $M < M_J$, pressure wins. This is exactly the result already quoted as an order-of-magnitude estimate in Equation (5.11) of Chapter 5 — here obtained as an *exact* result for a uniform, isothermal sphere.

Dispersion-relation method

The energy argument above is elegant, but treats the cloud as a single object and says nothing about *spatial scales*: why does collapse occur in pieces of a certain size, rather than in the whole cloud at once? To answer this — and to obtain the Jeans criterion independently, without using the Virial Theorem — we start from the hydrodynamic equations of a self-gravitating fluid: the continuity equation, the Euler equation (momentum conservation), and the Poisson equation for the gravitational potential,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0, \quad (7.7)$$

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} = -\frac{\nabla P}{\rho} - \nabla \phi, \quad (7.8)$$

$$\nabla^2 \phi = 4\pi G \rho. \quad (7.9)$$

We consider a background equilibrium state, homogeneous and static ($\rho_0, P_0, \vec{v}_0 = 0, \phi_0$ constant), and perturb it linearly: $\rho = \rho_0 + \delta\rho, P = P_0 + \delta P, \vec{v} = \delta\vec{v}, \phi = \phi_0 + \delta\phi$, with $\delta\rho, \delta P, \delta\vec{v}, \delta\phi \ll$ their unperturbed counterparts.

The Jeans swindle

Strictly, an infinite, homogeneous medium with $\rho_0 \neq 0$ cannot be in hydrostatic equilibrium: Equation (7.9) applied to ϕ_0 would require $\nabla^2 \phi_0 = 4\pi G \rho_0 \neq 0$ everywhere in space, incompatible with a null or uniform background field. The usual practice — adopted here — is to ignore ϕ_0 and retain only the perturbation $\delta\phi$ via the linearized Poisson equation. This formal inconsistency is accepted because the final result agrees, to within factors of order unity, with more careful treatments (an expanding background or finite geometry), and because we are interested in perturbations of wavelength much smaller than the actual (finite) size of the cloud, where the curvature of the background potential is negligible.

Linearizing Equations (7.7)–(7.9) (discarding quadratic terms), using $\delta P = c_s^2 \delta\rho$ [$c_s^2 \equiv (\partial P / \partial \rho)$] and seeking plane-wave solutions, $\delta\rho, \delta\vec{v}, \delta\phi \propto e^{i(kx - \omega t)}$ (so $\partial/\partial t \rightarrow -i\omega, \nabla \rightarrow ik\hat{x}$), the three equations become algebraic in $\delta\rho, \delta v$, and $\delta\phi$. Eliminating $\delta\phi$ (via Poisson) and δv

(via continuity) in favor of $\delta\rho$, one arrives at the **dispersion relation**:

$$\boxed{\omega^2 = c_s^2 k^2 - 4\pi G \rho_0}. \quad (7.10)$$

This is the central equation of the hydrodynamic Jeans criterion: it describes how the frequency ω of a perturbation of wavenumber k depends on the competition between the term $c_s^2 k^2$, of *acoustic* origin (pressure attempts to restore equilibrium, generating sound waves), and the term $-4\pi G \rho_0$, *gravitational* and independent of k (self-gravity tends to amplify the perturbation, not restore it).

Stable oscillation vs. collapse: the role of the sign of ω^2

If $\omega^2 > 0$, ω is real and the perturbation oscillates in time as $e^{\mp i\omega t}$ — an ordinary sound wave, propagating without growing. But if $\omega^2 < 0$, then $\omega = \pm i|\omega|$ is *purely imaginary*, and the solution $e^{-i\omega t} = e^{\mp|\omega|t}$ contains a branch that grows exponentially in time: the density perturbation $\delta\rho$ does not oscillate, it amplifies without bound (in the linear approximation) — gravitational collapse. The instability condition is, therefore, simply $\omega^2 < 0$ in Equation (7.10).

From the condition $\omega^2 < 0$ in Equation (7.10),

$$c_s^2 k^2 < 4\pi G \rho_0 \implies k < k_J \equiv \sqrt{\frac{4\pi G \rho_0}{c_s^2}}, \quad (7.11)$$

that is, perturbations of wavenumber *smaller* than the critical Jeans wavenumber k_J (wavelength *larger* than the **Jeans length**) are unstable:

$$\boxed{\lambda_J = \frac{2\pi}{k_J} = c_s \sqrt{\frac{\pi}{G \rho_0}}}. \quad (7.12)$$

For a cold molecular cloud, cooling by dust and molecular lines (see box above) is efficient enough for the temperature to remain approximately constant during the initial compression — collapse is, at this stage, well approximated as *isothermal*, so that the relevant sound speed in Equation (7.12) is the **isothermal** sound speed, $c_s^2 = k_B T / (\mu m_H)$ (and not the adiabatic one, $c_s^2 = \gamma k_B T / \mu m_H$, which would apply only to a collapse fast enough to prevent the escape of heat generated by compression).

To obtain the Jeans mass from the Jeans length, we take the mass contained within a sphere of diameter λ_J (the usual convention in the literature; other conventions, with slightly different numerical factors, also appear):

$$M_J = \frac{4}{3}\pi\rho_0 \left(\frac{\lambda_J}{2}\right)^3 = \frac{\pi}{6}\rho_0\lambda_J^3. \quad (7.13)$$

Substituting Equation (7.12) and $c_s^2 = k_B T / (\mu m_H)$:

$$M_J = \frac{\pi}{6}\rho_0 \left[\frac{k_B T}{\mu m_H} \cdot \frac{\pi}{G \rho_0} \right]^{3/2} = \frac{\pi^{5/2}}{6} \left(\frac{k_B T}{G \mu m_H} \right)^{3/2} \rho_0^{-1/2}. \quad (7.14)$$

Comparison of the two methods

Equations (7.6) and (7.14) have exactly the same functional dependence on T , μ , and ρ — both scale as $T^{3/2}\rho^{-1/2}\mu^{-3/2}$ —, differing only in the dimensionless numerical coefficient:

$$\frac{M_J^{\text{virial}}}{M_J^{\text{dispersion}}} = \frac{5^{3/2}\sqrt{3/(4\pi)}}{\pi^{5/2}/6} = \frac{6 \cdot 5^{3/2}\sqrt{3}}{2\pi^3} \approx \frac{5.46}{2.92} \approx 1.9. \quad (7.15)$$

The two methods agree to within a factor of ≈ 2 — as expected, since the energy method uses a *uniform, isolated* sphere at marginal equilibrium, while the dispersion relation describes a perturbation of infinite wavelength (the Jeans swindle) and the conversion $\lambda_J \rightarrow M_J$ carries its own choice of geometry. Order-unity factors like this are inherent to any Jeans mass estimate — the essential physics is the scaling $M_J \propto T^{3/2}\rho^{-1/2}$, not the exact coefficient.

Jeans mass of a typical prestellar core

Let us take the typical physical conditions of a prestellar core discussed in Section 7.2: $T = 10$ K, $n(\text{H}_2) = 10^4 \text{ cm}^{-3}$, and mean molecular weight $\mu \approx 2.3$ (molecular gas with traces of helium). The corresponding mass density is

$$\rho = n\mu m_{\text{H}} = 10^4 \text{ cm}^{-3} \times 2.3 \times 1.6726 \times 10^{-24} \text{ g} \approx 3.85 \times 10^{-20} \text{ g/cm}^3. \quad (7.16)$$

Using Equation (7.6) (energy method) with $k_B = 1.381 \times 10^{-16} \text{ erg/K}$ and $G = 6.674 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}$:

$$\frac{5k_B T}{G\mu m_{\text{H}}} \approx 2.69 \times 10^{16} \text{ g/cm}, \quad (7.17)$$

$$M_J^{\text{virial}} = \left(\frac{5k_B T}{G\mu m_{\text{H}}} \right)^{3/2} \left(\frac{3}{4\pi\rho} \right)^{1/2} \approx 1.10 \times 10^{34} \text{ g} \approx 5.5 M_{\odot}. \quad (7.18)$$

By the dispersion-relation method [Equation (7.14)], using the ratio ≈ 1.9 obtained above,

$$M_J^{\text{dispersion}} \approx \frac{5.5 M_{\odot}}{1.9} \approx 2.9 M_{\odot}. \quad (7.19)$$

Both methods agree, therefore, in predicting a Jeans mass of **a few solar masses** for a typical prestellar core — exactly the order of magnitude of the most populous stellar mass range in the initial mass function (IMF), which peaks (in number per logarithmic mass interval) between ~ 0.1 and $\sim 1 M_{\odot}$ and extends up to a few tens of $M_{\odot}$ (Hennebelle and Grudić, 2024). The result is also consistent with the median mass of $\sim 0.5 M_{\odot}$ directly observed in the prestellar cores of the *Herschel* Gould Belt Survey (André et al., 2010) — the discrepancy of a factor of a few is expected, since real cores are not uniform, isolated spheres (additional support from turbulence and magnetic field tends to reduce the mass that effectively collapses relative to the pure thermal Jeans mass calculated here).

The fragmentation problem

A typical GMC has a mass of 10^4 – $10^6 M_{\odot}$ (Heyer and Dame, 2015) — many orders of magnitude above the Jeans mass of a few solar masses calculated above. If the cloud collapsed as a single object, the result would be a single supermassive star, not the cluster of thousands of stars actually observed forming in regions like the Orion Nebula. The resolution lies in the fact that, as the cloud collapses, the Jeans mass itself *decreases*

($M_J \propto \rho^{-1/2}$): progressively smaller regions become individually unstable and break away dynamically, a hierarchical process called **fragmentation** that repeats in cascade, generating many cores of local mass $\sim M_J$ instead of a single supermassive object. ALMA observations, with resolution of ~ 1000 au, directly confirm this picture: massive cores fragment into subcores whose number and separation are consistent with the local Jeans length [Equation (7.12)] (Motte et al., 2018).

Fragmentation alone does not explain why the **star formation efficiency** — the fraction of a GMC’s mass that actually turns into stars, rather than being dispersed by turbulence, magnetic fields, and stellar feedback — is typically low, only a few percent per free-fall time (McKee and Ostriker, 2007): one of the central, still-open problems of modern star formation theory.

Further Reading and Current References

Updated observational review of molecular clouds in the Milky Way: Heyer and Dame (2015). Initial results of the *Herschel* Gould Belt Survey on prestellar cores and the initial mass function: André et al. (2010). Fragmentation and cluster formation via ALMA: Motte et al. (2018). Physical theory of star formation, turbulence, and conversion efficiency: McKee and Ostriker (2007).

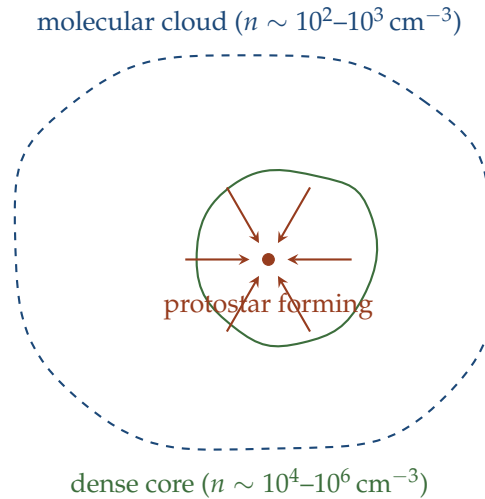


Figure 7.1: Schematic representation of a giant molecular cloud (dashed contour, density traced by CO) containing a dense prestellar core (inner contour) in gravitational collapse: as the density increases locally, the Jeans mass [Equation (7.6)] decreases, favoring the hierarchical fragmentation discussed in the text.

7.4 Collapse dynamics: inside-out collapse and the accretion rate

Once the Jeans mass is exceeded (Section 7.3), what happens in detail? The naive answer — **homologous collapse**, the whole cloud falling simultaneously with the same $t_{\text{ff}} \approx 1/(2\sqrt{G\bar{\rho}})$ [Equation (5.44)], density remaining uniform — is not what occurs: $t_{\text{ff}}(r) \propto [r^3/M(r)]^{1/2}$ depends on the internal profile, and a real core is not uniform. Its isothermal hydrostatic-equilibrium structure approaches the **singular isothermal sphere** (SIS), obtained from the

hydrostatic equilibrium equation with $P = c_s^2 \rho$ and $\rho = Ar^{-2}$:

$$\frac{dP}{dr} = c_s^2 \frac{d\rho}{dr} = -2Ac_s^2 r^{-3} \stackrel{!}{=} -\frac{GM(r)\rho}{r^2} = -\frac{G(4\pi Ar)(Ar^{-2})}{r^2} = -4\pi GA^2 r^{-3}, \quad (7.20)$$

where we used $M(r) = \int_0^r 4\pi r'^2 \rho(r') dr' = 4\pi Ar$. Equating the coefficients, $2c_s^2 = 4\pi GA$, that is,

$$\boxed{\rho_{\text{SIS}}(r) = \frac{c_s^2}{2\pi Gr^2}}, \quad (7.21)$$

a profile that diverges as $r \rightarrow 0$ ("singular"), with mass growing linearly, $M(r) = 2c_s^2 r/G$. Since $M(r) \propto r$, the local free-fall time $t_{\text{ff}}(r) \propto \sqrt{r^3/M(r)} \propto r$ is *smaller* at the center: the core "knows" it must collapse before the envelope. The result, formalized by Shu (1977), is **inside-out collapse**: the center enters free fall first, forming the protostar, while a **rarefaction front** propagates outward at $\sim c_s$. Interior to it ($r < c_s t$), the gas is already in free fall ($\rho \propto r^{-3/2}$, $v \propto r^{-1/2}$); exterior ($r > c_s t$), it remains in the original static structure — the information that the central pressure has dropped has not yet reached that far. This picture — extended infall, with a sharp boundary between collapsing and static gas — is exactly what is observed in Class 0 envelopes, and is why the model of Shu (1977) replaced homologous collapse as the standard paradigm of low-mass star formation (Stahler and Palla, 2004).

A remarkable consequence of the self-similar character of the Shu solution is that the **accretion rate** $\dot{M} \equiv dM_*/dt$ is *constant* in time and depends only on c_s and G : as the SIS [Equation (7.21)] introduces no independent mass or length scale, the only dimensional combination of c_s and G with dimensions of a mass rate is

$$\dot{M} \sim \frac{c_s^3}{G}, \quad (7.22)$$

since $[c_s^3/G] = (\text{cm}^3 \text{s}^{-3})/(\text{cm}^3 \text{g}^{-1} \text{s}^{-2}) = \text{g s}^{-1}$. The complete self-similar solution of Shu (1977) confirms this scaling and provides the exact numerical coefficient, for the non-magnetic, non-rotating case:

$$\boxed{\dot{M} = 0.975 \frac{c_s^3}{G}}. \quad (7.23)$$

Shu accretion rate for a typical core

Using the same reference temperature from Section 7.2 ($T = 10 \text{ K}$) and $\mu \approx 2.3$, the isothermal sound speed is

$$c_s = \sqrt{\frac{k_B T}{\mu m_H}} \approx 1.89 \times 10^4 \text{ cm/s} = 0.19 \text{ km/s}, \quad (7.24)$$

a typical value quoted for cold molecular clouds. Substituting into Equation (7.23):

$$\dot{M} = 0.975 \frac{(1.89 \times 10^4 \text{ cm/s})^3}{6.674 \times 10^{-8} \text{ cm}^3 \text{g}^{-1} \text{s}^{-2}} \approx 9.9 \times 10^{19} \text{ g/s} \approx 1.6 \times 10^{-6} M_\odot/\text{yr}. \quad (7.25)$$

Integrating over the characteristic timescale for the formation of a low-mass star, $t \sim 10^5$ – 10^6 yr (Stahler and Palla, 2004), we obtain a final accreted mass $M_* \sim \dot{M}t \sim 0.16$ – $1.6 M_\odot$ — exactly the order of magnitude of a solar-type star, a remarkable internal-consistency confirmation, since the timescale was not used as an input in the calculation of $\dot{M}$ (which depends only on c_s).

7.5 The angular momentum problem

The inside-out collapse model of the previous section is spherically symmetric — but real cores are not static: observations of molecular lines (NH_3) reveal systematic velocity gradients, interpreted as slow solid-body rotation, with typical gradients of $\sim 1\text{--}2 \text{ km s}^{-1} \text{ pc}^{-1}$ (Goodman et al., 1993) — numerically equal to Ω , which gives $\Omega \sim 3\text{--}6 \times 10^{-14} \text{ s}^{-1}$, consistent with the fiducial value $\Omega_0 \sim 10^{-14} \text{ s}^{-1}$ adopted in the literature. Rotations of this magnitude are small — $\beta \equiv T_{\text{rot}}/|E_g| \sim 0.02$ (Goodman et al., 1993), far from affecting the hydrostatic equilibrium of the core. The problem, as we shall see, is not rotation itself, but what happens to it during a collapse spanning several orders of magnitude in scale.

The angular momentum problem: a "naive" collapse is impossible

Consider a gas element at the edge of a typical core, with radius $R_0 \sim 0.1 \text{ pc} \approx 3.09 \times 10^{17} \text{ cm}$ and angular velocity $\Omega_0 \sim 10^{-14} \text{ s}^{-1}$. If this element fell *radially*, conserving its specific angular momentum $j = \Omega_0 R_0^2$ (in the absence of external torques — the same physics behind a figure skater spinning faster when pulling in their arms), its angular velocity upon reaching a stellar radius $R_* \sim R_\odot \approx 6.96 \times 10^{10} \text{ cm}$ would be

$$\Omega_* = \Omega_0 \left(\frac{R_0}{R_*} \right)^2 \approx 10^{-14} \text{ s}^{-1} \times \left(\frac{3.09 \times 10^{17}}{6.96 \times 10^{10}} \right)^2 \approx 0.20 \text{ s}^{-1}, \quad (7.26)$$

corresponding to an equatorial velocity $v_* = \Omega_* R_* \approx 1.4 \times 10^5 \text{ km/s}$ — nearly half the speed of light. Compare this with the **critical** (*breakup*) angular velocity, the limit above which the centrifugal force at the surface exceeds gravity and the star disintegrates:

$$\Omega_{\text{crit}} = \sqrt{\frac{GM_\odot}{R_\odot^3}} \approx 6.3 \times 10^{-4} \text{ s}^{-1} \implies v_{\text{crit}} = \Omega_{\text{crit}} R_\odot \approx 437 \text{ km/s}, \quad (7.27)$$

a value that correctly reproduces the equatorial *breakup* velocity of an object with solar mass and radius. The ratio between the two results,

$$\frac{\Omega_*}{\Omega_{\text{crit}}} \approx \frac{0.20}{6.3 \times 10^{-4}} \approx 320, \quad (7.28)$$

shows that naive radial collapse would produce a star spinning **more than two orders of magnitude above** the breakup velocity — physically impossible. This is the classic **angular momentum problem**: the specific angular momentum observed in prestellar cores is too large, by many orders of magnitude, to be carried all the way to the surface of a main-sequence star (in practice the problem is even more severe: young T Tauri stars typically rotate at a few percent of Ω_{crit} , not close to it — most of the angular momentum needs to be *removed*, not merely prevented from causing breakup).

The resolution is, at once, simple and central to this chapter: the collapse is *not* radial. Since the specific angular momentum $j = \Omega r^2$ of each shell is conserved during infall and *grows* with the initial radius of the shell, each gas element falls only as far as the radius where the centrifugal force associated with its own j balances the gravitational pull of the already-accreted central mass. This **centrifugal radius** R_c is obtained by equating the centrifugal

acceleration $v_\phi^2/r = j^2/r^3$ (with $v_\phi = j/r$) to the gravitational acceleration $GM_\star/r^2$:

$$\frac{j^2}{R_c^3} = \frac{GM_\star}{R_c^2} \implies \boxed{R_c = \frac{j^2}{GM_\star}}. \quad (7.29)$$

Since j grows with the initial radius of the shell, material coming from larger R_0 settles at progressively *larger* centrifugal radii. The result is not a star spinning absurdly, but a **circumstellar accretion disk**: matter accumulates in a thin, Keplerian-rotating disk, which stores the large majority of the system's angular momentum. Only the fraction of mass that loses enough angular momentum — through internal viscosity and other transport processes (box below) — crosses the disk and is accreted by the protostar. The detailed physics of this disk will be developed further on in this chapter.

Other angular momentum transport mechanisms

The formation of a disk solves the problem of *where* the angular momentum goes, but does not remove it from the system as a whole. The literature identifies several additional processes, of relative importance still under active research: **magnetic braking**, in which field lines anchored in the interstellar medium exert torques already at the scale of the core; **protostellar winds and jets**, launched magnetocentrally from the inner disk; and **gravitational torques** in massive, non-axisymmetric disks, which redistribute angular momentum without removing it. A review of these mechanisms: Li et al. (2014); evolution of angular momentum from core to disk in individual systems: Lee (2026).

Scale of the centrifugal radius

Let us estimate R_c for the outermost shell of a typical core ($R_0 = 0.1$ pc, $\Omega_0 = 10^{-14}$ s $^{-1}$), falling onto a protostar that already has $M_\star \approx 1 M_\odot$ of accreted central mass (a situation representative of more advanced stages of collapse, when most of the core has already been processed). The specific angular momentum of this shell is

$$j = \Omega_0 R_0^2 \approx 10^{-14} \text{ s}^{-1} \times (3.09 \times 10^{17} \text{ cm})^2 \approx 9.5 \times 10^{20} \text{ cm}^2/\text{s}. \quad (7.30)$$

By Equation (7.29),

$$R_c = \frac{j^2}{GM_\star} \approx \frac{(9.5 \times 10^{20})^2}{6.674 \times 10^{-8} \times 1.989 \times 10^{33}} \text{ cm} \approx 6.8 \times 10^{15} \text{ cm} \approx 460 \text{ au}. \quad (7.31)$$

This is the radius at which the outermost material (largest j) settles — comparable to the largest observed disks. Inner shells, with smaller R_0 , fall onto proportionally smaller radii: since $R_c \propto R_0^4$ [via $j \propto R_0^2$], a shell with half the initial radius settles $16\times$ closer, $R_c/16 \approx 29$ au — the scale of the Solar System out to Neptune. The disk formed by successive shell deposition therefore naturally covers the entire range of **tens to hundreds of au** observed in real protoplanetary disks.

Further Reading and Current References

Direct evidence for this section, via ALMA: the *eDisk* survey (resolution $\sim 0.04''$ in Class 0/I protostars) is in Ohashi et al. (2023); the first unambiguous detection of a Keplerian disk around a Class 0 protostar (system L1527 IRS, $\approx 0.2 M_\odot$), confirming the centrifugal-radius mechanism, is in Tobin et al. (2012). Evolution of the specific angular

momentum from core to disk: Lee (2026).

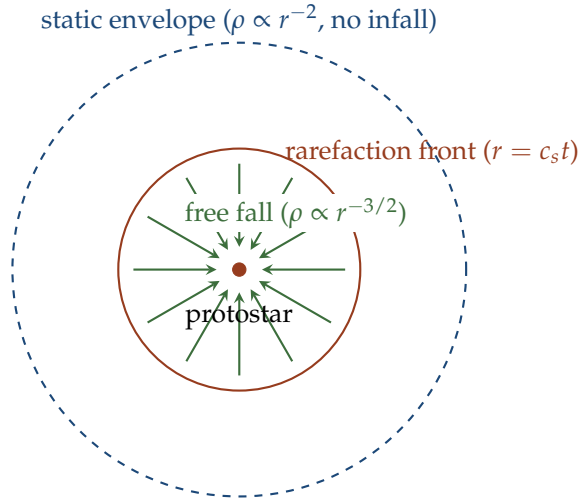


Figure 7.2: Inside-out collapse of Shu (1977): the rarefaction front, propagating at $r = c_s t$, separates the inner region already in free fall ($\rho \propto r^{-3/2}$, $v \propto r^{-1/2}$) from the outer envelope still static, with the original SIS profile [Equation (7.21)]. The protostar grows at an approximately constant rate [Equation (7.23)] while the front advances, progressively converting the core into infalling material.

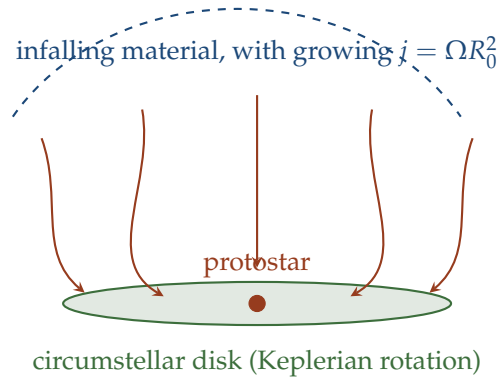


Figure 7.3: Disk formation by deposition of material with increasing j : each infall trajectory conserves its j and settles at the corresponding centrifugal radius [Equation (7.29)] — low- j material falls almost directly onto the protostar, while high- j material settles at larger radii, building a thin, Keplerian-rotating disk.

7.6 From protostar to pre-main sequence

As accretion proceeds and the rarefaction front sweeps through the entire original core (Section 7.4), the infall rate onto the protostar+disk system decreases, and the surrounding envelope — increasingly less massive — gradually becomes optically thin. The central protostar, previously obscured, becomes gradually visible in the optical and near infrared, still surrounded by a remnant accretion disk (Section 7.5) and, occasionally, by vestiges of the envelope. This object — a star whose mass is nearly fully assembled, still contracting

gravitationally, without stable nuclear hydrogen fusion in its core — is a **pre-main-sequence star** (PMS). Its evolution in the HR diagram, from the youngest stage to the ZAMS, follows a well-defined two-stage pattern: first the **Hayashi track**, then the **Heney track**.

The Hayashi track

A newly-revealed protostar is, in practically its entire mass, **fully convective**. The reason is directly tied to the Schwarzschild criterion [Equation (5.142), Chapter 5] and to the explicit form of the radiative gradient already derived in the same chapter,

$$\nabla_{\text{rad}} = \frac{P}{T} \frac{dT}{dP} = \frac{3}{16\pi acG} \cdot \frac{\kappa_{\text{rad}} l P}{m T^4} \quad (7.32)$$

[rewritten in logarithmic form from Equation (5.117)]: ∇_{rad} is directly proportional to the opacity κ_{rad} . In the relatively cold outer layers ($T \sim 4000\text{--}8000\text{ K}$) of a low-mass protostar or PMS star, the opacity is dominated by continuum absorption from the **negative hydrogen ion**, H^- , whose standard approximate form is

$$\kappa_{\text{H}^-} \approx 1.1 \times 10^{-25} Z^{1/2} \rho^{1/2} T^{7.7} \text{ cm}^2/\text{g}, \quad (7.33)$$

valid approximately over the range $4 \times 10^3 \lesssim T \lesssim 10^4\text{ K}$. Unlike the Kramers opacity of the deep interior (which *decreases* with T), κ_{H^-} *increases* rapidly with temperature: more T ionizes more electrons from residual metals, which attach to neutral H forming more H^- . Via $\nabla_{\text{rad}} \propto \kappa_{\text{rad}}$, ∇_{rad} rises abruptly just below the photosphere, exceeding $\nabla_{\text{ad}} = 2/5$ [Equation (5.155)] and triggering Schwarzschild convection throughout practically the entire star — not just in a thin surface layer, as in the present-day Sun.

A fully convective star has a remarkable property: convective transport is so efficient that the entire structure closely follows a single adiabat (an $n = 3/2$ polytrope), so that, for fixed mass M , the entire structure is determined by the boundary condition at the photosphere, where the H^- opacity connects the convective interior to the radiative atmosphere. Since κ_{H^-} [Equation (7.33)] depends so strongly on T , this condition admits solutions only within a narrow range of effective temperature, $T_{\text{eff}} \sim 4000\text{ K}$ (**Hayashi temperature**, weakly dependent on mass), *regardless of L* — the radius adjusts to keep T_{eff} fixed. In the HR diagram, this is a nearly **vertical** trajectory: the protostar contracts and L falls (losing gravitational energy, Section 5.7), with T_{eff} nearly fixed — the **Hayashi track** (Hayashi, 1961), which defines a **limit on the right** of the HR diagram: no star in quasi-hydrostatic equilibrium can be cooler than this line, at any L .

The Heney track

As the protostar contracts along the Hayashi track, its central temperature rises (effective negative heat capacity, Section 5.7). At some point — earlier for massive stars, whose interior is hotter — the internal temperature becomes high enough for the opacity to fall (the Kramers regime, $\kappa \propto \rho T^{-3.5}$, decreasing with T , unlike the previous H^- regime) and ∇_{rad} to fall back below ∇_{ad} in the interior [Equation (5.156)]: a **radiative core** forms and grows, surrounded by an increasingly thin convective envelope. The star leaves the Hayashi track and moves onto the **Heney track** (Heney et al., 1955): evolution at **nearly constant luminosity**, contracting and heating up — predominantly **horizontal** motion in the HR diagram — until T_c reaches $\sim 1\text{--}1.5 \times 10^7\text{ K}$, sufficient for stable hydrogen fusion, establishing the star on the **ZAMS**. The timescale of this radiative contraction is the **Kelvin-Helmholtz timescale** $\tau_{\text{KH}} \approx \frac{1}{2} GM^2 / (LR)$

[Equation (5.77)], the same mechanism that regulates the pre-nuclear contraction of the present-day Sun.

Contraction times to the ZAMS: strong mass dependence

Since $\tau_{\text{KH}} \propto M^2/(LR)$, and L and R grow with mass on the ZAMS, the contraction time decreases rapidly with M — but the exact exponent of $\tau_{\text{KH}} \propto M^n$ is sensitive to the adopted mass-luminosity/radius relations (n typically between -1.8 and -3). For this reason Equation (5.77) serves here only as an order-of-magnitude guide; the values below come from complete PMS tracks tabulated by Stahler and Palla (2004):

- $M = 0.5 M_{\odot}$: $\sim 1.6 \times 10^8$ years (160 million years) — the star never develops a significant radiative core before the ZAMS and remains almost entirely on the Hayashi track throughout the contraction;
- $M = 2 M_{\odot}$: $\sim 10^7$ years (10 million years) — the star leaves the Hayashi track early, after developing a radiative core while still at high luminosity, and most of the contraction time elapses already on the Henyey track.

The difference of more than one order of magnitude between the two times illustrates why young star clusters show such a heterogeneous stellar population: the most massive stars have already reached the main sequence and begun fusing hydrogen stably while those of solar mass or less are still in full pre-main-sequence contraction.

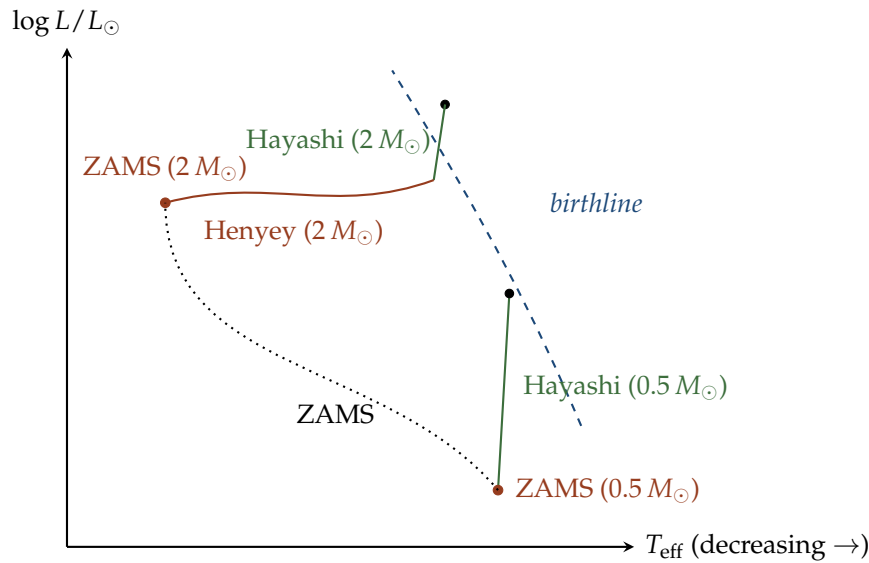


Figure 7.4: PMS evolution for two masses: for $0.5 M_{\odot}$ (green), the Hayashi track dominates the entire contraction ($\sim 1.6 \times 10^8$ years to the ZAMS); for $2 M_{\odot}$, a short Hayashi stage is followed by Henyey (orange, nearly constant L) until the ZAMS in only $\sim 10^7$ years. The dashed *birthline* marks where protostars become visible (Section 7.6).

7.7 Observational classification: Classes 0, I, II, and III

From an observational standpoint — before disks and envelopes could be spatially resolved individually — the evolutionary stage of a young stellar object (YSO) is inferred from the

shape of its **spectral energy distribution** (SED, νF_ν versus wavelength), quantified by the **infrared spectral index**

$$\alpha_{\text{IR}} \equiv \frac{d \log(\nu F_\nu)}{d \log \nu}, \quad (7.34)$$

typically measured between 2 and 10–25 μm (Lada, 1987). This scheme, refined by Greene et al. (1994) with the addition of the "flat-spectrum" class, defines four observational categories:

- **Class 0:** SED dominated by the dense, cold envelope, peaking in the submillimeter/FIR; almost no stellar light escapes. Originally defined by the ratio $L_{\text{smm}}/L_{\text{bol}} > 0.5\%$ (not by α_{IR} , which is undefined) (André et al., 1993);
- **Class I:** $\alpha_{\text{IR}} \geq 0.3$. Detectable protostar, disk, and still-substantial remnant envelope;
- **Class II:** $-1.6 \leq \alpha_{\text{IR}} < -0.3$. Envelope dissipated; thin, optically thick disk around an already visible classical T Tauri star (CTTS) — the stage detailed further on in the chapter;
- **Class III:** $\alpha_{\text{IR}} < -1.6$. Disk dissipated or residual; weak-line T Tauri star (WTTS), essentially photospheric SED.

(The range $-0.3 \leq \alpha_{\text{IR}} < 0.3$, "flat spectrum," occupies the Class I $\rightarrow$ II transition.)

Observational classes as a sequence of decreasing envelope mass

The SED classification reflects the physical ratio $M_{\text{env}}/M_\star$ between mass still infalling and mass already incorporated into the star+disk system: $\gtrsim 1$ in Class 0, $\lesssim 1$ but still relevant in Class I, $\ll 1$ (disk only) in Class II, and residual in Class III. The sequence $0 \rightarrow \text{I} \rightarrow \text{II} \rightarrow \text{III}$ is, in essence, one of decreasing envelope mass — consistent with inside-out collapse and disk formation (Sections 7.4 and 7.5) (André et al., 1993).

Evolutionary timescales for a $1 M_\odot$ protostar

Combining results from the *Spitzer* c2d survey (Evans et al., 2009) with the review by Dunham et al. (2014), the typical durations of each class, for a protostar of $\sim 1 M_\odot$, are approximately:

- **Class 0:** $\lesssim 3 \times 10^5$ years (often quoted as $\sim 10^4$ – 10^5 years in individual studies, but with a population upper limit of ~ 0.3 Myr);
- **Class I:** the combined Class 0 + Class I duration is ~ 0.44 – 0.5 Myr, so that Class I alone typically lasts $\sim (1\text{--}2) \times 10^5$ years;
- **Class II:** half-life of ~ 2 Myr (Evans et al., 2009), with individual protoplanetary disks persisting, in general, for ~ 1 – 10 Myr;
- **Class III:** final transition phase, without a detectable disk, extending until the star reaches the ZAMS — for $1 M_\odot$, the Sun itself, this total pre-main-sequence contraction time is $\tau_{\text{KH},\odot} \approx 1.5 \times 10^7$ years [already calculated in Chapter 5, Equation (5.79)], meaning the Class III phase itself can last for about 15 million years for a solar-type star — consistent with the much longer times obtained in the previous Example for $0.5 M_\odot$ and the much shorter time obtained for $2 M_\odot$.

Note the hierarchy: from Class 0 to Class II, the timescales grow by more than one order of magnitude at each stage — consistent with the picture of an envelope that dissipates

rapidly in the early phases, followed by a much longer-lived disk.

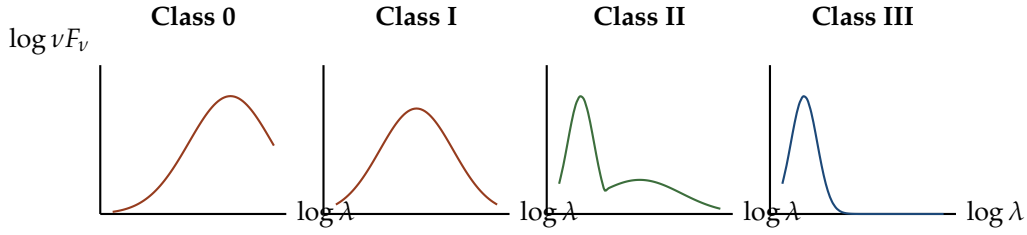


Figure 7.5: Schematic SEDs ($\log \nu F_\nu$ vs. $\log \lambda$) of the Class 0 $\rightarrow$ I $\rightarrow$ II $\rightarrow$ III sequence: the peak shifts from long wavelengths (cold envelope) to short ones (photosphere), as envelope and disk dissipate — the same sequence of decreasing $M_{\text{env}} / M_\star$ from the previous box.

Further Reading and Current References

Modern grids of pre-main-sequence evolutionary models: Baraffe et al. (2015). The JWST JOYS program (5–28 μm spectroscopy of jets, winds, and chemistry of Class 0/I protostars): van Dishoeck et al. (2025). Review of timescales, luminosity functions, and observational classes of protostars: Dunham et al. (2014).

7.8 Structure and evolution of protoplanetary disks

Section 7.5 showed that conservation of angular momentum forces the formation of an accretion disk around the protostar — the reservoir that feeds stellar growth throughout the life of Classes 0, I, and II, and also the physical cradle of planetary systems (next section). This section describes the equilibrium structure of the disk, its evolution/accretion mechanism, and the temperature profile that determines where different solids condense.

Disk geometry: Keplerian rotation and scale height

An accretion disk is, geometrically, *thin*: $H/r \ll 1$ at practically every radius, a consequence of the balance between two distinct forces — in the radial plane, gravity dominates strongly over pressure (dynamically negligible), and the gas orbits in nearly Keplerian orbits,

$$\Omega(r) \approx \Omega_K(r) = \sqrt{\frac{GM_\star}{r^3}}; \quad (7.35)$$

while in the vertical direction, it is gas pressure that supports the disk against stellar gravity.

To obtain $H(r)$, consider vertical hydrostatic equilibrium at a height $z \ll r$ above the midplane, where the vertical component of gravity is

$$g_z = \frac{GM_\star z}{(r^2 + z^2)^{3/2}} \approx \frac{GM_\star}{r^3} z = \Omega_K^2(r) z, \quad (7.36)$$

that is, vertical gravity behaves, for $z \ll r$, like a harmonic oscillator of frequency Ω_K . Vertical hydrostatic equilibrium,

$$\frac{dP}{dz} = -\rho g_z = -\rho \Omega_K^2 z, \quad (7.37)$$

closed with the local isothermal equation of state $P = \rho c_s^2$ [$c_s^2 = k_B T / (\mu m_H)$], the same ideal-gas relation used throughout Chapter 5, here applied to the local disk temperature $T(r)$], yields

$$c_s^2 \frac{d\rho}{dz} = -\rho \Omega_K^2 z \implies \frac{d\rho}{\rho} = -\frac{\Omega_K^2}{c_s^2} z dz. \quad (7.38)$$

Integrating,

$$\rho(r, z) = \rho_0(r) \exp\left(-\frac{z^2}{2H^2}\right), \quad H(r) \equiv \frac{c_s(r)}{\Omega_K(r)}, \quad (7.39)$$

a Gaussian vertical profile with scale height H . Dividing by r and using the Keplerian velocity $v_K = \Omega_K r$,

$$\frac{H}{r} = \frac{c_s}{\Omega_K r} = \frac{c_s}{v_K}, \quad (7.40)$$

confirming that the aspect ratio is essentially the ratio between the local sound speed (thermal agitation) and the orbital velocity. Since $c_s \ll v_K$ for a cold disk (verified numerically below), the disk is necessarily thin.

The disk is characterized, in addition to $H(r)$, by two radial profiles: the surface density $\Sigma(r) = \int \rho dz \approx \sqrt{2\pi} \rho_0(r) H(r)$ and the temperature $T(r)$ (both decreasing with r). The reference profile of the Minimum Mass Solar Nebula (MMSN), constructed by Hayashi (1981) from the mass and composition presently locked up in the planets of the Solar System, adopts

$$\Sigma(r) \approx 1700 \left(\frac{r}{1 \text{ au}}\right)^{-3/2} \text{ g cm}^{-2}, \quad T(r) \approx 280 \left(\frac{r}{1 \text{ au}}\right)^{-1/2} \text{ K}, \quad (7.41)$$

which we will use as a numerical reference throughout this section.

Aspect ratio of a protoplanetary disk at 1 au

For a $1 M_\odot$ protostar and the MMSN profile of Equation (7.41), evaluated at $r = 1 \text{ au}$: with $T = 280 \text{ K}$ and mean molecular weight $\mu \approx 2.3$ (molecular $\text{H}_2 + \text{He}$ gas), the sound speed is

$$c_s = \sqrt{\frac{k_B T}{\mu m_H}} \approx \sqrt{\frac{(1.381 \times 10^{-16})(280)}{(2.3)(1.673 \times 10^{-24})}} \approx 1.0 \times 10^5 \text{ cm/s} \approx 1.0 \text{ km/s}. \quad (7.42)$$

The Keplerian velocity, with $GM_\odot \approx 1.327 \times 10^{26} \text{ cm}^3/\text{s}^2$ and $r = 1 \text{ au} \approx 1.496 \times 10^{13} \text{ cm}$, is

$$v_K = \sqrt{\frac{GM_\odot}{r}} \approx 3.0 \times 10^6 \text{ cm/s} \approx 30 \text{ km/s} \quad (7.43)$$

(the same value as Earth's orbital velocity, as it should be, since $r = 1 \text{ au}$ is literally Earth's orbit). Hence, by Equation (7.40),

$$\left.\frac{H}{r}\right|_{1 \text{ au}} \approx \frac{1.0}{30} \approx 0.034, \quad (7.44)$$

that is, $H \approx 0.034 \text{ au} \approx 5 \times 10^{11} \text{ cm}$: the scale height is about 3% of the orbital radius, quantitatively confirming that the disk is thin. The corresponding midplane density is

$$\rho_0 = \frac{\Sigma}{\sqrt{2\pi} H} \approx \frac{1700}{\sqrt{2\pi} (5.0 \times 10^{11})} \approx 1.3 \times 10^{-9} \text{ g/cm}^3, \quad (7.45)$$

corresponding to a number density $n = \rho_0 / (\mu m_H) \approx 3.5 \times 10^{14} \text{ cm}^{-3}$ — this number will be used in the next subsection to estimate the molecular mean free path of the disk gas.

The viscosity problem and the α -disk model

A purely Keplerian disk, with no angular momentum transport, would be eternal: no mass would ever fall onto the protostar — which contradicts the observed accretion rates of classical T Tauri stars ($\dot{M}_* \sim 10^{-9}$ – $10^{-7} M_\odot/\text{yr}$), requiring a mechanism that transports mass inward and angular momentum outward.

The obvious candidate is the microscopic viscosity of the gas — the molecular friction between adjacent rings. The problem is that this viscosity is absurdly small: $\nu_{\text{mol}} \sim \ell c_s$, with $\ell = 1/(n\sigma)$ the mean free path ($\sigma \approx 2 \times 10^{-15} \text{ cm}^2$ for H_2). Using the values from the previous Example ($n \approx 3.5 \times 10^{14} \text{ cm}^{-3}$, $c_s \approx 1.0 \times 10^5 \text{ cm/s}$) at 1 au,

$$\ell = \frac{1}{n\sigma} \approx \frac{1}{(3.5 \times 10^{14})(2 \times 10^{-15})} \approx 1.4 \text{ cm}, \quad (7.46)$$

$$\nu_{\text{mol}} \sim \ell c_s \approx (1.4)(1.0 \times 10^5) \approx 1.4 \times 10^5 \text{ cm}^2/\text{s}. \quad (7.47)$$

To assess whether this value is dynamically relevant, we form the Reynolds number of the flow, $\text{Re} = vL/\nu$, using as the characteristic velocity and length of the orbital shear the Keplerian velocity itself and the disk radius, $v \sim v_K$, $L \sim r$:

$$\text{Re} = \frac{v_K r}{\nu_{\text{mol}}} \approx \frac{(3.0 \times 10^6)(1.5 \times 10^{13})}{1.4 \times 10^5} \approx 3 \times 10^{14}. \quad (7.48)$$

This is an absurdly large number: laboratory flows already turn turbulent for $\text{Re} \gtrsim 10^3$ – 10^4 . The associated viscous time, $t_{\nu, \text{mol}} \sim r^2/\nu_{\text{mol}} \approx 1.6 \times 10^{21} \text{ s} \approx 5 \times 10^{13} \text{ years}$ — about *four thousand times the age of the Universe* — makes it clear that molecular viscosity is physically irrelevant to observed accretion.

What is needed, therefore, is a much larger *effective* viscosity, typically identified with turbulence in the gas (turbulent transport works like microscopic viscosity, but with turbulent "fluid parcels" in place of molecules, and much larger effective free path/velocity). The classic parametrization is that of Shakura and Sunyaev (1973), who — without specifying the physical origin of the turbulence — writes the effective viscosity as

$$\nu = \alpha c_s H, \quad 0 < \alpha \lesssim 1, \quad (7.49)$$

the α -disk model: c_s is the characteristic velocity of the subsonic turbulent motion expected (supersonic motion would be damped by shocks) and H is the largest length scale available vertically, so that α measures the efficiency of turbulence in transporting angular momentum as a fraction of that upper limit.

Comparing Equation (7.49) with the estimate of $\nu_{\text{mol}} \approx 1.4 \times 10^5 \text{ cm}^2/\text{s}$ above: using $H \approx 5 \times 10^{11} \text{ cm}$ and $c_s \approx 1.0 \times 10^5 \text{ cm/s}$ at 1 au, even an α as small as 10^{-4} already corresponds to $\nu = \alpha c_s H \approx 5 \times 10^{12} \text{ cm}^2/\text{s}$ — about 3.6×10^7 times larger than ν_{mol} ; for $\alpha \sim 10^{-2}$, the amplification factor reaches $\sim 3.6 \times 10^9$. The central physical question then becomes: which hydrodynamic or magnetohydrodynamic mechanism generates this turbulence?

The mechanism now considered dominant in sufficiently ionized disks is the **magnetorotational instability** (MRI): a weak magnetic field renders a differentially (Keplerian) rotating disk linearly unstable, generating MHD turbulence with efficiency $\alpha \sim 10^{-3}$ – 10^{-1} in idealized simulations. The problem is that the MRI requires reasonable coupling between field and gas (minimum ionization) — insufficient in the cold, shielded midplane of the inner/intermediate regions (~ 1 – 20 au), the so-called **dead zones**, where the MRI is suppressed. Alternative hydrodynamic mechanisms, which do not depend on ionization — such as the

vertical shear instability (VSI), which exploits the disk's radial temperature gradient — then become relevant. A review of these mechanisms is in Turner et al. (2014).

Observationally, α values inferred from accretion rates and disk masses (ALMA) do not cluster around a single value: Rafikov (2017), in 26 disks, finds $\alpha \sim 10^{-4}$ –0.04, with no clear correlation with mass/size/stellar parameters — consistent with the coexistence of MRI-dominated and dead-zone regions. This range, $\alpha \sim 10^{-4}$ – 10^{-2} , will be used as a reference going forward.

Viscous evolution: accretion and disk spreading

Physically, the effective viscosity acts by exerting a torque between adjacent disk rings: the inner ring, rotating slightly faster, "drags" the outer ring forward (accelerating it), while the outer ring "brakes" the inner one. By Newton's third law, this pair of torques transports angular momentum from the inner to the outer part of the disk — without changing the total energy of the system through this mechanism alone — while mass, in order to conserve angular momentum upon losing part of it, migrates in the opposite direction, inward. The net result is exactly what observation requires: mass accretion onto the protostar, accompanied by radial spreading of the disk (most of the mass concentrates progressively closer to the center, while a shrinking fraction of the mass, carrying a growing fraction of the total angular momentum, moves out to large radii). Formally, this process is described by a diffusion equation for $\Sigma(r, t)$ with a diffusion coefficient proportional to ν (whose full derivation is beyond the scope of these notes, but whose qualitative behavior — diffusive evolution, with inner accretion and outer spreading — is exactly what the intuition above suggests).

The characteristic timescale of this diffusive evolution is the viscous time,

$$t_{\text{visc}}(r) \sim \frac{r^2}{\nu(r)} = \frac{r^2}{\alpha c_s(r) H(r)} = \frac{1}{\alpha \Omega_K(r)} \left(\frac{r}{H(r)} \right)^2, \quad (7.50)$$

where the last form uses $\nu = \alpha c_s H$ and $H/r = c_s/v_K$ [Equations (7.49) and (7.40)]. Since $(r/H)^2 \gg 1$ for a thin disk, t_{visc} is much longer than the local orbital period $2\pi/\Omega_K$ — viscous accretion is a slow process compared to orbital dynamics, which justifies a priori treating the disk, at each instant, as approximately Keplerian (Equation (7.35)).

Viscous time in the inner and outer disk regions

Comparing $r \sim 1$ au (where the MRI operates with $\alpha \sim 10^{-2}$, using $c_s \approx 1.0 \times 10^5$ cm/s, $H \approx 5.0 \times 10^{11}$ cm from the previous Example) with $r \sim 30$ au (a more neutral region, affected by the dead zone, $\alpha \sim 10^{-3}$; by the MMSN profile, $T \approx 51$ K, $c_s \approx 4.3 \times 10^4$ cm/s, $H \approx 3.5 \times 10^{13}$ cm, with $H/r \approx 0.08$ larger than at 1 au — the disk is *flaring*, its aspect ratio grows with radius):

r	$\nu = \alpha c_s H$	$t_{\text{visc}} \sim r^2/\nu$	
1 au ($\alpha = 10^{-2}$)	5.0×10^{14} cm ² /s	4.5×10^{11} s	$\approx 1.4 \times 10^4$ years
30 au ($\alpha = 10^{-3}$)	1.5×10^{15} cm ² /s	1.3×10^{14} s	$\approx 4 \times 10^6$ years

The inner region processes and accretes its mass rapidly; the outer region evolves over a few million years, in excellent order-of-magnitude agreement with the observed lifetime of protoplanetary disks (Section 7.8) — supporting the interpretation that it is the viscous evolution of the *outer*, slower regions that sets how long a disk survives as a gas reservoir.

Temperature profile: viscous heating vs. stellar irradiation

The $T(r)$ profile of a protoplanetary disk generally results from the competition between two energy sources with very different geometries.

In early phases and inner regions, where density and accretion rate are high, the viscous dissipation itself — the turbulent "friction" between rings — converts orbital energy into heat at a rate $\sim \nu \Sigma \Omega_K^2$ per unit area, re-radiated locally: the **internally heated disk** regime, formally identical to stellar radiative transport (Chapter 5), with T growing with the local viscous dissipation.

In outer regions and later phases (accretion already declining), on the other hand, the dominant source becomes **stellar irradiation**. Since the disk is optically thin there, each area element absorbs a fraction of the stellar light and re-emits it as a blackbody. The thin geometry is essential: since $H/r \ll 1$, light from a nearly point-like source in the midplane arrives at a *grazing* angle — not perpendicular —, so that only a fraction $\sim H/r$ of the stellar flux is intercepted per unit disk area.

The energy balance per unit area — stellar flux $L_*/(4\pi r^2)$ times the grazing geometric factor $\gamma \sim H/r$ (treated as constant, in the zeroth-order approximation), set equal to blackbody re-emission from the two faces, $2\sigma T^4$ —

$$\gamma \frac{L_*}{4\pi r^2} = 2\sigma T^4, \quad \gamma \approx \text{constant} \implies T^4 \propto \frac{1}{r^2} \implies \boxed{T(r) \propto r^{-1/2}}, \quad (7.51)$$

exactly the exponent of the MMSN profile [Equation (7.41)] — the same balance Hayashi (1981) used to calibrate $T(1 \text{ au}) \approx 280 \text{ K}$.

On the constancy of γ

This approximation is zeroth order. Since H/r grows slowly with r (the disk is *flaring*), a self-consistent treatment — in which $\gamma \sim H/r$ is computed from the very temperature profile it helps determine — leads to $T \propto r^{-3/7}$ (Chiang and Goldreich, 1997). The difference between $-1/2$ and $-3/7$ is small enough not to qualitatively affect the conclusions of this section, so we keep the simpler MMSN profile.

The snow line

The fact that $T(r)$ decreases monotonically outward has a direct consequence for the chemical composition of the solids that condense in the disk — and, later, of the planets that accumulate from them. Different species condense at different temperatures, defining for each one a radius in the disk below which they remain gaseous and above which they condense onto grains. The most important of these boundaries is the **snow line**, the radius r_{snow} where T falls below the condensation temperature of water — typically $T_{\text{snow}} \approx 170 \text{ K}$ (plausible range $\sim 145\text{--}170 \text{ K}$, depending on the local partial pressure) (Hayashi, 1981).

The snow line divides the disk into two regions of radically different composition: **inner** ($T > T_{\text{snow}}$), where only refractory materials (silicates, oxides, iron-nickel) condense and water remains gaseous; and **outer** ($T < T_{\text{snow}}$), where water also condenses as ice, increasing by a factor of $\sim 3\text{--}4$ the surface density of available solids — since oxygen (mostly H_2O) is several times more abundant, by mass, than all refractory elements combined.

Numerical position of the snow line

Using the MMSN temperature profile [Equation (7.41), with the same exponent $-1/2$ derived above in Equation (7.51)] and $T_{\text{snow}} = 170$ K,

$$T(r_{\text{snow}}) = 280 \left(\frac{r_{\text{snow}}}{1 \text{ au}} \right)^{-1/2} \text{ K} = 170 \text{ K} \implies r_{\text{snow}} = \left(\frac{280}{170} \right)^2 \text{ au} \approx 2.7 \text{ au}. \quad (7.52)$$

This value agrees very well with the inferred position of the snow line of the primitive Solar System, from meteorites and the distribution of spectral types in the asteroid belt ($\sim 2.7\text{--}3$ au) — close to the belt's own mean semi-major axis, consistent with the transition between S-type asteroids (rocky, inner) and C-type asteroids (hydrated, outer).

The position of the snow line in real disks only began to be measured directly in the past decade, with ALMA. In the disk of V883 Orionis, an FU Orionis-type accretion outburst pushes the snow line out to $\sim 40\text{--}42$ au, spatially resolved by a sharp discontinuity in dust continuum emission between two ALMA bands (Cieza et al., 2016) — confirming the mechanism of Equation (7.51), and bridging to the next section: giant gas planets form preferentially beyond the snow line precisely because of the jump in surface density of solids available there, the missing ingredient for inner rocky cores to reach the critical mass for gas accretion before disk dissipation.

Lifetime of protoplanetary disks

Infrared surveys of young clusters (ages via PMS isochrones, Section 7.6) measure the **disk fraction** — fraction of stars with infrared disk excess — as a function of age: $\sim 40\text{--}50\%$ at 2–3 Myr, $\lesssim 20\%$ at 5 Myr, $\lesssim 5\%$ between 8–10 Myr (Evans et al., 2009). This gives $\tau_{\text{disco}} \sim 2\text{--}4$ Myr, though individual disks vary from < 1 Myr to tens of Myr depending on stellar mass, initial disk mass, and environment — consistent with $t_{\text{visc}}(30 \text{ au}) \sim 4$ Myr (Section 7.8).

Dissipation results from the combination of three mechanisms: **viscous accretion**, discussed in this section, which drains mass onto the protostar over t_{visc} and spreads the remainder to larger radii; **photoevaporation**, UV/X-ray photons (from the protostar itself, or from nearby OB stars) that ionize the upper layers to the point of making them unbound, dominant in the final phases and responsible for the sharp disk/no-disk transition in late Class II/III; and **consumption by planet formation**, subdominant in mass but responsible for part of the fine structure (rings, gaps) observed in well-resolved disks.

High-resolution disk structure: the disk of HL Tauri

One of the most iconic results in the astronomy of protoplanetary disks is the image of the HL Tauri disk (Class I/II, ~ 1 Myr), obtained by ALMA in 2014 (ALMA Partnership et al., 2015): instead of the smooth disk expected from simple models, it revealed a pattern of **bright rings and dark, concentric gaps** in the millimeter dust emission (at least seven rings; deeper gaps at ~ 13 , ~ 32 , and ~ 74 au, with density reduced by $\gtrsim 10\times$). The most immediate interpretation is that forming planets clear dust-free orbits through tidal torques (the gap at ~ 32 au would be consistent with $\sim 0.3\text{--}0.4 M_{\text{Jup}}$), but the interpretation is not unique: dust traps at condensation fronts (H_2O , CO, NH_3 ice lines), gas hydrodynamic instabilities, and opacity effects also produce rings. Subsequent ALMA surveys showed that such substructures are, in fact, the norm — not the exception — among resolved disks, making their physical origin an active research topic.

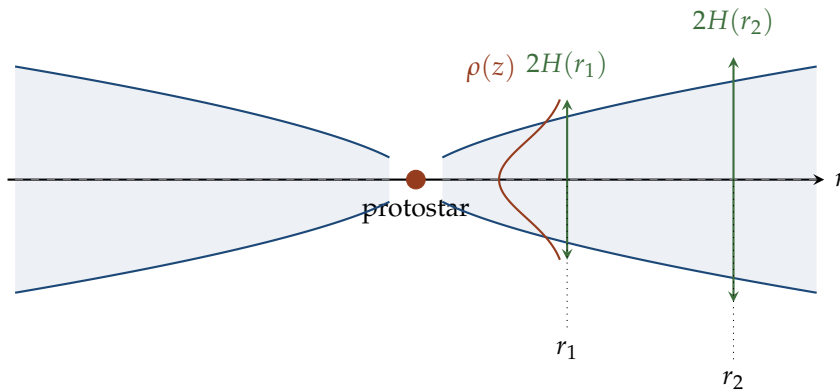


Figure 7.6: Vertical structure of a disk (not to scale): $H(r)$ grows with radius (a *flaring* disk, Section 7.8), and at each radius the density follows the Gaussian profile $\rho(z) \propto \exp(-z^2/2H^2)$ [Equation (7.39)], illustrated on the left at r_1 .

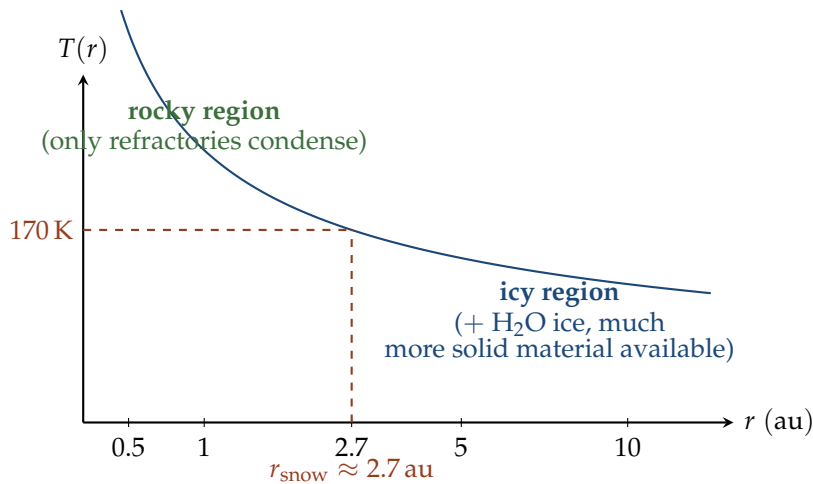


Figure 7.7: Profile $T(r) \propto r^{-1/2}$ [Equation (7.51), MMSN normalization] with the snow line ($T = 170$ K, $r_{\text{snow}} \approx 2.7$ au) separating the region where only refractories condense from the region where water ice also condenses.

Further Reading and Current References

Observed structure in protoplanetary disks (rings, gaps, spirals): Andrews (2020). Angular momentum transport in disks (MRI, dead zones, alternative hydrodynamic mechanisms): Turner et al. (2014). Original α -disk parametrization: Shakura and Sunyaev (1973). α values inferred from ALMA-resolved disks: Rafikov (2017).

7.9 Magnetospheric accretion and the T Tauri phase

Section 7.8 described how the viscous disk transports matter inward — but this hydrodynamic picture cannot hold all the way to the stellar surface: close to the protostar, the stellar magnetic field comes to dominate the gas dynamics. This section describes that transition, characterizes the resulting T Tauri phase, and closes, with winds/jets, the last angular momentum transport mechanism left open in Section 7.5.

From the viscous disk to magnetospheric accretion

T Tauri stars have surprisingly intense surface magnetic fields: spectropolarimetric measurements indicate average fields of 1–3 kG (up to ~ 6 kG locally) — orders of magnitude above the Sun (~ 1 –2 G). A field of this intensity truncates the viscous disk before it reaches the surface: inside this truncation radius (magnetospheric radius), magnetic pressure exceeds the ram pressure of the accreting gas, and the material is captured by the field lines instead of following Keplerian orbits.

To estimate this radius to order of magnitude, approximate the stellar field by a dipole, $B(r) = B_*(R_*/r)^3$, with associated magnetic pressure $P_{\text{mag}} = B(r)^2/(8\pi)$. On the gas side, the ram pressure of a nearly radial accretion flow, with rate $\dot{M}$, is $P_{\text{ram}} = \rho v^2$; using mass continuity ($\dot{M} = 4\pi r^2 \rho v$) to eliminate ρ , and approximating the infall velocity by free fall, $v \approx v_{\text{ff}}(r) = \sqrt{2GM_*/r}$ (the same form already used for core collapse in Section 7.4),

$$P_{\text{ram}}(r) = \frac{\dot{M}}{4\pi r^2} v_{\text{ff}}(r) = \frac{\dot{M}}{4\pi r^2} \sqrt{\frac{2GM_*}{r}}. \quad (7.53)$$

Equating $P_{\text{mag}} = P_{\text{ram}}$ and solving for r , one obtains the **truncation radius** (or magnetospheric radius) R_A :

$$\frac{B_*^2 R_*^6}{8\pi r^6} = \frac{\dot{M}}{4\pi r^2} \sqrt{\frac{2GM_*}{r}} \implies R_A = \left(\frac{B_*^4 R_*^{12}}{8GM_* \dot{M}^2} \right)^{1/7}. \quad (7.54)$$

The unusual exponent $1/7$ is the signature of this balance between a dipolar field ($\propto r^{-3}$, squared in the magnetic pressure) and a free-fall flow — the same result (up to an efficiency factor $\zeta \sim 0.5$ –1, here taken as 1) obtained from more complete treatments, including non-spherical geometry.

The second relevant radius is the **corotation radius** R_{co} , where the Keplerian angular velocity of the disk equals the star's own rotational angular velocity, $\Omega_K(R_{\text{co}}) = \Omega_* = 2\pi/P_*$:

$$R_{\text{co}} = \left(\frac{GM_*}{\Omega_*^2} \right)^{1/3} = \left(\frac{GM_* P_*^2}{4\pi^2} \right)^{1/3}. \quad (7.55)$$

The relative position of R_A and R_{co} determines the regime of star-disk interaction: when $R_A \lesssim R_{\text{co}}$ (the regime observed in the great majority of classical T Tauri stars, called *disk-locking*), the material is captured by the magnetic field already rotating sub-Keplerian relative to the star and falls toward it; when $R_A \gtrsim R_{\text{co}}$, the star rotates faster than the disk at that radius, and the material can be magneto-centrifugally ejected instead of accreted (the "propeller" regime).

Truncation and corotation radii of a typical T Tauri star

For a classical T Tauri star of $M_* = 0.5 M_\odot$, $R_* = 2 R_\odot \approx 1.39 \times 10^{11}$ cm, $B_* = 1$ kG = 10^3 G, and fiducial accretion rate $\dot{M} = 10^{-8} M_\odot/\text{yr} \approx 6.3 \times 10^{17}$ g/s (typical CTTS value, as we will see in the next subsection), Equation (7.54) gives

$$R_A = \left(\frac{(10^3)^4 (1.39 \times 10^{11})^{12}}{8 (6.674 \times 10^{-8}) (9.95 \times 10^{32}) (6.3 \times 10^{17})^2} \right)^{1/7} \approx 8.2 \times 10^{11} \text{ cm} \approx 5.9 R_*. \quad (7.56)$$

For the corotation radius, using a typical CTTS rotation period $P_* \approx 8$ days — close to the longer-period peak of the bimodal distribution observed by Herbst et al. (2002) for

longer-period T Tauri stars — Equation (7.55) gives

$$R_{\text{co}} = \left(\frac{(6.674 \times 10^{-8})(9.95 \times 10^{32})(6.91 \times 10^5)^2}{4\pi^2} \right)^{1/3} \approx 9.3 \times 10^{11} \text{ cm} \approx 6.7 R_*. \quad (7.57)$$

Both radii are, therefore, of order **a few stellar radii** and very close to one another, with R_A slightly smaller than R_{co} — the expected *disk-locking* regime, consistent with radii measured directly in well-studied systems (e.g. $\sim 8 R_*$ in TW Hya). The viscous disk of Section 7.8, which extends over tens to hundreds of au, is thus truncated at a distance thousands of times smaller than its own outer radius — a tiny magnetospheric "hole," but dynamically decisive, at the inner edge of the disk.

Accretion columns and hot spots

Inside the truncation radius, the gas leaves centrifugal equilibrium and is **channeled** by magnetic field lines (magnetic energy density dominates there). The material is accelerated in nearly free fall until it hits the surface at **high latitudes**, close to the magnetic poles. The supersonic impact against the photosphere produces an **accretion shock** that converts kinetic energy into heat, heating small regions — the **hot spots**.

Impact velocity and shock temperature

The free-fall velocity between the truncation radius and the stellar surface, using the values from the previous Example ($R_A \approx 8.2 \times 10^{11} \text{ cm}$, $R_* \approx 1.39 \times 10^{11} \text{ cm}$, $M_* = 0.5 M_\odot$), is

$$\begin{aligned} v_{\text{imp}} &= \sqrt{2GM_* \left(\frac{1}{R_*} - \frac{1}{R_A} \right)} \approx \sqrt{2(6.674 \times 10^{-8})(9.95 \times 10^{32})(5.97 \times 10^{-12})} \\ &\approx 2.8 \times 10^7 \text{ cm/s} \approx 280 \text{ km/s}, \end{aligned} \quad (7.58)$$

in excellent agreement with observed and modeled impact velocities in accretion columns of classical T Tauri stars, typically ~ 200 – 500 km/s (Bouvier et al., 2007; Hartmann et al., 2016). In the resulting strong shock, the Rankine-Hugoniot jump condition for a monatomic gas ($\gamma = 5/3$) converts the impact kinetic energy into a post-shock temperature

$$T_{\text{shock}} \approx \frac{3\mu m_H v_{\text{imp}}^2}{16 k_B}, \quad (7.59)$$

which, using $\mu \approx 0.6$ (post-shock ionized plasma),

$$T_{\text{shock}} \approx \frac{3(0.6)(1.673 \times 10^{-24})(2.8 \times 10^7)^2}{16(1.381 \times 10^{-16})} \approx 1.1 \times 10^6 \text{ K}, \quad (7.60)$$

of the same order of magnitude — a few million kelvin — as shock temperatures actually inferred spectroscopically (chiefly in X-rays and UV) in hot spots of classical T Tauri stars (Bouvier et al., 2007). The small numerical discrepancy relative to more detailed literature values ($\sim 2 \times 10^6 \text{ K}$) reflects additional post-shock structure effects (radiative reprocessing, non-uniform column geometry) omitted in this order-of-magnitude estimate.

Classical and weak-line T Tauri stars

Observationally, T Tauri stars — low-mass, already optically visible protostars, corresponding roughly to the Classes II and III already defined in Section 7.7 — split into two categories based directly on the presence or absence of active accretion:

- **Classical T Tauri stars (CTTS)**: active disk and magnetospheric accretion; strong H α emission (equivalent width $\gtrsim 10 \text{ \AA}$, the classic selection criterion) and UV/blue excess from the accretion shock, plus irregular photometric variability. They correspond, in essence, to Class II;
- **Weak-line T Tauri stars (WTTS)**: dissipated or residual disk, no detectable accretion; weak H α (chromospheric/coronal origin), no UV excess. They correspond to Class III.

Since H α width and UV excess scale with accretion luminosity, both serve as **quantitative diagnostics** of $\dot{M}_{\text{acc}}$, not merely qualitative indicators. The literature converges on $\dot{M}_{\text{acc}} \sim 10^{-11} - 10^{-7} M_{\odot}/\text{yr}$ (lower for low-mass/older objects, higher for young/massive ones undergoing vigorous accretion), decreasing with age and increasing with stellar mass (Hartmann et al., 2016).

Protostellar winds and jets

Section 7.5 already mentioned the **protostellar winds and jets** launched magnetocentrifugally from the inner disk — a mechanism we now physically situate in the same field-gas interaction region of this section: the field lines that channel part of the material inward also launch another part outward along the rotation axis, extracting angular momentum and energy.

Observationally two forms appear: **collimated jets** ($\sim 100\text{--}400 \text{ km/s}$, aligned with the axis), traced by **Herbig-Haro objects** (shock knots in [S II]/H α that recede over years-decades); and broader, slower, less collimated **disk winds**, launched from a larger range of radii. Both are observed from Class 0 through classical T Tauri stages, declining with the accretion reservoir — closing the point left open in Section 7.5 about how angular momentum actually *escapes* the star-disk system.

Arrival at the main sequence

The T Tauri phase ends, by definition, when the protostar reaches the ZAMS: the moment when stable hydrogen fusion alone provides the entire radiated luminosity. Formally, this is when ε_{nuc} in the energy conservation equation [Equation (5.183)] alone comes to balance $\partial l / \partial m$, without the gravitational contraction terms that dominated the Hayashi/Henyey tracks (Section 7.6).

This moment typically coincides with the final dissipation of the disk — not by coincidence, but because the mechanisms of Section 7.8 operate on timescales of a few million years, comparable to PMS contraction. The observational result is the transition from CTTS (Class II, with disk) to WTTS (Class III, without disk) — the final interval of contraction toward the ZAMS, already without the disk that characterized all of the preceding evolution.

Accretion luminosity vs. photospheric luminosity

The energy released by material falling from the truncation radius to the stellar surface appears, in large part, as radiation at the accretion shock itself (Section 7.9). Since $R_A \gg R_*$ (previous Example), the gravitational potential energy released per unit mass is

dominated by the last stretch of the fall, so that the **accretion luminosity** can be estimated, up to a geometric factor of order unity absorbed in the approximation $R_A \gg R_*$, by

$$L_{\text{acc}} \approx \frac{GM_* \dot{M}_{\text{acc}}}{R_*}. \quad (7.61)$$

For the same fiducial CTTS parameters used above ($M_* = 0.5 M_\odot$, $R_* = 2 R_\odot$, $\dot{M}_{\text{acc}} = 10^{-8} M_\odot/\text{yr}$),

$$L_{\text{acc}} \approx \frac{(6.674 \times 10^{-8})(9.95 \times 10^{32})(6.3 \times 10^{17})}{1.39 \times 10^{11}} \approx 3.0 \times 10^{32} \text{ erg/s} \approx 0.08 L_\odot. \quad (7.62)$$

Comparing with the "normal" photospheric luminosity of a $0.5 M_\odot$ protostar at this stage (tenths to a few $L_\odot$, Section 7.6), L_{acc} contributes here a minor but measurable fraction of the total brightness — part of the blue/UV excess and variability discussed above, the expected regime of a late-Class II CTTS.

The situation is radically different in the youngest phases: with the Shu accretion rate [Equation (7.23), $\dot{M} \approx 1.6 \times 10^{-6} M_\odot/\text{yr}$, ~ 160 times larger than the fiducial CTTS rate], the same Equation (7.61) scales linearly with $\dot{M}$ and gives $L_{\text{acc}}|_{\text{Class 0/I}} \approx 160 \times 0.08 L_\odot \approx 13 L_\odot$ — one to two orders of magnitude *larger* than any plausible photospheric luminosity for such a young object. This is the physical reason, already anticipated without quantitative justification in Sections 7.6 and 7.7, why in the more embedded phases (Class 0/I) the observed brightness is dominated by accretion, not the photosphere — only returning to the latter when $\dot{M}_{\text{acc}}$ falls, along the Class 0 $\rightarrow$ I $\rightarrow$ II $\rightarrow$ III evolution, in the weak-line T Tauri situation described above.

Episodic accretion: FUors and EXors

The picture of smooth, constant $\dot{M}_{\text{acc}}$ is a convenient idealization — the photometric variability of classical T Tauri stars shows that real accretion is far from smooth. At the extreme of this variability are the **FUors** (FU Orionis): outbursts in which luminosity rises $\gtrsim 4$ –6 magnitudes over decades, with $\dot{M}_{\text{acc}} \gtrsim 10^{-4} M_\odot/\text{yr}$ — orders of magnitude above even the Class 0/I rate —, contributing significantly to the final accreted mass; and the **EXors** (EX Lupi), smaller outbursts ($\gtrsim 2$ –3 mag), repeated, of shorter duration. The physical origin is still an active research topic (inner-disk instabilities, perturbations by companions), but evidence indicates that accretion is, in many systems, **episodic**: recent reviews suggest that most of the final mass may be accreted in outbursts, not smoothly, with implications for PMS evolution (Section 7.6) (Audard et al., 2014; Fischer et al., 2023).

Further Reading and Current References

Accretion in pre-main-sequence stars (diagnostics, rates, magnetospheric models): Hartmann et al. (2016). Detailed magnetospheric accretion model (columns, shocks, spectroscopic diagnostics): Bouvier et al. (2007). Accretion variability (FUors, EXors, *Kepler*/K2, TESS monitoring): Audard et al. (2014); Fischer et al. (2023).

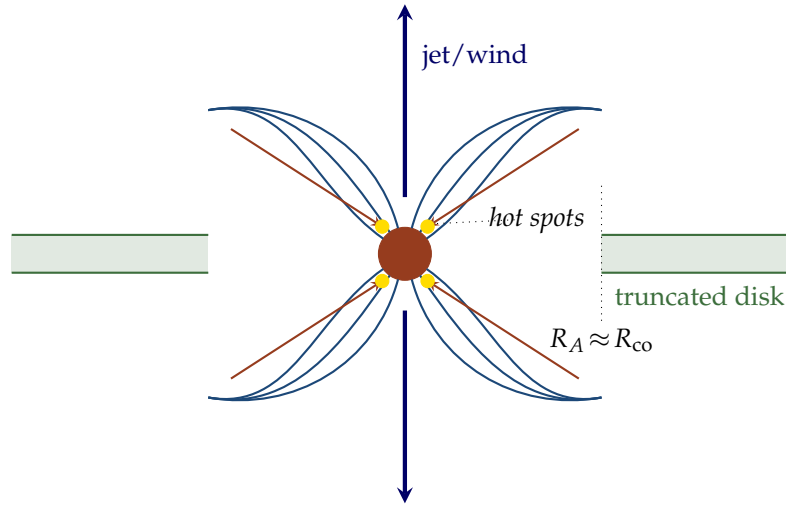


Figure 7.8: Magnetospheric accretion (not to scale): the viscous disk (Section 7.8) is truncated at R_A [Equation (7.54)], close to R_{co} [Equation (7.55)]; inside it, the gas is channeled in accretion columns toward *hot spots* (Section 7.9), while winds/jets are launched along the axis (Section 7.9).

7.10 Synthesis: from star formation to planet formation

The disk described in Section 7.8 — vertical structure, density/temperature profiles, and the snow line (Section 7.8) — is not only the mechanism that feeds stellar accretion: it is also the cradle of any planetary system. The same physics of this chapter (self-gravity vs. support, Section 7.3; angular momentum, Section 7.5; the viscous Keplerian disk, Section 7.8) remains relevant, but now competes with a qualitatively new process: the **collisional growth of solids**, which converts microscopic dust into macroscopic bodies and, eventually, planets. This final section traces that path — from dust to planetesimals, to planetary embryos, and finally to rocky planets and gas giants.

From gas to solid: dust and growth by coagulation

The disk inherits submicron dust grains from the interstellar medium (silicates, and, beyond the snow line, ice mantles over refractory cores), $a \sim 0.1 \mu\text{m}$. The first step toward planet formation is growth through successive collisions: at low relative velocity, van der Waals forces between the surfaces permanently bind the grains together (*sticking*) rather than bouncing off — the same mechanism behind the adhesion of household dust.

For the characteristic coagulation timescale, consider a grain of radius a and internal density $\rho_s \sim 1 \text{ g/cm}^3$, sweeping up smaller grains of density ρ_d (solids density in the disk) at relative velocity Δv . Through the geometric cross section πa^2 , the mass accretion rate is

$$\frac{dm}{dt} \sim \rho_d \pi a^2 \Delta v, \quad (7.63)$$

and, since $m = \frac{4}{3}\pi a^3 \rho_s$ implies $dm/dt = 4\pi a^2 \rho_s da/dt$, it follows that

$$\frac{da}{dt} \sim \frac{\rho_d \Delta v}{4\rho_s}, \quad (7.64)$$

a *constant* growth rate (independent of a !) — a direct result of the cross section growing as a^2 while the mass grows as a^3 . The radius therefore grows approximately *linearly* in time, and the characteristic time to reach a size a (starting from a microscopic initial grain, $a_0 \ll a$) is

$$t_{\text{growth}}(a) \sim \frac{a}{da/dt} \sim \frac{4\rho_s a}{\rho_d \Delta v}. \quad (7.65)$$

Coagulation growth time at 1 au

Using the gas density at 1 au (Section 7.8, $\rho_0 \approx 1.3 \times 10^{-9} \text{ g/cm}^3$) and a dust-to-gas ratio $\sim 1\%$ (interstellar-medium value), $\rho_d \approx 1.3 \times 10^{-11} \text{ g/cm}^3$. For subsonic turbulent stirring of centimeter-sized aggregates ($\alpha \sim 10^{-3}$ – 10^{-2} , Section 7.8), take $\Delta v \sim 10 \text{ cm/s} \ll c_s \approx 1.0 \times 10^5 \text{ cm/s}$. By Equation (7.65), with $\rho_s \approx 1 \text{ g/cm}^3$:

$$t_{\text{growth}}(1 \text{ cm}) \sim \frac{4(1)(1)}{(1.3 \times 10^{-11})(10)} \text{ s} \approx 3.1 \times 10^{10} \text{ s} \approx 1 \times 10^3 \text{ years}, \quad (7.66)$$

$$t_{\text{growth}}(1 \text{ m}) \sim 100 \times t_{\text{growth}}(1 \text{ cm}) \approx 1 \times 10^5 \text{ years}, \quad (7.67)$$

since $t_{\text{growth}} \propto a$. Both values are much smaller than the disk lifetime, $\tau_{\text{disco}} \sim 2$ – 4 Myr (Section 7.8) — simple coagulation is, indeed, an extremely efficient mechanism for bringing submicron grains to sizes of centimeters to meters, typically consuming less than 10 % of the disk lifetime.

The meter-size barrier

If simple coagulation brings grains to meter sizes so quickly, why does planet formation not simply proceed to planetesimals ($\gtrsim 1 \text{ km}$)? The answer is the **meter-size barrier**, a central and still active problem of planet formation theory (Weidenschilling, 1977): bodies from centimeters to tens of meters undergo rapid **radial drift**, being lost in the disk before coagulation brings them to the size at which self-gravity decouples them from the gas.

The physical origin is the same sub-Keplerianity of the gas introduced in Section 7.9: partially supported by pressure, the gas orbits at $v_{\text{gas}} = v_K(1 - \eta)$, with

$$\eta \equiv -\frac{1}{2} \left(\frac{H}{r} \right)^2 \frac{d \ln P}{d \ln r}, \quad (7.68)$$

typically $\eta \sim 10^{-3}$ for a thin disk. Solids do not feel the gas pressure and would tend to orbit exactly at v_K — but aerodynamic drag against the slower gas continuously brakes them, making them spiral inward. This drag is most efficient for intermediate-sized bodies (small enough to feel the drag, large enough to have considerable inertia), with maximum drift when the coupling time equals the orbital period (Stokes number $\text{St} \sim 1$) — the condition met by bodies of ~ 0.1 – 1 m at 1 au in an MMSN disk. In this regime,

$$v_{\text{drift}} \sim \eta v_K. \quad (7.69)$$

Radial drift velocity and time at 1 au

Using the MMSN profile [Equation (7.41)], with $\Sigma \propto r^{-3/2}$, $T \propto r^{-1/2}$, and, therefore, $H \propto r^{5/4}$ (since $H/r = c_s/v_K \propto T^{1/2}r^{1/2}$), the midplane density $\rho_{\text{mid}} = \Sigma/(\sqrt{2\pi}H)$ scales as $\rho_{\text{mid}} \propto r^{-11/4}$ and the pressure $P = \rho_{\text{mid}}c_s^2 \propto r^{-13/4}$, that is $d \ln P/d \ln r \approx -3.25$.

With $H/r \approx 0.034$ at 1 au (Example in Section 7.8),

$$\eta \approx \frac{1}{2}(0.034)^2(3.25) \approx 1.9 \times 10^{-3}. \quad (7.70)$$

Using $v_K \approx 3.0 \times 10^6$ cm/s at 1 au (same Example), Equation (7.69) gives

$$v_{\text{drift}} \approx (1.9 \times 10^{-3})(3.0 \times 10^6) \approx 5.6 \times 10^3 \text{ cm/s} \approx 56 \text{ m/s}, \quad (7.71)$$

in excellent agreement with the classic value ~ 50 m/s quoted for meter-sized bodies at 1 au (Weidenschilling, 1977). The characteristic radial drift time — the time for the body to spiral inward a distance of order the orbital radius itself — is then

$$t_{\text{drift}} \sim \frac{r}{v_{\text{drift}}} \approx \frac{1.5 \times 10^{13} \text{ cm}}{5.6 \times 10^3 \text{ cm/s}} \approx 2.7 \times 10^9 \text{ s} \approx 85 \text{ years}, \quad (7.72)$$

an *extraordinarily short* timescale — only tens to hundreds of orbits —, and of the same order of magnitude, ~ 100 – 1000 years, as the classic literature value at comparable disk radii (Weidenschilling, 1977).

The problem is, then, one of timescale: $t_{\text{drift}} \sim 10^2$ years is orders of magnitude smaller than $t_{\text{growth}}(1 \text{ m}) \sim 10^5$ years — the very coagulation process that brings grains to meter size is $\sim 1000\times$ slower than the drift that removes them from the disk as soon as they reach it. Forming kilometer-sized planetesimals by pure sequential coagulation is, therefore, not viable: a qualitatively different mechanism must cross this window quickly.

Proposed resolutions: streaming instability and pebble accretion

Modern research identifies two complementary mechanisms to overcome the meter-size barrier without relying on sequential coagulation:

- **Streaming instability:** particles partially coupled to the gas ($St \sim 0.01$ – 1 , pebbles of mm to tens of cm, before the peak of the barrier) spontaneously concentrate via unstable feedback between drag and gas motion — small concentrations reduce local drift, attracting more particles (a *runaway* process). When the local density of solids grows enough (requiring enrichment above the cosmic value $\sim 1\%$, favored near the snow line), the self-gravity of the concentration overcomes the velocity dispersion and collapses directly into planetesimals of tens to hundreds of km (Youdin and Goodman, 2005; Johansen et al., 2014);
- **Pebble accretion:** once planetesimals/larger embryos exist, they capture pebbles drifting through the disk not by ballistic collision (inefficient), but because aerodynamic drag around the embryo curves and captures pebbles within a capture radius much larger than its physical radius — orders of magnitude faster than accretion of solid planetesimals, allowing planet-mass cores to form even at large orbital distances (Johansen and Lambrechts, 2017).

These two mechanisms — concentration via streaming instability, followed by growth via pebble accretion — constitute the framework now favored for crossing the meter-size barrier, replacing the old pure sequential-coagulation model.

From planetesimals to planets: core accretion

Once kilometer-sized planetesimals have formed — large enough for mutual gravity, not gas drag, to dominate their dynamics —, growth proceeds in two phases: **runaway growth**, in which slightly larger bodies grow disproportionately faster (the gravitational cross section, enhanced by *gravitational focusing*, grows faster than linearly with mass), quickly producing a few much larger bodies; followed by **oligarchic growth**, in which these planetary embryos (up to Earth/super-Earth mass) compete for material in their feeding zones, growing more evenly until the local solid material is exhausted.

It is precisely here that the snow line (Section 7.8) becomes decisive: beyond it, the surface density of solids is $3\text{--}4\times$ larger than in the inner region, allowing embryos to grow larger and faster than with rocky material alone — the necessary condition for the next step, the formation of a gas giant.

Formation of gas giants: runaway gas accretion

A sufficiently massive rocky/icy embryo, still embedded in the disk, maintains a gaseous envelope in hydrostatic equilibrium — the same stellar physics of Chapter 5, now on a solid core. While the core is small, this equilibrium is stable and the envelope grows slowly by quasi-static radiative cooling (analogous to PMS contraction, Section 7.6). There is, however, a **critical core mass**, M_{crit} , above which the envelope can no longer be sustained: it collapses dynamically and the planet begins accreting gas at a runaway rate (*runaway gas accretion*), limited only by the disk's capacity to supply gas — forming a gas giant. The classic value, $M_{\text{crit}} \approx 10\text{--}12 M_{\oplus}$ (Mizuno, 1980), is only weakly sensitive to external conditions, though modern refinements point to $\sim 1\text{--}40 M_{\oplus}$ depending on the opacity and metal enrichment of the envelope.

The clock that limits this process is the disk lifetime (Section 7.8, $\tau_{\text{disco}} \sim 2\text{--}4 \text{ Myr}$): a core that only reaches M_{crit} after the gas has already dissipated is condemned to remain a massive rocky/icy core — the fate of cores that form too late or too far from the snow line. The competition between these two timescales is the determining factor for *where* and *when* gas giants form.

Alternative channel: gravitational instability of the disk

There is a second, physically distinct channel: instead of growing a solid core and only then accreting gas, the disk itself can fragment under its own self-gravity, collapsing directly into an object of planetary or brown-dwarf mass, with no pre-existing core. The local stability criterion of a thin, differentially rotating disk is the **Toomre criterion**,

$$Q \equiv \frac{c_s \Omega}{\pi G \Sigma}, \quad (7.73)$$

with fragmentation favored for $Q \lesssim 1$ (in practice, $Q \lesssim 1.3\text{--}2$ with finite radiative cooling). The analogy with the Jeans instability (Section 7.3) is direct: self-gravity ($G\Sigma$ vs. $G\rho$) competes with pressure support (c_s , in both cases), but here differential rotation Ω (via the epicyclic frequency $\kappa \approx \Omega$) plays the stabilizing role that pressure alone played in the non-rotating cloud — $Q \lesssim 1$ is, in essence, the same "local Jeans mass smaller than the perturbation" as the condition $M > M_J$, now for a 2D perturbation in a rotating disk.

The Toomre parameter of the MMSN disk is large — the disk is stable

Evaluating Q at $r = 50$ au using the MMSN profile [Equation (7.41)]: $\Sigma(50 \text{ au}) \approx 4.8 \text{ g/cm}^2$, $T(50 \text{ au}) \approx 40 \text{ K}$ ($c_s \approx 3.8 \times 10^4 \text{ cm/s}$), and $\Omega_K(50 \text{ au}) \approx 5.6 \times 10^{-10} \text{ s}^{-1}$ (by Equation (7.35)). By Equation (7.73),

$$Q \approx \frac{(3.8 \times 10^4)(5.6 \times 10^{-10})}{\pi(6.674 \times 10^{-8})(4.8)} \approx 21, \quad (7.74)$$

strongly stable ($Q \gg 1$): a disk with the "standard" MMSN surface density — the same numerical reference used throughout this chapter — is far from gravitational fragmentation even at 50 au. Fragmentation would require Σ about 20 times larger than the MMSN value at that radius, that is, a substantially more massive (and, typically, younger) disk than the reference value adopted in this chapter.

This result illustrates why gravitational instability is today a **secondary** channel compared to core accretion: disks with "standard" MMSN mass do not satisfy $Q \lesssim 1$ at any reasonable radius. The channel remains relevant in exceptionally massive and young disks, and especially at large radii (tens to hundreds of au, where small Ω gives self-gravity more "time" before being sheared) — conditions associated with the population of giants/brown dwarfs on wide orbits (~ 10 – 100 au) seen in direct imaging (Nielsen et al., 2019). The overall demographics of giant exoplanets — more numerous at close orbits, correlated with the host star's mass/metallicity, the signature of core accretion — are interpreted as evidence that this mechanism dominates in most systems, with gravitational instability contributing mainly to very wide-orbit companions (Nielsen et al., 2019; Johansen and Lambrechts, 2017).

Comparative synthesis: what is common and what is distinct

Having traced, in this chapter, both star formation and planet formation from the same collapse and the same disk, it is worth making the comparison between the two processes explicit.

What is common: both originate in the same physical event — the collapse of a prestellar core (Section 7.4) and the formation of the resulting disk (Section 7.5) —, not being, strictly speaking, independent phenomena, but stages of a single evolutionary chain. Both are governed by the competition between self-gravity and support (thermal pressure in the Jeans instability, Section 7.3; pressure and rotation in the Toomre criterion, Section 7.10); both depend on the conservation and transport of angular momentum, the same problem that forced disk formation reappearing on a smaller scale for any material orbiting within it; and both depend on analogous dissipative processes — the turbulent viscosity that transports mass and angular momentum (Section 7.8) is, formally, the same type of redistribution by local interactions that, on a microscopic scale and via grain-grain collisions, enables the growth and migration of solids in the disk.

What is distinct: star formation, from core collapse to the ZAMS (Section 7.9), is dominated from beginning to end by the continuous self-gravity of a **fluid**, evolving smoothly (Hayashi/Henyey tracks, Section 7.6) toward self-luminous hydrostatic equilibrium. Planet formation by core accretion, in contrast, is dominated by **collisional, discrete** physics of individual solid bodies — coagulation (Section 7.10), streaming instability and pebble accretion (Section 7.10), runaway/oligarchic growth (Section 7.10) — with no analogue in star formation; only at the *final* stage of a gas giant (Section 7.10) does hydrostatic fluid physics

reappear, now around a pre-existing solid core.

The **gravitational instability of the disk** (Section 7.10) is the only planetary channel genuinely *analogous* to star formation — the same class of linear instability (self-gravity vs. support, with the Toomre parameter Q playing the role of M/M_J) —, and perhaps it is no coincidence that it is precisely this channel, forming a giant planet "like a miniature star" without going through discrete collisional physics, that produces the objects with mass closest to the boundary between planets and stars (brown dwarfs on wide orbits).

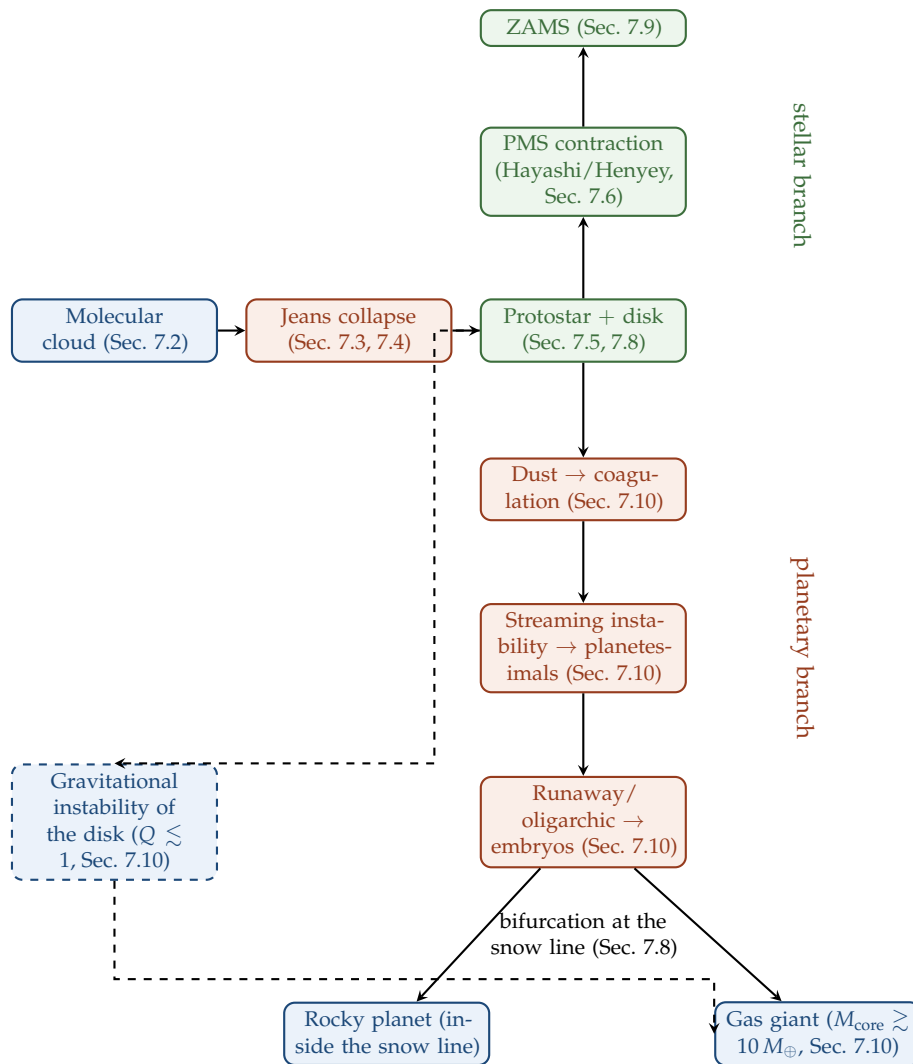


Figure 7.9: Flowchart summarizing the chapter: from the collapse of a prestellar core to the bifurcation between the stellar branch (PMS contraction to the ZAMS) and the planetary branch (dust to rocky planets or gas giants, with the snow line determining the outcome). The dashed channel (gravitational instability of the disk) bypasses the collisional physics of the main planetary branch — the only stage genuinely analogous to stellar physics (Section 7.10).

Further Reading and Current References

Formation of planetesimals via streaming instability and concentration of solids: Johansen et al. (2014). Pebble accretion and formation of cores/gas giants: Johansen and Lambrechts (2017). Original work on the streaming instability: Youdin and Goodman (2005). Radial drift and the meter-size barrier (original): Weidenschilling (1977). Critical core mass and

runaway gas accretion: Mizuno (1980). Observational demography of giant exoplanets on wide orbits (direct imaging): Nielsen et al. (2019).

Chapter 8

Stellar Equations of State

8.1 Introduction

Let us begin the analysis of the equations of state from a relatively simple case: that of an **ionized ideal gas**, in which we disregard quantum effects (degeneracy) and assume the material is completely ionized. In this situation, the total pressure in the stellar interior is the sum of the gas and radiation contributions,

$$P = P_{\text{gas}} + P_{\text{rad}}. \quad (8.1)$$

8.2 Ideal gas pressure

For the ideal gas, P_{gas} is given by the already familiar Clapeyron equation,

$$PV = \bar{n} R_u T, \quad \bar{n} = \frac{N}{N_A}, \quad (8.2)$$

where N is the number of particles, N_A Avogadro's number, and R_u the universal gas constant.

Notational collision: two different gas constants both called R

The symbol R is used, in different contexts, both for the *universal* gas constant in the Clapeyron equation [Equation (8.2)] and for the **specific gas constant** $R \equiv k/m_a$ (k = Boltzmann constant, m_a = atomic mass, 1 Dalton) — the same convention already used in Chapters 3 and 5 to write $P = \rho RT/\mu$ with μ dimensionless. To avoid ambiguity in this derivation, we use here R_u exclusively for the universal molar constant of Equation (8.2), and reserve R (with no subscript) for the specific constant $R \equiv k/m_a \approx 8.314 \times 10^7 \text{ erg g}^{-1} \text{ K}^{-1}$, consistent with the rest of the book.

Rewriting Equation (8.2) in terms of the volumetric particle number density, $n \equiv N/V$,

$$P = \frac{N}{VN_A} R_u T = n \frac{R_u}{N_A} T = nkT, \quad (8.3)$$

where we identify the Boltzmann constant $k \equiv R_u/N_A$.

The mass density ρ is related to n through the mean mass per particle, $\rho = n \mu m_a$, where μ is the mean molecular weight (dimensionless, in units of the atomic mass m_a). Hence $n = \rho/(\mu m_a)$, and Equation (8.3) becomes

$$P_{\text{gas}} = nkT = \frac{\rho (k/m_a) T}{\mu} \equiv \frac{\rho RT}{\mu}, \quad R \equiv \frac{k}{m_a}, \quad (8.4)$$

the usual form of the ideal gas pressure, already used in previous chapters. Combining this with the radiation pressure $P_{\text{rad}} = \frac{1}{3} a T^4$ [already established in Chapter 4], Equation (8.1) becomes

$$P = \frac{\rho RT}{\mu} + \frac{1}{3} a T^4. \quad (8.5)$$

8.3 Decomposition for a fully ionized gas

For a fully ionized gas, the gas pressure receives contributions from the free electrons and from each ionic species,

$$P_{\text{gas}} = P_{\text{electrons}} + P_{\text{ions}} = \sum_i (1 + Z_i) n_i kT, \quad (8.6)$$

where Z_i is the atomic number of element i (the number of electrons released per fully ionized atom) and the factor $(1 + Z_i)$ counts the ion plus its Z_i electrons as $1 + Z_i$ independent particles per original atom. The number density of ions of species i is

$$n_i = \frac{\rho_i X_i}{m_a \mu_i}, \quad (8.7)$$

where X_i is the mass fraction of element i and μ_i its atomic weight. The notes consider, as a simplified example, a composition of hydrogen and helium alone, with $X_{\text{H}} = 0.7$ and $X_{\text{He}} = 0.3$.

8.4 Mean molecular weight of a fully ionized gas

Substituting Equation (8.7) into Equation (8.6), with $\rho_i = \rho$ (the density does not vary between species, only the mass fraction),

$$P_{\text{gas}} = \sum_i (1 + Z_i) \frac{\rho X_i}{m_a \mu_i} kT = \rho RT \sum_i (1 + Z_i) \frac{X_i}{\mu_i}, \quad R = \frac{k}{m_a}. \quad (8.8)$$

Comparing directly with the general ideal gas form [Equation (8.4), $P_{\text{gas}} = \rho RT / \mu$], we obtain the classic stellar-structure result for the mean molecular weight of a fully ionized gas:

$$\boxed{\frac{1}{\mu} = \sum_i (1 + Z_i) \frac{X_i}{\mu_i}}. \quad (8.9)$$

This formula connects directly with the estimate $\mu = 0.50$ used in Chapter 5 [Equation (5.29)] for the calculation of the central temperature of the Sun: for pure fully ionized hydrogen ($X_{\text{H}} = 1$, $Z_{\text{H}} = 1$, $\mu_{\text{H}} = 1$), Equation (8.9) gives exactly $1/\mu = (1 + 1) \cdot 1/1 = 2 \Rightarrow \mu = 0.50$ — the value used in that chapter as an initial approximation.

Mean molecular weight: pure H vs. realistic composition

For pure fully ionized hydrogen ($X_{\text{H}} = 1$, $Z_{\text{H}} = 1$, $\mu_{\text{H}} = 1$),

$$\frac{1}{\mu} = (1 + 1) \cdot \frac{1}{1} = 2 \quad \Rightarrow \quad \mu = 0.50, \quad (8.10)$$

the value used in Chapter 5 to estimate T_c for the Sun.

Now for the simplified H and He mixture used in the notes for this chapter ($X_{\text{H}} = 0.7$, $X_{\text{He}} = 0.3$, with $Z_{\text{He}} = 2$ and $\mu_{\text{He}} = 4$),

$$\frac{1}{\mu} = (1 + 1) \frac{0.7}{1} + (1 + 2) \frac{0.3}{4} = 1.4 + 0.225 = 1.625, \quad (8.11)$$

$$\mu = \frac{1}{1.625} \approx 0.615. \quad (8.12)$$

The realistic composition (with $\approx 30\%$ helium by mass, close to the solar composition) raises the mean molecular weight from 0.50 (pure H) to ≈ 0.62 — in good agreement with the value $\mu \approx 0.6$ already used in Chapter 3 for the solar core plasma. This increase makes physical sense: helium atoms are 4 times more massive than hydrogen but release only 2 electrons (instead of 1), so each fully ionized He atom contributes *fewer* particles per unit mass than the same total mass in hydrogen — raising the effective mean molecular weight of the mixture.

Why the mean molecular weight matters so much

The mean molecular weight μ appears directly in the gas pressure [Equation (8.4)] and, as a consequence, in practically every estimate of the central temperature and pressure of a star (as we saw in Chapter 5). Since μ depends on the local chemical composition — which changes over the course of stellar evolution as hydrogen is converted into helium and heavier elements by nuclear fusion — the core of an evolved star has a systematically higher μ than that of a young star of the same mass. This is, in fact, one of the physical reasons why stars become more luminous over the course of the main sequence: for a given M and R , a larger μ in the core requires a higher central temperature to sustain the same pressure [see Equation (5.32) of Chapter 5], which accelerates the nuclear reactions and increases L .

Further Reading and Current References

For the complete treatment of stellar equations of state, including the regimes of electron degeneracy (partial and total, relativistic and non-relativistic) covered in the next block of this chapter, see Kippenhahn et al. (2012), chaps. 13-14, and Chandrasekhar (1939) for the classic treatment of degeneracy. On the precise determination of the mean molecular weight and its evolution with the star's chemical composition, see also Kippenhahn et al. (2012), chap. 3.

8.5 Degenerate conditions in the stellar interior

To better understand the existence of certain extremely hot compact stars (radius $R_\star \sim R_\oplus$, with $T_{\text{eff}} \sim 7000$ K), we must study the electron gas as a **Fermi–Dirac gas**. Recall that fermions are particles of half-integer spin, such as electrons, and obey the Pauli exclusion principle — two identical fermions cannot occupy the same quantum state.

8.5.1 Motivation: an electron gas confined in 1D

Let us first consider a simplified situation in one dimension, at $T = 0$ K: a single electron confined to a region of size L . By Heisenberg's uncertainty principle,

$$\Delta x \Delta p \gtrsim \frac{h}{2}. \quad (8.13)$$

Since $\Delta x \equiv L$,

$$\Delta p \gtrsim \frac{h}{2L}. \quad (8.14)$$

At $T = 0$, the mean thermal velocity (and momentum) is zero, so the maximum momentum of a single particle is given entirely by the quantum uncertainty,

$$p_{\max} = p(T) + \Delta p = 0 + \Delta p = \Delta p = \frac{h}{2L}. \quad (8.15)$$

If we add more electrons, the Pauli exclusion principle prevents two electrons from occupying the same quantum state — but two electrons of opposite spin *can* occupy the same momentum level. Filling the levels successively:

Electron	$p_{\max}$	Spin
1 st	$h/2L$	$\uparrow$
2 nd	$h/2L$	$\downarrow$
3 rd	$h/2L + h/2L = h/L$	$\uparrow$
4 th	h/L	$\downarrow$
5 th	$h/L + h/2L = \frac{3}{2} h/L$	$\uparrow$
6 th	$\frac{3}{2} h/L$	$\downarrow$

Each *pair* of electrons occupies a new level of width $h/2L$ in momentum space. Generalizing to N electrons,

$$p_{\max} = \frac{N}{2} \cdot \frac{h}{2L} \implies \Delta p = \frac{2 p_{\max}}{N}. \quad (8.16)$$

8.5.2 Generalization to 3D

In the three-dimensional case, the same gradual filling occurs independently in each direction,

$$\Delta p_x = \Delta p_y = \Delta p_z = \frac{h}{2L}, \quad (8.17)$$

so that, by analogy with Equation (8.16),

$$p_{\max,x} = \Delta p_x, \quad p_{\max,y} = \Delta p_y, \quad p_{\max,z} = \Delta p_z, \quad \text{and} \quad N = \frac{2 p_{\max}}{\Delta p}. \quad (8.18)$$

Since in the 1D case we found $N = 2p_{\max}/\Delta p$, we can infer that in three dimensions the number of available states inside a sphere of radius $p_{\max}$ in momentum space is

$$N = 2 \cdot \left(\frac{4\pi p_{\max}^3/3}{8} \right) \cdot \frac{1}{\Delta p^3}, \quad (8.19)$$

where the term $4\pi p_{\max}^3/3$ describes the volume of the momentum field propagating in a spherical shell.

Where does the factor 1/8 come from?

The factor 1/8 in Equation (8.19) arises for a purely geometric reason, linked to how the momentum levels are counted. Just as in the 1D case the electrons fill only *positive* momenta (the quantization of a particle confined to a box of size L is indexed by positive quantum numbers, not by signed momenta), in 3D each component p_x, p_y, p_z likewise takes only positive values — that is, the occupied states fill only the **positive octant** of the momentum sphere, not the full sphere. Since the octant corresponds to exactly 1/8 of the total volume of the sphere $4\pi p_{\max}^3/3$, this factor of 1/8 enters multiplying the spherical

volume. The factor 2 multiplying everything, on the other hand, is the spin degeneracy (two electrons, with opposite spins, per spatial state). Thus,

$$N = 2 \left(\frac{4\pi p_{\max}^3/3}{8} \right) \left(\frac{1}{h/2L} \right)^3 = 2 \left(\frac{4\pi p_{\max}^3/3}{8} \right) \frac{8L^3}{h^3} = \frac{8\pi}{3} \frac{p_{\max}^3 L^3}{h^3}, \quad (8.20)$$

where the factor 8 coming from $(2L/h)^3 = 8L^3/h^3$ exactly cancels the factor $1/8$ from the octant, leaving only the factor 2 from the spin degeneracy multiplying the full spherical volume — the net result is the same as one would obtain by counting the *entire* momentum sphere with weight $2/h^3$ per volume element d^3p (the more common way of presenting this result in the literature, and the one we will use from here on). The factor $1/8$ is correct, but its partial cancellation with the factor 8 from the unit change makes the final result simpler than the intermediate step suggests.

Dividing Equation (8.20) by L^3 , we obtain the **electron number density** of the gas,

$$n_e \equiv \frac{N}{L^3} = \frac{8\pi}{3} \frac{p_{\max}^3}{h^3}. \quad (8.21)$$

We can express this result in terms of the **Fermi momentum**, using $\hbar \equiv h/2\pi$:

$$p_F = \left(3\pi^2 \hbar^3 n_e \right)^{1/3} = \hbar (3\pi^2 n_e)^{1/3}. \quad (8.22)$$

Thus, an electron gas at $T = 0$ has a well-defined maximum momentum. Assuming the gas is non-relativistic, $E = p^2/2m_e$, the non-relativistic **Fermi energy** is

$$E_F^{\text{NR}} = \frac{\hbar^2}{2m_e} (3\pi^2 n_e)^{2/3}. \quad (8.23)$$

From the quantum box to white dwarfs

The initial motivation for this section — compact stars with $R_\star \sim R_\oplus$ — is no coincidence: it is exactly the **electron degeneracy pressure** we are deriving here that sustains **white dwarfs** against gravitational collapse, without depending on the temperature of the gas (unlike the thermal pressure of an ordinary ideal gas). This is why white dwarfs can cool for billions of years without shrinking significantly: their structure is sustained by a purely quantum effect. The future chapter on White Dwarfs will revisit this result in depth, including the characteristic mass-radius relation of these objects.

8.6 Occupation distribution and electron density

We can study the occupation probability of an energy state E , defined by $f(E)$. Assuming the gas is an electron gas, the occupation probability distribution as a function of energy has the shape shown in Figure 8.1: at $T = 0$, $f(E) = 1$ for every state with $E \leq E_F$ (maximum occupation probability) and $f(E) = 0$ for $E > E_F$; at finite temperatures, the transition at E_F smooths out and the distribution decays as $f(E) \propto e^{-E/kT}$ for $E \gg E_F$.

In high-density regimes (total degeneracy), all energy levels up to a maximum value are occupied. Since the total particle density is finite, the states are occupied up to a certain

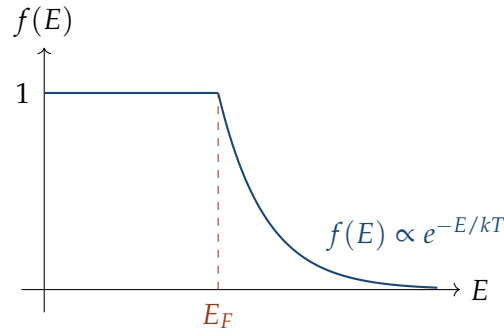


Figure 8.1: Occupation probability $f(E)$ of an energy state E for an electron gas. At $T = 0$ (total degeneracy), $f(E)$ is a plateau at 1 up to E_F and then drops abruptly to 0; at finite temperatures, the distribution decays smoothly as $e^{-E/kT}$ above E_F .

momentum value p_0 , such that

$$n_e(p) = \begin{cases} \frac{2}{h^3} 4\pi p^2, & p \leq p_0, \\ 0, & p > p_0, \end{cases} \quad (8.24)$$

where, for $p > p_0$, the density no longer includes electrons beyond this point — this limit must be the lowest available kinetic-energy state for the electron gas, since all lower-energy states are already occupied.

Since the total particle density is the sum (integral) over all occupied momenta,

$$n_e = \int_0^\infty n_e(p) dp = \frac{8\pi}{3h^3} p_0^3, \quad (8.25)$$

or, in terms of the maximum momentum,

$$p_0 = \left(\frac{3h^3}{8\pi} n_e \right)^{1/3} = \left(3\pi^2 \hbar^3 n_e \right)^{1/3}. \quad (8.26)$$

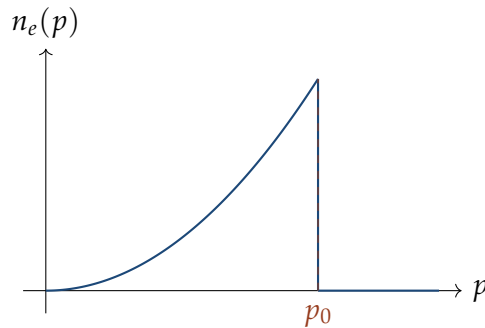


Figure 8.2: Electron density per momentum interval, $n_e(p)$, for a fully degenerate gas at $T = 0$: it grows $\propto p^2$ up to p_0 and then drops abruptly to zero above that value — all states up to p_0 are occupied, and none beyond it.

Distinction from the Maxwell–Boltzmann distribution

The distribution actually used in this section is the **Fermi–Dirac distribution at $T = 0$** [Equation (8.24)], with an abrupt cutoff at p_0 (Figure 8.2), qualitatively very different from the Maxwell–Boltzmann distribution, $\propto v^2 e^{-mv^2/2kT}$ (smooth, without a cutoff, valid for a *classical non-degenerate* gas in thermal equilibrium). The contrast is worth noting: in the opposite limit from the degenerate one (low densities, high temperatures), the velocity distribution would be the classical Maxwell–Boltzmann one, not the Fermi–Dirac distribution with its abrupt cutoff. No numerical result in this section uses the Maxwell–Boltzmann formula — all of them start from $n_e(p)$ [Equation (8.24)].

8.7 Momentum flux and mechanical pressure

We know that the microscopic source of pressure in a perfect gas is the bombardment of particles against the walls of the container. In a perfect (ideal) gas, there is no interaction between the constituent particles of the gas — a criterion that is never actually fully attainable, but is a good approximation when the thermal energy is much larger than the interaction energy between particles.

Using Newton’s second law, $F = dp/dt$, and the relation between work and pressure,

$$P = \frac{F}{A} = \frac{F \cdot d}{A \cdot d} = \frac{W}{V} \implies W = P V, \quad (8.27)$$

or, in differential form,

$$dW = P dV. \quad (8.28)$$

Consider the geometry of a conical section of angle θ around the normal $\hat{n}$ to the wall (Figure 8.3). When particles strike the surface, they transfer momentum to it; assuming specular reflection, the reaction momentum along the normal direction is

$$\cos \theta = \frac{\Delta p}{2p_x} = \frac{\Delta p}{2p} \implies \Delta p = 2p \cos \theta, \quad (8.29)$$

since the change in the initial momentum is the normal component, $2p_x$, with $p_x = p \cos \theta$.

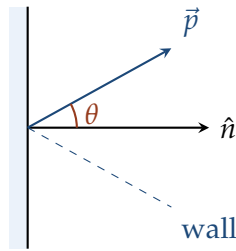


Figure 8.3: Geometry used in deriving the kinetic pressure: a particle of momentum $\vec{p}$ collides with the wall (normal $\hat{n}$) at an angle θ from the normal; specular reflection transfers momentum $\Delta p = 2p \cos \theta$ to the wall.

Let $F(\theta, p)$ be the number of particles with momentum p in the interval dp colliding with the wall, per unit area, time, and solid angle, in the direction θ . The total pressure contribution is

$$dP = 2p \cos \theta F(\theta, p) dp d\theta, \quad (8.30)$$

so that the total pressure is obtained by integrating,

$$P = \int_0^{\pi/2} \int_0^\infty 2p \cos \theta F(\theta, p) d\theta dp. \quad (8.31)$$

To relate $F(\theta, p)$ to the number of particles per unit volume, we consider the volume V swept out by particles crossing a surface element dA in a time interval dt ,

$$V = (n_x \cos \theta) dA \cdot v dt = v \cos \theta dt dA, \quad (8.32)$$

where $n_x \cos \theta$ is the tangential component of the surface element along the direction of propagation. From this, we obtain

$$F(\theta, p) d\theta dp = v \cos \theta n(\theta, p) d\theta dp, \quad (8.33)$$

where $n(\theta, p)$ is the number of particles per unit volume in the pressure cone. For an isotropic gas,

$$\frac{n(\theta, p) d\theta dp}{n(p) dp} = \frac{2\pi \sin \theta}{4\pi} = \frac{1}{2} \sin \theta, \quad (8.34)$$

that is,

$$d_\theta n(\theta, p) = \frac{1}{2} \sin \theta d_\theta n(p) dp. \quad (8.35)$$

Substituting Equation (8.35) into Equation (8.33),

$$F(\theta, p) d\theta dp = v \cos \theta \cdot \frac{1}{2} \sin \theta d_\theta n(p) dp, \quad (8.36)$$

and substituting into Equation (8.31),

$$P = \int_0^{\pi/2} \int_0^\infty 2p \cos \theta \cdot v \cos \theta \cdot \frac{1}{2} \sin \theta n(p) d\theta dp = \int_0^{\pi/2} \int_0^\infty n(p) v p dp \cos^2 \theta \sin \theta d\theta. \quad (8.37)$$

Integrating over θ , with the substitution $\mu = \cos \theta$, $d\mu = -\sin \theta d\theta$,

$$\int_0^{\pi/2} \cos^2 \theta \sin \theta d\theta = - \int_1^0 \mu^2 d\mu = \frac{1}{3}, \quad (8.38)$$

we finally arrive at the **general formula for kinetic pressure**, valid for any gas — classical or degenerate, relativistic or not:

$$P = \frac{1}{3} \int_0^\infty p v n(p) dp. \quad (8.39)$$

With this relation, we pass from a microscopic description (the distribution of particles in momentum) to a macroscopic description (the gas pressure).

8.8 Total non-relativistic degeneracy

In a case where the energy associated with the momentum p_0 is much smaller than the electron's rest energy, that is, $v \ll c$, the treatment can be described classically: $p = m_e v$, so

$$v = \frac{p}{m_e}. \quad (8.40)$$

We can then determine the pressure exerted by the electrons on the gas, using Equation (8.39) with $v = p/m_e$ [Equation (8.40)] and $n_e(p)$ [Equation (8.24)]:

$$P = \frac{1}{3} \int_0^\infty p \cdot \frac{p}{m_e} \cdot \frac{8\pi p^2}{h^3} dp = \frac{8\pi}{3m_e h^3} \int_0^{p_0} p^4 dp, \quad (8.41)$$

where the upper limit of the integral becomes p_0 (instead of ∞), since $n_e(p) = 0$ for $p > p_0$ [Equation (8.24)]. Integrating,

$$P = \frac{8\pi}{3m_e h^3} \cdot \frac{p_0^5}{5}. \quad (8.42)$$

Substituting $p_0 = (3h^3 n_e / 8\pi)^{1/3}$ [Equation (8.26)],

$$P = \frac{8\pi}{3m_e h^3} \cdot \frac{1}{5} \left(\frac{3h^3}{8\pi} n_e \right)^{5/3} = \frac{1}{20} \left(\frac{3}{\pi} \right)^{2/3} \frac{h^2}{m_e} n_e^{5/3}. \quad (8.43)$$

Using $n_e = N_A \rho / \mu_e$ (μ_e the mean molecular weight per electron), the **equation of state of a fully degenerate, non-relativistic Fermi gas** ($T = 0$) is

$$P = \frac{1}{20} \frac{h^2}{m_e} \left(\frac{3}{\pi} \right)^{2/3} \left(\frac{N_A}{\mu_e} \right)^{5/3} \rho^{5/3}, \quad P \propto \rho^{5/3}. \quad (8.44)$$

Numerically, this equation of state amounts to

$$P \sim 1.00 \times 10^{13} \left(\frac{\rho}{\mu_e} \right)^{5/3} \text{ dyn/cm}^2, \quad (8.45)$$

valid in the non-relativistic regime, approximately when

$$\frac{\rho}{\mu_e} > 2.4 \times 10^{-8} T^{3/2} \text{ g/cm}^3. \quad (8.46)$$

Numerical coefficient of Equation (8.45)

Calculating directly the coefficient $\frac{1}{20} (3/\pi)^{2/3} (h^2/m_e) N_A^{5/3}$ in cgs units, one obtains $\approx 1.0036 \times 10^{13} \text{ dyn cm}^{-2}$ — in excellent agreement with the standard value quoted in the stellar-structure literature (Kippenhahn et al., 2012; Chandrasekhar, 1939). We adopt here the value 1.00×10^{13} , consistent with this direct calculation.

The physical origin of the polytropic index $n = 3/2$

This is one of the most elegant results in this chapter: Equation (8.44), $P \propto \rho^{5/3}$, is **exactly** the polytropic relation of case (i) in the table from Chapter 6 ($\gamma = 5/3$, $n = 3/2$, “non-relativistic degenerate gas”). This shows that this particular polytropic index is not an arbitrary or purely mathematical choice: it has a direct physical origin in the quantum mechanics of a fully degenerate Fermi gas — the exponent $5/3$ arises directly from the combination of state counting in momentum space (which gives $n_e \propto p_0^3$) and the integration of the kinetic pressure (which gives $P \propto p_0^5$), with no thermodynamic input whatsoever. This is the underlying physical reason why this polytrope describes so well low-mass white dwarfs and the cores of red giants — both environments where the electron gas is fully degenerate but still non-relativistic.

Central pressure of a C/O white dwarf

Using Equation (8.45) with a typical white dwarf central density, $\rho_c \sim 10^6\text{--}10^7 \text{ g/cm}^3$, and $\mu_e \approx 2$ (carbon- and oxygen-rich composition, fully ionized, with $Z/\mu_i \approx 1/2$ for both):

$$\rho_c \sim 10^6 \text{ g/cm}^3 : P_c \sim 1.00 \times 10^{13} \left(\frac{10^6}{2} \right)^{5/3} \approx 3.1 \times 10^{22} \text{ dyn/cm}^2, \quad (8.47)$$

$$\rho_c \sim 10^7 \text{ g/cm}^3 : P_c \sim 1.00 \times 10^{13} \left(\frac{10^7}{2} \right)^{5/3} \approx 1.5 \times 10^{24} \text{ dyn/cm}^2. \quad (8.48)$$

These values are in good order-of-magnitude agreement with detailed white dwarf models ($P_c \sim 10^{22}\text{--}10^{24} \text{ dyn/cm}^2$ for typical masses, $0.5\text{--}1.0 M_\odot$), confirming that the non-relativistic degenerate equation of state [Equation (8.44)] correctly captures the central physics of these objects in this density range.

From non-relativistic to relativistic in white dwarfs

The non-relativistic validity condition [Equation (8.46)] shows that, for an already-cooled white dwarf (low internal temperature), most of the relevant density range is indeed in the non-relativistic regime described in this section. However, at higher central densities — more massive white dwarfs, approaching the Chandrasekhar limit — the electron gas becomes progressively more relativistic, which *softens* the equation of state (the effective exponent of $P(\rho)$ decreases from $5/3$ to $4/3$, as we will see next). It is precisely this transition that imposes an upper limit on the mass that degeneracy pressure can sustain.

8.9 Total ultra-relativistic degeneracy

Analogously to the case of total non-relativistic degeneracy, we can determine the pressure exerted by the **electrons**, now assuming an ultra-relativistic case, that is, $v \approx c$. This treatment makes the pressure analysis even simpler, since the speed of light is constant:

$$P = \frac{1}{3} \int_0^\infty p v n(p) dp = \frac{1}{3} c \cdot \frac{8\pi}{h^3} \int_0^{p_0} p^3 dp = \frac{2\pi c}{3h^3} p_0^4. \quad (8.49)$$

This is the electron pressure, not the photon pressure

The entire integration above uses $n(p)$ for the degenerate **electron gas**, already established in previous blocks [Equation (8.24)], not a photon distribution — this is the pressure exerted by ultra-relativistic **electrons** (the same formalism from Section 8.7 applied to $v = c$). It is worth noting the distinction from radiation (photon) pressure, already established in an earlier chapter as $P_{\text{rad}} = \frac{1}{3} a T^4$ [Equation (8.5)] — a completely different expression from this kinetic derivation.

Substituting $p_0 = (3h^3 n_e / 8\pi)^{1/3}$ [Equation (8.26)] into Equation (8.49),

$$P = \frac{2\pi c}{3h^3} \left(\frac{3h^3}{8\pi} n_e \right)^{4/3} = \left(\frac{3}{8\pi} \right)^{1/3} \frac{hc}{4} n_e^{4/3}, \quad (8.50)$$

or, in terms of the density, using $n_e = N_A \rho / \mu_e$,

$$P = \left(\frac{3}{8\pi} \right)^{1/3} \frac{hc}{4} \left(\frac{N_A}{\mu_e} \right)^{4/3} \rho^{4/3}, \quad \text{that is,} \quad P \propto \rho^{4/3} \propto n_e^{4/3}. \quad (8.51)$$

Note that the values of P and ρ/μ_e in this equation are exactly the same as in the non-relativistic case [Equation (8.44)] — the only difference is the exponent, $4/3$ instead of $5/3$. This is the **first route** by which we arrive at the ultra-relativistic equation of state; Section 8.10 below reaches the same result by an independent path, as a cross-check. Note also that we are still considering $T = 0$ K, constant temperature, and the pressure being a function of density alone — that is, we remain in the polytropic case.

8.10 Total relativistic degeneracy

As the electron density in a star's core increases, the maximum momentum of the gas also increases through the agitation of the particles, and the most energetic electrons become relativistic. In this case, the classical form for momentum can no longer be used — we must use the relativistic formula,

$$p = \frac{m_e v}{(1 - (v/c)^2)^{1/2}}. \quad (8.52)$$

To write v as a function of p , we square both sides,

$$p^2 = \frac{m_e^2 v^2}{1 - v^2/c^2} \implies \frac{p^2}{m_e^2} = \frac{v^2}{1 - v^2/c^2}, \quad (8.53)$$

and, using $v^2 = p^2/m_e^2 - (p^2/m_e^2)(v^2/c^2)$, we factor out v^2 :

$$v^2 \left(1 + \frac{p^2}{m_e^2 c^2} \right) = \frac{p^2}{m_e^2} \implies v = \frac{p/m_e}{\left(1 + \frac{p^2}{m_e^2 c^2} \right)^{1/2}}. \quad (8.54)$$

Substituting Equation (8.54) into the general formula for kinetic pressure [Equation (8.39)], to determine the pressure of a fully degenerate gas in any relativistic regime,

$$P = \frac{1}{3} \int_0^{p_0} p \cdot \frac{p/m_e}{\left(1 + \frac{p^2}{m_e^2 c^2} \right)^{1/2}} \cdot \frac{8\pi p^2}{h^3} dp = \frac{8\pi}{3m_e h^3} \int_0^{p_0} \frac{p^4 dp}{\left(1 + \left(\frac{p}{m_e c} \right)^2 \right)^{1/2}}. \quad (8.55)$$

Defining the dimensionless variable $Y \equiv p/m_e c$ (so $dY = dp/m_e c$),

$$P = \frac{8\pi m_e^4 c^5}{3h^3} \int_0^X \frac{Y^4 dY}{(1 + Y^2)^{1/2}}, \quad X \equiv \frac{p_0}{m_e c}. \quad (8.56)$$

This integral can be solved with the trigonometric substitution $Y = \tan u$, $dY = \sec^2(u) du$; after some manipulation, one obtains the closed form

$$\int_0^X \frac{Y^4 dY}{(1 + Y^2)^{1/2}} = \frac{1}{8} X(2X^2 - 3)(X^2 + 1)^{1/2} + 3 \sinh^{-1}(X). \quad (8.57)$$

Substituting back into Equation (8.56),

$$P = \frac{8\pi m_e^4 c^5}{3h^3} \cdot \frac{1}{8} F(X) = \frac{\pi m_e^4 c^5}{3h^3} F(X), \quad F(X) = X(2X^2 - 3)(X^2 + 1)^{1/2} + 3 \sinh^{-1}(X). \quad (8.58)$$

X: the Fermi momentum in units of $m_e c$

Just as $Y = p/m_e c$ is the dimensionless momentum of any given particle, $X \equiv p_0/m_e c$ is the Fermi momentum (the maximum momentum of the degenerate gas) in the same unit — using $p_0 = (3h^3 n_e / 8\pi)^{1/3}$ [Equation (8.26)],

$$X = \frac{p_0}{m_e c} = \frac{h}{m_e c} \left(\frac{3n_e}{8\pi} \right)^{1/3}. \quad (8.59)$$

Equation (8.58) can be regarded as the **general form of the pressure of a fully degenerate electron gas** in the stellar interior, valid for any degree of relativity, always at $T = 0$ K.

8.10.1 Ultra-relativistic limit: cross-check

We can consider the ultra-relativistic case, where $X \gg 1$. In this case, the lower-order terms in Equation (8.58) can be neglected in comparison with the higher-order terms, so that

$$F(X) \approx 2X^4 - 2X^2 + \dots, \quad (8.60)$$

and, taking only the leading term of the approximation,

$$P \approx \frac{\pi m_e^4 c^5}{3h^3} \cdot 2X^4 = \frac{2\pi m_e^4 c^5}{3h^3} \left[\frac{h}{m_e c} \left(\frac{3n_e}{8\pi} \right)^{1/3} \right]^4. \quad (8.61)$$

Simplifying the exponents ($4/3 - 1 = 1/3$, and $8^{4/3} \equiv 2^{3 \cdot 4/3} = 2^4$),

$$P = \left(\frac{3}{8\pi} \right)^{1/3} \frac{hc}{4} n_e^{4/3}, \quad (8.62)$$

which reproduces **exactly** the pressure obtained in the ultra-relativistic case seen in Equation (8.50) — a cross-check between two independent derivations: the direct substitution $v = c$ (Section 8.9) and the limit $X \gg 1$ of the general relativistic formula derived here.

The opposite limit, $X \ll 1$, recovers the non-relativistic case ($v \ll c$, $p \approx m_e v$) from the previous block [Equation (8.44)].

The coefficient $hc/4$ in Equation (8.50)

This form is algebraically identical to the more common form found in the literature, $P = (3/\pi)^{1/3} (hc/8) n_e^{4/3}$, since $(3/8\pi)^{1/3} / 4 = (3/\pi)^{1/3} / 8$ — just two equivalent ways of grouping the same constant.

The physics of the Chandrasekhar mass

Now that we have derived, from first principles, **both** regimes of electron degeneracy — $P \propto \rho^{5/3}$ in the non-relativistic case [Equation (8.44)] and $P \propto \rho^{4/3}$ in the ultra-relativistic case [Equation (8.51)] — we can understand the physical origin of the **Chandrasekhar mass**. As the mass of a white dwarf increases, its interior becomes denser and the electrons more relativistic, softening the equation of state from $\gamma = 5/3$ ($n = 3/2$) to $\gamma = 4/3$ ($n = 3$). Solving the internal structure equation with this pressure leads back to the Lane-Emden equation from Chapter 6, now with $n = 3$ — the second important polytropic case in the table of that chapter, now with a complete physical derivation from quantum mechanics. The crucial point (to be derived in detail in the future chapter on White

Dwarfs) is that the index $n = 3$ produces, in the Lane–Emden equation, a total mass that is *independent of the radius* — which implies the existence of a maximum mass, $M_{\text{Ch}} \approx 1.4 M_{\odot}$, above which no hydrostatic-equilibrium configuration sustained by electron degeneracy is possible (Chandrasekhar, 1935, 1939).

8.11 Partial degeneracy

For $T \neq 0$, not all electrons will be as close as possible to the origin in momentum space. As T increases, electrons move away from the origin ($E = 0$) and populate states toward the Fermi surface. The statistical function describing this case is the **Fermi–Dirac distribution**,

$$F(p) = n_e(p) \cdot \frac{1}{1 + e^{\varepsilon/kT - \psi}}, \quad (8.63)$$

where $n_e(p) = 8\pi p^2/h^3$ [the differential form of Equation (8.24)], $\varepsilon = \varepsilon(p)$ is the particle energy, and ψ is the **degeneracy parameter**. In the limit $\psi \rightarrow -\infty$, the exponential term dominates and we recover the non-degenerate (classical) case.

Sign convention for ψ

We consistently adopt the convention with “ $-\psi$ ” [Equation (8.63)], consistent with the standard definition of the degeneracy parameter in the literature.

From $F(p)$, we can determine n_e ,

$$n_e = \int_0^\infty F(p) dp = \int_0^\infty \frac{8\pi p^2}{h^3} \frac{1}{1 + e^{\varepsilon/kT - \psi}} dp. \quad (8.64)$$

8.11.1 Non-relativistic case

For the Newtonian case, $\varepsilon = p^2/2m_e$, that is, $p = (2m_e\varepsilon)^{1/2}$ and $dp = m_e d\varepsilon/(2m_e\varepsilon)^{1/2}$. Substituting into Equation (8.64),

$$n_e = \frac{8\pi}{h^3} \int_0^\infty (2m_e\varepsilon) \cdot m_e \cdot \frac{1}{1 + e^{\varepsilon/kT - \psi}} \cdot \frac{d\varepsilon}{(2m_e\varepsilon)^{1/2}} = \frac{8\pi}{h^3} (2m_e)^{1/2} m_e \int_0^\infty \frac{\varepsilon^{1/2}}{1 + e^{\varepsilon/kT - \psi}} d\varepsilon. \quad (8.65)$$

Making the change of variable $u = \varepsilon/kT$, $du = d\varepsilon/kT$,

$$n_e = \frac{8\pi}{h^3} (2m_e)^{1/2} m_e (kT)^{3/2} \int_0^\infty \frac{u^{1/2}}{1 + e^{u - \psi}} du = \frac{4\pi}{h^3} (2m_e kT)^{3/2} F_{1/2}(\psi), \quad (8.66)$$

where we define the **generalized Fermi–Dirac function**,

$$F_v(\psi) \equiv \int_0^\infty \frac{u^v}{1 + e^{u - \psi}} du, \quad (8.67)$$

where v describes the exponent of u^v and, depending on the case under analysis, can take different values.

Repeating the procedure (the same change of variables) to determine the electron pressure, starting from the general formula for kinetic pressure [Equation (8.39)] with $v(p) = p/m_e$,

$$P = \frac{1}{3} \int_0^\infty \frac{8\pi p^2}{h^3} \cdot p \cdot \frac{p}{m_e} \cdot \frac{1}{1 + e^{\varepsilon/kT - \psi}} dp = \frac{8\pi}{3m_e h^3} \int_0^\infty \frac{p^4}{1 + e^{\varepsilon/kT - \psi}} dp, \quad (8.68)$$

and, again using $\varepsilon = p^2/2m_e$ and $u = \varepsilon/kT$,

$$P = \frac{8\pi kT}{3h^3} (2m_e kT)^{3/2} \int_0^\infty \frac{u^{3/2}}{1 + e^{u-\psi}} du = \frac{8\pi kT}{3h^3} (2m_e kT)^{3/2} F_{3/2}(\psi), \quad (8.69)$$

the **pressure of a partially degenerate Fermi gas**.

The functions $F_v(\psi)$ are, fortunately, tabulated, since their analytical solution is difficult (there is no general closed form). It can be shown that

$$P = n_e kT \left[\frac{2 F_{3/2}(\psi)}{3 F_{1/2}(\psi)} \right], \quad (8.70)$$

where the factor in brackets measures the deviation of the electron pressure from the non-degenerate ideal gas ($P = n_e kT$, recovered when this factor tends to 1). Note that ions are normally non-degenerate, since their available phase space is much larger than that of the electron (the ion mass, thousands of times larger than m_e , makes the ionic degeneracy temperature much higher than the electronic one, for the same density).

We have that the thermal energy and the (non-relativistic) kinetic energy are described by

$$E_t = \frac{3}{2} kT, \quad E_c = \frac{1}{2} m_e v^2, \quad (8.71)$$

so the total gas pressure is described by

$$P_{\text{gas}} = P_e + \frac{N_A k}{\mu_i} \rho T, \quad (8.72)$$

combining the partially degenerate electron contribution [Equation (8.70)] with the ionic contribution, treated as a classical ideal gas.

8.12 Density–temperature diagram

The density–temperature diagram for the regimes described throughout this chapter has the schematic form shown in Figure 8.4: at low densities and high temperatures, radiation pressure dominates over gas pressure; at higher densities, gas pressure dominates, which in turn is split into a non-degenerate region (classical ideal gas) and a degenerate region (where the electron degeneracy pressure derived in this chapter comes to dominate); near $T \sim 10^4$ K, a horizontal band separates neutral hydrogen (below) from ionized hydrogen (above); and, at very high densities, a vertical boundary separates non-relativistic electrons (to the left) from relativistic ones (to the right) — the boundary that motivated Section 8.10.

Locating real stars on the diagram

We can locate real astrophysical objects on this diagram. The **solar core** ($\rho_c \sim 150 \text{ g/cm}^3$, $T_c \sim 1.5 \times 10^7 \text{ K}$) sits well within the non-degenerate ideal gas region — consistent with the ideal-gas approximation used in Chapter 5 to estimate the solar T_c . The **core of a typical white dwarf** ($\rho \sim 10^6 \text{ g/cm}^3$, $T \sim 10^7 \text{ K}$) already lies deep in the degenerate region — consistent with the discussion in Section 8.8. And the **degenerate helium core** of a red giant, before the *helium flash* (also with $\rho \sim 10^5\text{--}10^6 \text{ g/cm}^3$ but lower T than in the previous case), occupies a region similar to that of the white dwarf — the physical origin of the helium flash itself is precisely the fact that the pressure there is degenerate

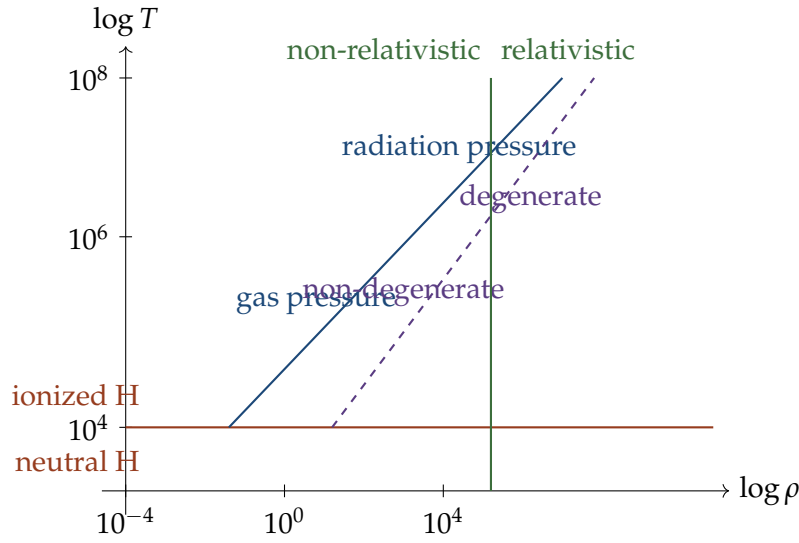


Figure 8.4: Schematic density–temperature diagram showing the main boundaries discussed in this chapter: radiation pressure vs. gas pressure dominance (diagonal solid line); non-degenerate vs. degenerate gas (diagonal dashed line); neutral vs. ionized hydrogen (horizontal line near $T \sim 10^4$ K); and non-relativistic vs. relativistic electrons (vertical line at high density).

(practically independent of T), which prevents the normal thermal self-regulation and allows the triple-alpha reaction rate to run away before the core can expand and cool.

We can also include radiation pressure for a more complete description,

$$P_{\text{gas}} = P_e + \frac{N_A k}{\mu_i} \rho T + P_{\text{rad}}. \quad (8.73)$$

Note that the partial degeneracy described so far covers the non-relativistic case. We can also obtain the pressures for the ultra-relativistic and general relativistic cases, following the same methodology. For example, for the ultra-relativistic case, where $v \approx c$, we have $p = E/c$; repeating the same treatment,

$$n_e = 8\pi \left(\frac{kT}{hc} \right)^3 F_2(\psi), \quad P_e = \frac{8\pi}{3h^3 c^3} (kT)^4 F_3(\psi). \quad (8.74)$$

For cases of partial degeneracy and moderately relativistic gas (neither $X \ll 1$ nor $X \gg 1$), we do not have simple analytical solutions for the integrals, since we cannot separate the energy ε from the momentum p in the Fermi–Dirac distribution, where

$$n_e = \frac{8\pi}{h^3} \int_0^\infty \frac{p^2 dp}{1 + e^{\varepsilon/kT - \psi}}, \quad \varepsilon = \left(m_e^2 c^4 + (pc)^2 \right)^{1/2} - m_e c^2. \quad (8.75)$$

In this case, more elaborate numerical models must be used.

8.13 Total equation of state

For an ordinary star, the total equation of state is described by the sum of the pressures acting in the star. Considering the ion, electron, and radiation pressures,

$$P = P_{\text{ion}} + P_e + P_{\text{rad}}, \quad (8.76)$$

where, in a partially degenerate case, the ion pressure can be expressed as that of an ideal (non-degenerate) gas, for the reason discussed in Section 8.11, and the electron pressure by the complete integral form that encompasses any degree of degeneracy and relativity:

$$P = \frac{R}{\mu_i} \rho T + \frac{8\pi}{3h^3} \int_0^\infty \frac{p^3 v(p)}{1 + e^{\varepsilon/kT - \psi}} dp + \frac{1}{3} a T^4, \quad v(p) = \frac{\partial \varepsilon}{\partial p}, \quad (8.77)$$

where $R \equiv k/m_a$ is the specific gas constant [the same convention as in Equation (8.4)].

From this relation, we can also estimate the total internal energy of the gas. For the ions (always a classical non-relativistic gas) and for radiation (always ultra-relativistic), the ratio between internal energy and pressure is exact and fixed,

$$U_{\text{ion}} = \frac{3}{2} P_{\text{ion}}, \quad U_{\text{rad}} = 3 P_{\text{rad}}. \quad (8.78)$$

For the electrons, on the other hand, the same proportionality $U_e = 3P_e$ is exact only in the ultra-relativistic limit [where $\varepsilon(p) = p v(p)$, since $v \rightarrow c$]; in the non-relativistic limit, instead, $U_e = \frac{3}{2} P_e$ holds [the same relation as for the ions, Equation (8.71)], and in the intermediate regime there is no fixed ratio between U_e and P_e — the electronic internal energy would in general require the integral of $\varepsilon(p) n(p)$ directly, which does not reduce to a simple multiple of the pressure integral already obtained. Adopting here the ultra-relativistic approximation $U_e \approx 3P_e$ — reasonable in the higher-density regime, where the electronic contribution to the internal energy is more relevant —, we obtain the estimate

$$u \equiv \frac{U_{\text{ion}} + U_e + U_{\text{rad}}}{\rho} \approx \frac{3}{2} \frac{R}{\mu_i} T + \frac{8\pi}{h^3 \rho} \int_0^\infty \frac{p^3 v(p)}{1 + e^{\varepsilon/kT - \psi}} dp + \frac{a T^4}{\rho}, \quad (8.79)$$

where the density, in turn, is defined in terms of the complete Fermi–Dirac distribution,

$$\rho = \frac{4\pi}{h^3} (2m_e)^{3/2} \mu_e m_a \int_0^\infty \frac{\varepsilon^{1/2}}{1 + e^{\varepsilon/kT - \psi}} d\varepsilon. \quad (8.80)$$

We thus have the equation of state that describes, in general, the interaction of particles in the stellar interior. Unfortunately, there are no analytical solutions for these expressions in terms of the conventional thermodynamic quantities, such as c_P , c_V , χ , δ , ∇_{ad} — only numerical solutions exist. Note that, in approximate cases, it may be possible to determine analytical solutions (see Section 16.2.3 of Kippenhahn et al., 2012).

8.14 Crystallization and neutronization

In real gases, the presence of interactions between particles makes the equation of state more complex, and processes such as the formation of crystal lattices, superfluid states, and neutronization can occur. Let us look at two examples.

8.14.1 Crystallization

Crystal lattices can form when there are deviations from the ideal gas equation of state and the ratio between the Coulomb interaction energy and the thermal energy of the gas is about 175 times larger than the latter, where

$$\Gamma \equiv \frac{(Ze)^2}{r_{\text{ion}} kT}, \quad \Gamma_c \approx 175, \quad (8.81)$$

with Γ_c the critical ratio (reference value for a classical one-component plasma) above which crystallization occurs. This condition can also be obtained as a function of temperature,

$$T_m \approx \frac{(Ze)^2}{\Gamma_c k} \left(\frac{4\pi\rho}{3\mu_i m_u} \right)^{1/3}, \quad (8.82)$$

where m_u is the atomic mass unit. Note that the numerical coefficient depends explicitly on the composition, through Z and μ_i : for a pure carbon white dwarf ($Z = 6$, $\mu_i = 12$, representative of the interior composition of these objects),

$$T_m \approx 2.0 \times 10^4 \rho^{1/3} \text{ K}, \quad (8.83)$$

which, for $\rho \sim 10^6 \text{ g/cm}^3$ (typical white dwarf central density), gives $T_m \sim 2 \times 10^6 \text{ K}$ — of the same order of magnitude as the central temperatures at which the onset of crystallization is indeed observed in real white dwarfs. If the temperature is of this order of magnitude, crystallization can occur. This phenomenon can occur in cooling white dwarfs, where the temperature is decreasing while the density remains high.

Observational evidence of crystallization with *Gaia*

The theoretical prediction of crystallization in white dwarfs [Equation (8.82)] received direct observational confirmation with data from the *Gaia* mission: the latent heat released during the phase transition (liquid to solid) releases extra energy, slowing down the star's cooling. This produces an **observable pile-up** of white dwarfs in a specific region of the color-magnitude HR diagram, corresponding to stars that spend more time crystallizing at that luminosity than expected from simple cooling without a phase transition — a statistically detected excess in the population of white dwarfs of the Galactic disk (Tremblay et al., 2019).

8.14.2 Neutronization

The gas analysis carried out in this chapter considers only electrons; however, particle collisions can lead to the formation of neutrons (electron capture, $e^- + p \rightarrow n + \nu_e$). In these cases, a more complete expression for the momentum must be used,

$$p = \frac{1}{c} \left(E^2 - m_e^2 c^4 \right)^{1/2}. \quad (8.84)$$

In cases where the neutron density is very high — in very massive and hot stars — the neutron pressure must also be taken into account,

$$P = P_{\text{ion}} + P_e + P_{\text{rad}} + P_{\text{neutron}}. \quad (8.85)$$

From neutronization to neutron stars

The neutronization discussed here — inverse electron capture, $e^- + p \rightarrow n + \nu_e$, which becomes energetically favorable when the electron Fermi energy exceeds the neutron-proton mass difference — is the same physical process that becomes dominant beyond the Chandrasekhar limit, when relativistic electron degeneracy pressure can no longer sustain the object against gravitational collapse. It is this process that converts the remaining stellar core into a **neutron star**, then sustained by the degeneracy pressure of the neutrons

themselves — the final result, $P = P_{\text{ion}} + P_e + P_{\text{rad}} + P_{\text{neutron}}$ [Equation (8.85)], already anticipates exactly this physics, to be revisited in future chapters on compact objects.

With crystallization and neutronization, we conclude the study of stellar equations of state. We started from the simple ionized ideal gas (Section 8.2), moved through total electron degeneracy — non-relativistic (Section 8.8) and relativistic, with its ultra-relativistic limit (Section 8.10) —, through partial degeneracy via the Fermi–Dirac integrals (Section 8.11), through the total equation of state (Section 8.13), and ended with particle-interaction effects (crystallization and neutronization). The relativistic electron degeneracy derived here is exactly the physical ingredient needed for the next chapter, on **White Dwarfs**, and neutronization anticipates the physics of **neutron stars**, the subject of still later chapters.

Further Reading and Current References

For the complete derivation of ultra-relativistic degeneracy and its connection to the **Chandrasekhar limit**, see the original work Chandrasekhar (1935) and the classic treatment in Chandrasekhar (1939). For a modern and comprehensive reference on equations of state of degenerate matter in compact objects, including neutronization and the transition to neutron stars, see Shapiro and Teukolsky (1983). For the approximate analytical solutions of the general equation of state in limiting cases, see Section 16.2.3 of Kippenhahn et al. (2012). On the observational evidence of crystallization in white dwarfs from *Gaia*, see Tremblay et al. (2019); on more recent observational tests of the white dwarf mass-radius relation, see the review by Tremblay et al. (2024) — both topics will be revisited in detail in the next chapter, on White Dwarfs. On the Coulomb coupling parameter Γ in dense plasmas, see Horedt (2004), which also covers applications of polytropes to related astrophysical condensed-matter systems.

Chapter 9

White Dwarfs

9.1 Introduction

White dwarf stars are the final product of the stellar evolution of low- or intermediate-mass stars. This will be the fate of our Sun, for example. One of the characteristics of white dwarfs is that their core is a degenerate state of electrons, which simplifies the analysis in some respects — it is exactly the physics derived in Chapter 8 that we will use here.

The structure of a white dwarf follows the scheme in Figure 9.1: an outer layer of degenerate electron states, a helium layer, and a carbon and oxygen (C/O) core, where the crystallization discussed in Section 8.14 of the previous chapter takes place.

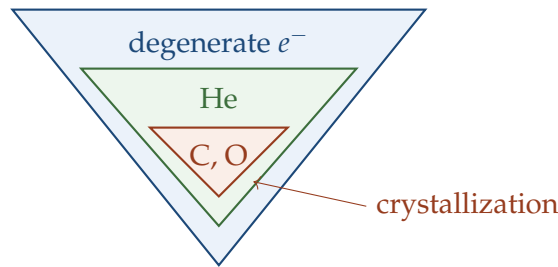


Figure 9.1: Schematic cross section of the layered structure of a white dwarf: outer layer of degenerate electrons, helium layer, and carbon and oxygen core, where the crystallization discussed in Chapter 8 takes place.

9.2 Chandrasekhar's theory

So far we have seen that the particle number density and electron pressure in a fully relativistic degenerate gas are

$$n_e(p) = \int F(p) dp, \quad P = \frac{1}{3} \int_0^\infty F(p) p v(p) dp, \quad (9.1)$$

where

$$P = \frac{\pi m_e^4 c^5}{3h^3} F(x), \quad F(x) = x(2x^2 - 3)(x^2 + 1)^{1/2} + 3 \sinh^{-1}(x), \quad x \equiv \frac{p_0}{m_e c}, \quad (9.2)$$

a result already derived in detail in Chapter 8 [Equations (8.58) and (8.59)].

On the sign inside the square root in $F(x)$

The sign inside the first square root of $F(x)$ must be positive, $(x^2 + 1)^{1/2}$, not $(x^2 - 1)^{1/2}$ — with a negative sign, the expression is undefined (square root of a negative number) for all $x < 1$, which makes no physical sense, since $x = p_0/m_e c$ must vary continuously

starting from 0. It is also worth noting the identity $\sinh^{-1}(x) = \ln[x + (1 + x^2)^{1/2}]$, with the square root inside the logarithm — the form used here [Equation (9.2)] is the one already derived and verified in Chapter 8.

The electron density n_e

We use here, consistently, the electron number density already established in Chapter 8 [Equation (8.25)], $n_e = 8\pi p_0^3/(3h^3)$ — the result of the integration $n_e = \int_0^{p_0} (8\pi p^2/h^3) dp$.

Using these equations of state together with the hydrostatic equilibrium equations, we can determine the radius and mass of a white dwarf star.

9.3 Combining the equation of state with hydrostatic equilibrium

Recalling the equations already established in Chapter 6,

$$\frac{dP}{dr} = -\rho \frac{d\Phi}{dr}, \quad \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right) = 4\pi G \rho, \quad (9.3)$$

substituting $d\Phi/dr = -(1/\rho) dP/dr$ (isolated from the first equation) into the second, we eliminate Φ and obtain a single second-order ODE involving only ρ and P :

$$\frac{1}{r^2} \frac{d}{dr} \left[r^2 \left(-\frac{1}{\rho} \frac{dP}{dr} \right) \right] = 4\pi G \rho. \quad (9.4)$$

Let us consider that the equation of state of a relativistic Fermi–Dirac gas is

$$P = c_1 F(x), \quad \rho = c_2 x^3, \quad x \equiv \frac{p_0}{m_e c}, \quad (9.5)$$

where $c_1 \equiv \pi m_e^4 c^5 / 3h^3$ and $c_2 \equiv 8\pi \mu_e m_a (m_e c)^3 / 3h^3$ gather the constant physical prefactors [comparing with Equation (9.2) and the relation $\rho = n_e \mu_e m_a = (8\pi p_0^3 / 3h^3) \mu_e m_a$]. Substituting into Equation (9.4),

$$-\frac{c_1}{c_2} \frac{1}{r^2} \frac{d}{dr} \left[\frac{r^2}{x^3} \frac{d}{dr} (F(x)) \right] = 4\pi G c_2 x^3. \quad (9.6)$$

9.4 Chandrasekhar's differential equation

The term $dF(x)/dr$ can be obtained by differentiating directly the closed-form integral already solved in Chapter 8 [Equation (8.57)],

$$\int_0^x \frac{Y^4 dY}{(1+Y^2)^{1/2}} = \frac{1}{8} \left[X(2X^2 - 3)(X^2 + 1)^{1/2} + 3 \sinh^{-1}(X) \right] = \frac{1}{8} F(X). \quad (9.7)$$

Differentiating $F(x) = x(2x^2 - 3)(x^2 + 1)^{1/2} + 3 \sinh^{-1}(x)$ with respect to x , with $z \equiv (x^2 + 1)^{1/2}$ (so $dz/dx = x/z$ and $d[\sinh^{-1}(x)]/dx = 1/z$),

$$\frac{dF}{dx} = (6x^2 - 3)z + (2x^3 - 3x) \frac{x}{z} + \frac{3}{z} = \frac{(6x^2 - 3)z^2 + 2x^4 - 3x^2 + 3}{z} = \frac{8x^4}{z}, \quad (9.8)$$

since $(6x^2 - 3)z^2 = (6x^2 - 3)(x^2 + 1) = 6x^4 + 3x^2 - 3$, which, added to $2x^4 - 3x^2 + 3$, gives exactly $8x^4$. Since $8x^4/z = 8x^3 \cdot (x/z) = 8x^3 (dz/dx)$, we obtain the elegant identity

$$\frac{1}{x^3} \frac{dF(x)}{dx} = 8 \frac{dz}{dr}, \quad z \equiv (x^2 + 1)^{1/2}. \quad (9.9)$$

Substituting Equation (9.9) into Equation (9.6),

$$-\frac{8c_1}{c_2} \frac{1}{r^2} \frac{d}{dr} \left[r^2 \frac{dz}{dr} \right] = 4\pi G c_2 x^3 \implies \frac{1}{r^2} \frac{d}{dr} \left[r^2 \frac{dz}{dr} \right] = -\frac{\pi G c_2^2}{2c_1} x^3. \quad (9.10)$$

Inverting the definition of z , $x = (z^2 - 1)^{1/2}$, and substituting back,

$$\frac{1}{r^2} \frac{d}{dr} \left[r^2 \frac{dz}{dr} \right] = -\frac{\pi G c_2^2}{2c_1} (z^2 - 1)^{3/2}. \quad (9.11)$$

We now define the dimensionless variables

$$\psi \equiv \frac{z}{z_c}, \quad y \equiv \frac{r}{\alpha}, \quad \alpha^2 \equiv \frac{2c_1}{\pi G c_2^2 z_c^2}, \quad (9.12)$$

where z_c is the central value of z (related to the central Fermi momentum, and thus to the star's central density). Note explicitly the structural similarity with the variables defined for the Lane–Emden equation in Chapter 6 [Equation (6.27)] — the same type of nondimensionalization, now applied to the equation of state of the relativistic degenerate gas.

Confirming the numerical coefficient 2 in α^2

It is worth checking that the numerical coefficient in Equation (9.10) and in the numerator of α^2 [Equation (9.12)] is indeed 2, not z_c : substituting $z = z_c \psi$ and $r = \alpha y$ into Equation (9.11) and requiring that the result reduce exactly to the final form [Equation (9.13)] below, with no numerical factor left over, the numerical coefficient is necessarily 2 — only this choice cancels all the factors of z_c and leaves the final equation in the expected clean form. We consistently use “ $2c_1$ ” (not “ $z_c c_1$ ”) in Equation (9.10) and in the definition of α^2 [Equation (9.12)], and later on in Equation (9.20).

Substituting $z = z_c \psi$, $r = \alpha y$ into Equation (9.11) — using $dz/dr = (z_c/\alpha) d\psi/dy$ and $r^2 dz/dr = \alpha z_c y^2 d\psi/dy$ — and using the definition of α^2 to cancel the common factors, we arrive at the final form:

$$\frac{1}{y^2} \frac{d}{dy} \left(y^2 \frac{d\psi}{dy} \right) = - \left(\psi^2 - \frac{1}{z_c^2} \right)^{3/2}, \quad \text{or, in expanded form,} \quad (9.13)$$

$$\left[\psi'' + \frac{2}{y} \psi' + \left(\psi^2 - \frac{1}{z_c^2} \right)^{3/2} = 0 \right], \quad (9.14)$$

Chandrasekhar's differential equation for the internal structure of a white dwarf star.

The unification of the two degenerate regimes

Note that Equation (9.13) resembles the Lane–Emden equation [Equation (6.30)] for a polytrope. In fact, it reduces exactly to the two cases derived independently in Chapter 8 in the appropriate limits: when $z_c \rightarrow \infty$ ($x \rightarrow \infty$), the term $1/z_c^2$ vanishes and we recover the Lane–Emden equation with $n = 3$ ($\gamma = 4/3$, the ultra-relativistic case); when $z_c \rightarrow 1$

($x \rightarrow 0$), we recover index $n = 3/2$ ($\gamma = 5/3$, the non-relativistic case). This is the great unification of this chapter: the two degenerate regimes derived separately in Chapter 8 are, in fact, two limits of a single master equation, parametrized by z_c — a quantity directly tied to the central Fermi momentum and, therefore, to the star's central density.

9.5 Boundary conditions, mass, and radius

The central boundary conditions, analogous to those of the Lane–Emden equation, are

$$y = 0, \quad \psi(0) = 1, \quad \psi'(0) = 0, \quad (9.15)$$

regularity at the center of the star. These conditions, together with Equation (9.13), fully describe the central region of a white dwarf. In terms of $\psi(y)$, the density is

$$\rho = c_2 x^3 = c_2 (z^2 - 1)^{3/2} = c_2 z_c^3 \left(\psi^2 - \frac{1}{z_c^2} \right)^{3/2}. \quad (9.16)$$

We can determine the mass of the white dwarf from the mass conservation equation,

$$M = \int_0^R 4\pi r^2 \rho \, dr = 4\pi \alpha^3 z_c^3 c_2 \int_0^{y_1} y^2 \left(\psi^2 - \frac{1}{z_c^2} \right)^{3/2} dy, \quad (9.17)$$

where y_1 is the value of y at the surface of the star (where the density vanishes, $\psi(y_1)^2 = 1/z_c^2$). Comparing the term inside the integral with Equation (9.13) itself — the same algebraic trick already used for the Lane–Emden mass equation in Chapter 6 [Equation (6.38)] —,

$$-\frac{d}{dy} \left[y^2 \frac{d\psi}{dy} \right] = y^2 \left(\psi^2 - \frac{1}{z_c^2} \right)^{3/2}, \quad (9.18)$$

so that the integral is an exact derivative,

$$M = -4\pi \alpha^3 z_c^3 c_2 \int_0^{y_1} \frac{d}{dy} \left[y^2 \frac{d\psi}{dy} \right] dy = \boxed{-4\pi \alpha^3 z_c^3 c_2 \left(y^2 \frac{d\psi}{dy} \right)_{y_1}}. \quad (9.19)$$

We can easily obtain the radius of the white dwarf using the definition $y = r/\alpha$: at $r = R$, $y = y_1$, so $R = \alpha y_1$. Substituting α explicitly [Equation (9.12)],

$$R = \sqrt{\frac{2c_1}{\pi G}} \frac{1}{z_c c_2} y_1. \quad (9.20)$$

9.6 Limits, the mass-radius relation, and the Chandrasekhar mass

In the limits where $x \rightarrow \infty$ ($z_c \rightarrow \infty$), we have the ultra-relativistic case; when $x \rightarrow 0$ ($z_c \rightarrow 1$), we return to the classical non-relativistic case.

The **mass-radius relation** of white dwarf stars follows the behavior sketched in Figure 9.2: a decreasing curve in the R vs. M plane. For a white dwarf at any point *on* the curve, hydrostatic equilibrium is satisfied ($F_{\text{pr}} = F_{\text{gr}}$, degenerate pressure force balancing the gravitational force). For any point *off* the curve, the hydrostatic regime breaks down, driving the star toward instability: below / inside the curve degenerate pressure dominates ($F_{\text{pr}} > F_{\text{gr}}$), above it gravity dominates ($F_{\text{gr}} > F_{\text{pr}}$), leading to collapse.

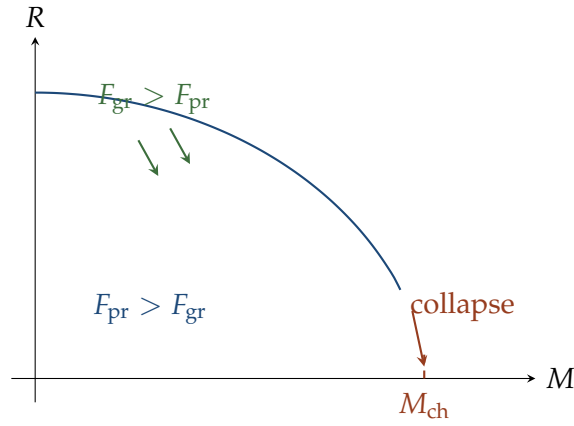


Figure 9.2: Schematic mass-radius relation of white dwarfs: decreasing curve separating the stable region ($F_{\text{pr}} > F_{\text{gr}}$, supported by electron degeneracy) from the unstable region ($F_{\text{gr}} > F_{\text{pr}}$), with collapse occurring as the star approaches the Chandrasekhar mass, M_{ch} , on the horizontal axis.

Diagram nomenclature

The axes of this diagram are R (radius) and M (mass) — the content is the **mass-radius relation** of white dwarfs, standard nomenclature in the literature for this type of diagram, and the name used here in the text.

The limiting mass that a white dwarf can reach is called the **Chandrasekhar mass**,

$$M_{\text{ch}} = \left(\frac{2}{\mu_e} \right)^2 \times 1.459 M_{\odot}, \quad (9.21)$$

obtained for the case where $z_c \rightarrow \infty$, which coincides with a polytrope with $n = 3$. White dwarf stars that exceed this mass limit become unstable and can explode as a Type Ia supernova.

M_{ch} for different compositions

For a typical fully ionized He/C/O composition, where $Z_i/\mu_i \approx 1/2$ for all the relevant elements (He, C, O), $\mu_e \approx 2$, and

$$M_{\text{ch}} = \left(\frac{2}{2} \right)^2 \times 1.459 M_{\odot} = 1.459 M_{\odot}, \quad (9.22)$$

recovering the classical value $M_{\text{ch}} \approx 1.4 M_{\odot}$. Fully ionized O, Ne, and Mg nuclei have exactly the same ratio $Z_i/\mu_i \approx 1/2$ (they are all $Z = N$ nuclei), so that O/Ne/Mg white dwarfs also have $\mu_e \approx 2$ — practically the same M_{ch} as the C/O case. On the other hand, for a hypothetical composition much richer in iron-peak elements (^{56}Fe : $Z = 26$, $\mu_i = 56$, $Z_i/\mu_i \approx 0.464$, so $\mu_e = 1/(Z_i/\mu_i) \approx 2.15$),

$$M_{\text{ch}} = \left(\frac{2}{2.15} \right)^2 \times 1.459 M_{\odot} \approx 1.26 M_{\odot}, \quad (9.23)$$

a noticeably lower limiting mass — showing that M_{ch} is not an absolutely universal number, but depends on chemical composition through μ_e , which increases as the degenerate matter becomes richer in neutrons (lower Z/A).

Sirius B: the first confirmed white dwarf

The first object identified as a white dwarf was Sirius B, the companion of the brightest star in the night sky, Sirius A. With a mass $\approx 1.0 M_{\odot}$ and a radius comparable to Earth's, Sirius B is exactly the kind of object that motivated the opening of Chapter 8 — a compact and extremely hot star ($R_{\star} \sim R_{\oplus}$) that requires treating the electron gas as a degenerate Fermi–Dirac gas. The determination of its mass and radius, in the early twentieth century, together with the gravitational redshift detected in its spectrum, provided some of the first observational tests of the electron degeneracy physics derived in this chapter.

The real composition of white dwarfs

Most observed white dwarfs have carbon and oxygen (C/O) cores, as in the scheme of Figure 9.1 — remnants of stars that finished core helium burning without reaching a temperature and density sufficient for carbon ignition. White dwarfs originating from more massive progenitor stars (which do burn carbon and neon, but not enough to trigger core collapse) can have oxygen, neon, and magnesium (O/Ne/Mg) cores — with $\mu_e \approx 2$, essentially the same as the C/O case, since O, Ne, and Mg are all $Z = N$ nuclei. Even so, real O/Ne/Mg white dwarfs tend to collapse into a neutron star at a mass slightly below the classical Chandrasekhar mass ($\approx 1.38 M_{\odot}$, versus $\approx 1.4 M_{\odot}$) — not because of a difference in *average* composition like the one illustrated in the example above, but because, at sufficiently high central densities, electron capture by ^{24}Mg and ^{20}Ne removes degenerate electrons from the gas, locally reducing the supporting pressure before the classical limit is reached — a density-dependent neutronization mechanism, distinct from the fixed-composition effect discussed above, and one that will be revisited in future chapters on compact objects.

Type Ia supernovae as standard candles

The physical uniformity of M_{ch} as the trigger for the runaway thermonuclear ignition of carbon is the physical basis behind the relative uniformity of the peak luminosity of Type Ia supernovae — which makes them extremely useful **standard candles** in observational cosmology. It was precisely the use of SNe Ia as distance indicators that led to the discovery of the accelerating expansion of the universe, recognized with the 2011 Nobel Prize in Physics (Perlmutter, Schmidt, and Riess).

Not every white dwarf is close to M_{ch}

It is important to note that the observed mass distribution of isolated white dwarfs peaks around $\sim 0.6 M_{\odot}$, well below the Chandrasekhar limit. For a white dwarf to reach M_{ch} and detonate as an SN Ia, a mass-accretion mechanism is required: the *single-degenerate* scenario (accretion of matter from a non-degenerate companion in a binary system) or the *double-degenerate* scenario (merger of two white dwarfs). Which of these channels dominates the actual production of observed SNe Ia — or whether both contribute comparably — is still an open question, actively debated in the current literature, with no definitive consensus (Ruiter and Seitenzahl, 2025).

Further Reading and Current References

For the original work on this derivation, see Chandrasekhar (1935) (reinforcing the citation already made in Chapter 8) and the classic treatment in Chandrasekhar (1939). For a contemporary, comprehensive review of the debate between the *single-degenerate* and

double-degenerate scenarios as progenitors of Type Ia supernovae, see Ruiter and Seitenzahl (2025). For a complete didactic treatment of white dwarfs, including internal structure and the mass-radius relation, see the dedicated chapter in Hansen et al. (2004). On modern observational tests of the theoretical mass-radius relation, combining astrometry and photometry from *Gaia*, see Tremblay et al. (2024).

Chapter 10

Energy Generation Processes (Nucleosynthesis)

10.1 Introduction: stellar nucleosynthesis

Stellar nucleosynthesis is the main mechanism for the generation of heavier chemical elements in the present-day universe. The proton-proton chain, the CNO cycle, and the triple-alpha process are the main nucleosynthetic processes in stars — the first two will be detailed in later sections of this chapter.

In primordial (Big Bang) nucleosynthesis, H, He, and Li were created; all other elements were produced by supernova explosions or in the core or atmosphere of evolved stars.

When we study chemical elements, we also encounter many **isotopes**, which are chemical elements with the same atomic number Z but different mass, that is, different numbers of neutrons. For example: $^{13}_6\text{C}$ has 6 protons and $13 - 6 = 7$ neutrons. More stable elements tend to have a similar number of protons and neutrons.

The valley of nuclear stability

The proton-neutron ratio of stable nuclei is the basis of the so-called **valley of nuclear stability**: when all known nuclei are plotted in the (Z, N) plane, stable nuclei form a narrow, continuous band — for light elements, close to the line $N = Z$; for heavier elements, with a growing neutron excess, since the Coulomb repulsion between protons requires more neutrons (which do not feel this repulsion) to keep the nucleus bound by the strong force. Nuclei outside this valley are unstable and decay (by β^- , β^+ emission, or other processes) until they reach a stable or quasi-stable isotope of the same mass number A . This stability map is, in essence, the route that stellar nucleosynthesis follows: fusion reactions drive light nuclei toward more stable configurations (higher binding energy per nucleon), releasing energy in the process — up to the stability peak around iron and nickel.

10.2 Formalism of nuclear reactions

Let us define a few concepts to understand the process of energy generation. When we studied the equations of internal structure in Chapter 5, we saw that

$$\frac{\partial X_i}{\partial t} = \frac{m_i}{\rho} \left(\sum_j r_{ji} - \sum_k r_{ik} \right), \quad (10.1)$$

or, in the equivalent form in terms of energy per transition [Equation (5.198)]. From these relations, we can determine the energy and rate of the atomic transitions taking place in the core.

The notation used to describe a nuclear reaction is

$$X + a \rightarrow Y + b, \quad \text{or} \quad X(a, b)Y, \quad (10.2)$$

where the amount of energy released in each reaction is

$$Q = (M_X + M_a - M_Y - M_b) c^2, \quad \begin{cases} Q > 0, & \text{exothermic process,} \\ Q < 0, & \text{endothermic process.} \end{cases} \quad (10.3)$$

One of the greatest difficulties in the study of stellar nucleosynthesis is the determination/measurement of the **cross section** of each reaction.

10.3 Cross section and reaction rate

In the quantum-mechanical treatment, the cross section follows the equation

$$\sigma = \pi \lambda^2, \quad \lambda = \left(\frac{m_p + m_t}{m_t} \right)^{1/2} \frac{h}{(2m_p E)^{1/2}}, \quad (10.4)$$

where m_p is the mass of the incident particle (projectile) and m_t the total (target) mass.

The cross section varies strongly from reaction to reaction. For transitions governed by the weak force, such as $p(p, e^+ \nu)d$ (the first step of the proton-proton chain), the cross section is enormously smaller than for cases governed by the strong force, such as $^{15}\text{N}(p, \alpha)^{12}\text{C}$ — the difference of many orders of magnitude between the two cases is the central pedagogical point here.

Units of the numerical cross section values

Nuclear cross sections are typically measured in **barns** (1 barn = 10^{-24} cm²), the natural scale of the size of an atomic nucleus. The fairest way to compare the intrinsic “strength” of two reactions mediated by different interactions is through the **astrophysical S-factor** $S(E)$ [Equation (10.16), further below], which already removes the Coulomb suppression common to both: $S_{pp}(0) \approx 4 \times 10^{-25}$ MeV barn for $p(p, e^+ \nu)d$ (weak force) versus $S(0) \approx 70$ MeV barn for $^{15}\text{N}(p, \alpha)^{12}\text{C}$ (strong force, Iliadis, 2015) — a difference of ~ 26 orders of magnitude that does not depend on the energy chosen for the comparison.

The **actual cross section** $\sigma(E)$, which already includes the Gamow suppression, is even smaller and varies strongly with energy. At the solar Gamow peak of the pp chain ($E_0 \approx 6$ keV, see Section 10.5), $2\pi\eta \approx 9$ and $\sigma_{pp}(E_0) \approx 8 \times 10^{-27}$ barn ($\approx 8 \times 10^{-51}$ cm²) — many orders of magnitude below the typical nuclear scale of the barn, reflecting the extreme rarity of a p - p collision resulting in fusion at such low energies. By contrast, $^{15}\text{N}(p, \alpha)^{12}\text{C}$, near its strong resonances ($E_p \approx 429$ and 897 keV), reaches $\sigma \sim 0.5$ barn ($\approx 5 \times 10^{-25}$ cm²) — but this direct comparison of σ mixes two very different energies (the Gamow peak of the pp chain, ~ 6 keV, versus a resonance at hundreds of keV), so the contrast in $S(0)$ above is the pedagogically cleaner number for comparing the two interactions.

The main parameter of this description is the nuclear reaction rate r_{ij} , which can be defined as

$$r_{ij} = N_i N_j v \sigma(v), \quad (10.5)$$

where N_i, N_j are the number densities of particles i and j per cm^3 , v is the relative velocity between i and j , and σ is the cross section. In some textbooks, one finds the alternative form

$$r_{ij} = \frac{1}{1 + \delta_{ij}} N_i N_j \langle\sigma v\rangle, \quad \delta_{ij} = \begin{cases} 0, & i \neq j, \\ 1, & i = j, \end{cases} \quad (10.6)$$

which avoids double-counting collisions when $i = j$ (identical particles).

In a stellar gas, the velocity distribution follows a Maxwellian, so that

$$r_{ij} = N_i N_j \langle\sigma v\rangle, \quad \langle\sigma v\rangle = \int_0^\infty \phi(v) v \sigma(v) dv. \quad (10.7)$$

10.4 Derivation of $\langle\sigma v\rangle$ and the three physical factors

The Maxwell–Boltzmann velocity distribution is

$$\phi(v) = 4\pi v^2 \left(\frac{m}{2\pi kT}\right)^{3/2} e^{-mv^2/2kT}, \quad m \equiv \frac{m_i m_j}{m_i + m_j} \text{ (reduced mass)}. \quad (10.8)$$

Knowing that $E = \frac{1}{2}mv^2$ (hence $v = (2E/m)^{1/2}$), we can obtain the term $\langle\sigma v\rangle$ as

$$\langle\sigma v\rangle = \int_0^\infty 4\pi v^2 \left(\frac{m}{2\pi kT}\right)^{3/2} e^{-mv^2/2kT} v \sigma(v) dv, \quad (10.9)$$

and, substituting $v = (2E/m)^{1/2}$, $v^2 = 2E/m$, and $dv = dE/(2mE)^{1/2}$,

$$\langle\sigma v\rangle = \int_0^\infty 4\pi \frac{2E}{m} \left(\frac{m}{2\pi kT}\right)^{3/2} e^{-E/kT} \sigma(E) \left(\frac{2E}{m}\right)^{1/2} \frac{dE}{(2mE)^{1/2}}. \quad (10.10)$$

Collecting the E -dependence, $(2E/m) \cdot (2E/m)^{1/2} \cdot (2mE)^{-1/2} = 2E/m^2$, and simplifying the remaining numerical factors of m and π ,

$$4\pi \cdot \frac{2}{m^2} \left(\frac{m}{2\pi kT}\right)^{3/2} = \frac{8\pi m^{3/2}}{m^2 (2\pi kT)^{3/2}} = \frac{8\pi}{m^{1/2} (2\pi)^{3/2} (kT)^{3/2}} = \left(\frac{8}{\pi m}\right)^{1/2} \frac{1}{(kT)^{3/2}}, \quad (10.11)$$

arriving at the **general form of the cross section as a function of particle energy**:

$$\boxed{\langle\sigma v\rangle = \left(\frac{8}{\pi m}\right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty \sigma(E) E e^{-E/kT} dE}. \quad (10.12)$$

The value of $\langle\sigma v\rangle$ depends on three physical factors:

- the probability of overcoming the Coulomb barrier;
- the probability of quantum interaction;
- whether the reaction occurs near a resonant nucleus.

The Coulomb barrier, of height

$$V_c = \frac{Z_i Z_j e^2}{r}, \quad (10.13)$$

is illustrated in Figure 10.1: a short-range attractive nuclear well (strong force, within the nuclear radius) and a long-range repulsive barrier (electromagnetic repulsion between the positively charged nuclei), with the projectile's energy typically well below the top of the barrier — hence the need for the quantum tunneling discussed in the next section.

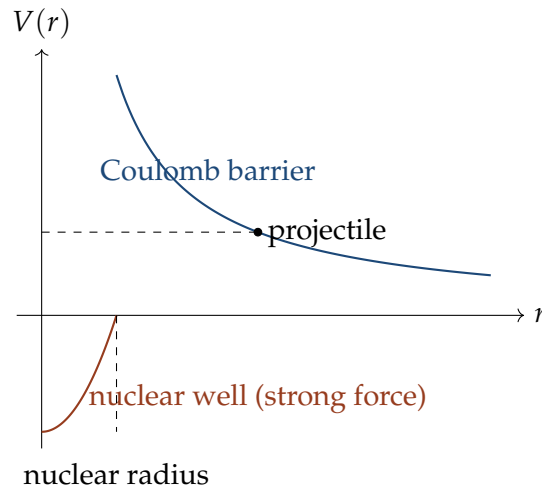


Figure 10.1: Coulomb barrier: potential energy as a function of the distance r between the nuclei. Inside the nuclear radius, the strong force dominates, forming an attractive well; outside it, the Coulomb repulsion between the positive charges dominates, forming the barrier. The typical kinetic energy of the projectile (dashed line) lies well below the top of the barrier, requiring quantum tunneling for the reaction to occur.

10.5 Gamow factor and the Gamow peak

The probability that a particle (transition) overcomes the Coulomb barrier is given by

$$P = e^{-2\pi\eta}, \quad \eta = \frac{Z_i Z_j e^2}{\hbar v}, \quad (10.14)$$

where η is the **Sommerfeld parameter**, or, in a practical numerical form,

$$2\pi\eta = 31.29 Z_i Z_j \left(\frac{A}{E} \right)^{1/2}, \quad (10.15)$$

where A is the reduced mass (in atomic mass units) and E is the particle energy in keV.

Unit of E in Equation (10.15)

The numerical coefficient 31.29 agrees with the literature (the standard form uses 31.335, with a small difference due to updated values of the fundamental constants — see Iliadis, 2015), and the energy unit associated with this coefficient is keV: this is the usual convention in low-energy stellar nuclear physics, since the typical thermal energies of the stellar interior (a few keV) are much smaller than 1 MeV. We use E in keV consistently from here on.

In the absence of resonances between the particles, we can define the **astrophysical cross-section factor** $S(E)$ (or astrophysical S-factor), where

$$\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta}, \quad (10.16)$$

which isolates the nuclear physics (which varies smoothly with E , contained in $S(E)$) from the dominant, rapidly varying effect of the Coulomb barrier (contained in the exponential factor). Substituting into Equation (10.12),

$$\langle \sigma v \rangle = \left(\frac{8}{\pi m} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty S(E) \exp\left(-\frac{E}{kT} - \frac{b}{E^{1/2}}\right) dE, \quad (10.17)$$

where

$$b = \frac{(2m)^{1/2} \pi e^2 Z_i Z_j}{\hbar}. \quad (10.18)$$

We have a distribution in which the decreasing Maxwellian tail, $e^{-E/kT}$, competes with the increasing tunneling probability, $e^{-bE^{-1/2}}$ — most reactions occur between the two functions, in a narrow peak known as the **Gamow peak** (Figure 10.2). The energy of this peak is defined by

$$E_0 = \left(\frac{bkT}{2} \right)^{2/3}. \quad (10.19)$$

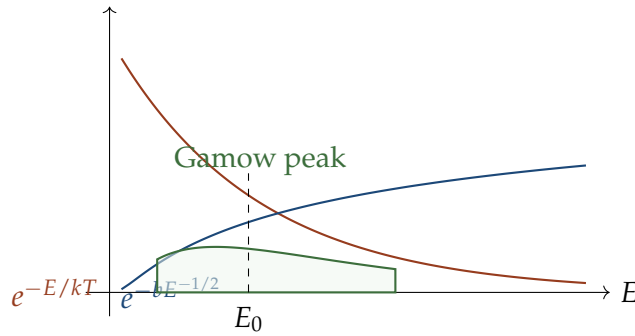


Figure 10.2: The Gamow peak: the tunneling probability ($e^{-bE^{-1/2}}$, increasing with E , blue curve) competes with the Maxwellian tail ($e^{-E/kT}$, decreasing, red curve); the product of the two (shaded region) forms a narrow peak at E_0 , where most nuclear reactions in the stellar interior occur.

Numerical range of E_0 and kT

The central temperature of the Sun, $T_c \approx 1.5 \times 10^7$ K, corresponds to $kT_c \approx 1.3$ keV, within the characteristic range $kT \sim 1\text{--}3$ keV of typical stellar plasmas. The corresponding Gamow peak lies at $E_0 \sim 15\text{--}30$ keV for reactions with moderate Z (such as those of the CNO cycle; the literature confirms, for example, $E_0 \approx 20\text{--}35$ keV for the reaction $^{14}\text{N}(p, \gamma)^{15}\text{O}$ at solar temperatures). For the pp chain itself ($Z_i = Z_j = 1$), the Gamow peak lies in an even lower range, typically 5–10 keV (Iliadis, 2015).

Historical origin: Gamow, Atkinson, and Houtermans

The application of quantum tunneling to stellar nuclear reactions was proposed by George Gamow (1928), and soon after applied to the stellar context by Atkinson and Houtermans (1929). Before this insight, classical physics could not explain how the Sun could shine: classically, the thermal energy of the solar core ($kT_c \approx 1.3$ keV) is hundreds of times smaller than the height of the Coulomb barrier between two protons (of order MeV) — without quantum tunneling, nuclear fusion would be classically forbidden at typical stellar temperatures. The recognition that particles can cross this barrier even without enough energy to classically overcome it was one of the first great successes of quantum mechanics applied to astrophysics.

Why the Sun burns slowly: a quasi-anthropic point

The extremely small cross section of the reaction $p + p \rightarrow d + e^+ + \nu$ [Section 10.3], mediated by the weak force, is the direct physical reason why the Sun (and similar stars)

burn hydrogen slowly and steadily over billions of years, rather than fusing all their fuel almost instantaneously. This is a nearly “anthropic” point: if the cross section of this reaction were orders of magnitude larger, stars like the Sun would have much shorter lifetimes — possibly incompatible with the time needed for the evolution of complex life on Earth (~ 4 billion years). The same type of argument appears in the discussion of the fine-tuning of the triple-alpha process, which will be mentioned when that process is detailed.

The Gamow peak in the solar core

For the Sun’s core, $T_c \approx 1.5 \times 10^7$ K, hence

$$kT_c \approx 1.3 \text{ keV}. \quad (10.20)$$

Placing this value in Figure 10.2: the Maxwellian tail already decays strongly above a few keV, but the tunneling probability for the $p + p$ reaction (with $Z_i = Z_j = 1$) only becomes appreciable at somewhat higher energies — the resulting Gamow peak for the pp chain in the Sun typically falls between 5 and 10 keV, that is, a few times kT_c . This illustrates concretely why only a small fraction of the particles in the solar core (those at the high-energy tail of the Maxwellian distribution, combined with tunneling) actually take part in fusion at any given instant — reinforcing the slow-burning point discussed in the context box above.

10.6 Electron screening

For nuclear reactions to occur, we must take into account **electron screening**. The electron shells in the outer part of atoms act as a shield: these electrons alter the potential energy of each ion and can increase the reaction rate of certain transitions.

To estimate the magnitude of this effect, let us consider the Boltzmann equation,

$$\frac{n}{\bar{n}} = e^{-E/kT}, \quad (10.21)$$

where n and $\bar{n}$ are the densities in different states and E is the energy difference between them.

In the presence of an electrostatic potential ϕ , the particle density relates to the reference density $\bar{n}$ by

$$n = \bar{n} e^{-Ze\phi/kT}, \quad (10.22)$$

with $\bar{n}$ multiplying the exponential.

For “normal” stars, we have $Ze\phi \ll kT$, so we can linearize the exponential,

$$n = \bar{n} e^{-Ze\phi/kT} \approx \bar{n} \left(1 - \frac{Ze\phi}{kT} \right). \quad (10.23)$$

We can then separate ions from electrons, where

$$n_i = \bar{n}_i \left(1 - \frac{Z_i e \phi}{kT} \right), \quad n_e = \bar{n}_e \left(1 + \frac{e \phi}{kT} \right). \quad (10.24)$$

We can define the charge density as

$$\rho_{\text{carga}} = Ze n_x, \quad (10.25)$$

which, as a function of the potential, takes the form

$$\rho_{\text{carga}} = \sum_i \bar{n}_i \left(Z_i e - \frac{Z_i^2 e^2 \phi}{kT} \right) - \bar{n}_e \left(1 + \frac{e^2 \phi}{kT} \right) e. \quad (10.26)$$

Note that, for a neutral gas with $\phi = 0$, we have $\bar{\rho}_{\text{carga}} = 0$, hence

$$\bar{\rho}_{\text{carga}} = \sum_i (Z_i e) \bar{n}_i - e \bar{n}_e = 0. \quad (10.27)$$

For $\phi \neq 0$, substituting n_i and n_e [Equation (10.24)] and keeping only the potential-dependent terms,

$$\rho_{\text{carga}} = - \sum_i \frac{Z_i^2 e^2 \phi}{kT} \bar{n}_i - \frac{e^2 \phi}{kT} \bar{n}_e = - \frac{e^2 \phi}{kT} \left(\sum_i Z_i^2 \bar{n}_i + \bar{n}_e \right). \quad (10.28)$$

Recalling that we can relate the ion density to the mass fraction and the density,

$$\bar{n}_i = \frac{\rho N_A X_i}{A_i}, \quad (10.29)$$

where X_i is the mass fraction, N_A Avogadro's number, ρ the density, and A_i the atomic weight of each particle. The electron density, in turn, for a fully ionized gas, relates to the same sum through an ionization factor F_i (the fraction of electrons released per ion i relative to the total possible, Z_i):

$$\bar{n}_e = \sum_i \rho N_A F_i \frac{Z_i X_i}{A_i}. \quad (10.30)$$

Substituting Equations (10.29) and (10.30) into Equation (10.28),

$$\rho_{\text{carga}} = - \frac{e^2 \phi}{kT} \left\{ \sum_i \rho N_A \frac{Z_i^2 X_i}{A_i} + \sum_i \rho N_A F_i \frac{Z_i X_i}{A_i} \right\}, \quad (10.31)$$

and, factoring the constants out of the sum,

$$\rho_{\text{carga}} = - \frac{e^2 \phi}{kT} \rho N_A \sum_i \frac{Z_i (Z_i + F_i) X_i}{A_i}. \quad (10.32)$$

The complete-ionization hypothesis, $F_i = 1$

The ionization factor F_i [Equation (10.30)] describes the fraction of electrons released per ion i relative to the total possible, Z_i . Assuming **complete ionization**, $F_i = 1$ — that is, each ion i contributes exactly Z_i free electrons to the plasma, the same hypothesis already used throughout Chapter 8 —, Equation (10.32) becomes

$$\rho_{\text{carga}} = - \frac{e^2 \phi}{kT} \rho N_A \sum_i \frac{Z_i (Z_i + 1) X_i}{A_i}. \quad (10.33)$$

We can define

$$\zeta \equiv \sum_i \frac{Z_i (Z_i + 1) X_i}{A_i}, \quad (10.34)$$

so that

$$\rho_{\text{carga}} = - \frac{e^2 \phi}{kT} \rho N_A \zeta, \quad (10.35)$$

the final charge density resulting from the interaction of the electrons with the atomic ions.

With this charge density, we can substitute into Poisson's equation in spherical symmetry [already established in Chapters 6 and 9] and obtain

$$\nabla^2 \phi = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = -4\pi \rho_{\text{carga}}. \quad (10.36)$$

Solving this ODE leads us to

$$r_D = \left(\frac{kT}{4\pi e^2 \rho N_A \zeta} \right)^{1/2}, \quad \phi = \frac{Ze}{r} e^{-r/r_D}, \quad (10.37)$$

where r_D is known as the **Debye-Hückel length**, and ϕ is the solution of the ODE. We can interpret r_D as the radius of an “electron cloud” surrounding and “screening” the nucleus from incoming particles — an effect that **increases** the nuclear reaction rate by reducing the effective Coulomb barrier felt by an incident nucleus.

Weak vs. strong screening

The Debye-Hückel derivation presented here holds in the *weak*-screening regime, appropriate for “normal,” non-degenerate stellar plasmas, as assumed throughout this section ($Ze\phi \ll kT$). In very dense, degenerate matter, such as the interior of white dwarfs (Chapter 9), the *strong*-screening regime requires more sophisticated treatments, going beyond the simple Debye-Hückel result — the Coulomb interaction energy between neighboring ions ceases to be a small thermal perturbation (it is exactly this same competition, in the opposite sense, that leads to the crystallization discussed in Section 8.14). The original formalism for electron screening in stellar plasmas was developed by Salpeter (1954), who treated both the weak- and strong-screening limits.

Further Reading and Current References

For the modern, complete treatment of nuclear reaction rates, from the astrophysical S-factor $S(E)$ to the Gamow peak, see Iliadis (2015) — the current standard reference in stellar nuclear physics. For the original work on electron screening in stellar plasmas, see Salpeter (1954). On the general formalism of reaction rates and nucleosynthesis, see also Clayton (1983), already cited in an earlier chapter. For modern measurements of astrophysical S-factors $S(E)$ in underground laboratories — which measure nuclear cross sections directly at energies close to stellar ones, reducing the need for theoretical extrapolation —, see the review of the LUNA experiment (Gran Sasso) in Broggini et al. (2010).

10.7 Nuclear reaction network formalism

Level of detail in this chapter

As already noted (this material draws on a lecture with slides not reproduced in writing): the complete physical description of energy generation processes involves ODEs, statistical mechanics, and quantum mechanics. This course does **not** cover these processes in detail, as they fall somewhat outside its scope — only a superficial treatment is presented here. The reader should calibrate expectations accordingly: what follows is the general

formalism of the reaction network and one fully worked example (the proton-proton chain), not a complete treatise on stellar nucleosynthesis.

Every energy generation process is primarily described by the 5th equation of stellar structure [already established in Chapter 5, Equation (5.198)].

We can also consider a case in which we have a first reaction between two nuclei, 0 and 1. The reaction rate is related to the mean lifetime of nucleus 0 in the plasma; thus, we can express the transition as

$$\left(\frac{dN_0}{dt}\right) = -\lambda_1 N_0 = -\frac{N_0}{\tau_1(0)}, \quad (10.38)$$

where N_0 is the density of nuclei 0, $\tau_1(0)$ describes the mean lifetime of nuclei 0 against reaction with species 1 (initial toward final state), and $\lambda \equiv 1/\tau$ is the decay constant (or rate).

From Equation (10.5) (Section 10.3), we can write

$$\left(\frac{dN_0}{dt}\right) = -(1 + \delta_{01}) r_{01} = -(1 + \delta_{01}) \frac{N_0 N_1 \langle \sigma v \rangle_{01}}{1 + \delta_{01}} = -N_0 N_1 \langle \sigma v \rangle_{01}. \quad (10.39)$$

Since the right-hand side of Equation (10.38) equals that of Equation (10.39), we have

$$\lambda_1(0) = N_1 \langle \sigma v \rangle_{01} \implies \tau_1(0) = \frac{1}{N_1 \langle \sigma v \rangle_{01}}. \quad (10.40)$$

Recalling that the mass fraction relates to the nuclear number density by

$$X_i = \frac{N_i A_i}{\rho N_A}, \quad (10.41)$$

where N_i is the nuclear number density and A_i is the **atomic mass** of species i , we have

$$\tau_1(0) = \frac{1}{\langle \sigma v \rangle_{01}} \cdot \frac{1}{\frac{X_1 \rho N_A}{A_1}} = \left(\rho \frac{X_1}{A_1} N_A \langle \sigma v \rangle_{01} \right)^{-1}, \quad (10.42)$$

and, since $1/\tau_1(0) = \lambda_1(0)$,

$$\lambda_1(0) = \rho \frac{X_1}{A_1} N_A \langle \sigma v \rangle_{01} = N_1 \langle \sigma v \rangle_{01}. \quad (10.43)$$

Notation $X_i = N_i A_i / (\rho N_A)$

In Equation (10.41), A_i is the **atomic mass** of species i (numerically equal to the atomic weight in atomic mass units, or to the molar mass in g/mol) — not to be confused with X_i , on the left-hand side of the same equation, which is the mass fraction. This notation is consistent with the usage already established in earlier chapters [e.g. Equation (8.7), Chapter 8].

10.8 Example: destruction of ^{25}Al

Mean proton-capture time for ^{25}Al

In an astrophysical plasma, the nucleus ^{25}Al can be destroyed by proton capture and transformed into Si, where $^{25}\text{Al}(p, \gamma)^{26}\text{Si}$. Neglecting other processes, determine the mean reaction time, given that

$$\begin{aligned} N_A \langle \sigma v \rangle &= 1.3 \times 10^3 \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}, \\ \rho &= 10^1 \text{ g/cm}^3, \\ X_H &= 0.7. \end{aligned}$$

Using Equation (10.42),

$$\tau_p(^{25}\text{Al}) = \left(\rho \frac{X_H}{A_H} N_A \langle \sigma v \rangle \right)^{-1}. \quad (10.44)$$

Value of ρ used in the calculation

The problem statement gives $\rho = 10^1 \text{ g/cm}^3$, but the calculation shown just below it literally uses 0.1 (that is, 10^{-1}) in place of ρ : “ $\tau_p(^{25}\text{Al}) = (0.1 \cdot 0.7 / 1.0078 \cdot 1.3 \times 10^3)^{-1} = 0.08 \text{ s}$ ”. There is, therefore, a sign inconsistency in the exponent between the statement and the calculation. Recalculating directly with $\rho = 10^{-1} \text{ g/cm}^3$ (the value actually used in the calculation shown), $X_H = 0.7$, $A_H \approx 1.0078 \text{ g/mol}$, and $N_A \langle \sigma v \rangle = 1.3 \times 10^3 \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$,

$$\tau_p(^{25}\text{Al}) = \left(0.1 \times \frac{0.7}{1.0078} \times 1.3 \times 10^3 \right)^{-1} \approx \frac{1}{90.3} \text{ s} \approx 0.011 \text{ s}. \quad (10.45)$$

This recalculation confirms that $\rho = 10^{-1} \text{ g/cm}^3$ (not 10^1) is the value physically consistent with the calculation shown — we adopt this correction. Even so, a small residual discrepancy remains between our recalculation ($\approx 0.011 \text{ s}$) and the final value recorded above (0.08 s): same order of magnitude, but not identical, possibly due to intermediate rounding in the original calculation. Since it is not possible to reconstruct with absolute certainty the exact path to 0.08 s , we flag this residual discrepancy explicitly rather than forcing an exact match.

The cross section, although a parameter that is difficult to determine theoretically, can be measured in the laboratory.

Let us continue analyzing an example with Al. Assuming that ^{25}Al can be created or destroyed by several processes (fusion reactions, photodisintegration, β decay, among others) during its evolution, we would need to introduce all these transitions to define the abundance of this element in the star. Figure 10.3 shows all the reactions possibly involving ^{25}Al , and Figure 10.4 summarizes the general scheme of the four types of transitions used: proton capture (p, γ), alpha-particle capture (α, p), photodisintegration with proton emission (γ, p), and β^+ decay with neutrino emission ($\beta^+ \nu$).

Using the diagram, we can define that all transitions producing ^{25}Al are

$$^{25}\text{Si}(\beta^+ \nu) ^{25}\text{Al}, \quad ^{26}\text{Si}(\gamma, p) ^{25}\text{Al}, \quad ^{24}\text{Mg}(p, \gamma) ^{25}\text{Al}, \quad ^{22}\text{Mg}(\alpha, p) ^{25}\text{Al}, \quad (10.46)$$

while those that destroy it are

$$^{25}\text{Al}(p, \gamma) ^{26}\text{Si}, \quad ^{25}\text{Al}(\alpha, p) ^{28}\text{Si}, \quad ^{25}\text{Al}(\beta^+ \nu) ^{25}\text{Mg}, \quad ^{25}\text{Al}(\gamma, p) ^{24}\text{Mg}. \quad (10.47)$$

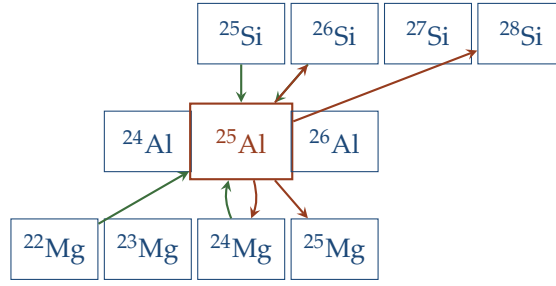


Figure 10.3: Fragment of the chart of nuclides centered on ^{25}Al . Green arrows indicate reactions that produce ^{25}Al ; red arrows indicate reactions that destroy it.

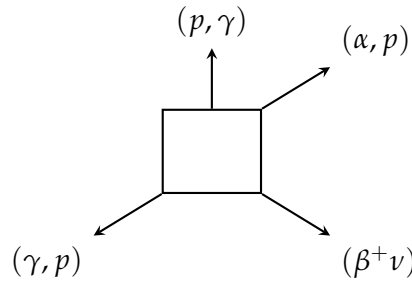


Figure 10.4: General scheme of the four types of transition connecting neighboring species on the chart of nuclides: proton capture, alpha particle capture, photodisintegration with proton emission, and β^+ decay.

Mass balance in $^{22}\text{Mg}(\alpha, p)^{25}\text{Al}$

The mass-number balance of this reaction closes exactly: $22 + 4 - 1 = 25$, consistent with the rest of the diagram, which lists exactly this reaction as one of the four that produce ^{25}Al [Equation (10.46)].

With this, to write the variation of the total density of ^{25}Al nuclei, we must have

$$\begin{aligned} \frac{dN_{^{25}\text{Al}}}{dt} = & N_{^{25}\text{Si}} \lambda_{^{25}\text{Si}(\beta^+\nu)} + N_{^{26}\text{Si}} \lambda_{^{26}\text{Si}(\gamma,p)} + N_H N_{^{24}\text{Mg}} \langle \sigma v \rangle_{^{24}\text{Mg}(p,\gamma)} + N_\alpha N_{^{22}\text{Mg}} \langle \sigma v \rangle_{^{22}\text{Mg}(\alpha,p)} \\ & - N_{^{25}\text{Al}} \lambda_{^{25}\text{Al}(\beta^+\nu)} - N_{^{25}\text{Al}} \lambda_{^{25}\text{Al}(\gamma,p)} - N_H N_{^{25}\text{Al}} \langle \sigma v \rangle_{^{25}\text{Al}(p,\gamma)} - N_\alpha N_{^{25}\text{Al}} \langle \sigma v \rangle_{^{25}\text{Al}(\alpha,p)}, \end{aligned} \quad (10.48)$$

where the capture terms (with a cross section, involving collision with N_H or N_α) describe fusion reactions, and the terms without a cross section describe decay reactions (β decay or photodisintegration). If other creation or annihilation channels existed, they would need to be included in this equation to determine the rate of occurrence and the abundance of this element.

Thus, in general, we can describe the generation rate of any particle by the summation, where

$$\begin{aligned} \frac{dN_i}{dt} = & \left[\sum_{j,k} N_j N_k \langle \sigma v \rangle_{jk \rightarrow i} + \sum_l \lambda_{\beta, l \rightarrow i} N_l + \sum_m \lambda_{\gamma, m \rightarrow i} N_m \right] \\ & - \left[\sum_n N_n N_i \langle \sigma v \rangle_{ni} + \sum_o \lambda_{\beta, i \rightarrow o} N_i + \sum_p \lambda_{\gamma, i \rightarrow p} N_i \right], \end{aligned} \quad (10.49)$$

the **master equation of the reaction network**. The first, second, and third terms of the positive bracket describe the sum of all reactions producing nucleus i from collisions, the second

term describes elements created by β decays, and the third describes photodisintegration; the negative bracket describes the analogous processes that destroy nucleus i .

10.9 Proton-Proton Chain

As mentioned several times, the proton-proton chain is the most fundamental form of stellar energy generation. It follows three branches, as follows:

PP I	PP II	PP III
$p(p, e^+ \nu)d$	${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$	${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$
$d(p, \gamma){}^3\text{He}$	${}^7\text{Be}(e^-, \nu){}^7\text{Li}$	${}^7\text{Be}(p, \gamma){}^8\text{B}$
${}^3\text{He}({}^3\text{He}, 2p)\alpha$	${}^7\text{Li}(p, \alpha)\alpha$	${}^8\text{B}(\beta^+ \nu){}^8\text{Be}$
		${}^8\text{Be}(\alpha)\alpha$

Using the same idea as in the ${}^{25}\text{Al}$ example, we can express the time evolution of the elements, that is, their evolution. We first have, for deuterium (produced by $p(p, e^+ \nu)d$ and destroyed by $d(p, \gamma){}^3\text{He}$, using the notation $H \equiv N_H$, $D \equiv N_D$ to avoid confusion with hydrogen),

$$\frac{dD}{dt} = \Gamma_{pp} - (1 + \delta_{pd}) \Gamma_{dp} = \frac{N_H N_H \langle \sigma v \rangle_{pp}}{1 + \delta_{pp}} - (1 + \delta_{dp}) \frac{N_H N_D \langle \sigma v \rangle_{dp}}{1 + \delta_{dp}}, \quad (10.50)$$

so,

$$\boxed{\frac{dD}{dt} = \frac{H^2}{2} \langle \sigma v \rangle_{pp} - HD \langle \sigma v \rangle_{dp}}, \quad (10.51)$$

where the factor $1/2$ comes from $\delta_{pp} = 1$ (identical particles, $p + p$) and $\delta_{dp} = 0$ ($d \neq p$).

For ${}^3\text{He}$ (produced by the same reaction $d(p, \gamma){}^3\text{He}$ obtained above, and destroyed by ${}^3\text{He}({}^3\text{He}, 2p)\alpha$),

$$\frac{d{}^3\text{He}}{dt} = \Gamma_{dp} - (1 + \delta_{{}^3\text{He}{}^3\text{He}}) \Gamma_{{}^3\text{He}{}^3\text{He}} = HD \langle \sigma v \rangle_{dp} - (1 + \delta_{{}^3\text{He}{}^3\text{He}}) \frac{({}^3\text{He})^2 \langle \sigma v \rangle_{{}^3\text{He}{}^3\text{He}}}{1 + \delta_{{}^3\text{He}{}^3\text{He}}}, \quad (10.52)$$

so,

$$\boxed{\frac{d{}^3\text{He}}{dt} = HD \langle \sigma v \rangle_{dp} - ({}^3\text{He})^2 \langle \sigma v \rangle_{{}^3\text{He}{}^3\text{He}}}. \quad (10.53)$$

We now have the evolution of the particles as a function of their cross section. We can analyze a case in which all the deuterium has been converted into other particles, so that, currently, $dD/dt = 0$. With this, we have

$$\frac{H^2}{2} \langle \sigma v \rangle_{pp} = HD \langle \sigma v \rangle_{dp} \implies \frac{D}{H} = \frac{\langle \sigma v \rangle_{pp}}{2 \langle \sigma v \rangle_{dp}} = \frac{N_A \langle \sigma v \rangle_{pp}}{2 N_A \langle \sigma v \rangle_{dp}}. \quad (10.54)$$

Using Equation (10.42), we can rewrite this ratio in terms of reaction times,

$$\frac{D}{H} = \frac{\tau_p(p)}{2 \tau_p(d)}, \quad (10.55)$$

where $\tau_p(X)$ denotes the **mean lifetime of species X against proton capture** — the same notation and interpretation already introduced in Equation (10.42) (Section 10.7) and used

in the ^{25}Al example: $\tau_p(p)$ is the mean time until a proton reacts with another proton (the reaction $p(p, e^+ \nu)d$ itself), and $\tau_p(d)$ is the mean time until a deuteron captures a proton [$d(p, \gamma)^3\text{He}$]. With this, it is possible to determine the decay time of D.

From this idea, it is possible to construct the evolutionary differential equations for all possible transitions. For example, the PP II, PP III chains, the CNO cycle, and so on. Note that these equations are differential in time, so we can integrate them and know the chemical composition of a given species as a function of time. The energy released in each transition is a function of the cross section. The chemical enrichment of the galaxy can be explained from these reactions.

Exercise proposed by the professor

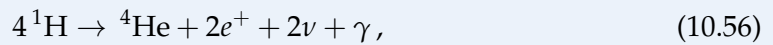
This material ends with an explicit invitation from the professor: “as additional practice exercises, write the differential equations for the CNO I, II, III, and IV cycles and for the PP II and PP III chains”. We preserve this invitation here as a pedagogical record — deliberately **not** solving this exercise in the main body of the text, leaving it as a practice exercise for the reader, exactly as the professor proposed.

10.10 Comparative overview of energy generation processes

As the professor himself pointed out (Section 10.7), this course covers energy generation processes only superficially. We close the chapter with a comparative overview of the three main processes, complementing the pp chain already detailed above with a general view of the CNO cycle and the triple-alpha process — enrichment material, outside the original scope of this material.

The full pp chain: branches and rates

The proton-proton chain detailed above [Section 10.9] has three branches that compete with one another depending on the local abundance of ^3He and ^4He . Under solar core conditions, approximately 86% of the reactions follow PP I, $\sim 14\%$ PP II, and only $\sim 0.1\%$ PP III — the rarest branch, but the most historically relevant, since it is the source of the highest-energy neutrinos (from ^8B), the first to be detected in the pioneering solar neutrino experiments (Acharya et al., 2025). The net reaction of all branches is



releasing $\approx 26.7\text{ MeV}$ (the total Q -value) per helium nucleus formed. A small fraction of this energy ($\sim 2\%$, somewhat larger in the PP II/PP III branches, whose neutrinos carry higher energy) escapes as neutrinos and is not thermally deposited in the star, so that the net energy deposited is $\approx 26.2\text{ MeV}$ per helium nucleus formed. The pp chain dominates energy generation in low-mass main-sequence stars like the Sun, at central temperatures in the range $T \sim 4\text{--}15 \times 10^6\text{ K}$.

The CNO cycle: catalysis and strong T -dependence

The CNO cycle uses carbon, nitrogen, and oxygen as **catalysts** — these nuclei take part in the intermediate reactions but are not net-consumed, only cyclically processed through four branches (CNO I–IV), with CNO I dominant under most stellar conditions. Unlike the pp chain, the CNO cycle has an extremely strong temperature dependence, $\varepsilon_{\text{CNO}} \propto T^{15\text{--}20}$

— already discussed in Chapter 5, where this extreme sensitivity was identified as the physical cause of convective cores in massive stars (energy generation is concentrated in such a small volume that the required radiative gradient violates the Schwarzschild criterion). The CNO cycle dominates energy generation in stars more massive than the Sun, typically with $T \gtrsim 1.7 \times 10^7$ K. A direct and relatively recent measurement of solar neutrinos from the CNO cycle, obtained by the Borexino experiment, confirmed that this cycle is already operating in the present-day Sun — though subdominant, accounting for only $\sim 1\%$ of the solar energy output — opening a new observational avenue for directly measuring the metallicity of the solar core (Borexino Collaboration, 2020).

The triple-alpha process and the Hoyle resonance

The triple-alpha process, $3\ ^4\text{He} \rightarrow\ ^{12}\text{C}$, proceeds in two steps: first, two helium nuclei fuse to form unstable ^8Be (half-life $\sim 10^{-16}$ s); before it can decay back, a small fraction captures a third helium nucleus, forming ^{12}C . This process is only efficient because ^{12}C has a resonant excited state at ≈ 7.65 MeV above the ground state — very close to the combined energy of $^8\text{Be} +\ ^4\text{He}$ at the temperatures typical of helium burning. Fred Hoyle predicted the existence of this resonance in 1954 using a quasi-anthropocentric argument: the observed abundance of carbon in the universe would be impossible to explain without it, since without the resonance the rate of the triple-alpha process would be far too low to produce appreciable amounts of carbon (Hoyle, 1954). The resonance was confirmed experimentally shortly afterward — one of the most cited examples of a successful theoretical prediction guided by astrophysical reasoning in nuclear physics. The triple-alpha process occurs in helium-burning cores, $T \sim 10^8$ K, and is central to the helium flash in red giants and to steady helium burning in horizontal-branch and AGB stars.

Comparative table of the three processes

Process	Reactants	Net product	Typical T	Dominant in
pp chain	$4\ ^1\text{H}$	^4He	$4\text{--}15 \times 10^6$ K	low-mass (Sun-like) stars
CNO cycle	$4\ ^1\text{H}$ (C/N/O catalyze)	^4He	$\gtrsim 1.7 \times 10^7$ K	massive stars ($M \gtrsim 1.3 M_\odot$)
Triple-alpha	$3\ ^4\text{He}$	^{12}C	$\sim 10^8$ K	He-burning cores (RGB/HB/AGB)

The approximate temperature dependence also distinguishes the three processes: $\varepsilon_{\text{pp}} \propto T^4$, $\varepsilon_{\text{CNO}} \propto T^{15-20}$, and $\varepsilon_{3\alpha} \propto T^{40}$ (this last one, among the most extreme dependences in all of stellar astrophysics, reflecting the need for three bodies to collide in rapid succession).

Further Reading and Current References

For the complete, modern treatment of the pp chain, the CNO cycle, and the triple-alpha process, including all branches and associated nuclear uncertainties, see Iliadis (2015). For a recent and comprehensive review specifically of the pp chain and the nuclear physics of hydrogen-burning stars, including the PP I/II/III branching ratios used above, see Acharya et al. (2025). For Hoyle’s original work on nucleosynthesis in massive stars and the prediction of the ^{12}C resonance, see Hoyle (1954). For the first direct detection of solar

neutrinos from the CNO cycle, see Borexino Collaboration (2020).

Chapter 11

Stellar Evolution with MIST Tracks and MESA Simulations: A Didactic Approach

Origin of this chapter

Unlike the previous chapters, this chapter is an **adaptation of a complete, already-published scientific article**, authored by Garcia, Carvalho, Oliveira, Porto, Santos, Santos, Sousa, and Souto (Department of Physics, UFS), entitled “*Evolução estelar com trilhas MIST e simulações MESA em uma abordagem didática para o ensino de Astrofísica*” (in Portuguese). The original scientific text has been preserved faithfully; only the editorial structure (title, article-style abstract, acknowledgments section) was adapted to the book-chapter format, and context, example, and note boxes were added (following the same conventions already used throughout the rest of the book) to connect the content explicitly to the previous chapters. Bibliographic references and figure files were integrated into the project’s unified collection.

This chapter presents a didactic-computational study of stellar evolution using: (i) public evolutionary tracks and isochrones from the MIST 1.2 project (0.3–40 $M_{\odot}$); and (ii) a simulation of a 50 $M_{\odot}$ star with the MESA code (version r23.05.1). The goal is to integrate, in a single reproducible study, the analysis of evolutionary tracks, isochrones, the time evolution of internal parameters, and nucleosynthesis, serving as a quantitative synthesis of everything built in the previous chapters of this book — from the stellar structure equations (Chapter 5) to the degenerate equations of state (Chapter 8), from the Chandrasekhar mass (Chapter 9) to the energy-generation processes (Chapter 10) — now illustrated with public data and a real computational simulation.

11.1 Introduction

The Hertzsprung-Russell (HR) diagram is a fundamental pillar for the study of stellar evolution, allowing observable properties such as luminosity and effective temperature to be correlated with the intrinsic physical parameters of stars. Two theoretical tools are particularly important for its interpretation: evolutionary tracks and isochrones.

The tracks shown in the diagram of Figure 1 correspond to the evolutionary trajectory followed by stars of different solar masses throughout their lives, from the formation of their protostar, before entering the main sequence (MS), through to the final evolutionary stages. This serves to illustrate how their internal structure and surface properties respond to nuclear fusion processes and changes in chemical composition. Isochrones, on the other hand, represent the locus occupied, at a given instant, by stars of the same age and metallicity but different masses, serving as a fundamental theoretical tool for dating stellar clusters, such as globular clusters, by comparing their shape and position in the Hertzsprung-Russell diagram

with the observed distribution of stars in the cluster. The precision of the dating depends crucially on the match between the metallicity of the isochrones and that of the cluster under study. The comparison between theoretical isochrones and observed stellar populations, such as those of clusters, allows us to determine ages and to test stellar evolution models.

A star's evolutionary trajectory is primarily governed by its initial mass and metallicity. Low-mass stars evolve slowly, remaining on the main sequence for billions of years, while massive stars consume their nuclear fuel at an accelerated pace, culminating in explosive or collapsing final stages. Metallicity, in turn, influences the opacity of the stellar envelope and the efficiency of fusion processes, affecting both the position on the HR diagram and the duration of each evolutionary phase.

This work aims to study and graphically visualize the dependence of stellar evolution on mass, age, and metallicity, using the theoretical models of the MIST project (Dotter, 2016) (MESA *Isochrones & Stellar Tracks*)¹. The methodology consists of the analysis of HR diagrams and of effective-temperature-versus-surface-gravity diagrams (Kiel diagrams), generated from evolutionary tracks and isochrones. In addition, a time-dependent analysis of fundamental structural and nuclear parameters — such as radius, temperature and central density, and the energy production via the proton-proton chain, the CNO cycle, and the triple- α process — is carried out for stars of different masses, so that, instead of limiting ourselves to static snapshots, this approach allows us to analyze and track how these physical and chemical quantities vary over the star's lifetime. It is thus possible to identify these critical transitions, such as entry into and exit from the main sequence, the helium *flash*, the correlation between changes in internal structure, and also to observe graphically the production of the heavier elements of the Universe. In addition, we chose to analyze the mass-loss rate in these stars, as an additional parameter of great relevance in massive stars.

The question guiding this work is: how can public evolutionary tracks (MIST) and our own MESA-code simulation be combined to illustrate, quantitatively and didactically, the effects of mass, metallicity, and rotation on stellar evolution and on nucleosynthesis? Through this approach, we seek to develop a quantitative and physically grounded understanding of the processes that govern the stellar life cycle, presenting the numerical results in dialogue with the principles of stellar evolution theory established in the literature. In particular, we follow the formulation presented by Maeder in *Physics, Formation and Evolution of Rotating Stars* (Maeder, 2009), whose theoretical framework provides the physical basis for interpreting the evolution of structure, transport mechanisms, and relevant nuclear processes. It should be emphasized that the contribution of this article is not the derivation of novel astrophysical results, but rather the didactic-computational synthesis: the integration, in a single study guided by public data (MIST) and by our own simulation (MESA), of evolutionary tracks, isochrones, time-dependent structural parameters, and nucleosynthesis, so as to offer a quantitative support material for teaching stellar evolution.

For the data analysis and figure creation, we used the *Python* programming language, together with its libraries. The main tools used were pandas (McKinney, 2010), numpy (Harris et al., 2020), and bokeh² for generating the plots. The pandas library was essential for the manipulation and analysis of the stellar evolutionary data, while numpy provided support for efficient numerical operations.

¹All data were obtained from the (MIST) project: <https://waps.cfa.harvard.edu/MIST/>.

²Bokeh is an interactive visualization library for *Python*, ideal for building dashboards and *web* applications. We chose Bokeh over the traditional matplotlib because of its ability to generate interactive and more aesthetically pleasing visualizations, facilitating the exploration of stellar evolutionary data. (Team, 2018).

We also made use of MESA (*Modules for Experiments in Stellar Astrophysics*), the main simulation code, and Tulips, the (*Python*) package for visualizing the simulations computationally, to compile and build an evolutionary model of a star, taking into account some important parameters such as: luminosity ($L/L_\odot$), temperature (Kelvin), time (Myr), radius ($R_\odot$), and stellar mass ($M_\odot$).

11.2 Evolutionary Trajectories for Low- and High-Mass Stars

Figures 11.1 and 11.2 present the evolutionary trajectories extracted from the MIST project for low-mass ($0.3\text{--}1\ M_\odot$), intermediate-mass ($3\text{--}8\ M_\odot$), and high-mass ($10\text{--}40\ M_\odot$) stars. In the diagrams, the horizontal axis represents the logarithm of the effective temperature, $\log T_{\text{eff}}$, with the scale reversed (higher temperatures to the left), following the HR-diagram convention. The vertical axis shows the logarithm of the bolometric luminosity in solar units, $\log(L/L_\odot)$.

For all curves, ages below $\sim 10^6$ years were excluded — the initial pre-main-sequence (pre-MS) phase, during which the hydrostatic structure has not yet stabilized and internal parameters such as luminosity, radius, and effective temperature vary abruptly, making analysis of the main sequence difficult. This physical criterion is consistent with the pre-processing filter documented in Section 2: for high-mass stars, 10^6 years corresponds to an instant still prior to the ZAMS³; for low-mass stars (for example, $0.5\text{--}1\ M_\odot$, with ZAMS at $\sim 157\text{--}42$ Myr), the remaining pre-MS phase is still represented in the curves and is identified in the corresponding figures. In addition, it should be emphasized that, in a first analysis, the rotation of these stars was disregarded, that is, the evolutionary tracks were obtained with $v/v_{\text{crit}} = 0$.

The data come from the MIST 1.2 project (Dotter, 2016) (*MESA Isochrones & Stellar Tracks*, MESA version r7503). The grid used covers masses from 0.3 to $40\ M_\odot$ at solar metallicity ($[\text{Fe}/\text{H}] = 0.00$, $Z_{\text{init}} = 0.01429$, $Y_{\text{init}} = 0.2703$) and without rotation ($v/v_{\text{crit}} = 0.00$). For the rotational comparisons (Section 2.4), additional tracks with $v/v_{\text{crit}} = 0.40$ for the same masses and metallicity were used; for the metallicity analysis (Section 4.1), grids with $[\text{Fe}/\text{H}] \in \{0.00; +0.02; +0.20\}$ at $v/v_{\text{crit}} = 0.40$.

The `.track.eep` files contain 77 columns; those used in this work were: `star_age` (col. 1, age in years), `log_Teff` (12), `log_L` (7), `log_R` (14), `log_center_T` (46), `log_center_Rho` (47), `center_h1` (52, central hydrogen mass fraction), and `pp/cno/tri_alfa` (cols. 60–62, logarithm of the nuclear luminosities in solar units).

The MIST equivalent-evolutionary-point (EEP) scheme ensures physical correspondence between tracks of different masses. The points relevant to this work are: EEP 1 (line 1) = start of the pre-MS; EEP 2 (line 202) = ZAMS, with $X_c \approx 0.714$ for the whole grid; EEP 4 (line 454) = TAMS⁴, defined by the exhaustion of central hydrogen ($X_c \rightarrow 0$). The adopted main-sequence lifetime is $\Delta t_{\text{MS}} = t_{\text{TAMS}} - t_{\text{ZAMS}}$, computed from the `star_age` field at these two points. The $0.3\ M_\odot$ track does not reach the TAMS in the MIST grid; its available evolution ends while still on the main sequence ($X_c \approx 0.32$). For the time-dependent plots (Section 3), the curves

³From the English *Zero-Age Main Sequence*, or zero-age main sequence. The ZAMS corresponds to the stage at which the star begins stable hydrogen burning in its core and joins the main sequence in the Hertzsprung–Russell diagram.

⁴From the English *Terminal-Age Main Sequence*, or terminal-age main sequence. The TAMS corresponds to the stage at which the star reaches the end of the main sequence, after the hydrogen available for nuclear fusion in its central region has been practically exhausted.

were filtered for ages $> 10^6$ years, so as to exclude the earliest pre-MS phases; the remaining pre-ZAMS phases are identified in the respective figures.

11.2.1 Low-Mass Stars

Figure 11.1 shows trajectories for stars with masses of 0.3, 0.5, 0.8, 1, 3, and 5 $M_{\odot}$. We can observe three distinct regimes:

- **Very low masses:** These stars occupy the lower-right region of the HR diagram, with low luminosities and low effective temperatures. They are fully convective stars, which homogenizes their chemical composition and results in nearly horizontal trajectories (little variation in luminosity) over extremely long timescales, comparable to the age of the Universe. Their evolution is dominated by hydrogen burning via the proton-proton chain, and they have main-sequence lifetimes on the order of trillions of years — a theoretical prediction of stellar evolution models (these times exceed the age of the Universe), with indirect observational support from the existence of essentially unevolved, very-low-mass stars.
- **Solar mass and nearby masses:** These lie on the main sequence at $\log(L/L_{\odot}) \approx 0$ and $\log T_{\text{eff}} \approx 3.75$ (solar star). They show typical evolution: after a few billion years, they exhaust the hydrogen in the core, leave the main sequence (turnoff point), and expand their envelope, moving toward the giant branch (increasing luminosity and decreasing effective temperature). The variation in luminosity is moderate (up to about 2 orders of magnitude) compared with more massive stars.
- **Intermediate masses:** These are more luminous and hotter on the main sequence (positioned to the upper left in the diagram). They evolve quickly (main-sequence lifetimes of hundreds of millions of years) and, after exhausting the central hydrogen, expand significantly, becoming giants with very expanded, cold envelopes. A movement toward the upper right is seen in the diagram, indicating increasing luminosity and surface cooling.

For these stars, we chose not to limit the axes, since the conventionally expected values do not match what we find for this mass range. We therefore adjusted the values for a better visualization of the HR diagram.

11.2.2 High-Mass Stars

Figure 11.2 shows the trajectories for stars of 8, 10, 15, 20, 25, 30, and 40 $M_{\odot}$. These stars exhibit striking characteristics:

- **Main Sequence:** They occupy the upper-left region of the HR diagram, with very high luminosities ($\log(L/L_{\odot}) > 3$) and effective temperatures ($\log T_{\text{eff}} > 4.2$). The main sequence for these masses is very steep, highlighting the mass–luminosity relation ($L \propto M^{3.5}$).
- **Post-Main-Sequence Evolution:** After a short time on the main sequence (from a few million to a few tens of millions of years), these stars evolve into the giant and supergiant region. Their envelopes expand enormously, leading to a large increase in luminosity and expansion of the radius, resulting in a decrease of the effective temperature. They

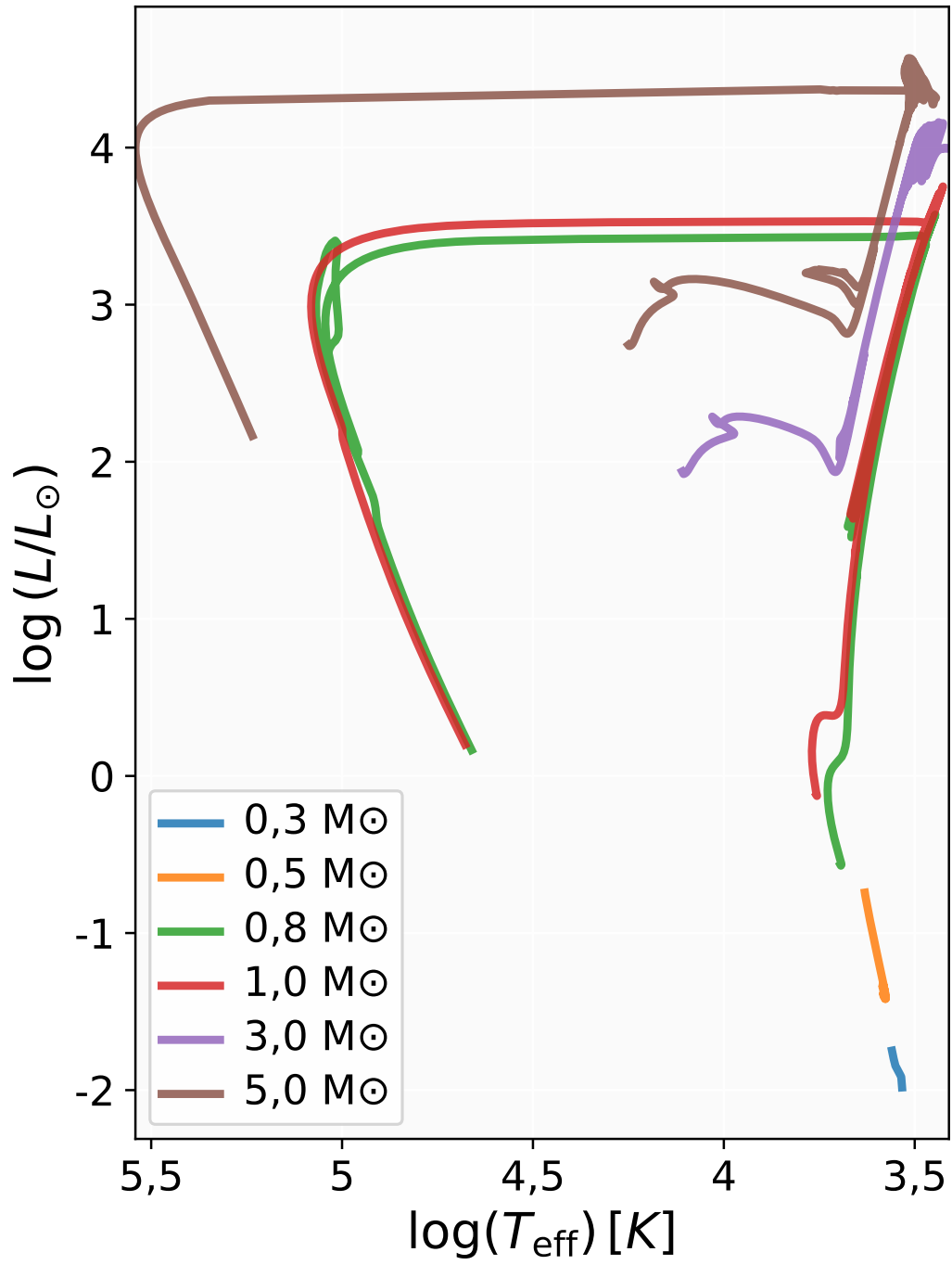


Figure 11.1: HR diagram for low-mass stars ($0.3\text{--}5 M_\odot$).

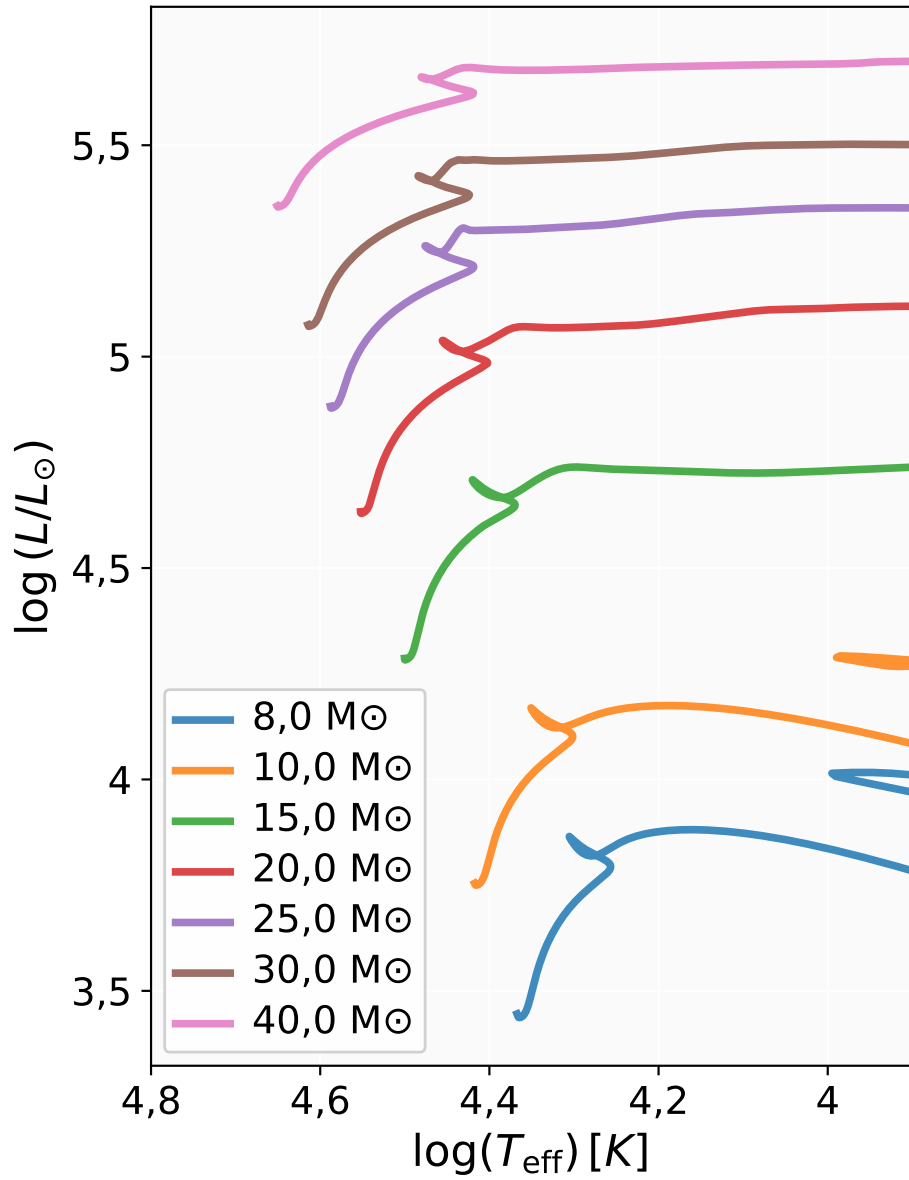


Figure 11.2: HR diagram for high-mass stars (8–40 $M_{\odot}$).

move to the right in the diagram, possibly reaching the red-supergiant region (low temperature, high luminosity) or, in some cases, moving back to the left (*blue loops*) during helium burning.

- **Evolutionary Endpoints:** Massive stars evolve rapidly toward advanced nuclear-burning stages (helium, carbon, etc.) and end their lives as supernovae (Type II, Ib, or Ic), leaving behind black holes or neutron stars. The trajectory in the HR diagram reflects the complex interplay between shell nuclear burning, mass loss through stellar winds (especially for stars between 25 and 30 $M_{\odot}$), and possible eruptions such as luminous blue variables, which can significantly alter the evolution.

11.2.3 Key Points

The evolution of stars in both mass ranges is governed by well-established principles of modern stellar evolution theory. These foundations — the mass–luminosity relation, nuclear lifetime, dependence on structural parameters, and composition effects — are discussed in detail in classic works on stellar structure (Kippenhahn et al., 2012; Hansen et al., 2004).

1. **Mass–Luminosity Relation:** On the main sequence, luminosity increases strongly with mass, approximately following $L \propto M^{\alpha}$, with $\alpha \sim 3–4$ depending on the opacity regime and the nuclear energy law. This behavior emerges directly from the hydrostatic-equilibrium equations and the strong increase of the nuclear rate with central temperature, as presented in Kippenhahn et al. (2012). This relation explains why massive stars are much more luminous and occupy the upper-left part of the HR diagram.
2. **Main-Sequence Lifetime:** The characteristic timescale for hydrogen burning, $t_{\text{MS}} \propto M/L$, decreases rapidly with mass. This implies that massive stars consume their fuel in just a few million years, while low-mass stars — with much lower luminosities — can live for tens of billions of years (Prialnik, 2000).
3. **The Stefan–Boltzmann Law and envelope expansion:** The relation $L = 4\pi R^2 \sigma T_{\text{eff}}^4$ directly connects internal parameters and observables. During post-main-sequence phases, the envelope expands, increasing the radius and shifting the star toward regions of lower effective temperature and higher luminosity.
4. **Opacity, chemical composition, and mass loss:** Metallicity modifies the opacity and, therefore, the temperature gradient in the envelope, shifting evolutionary trajectories in the HR diagram. In massive stars, intense winds — dependent on metallicity — remove outer layers and profoundly alter the structure and evolution (Langer, 2012). The combination of these factors helps explain the diversity of evolutionary trajectories observed, especially in populations with different Z .

In conclusion, the comparison between low- and high-mass stars reveals that the initial mass is the main parameter determining luminosity, lifetime, and evolutionary path, with the initial chemical composition (especially metallicity) being a crucial modifying factor. The HR diagram allows these differences to be visualized directly, functioning as an essential tool for diagnosing stellar populations and reconstructing evolutionary histories.

Table 11.1: ZAMS validation parameters and main-sequence lifetime for the MIST 1.2 models ($[\text{Fe}/\text{H}] = 0.00$, $v/v_{\text{crit}} = 0.00$, $Z_{\text{init}} = 0.01429$, MESA r7503). t_{ZAMS} (EEP 2, line 202) and t_{TAMS} (EEP 4, line 454): cumulative ages since the start of the pre-MS; MS lifetime: $\Delta t = t_{\text{TAMS}} - t_{\text{ZAMS}}$. Remaining parameters evaluated at EEP 2 (ZAMS). “Mec.”: dominant nuclear mechanism at the ZAMS (pp : proton-proton chain; CNO: CNO cycle). The $0.3 M_{\odot}$ track does not reach the TAMS in the MIST grid; its available evolution ends while still on the main sequence ($X_c \approx 0.32$). The tabulated parameters are evaluated at the ZAMS, as for the other masses. Source: `.track.eep` files from the MIST 1.2 project (Dotter, 2016); columns: `star_age` (1), `log_Teff` (12), `log_L` (7), `log_R` (14), `log_center_T` (46), `log_center_Rho` (47), `center_h1` (52), `pp/cno` (60–61).

M_{ini} [$M_{\odot}$]	t_{ZAMS} [yr]	t_{TAMS} [yr]	T_{eff} [K]	$\log(L/L_{\odot})$	$R/R_{\odot}$	T_c [K]	ρ_c [g cm^{-3}]	X_c	Mec.
0.3	4.70×10^8	—	3410	−1.98	0.29	7.57×10^6	99.3	0.715	pp
0.5	1.57×10^8	9.58×10^{10}	3800	−1.40	0.46	9.06×10^6	85.6	0.714	pp
0.8	6.69×10^7	2.24×10^{10}	4940	−0.57	0.71	1.18×10^7	79.4	0.715	pp
1.0	4.19×10^7	9.92×10^9	5710	−0.13	0.88	1.37×10^7	79.0	0.715	pp
3.0	3.48×10^6	3.59×10^8	12800	1.93	1.87	2.49×10^7	45.2	0.714	CNO
5.0	1.00×10^6	1.00×10^8	17750	2.74	2.47	2.84×10^7	24.0	0.714	CNO
8.0	3.16×10^5	3.57×10^7	23240	3.44	3.23	3.16×10^7	13.7	0.714	CNO
10.0	1.87×10^5	2.36×10^7	26110	3.75	3.67	3.31×10^7	10.7	0.714	CNO
15.0	8.59×10^4	1.24×10^7	31650	4.29	4.62	3.57×10^7	7.1	0.714	CNO
25.0	4.04×10^4	6.93×10^6	38720	4.88	6.13	3.88×10^7	4.6	0.714	CNO
40.0	2.39×10^4	4.74×10^6	44740	5.36	7.92	4.13×10^7	3.3	0.714	CNO

11.2.4 Effects of Stellar Rotation on the Evolutionary Trajectories

To complement the analysis of the evolutionary trajectories, additional HR diagrams were generated considering stars with initial rotation $v/v_{\text{crit}} = 0.4$, where v_{crit} is the critical rotation velocity. Figures 11.3 and 11.4 present the results for low- and high-mass stars, respectively, allowing a direct comparison with the previously discussed non-rotating models.

As shown in Tables 11.2 and 11.3, the effect of rotation on the main-sequence lifetime is modest and mass-dependent. The largest consistently measured increases are $\approx 8\%$ for $8 M_{\odot}$ and $\approx 6\%$ for $40 M_{\odot}$, while stars of $0.5\text{--}1 M_{\odot}$ show negligible variation ($\lesssim 0.1\%$).

Physical Mechanisms of Rotation Rotation profoundly modifies the structure and evolution of a star, introducing anisotropies, new forms of internal transport, and changes to the hydrostatic equilibrium. These effects become especially important in intermediate- and high-mass stars, in which the initial rotation is generally high (Maeder, 2009). Below we highlight the central mechanisms involved.

- **Rotational Flattening and the von Zeipel Theorem:** Rapid rotation leads to a flattening of the stellar body due to the increased centrifugal force at the equator. As a consequence, the effective gravity g_{eff} becomes larger at the poles and smaller at the equator. By the von Zeipel theorem (von Zeipel, 1924), in stars whose envelope is dominated by radiative transport, the surface effective temperature is related to the effective gravity by

$$T_{\text{eff}} \propto g_{\text{eff}}^{1/4}, \quad (11.1)$$

implying that the poles are hotter and brighter than the equator — a phenomenon known as gravity darkening. More recent work, such as Espinosa Lara and Rieutord

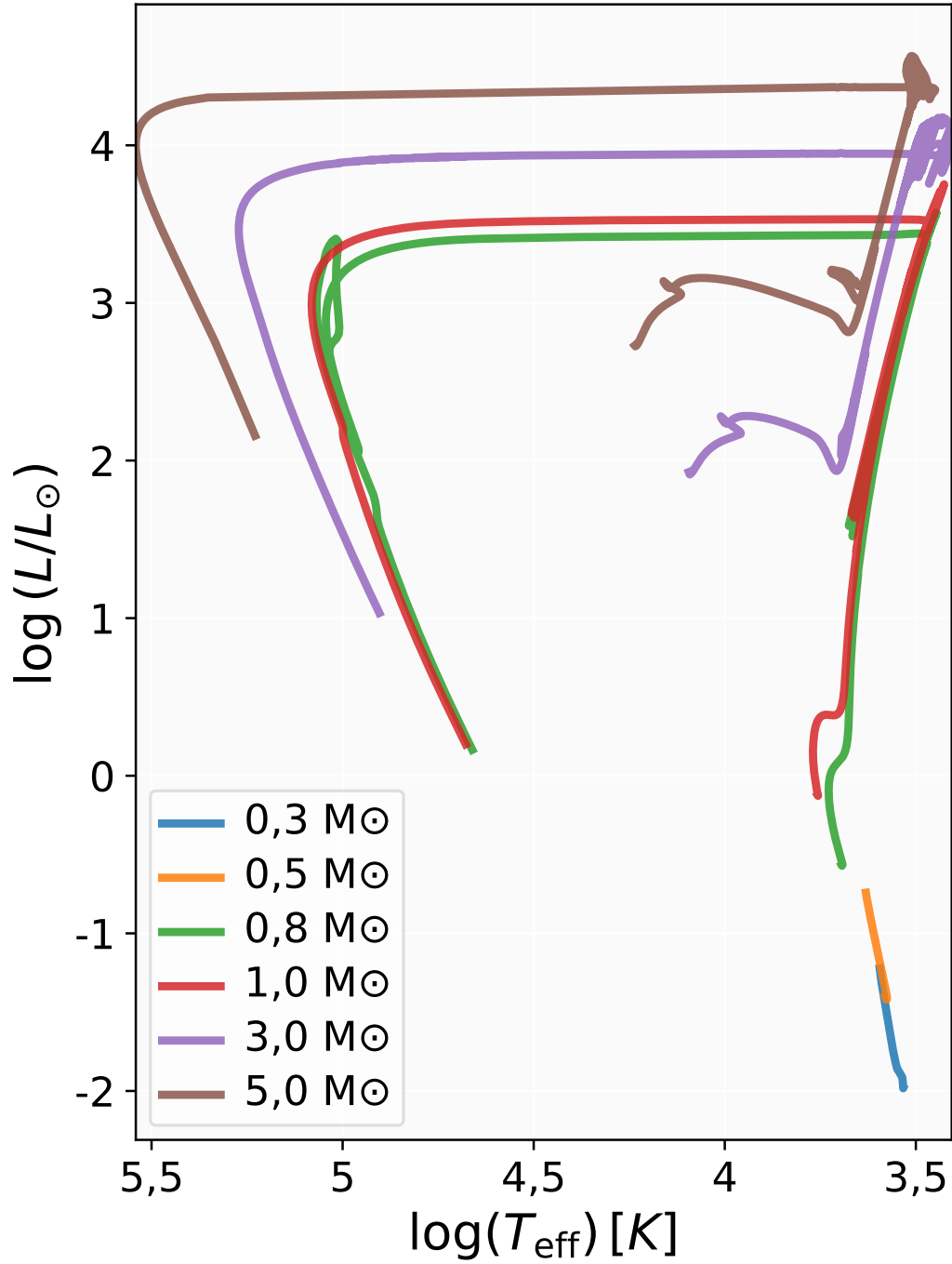


Figure 11.3: HR diagram for low-mass stars ($0.3\text{--}5 M_{\odot}$) with initial rotation $v/v_{\text{crit}} = 0.4$. Compared with Figure 11.1, a shift toward lower effective temperatures is observed, accompanied by adjustments in luminosity, due to the effects of rotational mixing and gravity darkening.

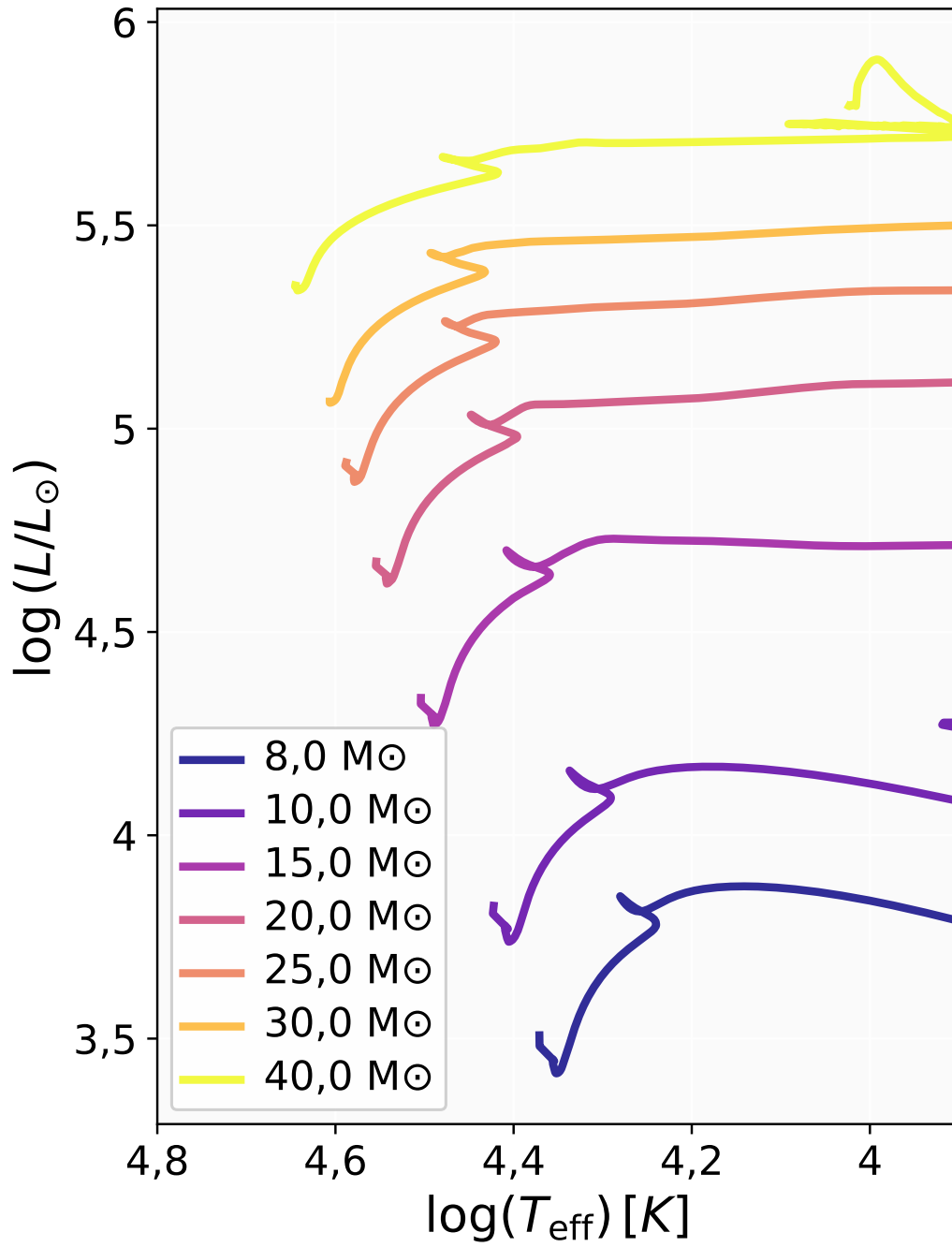


Figure 11.4: HR diagram for high-mass stars (8–40 $M_{\odot}$) with initial rotation $v/v_{\text{crit}} = 0.4$. The effects of rotation are more pronounced than in low-mass stars, with significant changes in the evolutionary trajectory, particularly in the supergiant phases, including modifications to mass loss and possibly to the occurrence of *blue loops*.

Table 11.2: Comparison of main-sequence lifetimes (TAMS, EEP4 of the MIST grid) for low-mass stars: models without and with rotation ($v/v_{\text{crit}} = 0.4$). The $0.3 M_{\odot}$ star was excluded because the non-rotating model does not reach the TAMS in the MIST grid (minimum $X_c \approx 0.32$), preventing a consistent comparison.

Mass	Lifetime (years)		Difference	Variation
	Without Rotation	With Rotation		
$0.5 M_{\odot}$	9.58×10^{10}	9.59×10^{10}	$+1 \times 10^8$	+0.1%
$0.8 M_{\odot}$	2.48×10^{10}	2.48×10^{10}	0	0%
$1 M_{\odot}$	1.15×10^{10}	1.15×10^{10}	0	0%
$3 M_{\odot}$	4.59×10^8	4.68×10^8	$+9 \times 10^6$	+2%
$5 M_{\odot}$	1.16×10^8	1.19×10^8	$+3 \times 10^6$	+2.6%

(2011), shows that the classical von Zeipel relation can be modified by the realistic three-dimensional structure of the outer layers, refining predictions for stars in extreme rotation.

- **Rotational Mixing:** Rotation triggers a series of angular and chemical transport processes. Among the main ones are:
 - meridional circulation, which has its origin in the paradox of von Zeipel (1924): rotation makes it impossible to have simultaneous thermal equilibrium in radiative layers (the equipotential surfaces and isobars do not coincide), so that axisymmetric currents arise to redistribute heat and angular momentum;
 - *shear* instability, produced by differential-velocity gradients;
 - horizontal turbulence, which promotes chemical homogenization.

These mechanisms increase the supply of hydrogen to the core, prolong the main-sequence lifetime, and can profoundly modify the star's surface composition (for example, nitrogen enrichment).

- **Changes in Hydrostatic Equilibrium:** The presence of the centrifugal force reduces the effective gravity, altering the classical hydrostatic equilibrium. The centrifugal term contributes to the required pressure gradient:

$$\frac{dP}{dr} = -\rho g_{\text{eff}} \quad (11.2)$$

with $g_{\text{eff}} = g - \Omega^2 r_{\perp}$. This influences the internal structure, decreasing the central temperature and affecting the nuclear fusion rates. At high masses, rotation can even suppress or delay the formation of a deeper convective core.

Comparative Analysis of the Effects of Rotation on Lifetimes Tables 11.2 and 11.3 present a quantitative comparison between the lifetimes of the models without and with rotation. The analysis reveals complex, mass-dependent patterns:

- **Variable Effect by Mass Range:** Rotation does not affect all stars uniformly. For stars between 0.5 – $1 M_{\odot}$, the differences are negligible (0–0.1%). For 3 – $5 M_{\odot}$, there is a modest

Table 11.3: Comparison of main-sequence lifetimes (TAMS, EEP4 of the MIST grid) for high-mass stars: models without and with rotation ($v/v_{\text{crit}} = 0.4$). Values were recalculated consistently using the equivalent TAMS point for both models.

Mass	Lifetime (years)		Difference	Variation
	Without Rotation	With Rotation		
8 $M_{\odot}$	4.01×10^7	4.32×10^7	$+3.1 \times 10^6$	+7.7%
10 $M_{\odot}$	2.63×10^7	2.70×10^7	$+7 \times 10^5$	+2.7%
15 $M_{\odot}$	1.38×10^7	1.42×10^7	$+4 \times 10^5$	+2.9%
20 $M_{\odot}$	9.62×10^6	9.91×10^6	$+2.9 \times 10^5$	+3%
25 $M_{\odot}$	6.93×10^6	7.19×10^6	$+2.6 \times 10^5$	+3.7%
30 $M_{\odot}$	5.89×10^6	6.14×10^6	$+2.5 \times 10^5$	+4.2%
40 $M_{\odot}$	4.74×10^6	5.03×10^6	$+2.9 \times 10^5$	+6.0%

increase of 2–2.5%. Comparison for $0.3 M_{\odot}$ is not possible because the non-rotating model does not reach the TAMS in the MIST grid.

- **Behavior in Massive Stars:** In high-mass stars, the effects are likewise variable. The $8 M_{\odot}$ star shows the largest relative increase (7.7%), while for masses between 10 – $30 M_{\odot}$ the increase is 2.7–4.2%. For $40 M_{\odot}$, the increase is +6.0%, a result of rotational mixing combined with intensified mass loss.
- **Nonlinearity of the Effects:** The data show that the influence of rotation does not follow a linear trend with mass, suggesting that different physical mechanisms dominate in different mass regimes.

Physical Interpretation of the Observed Patterns The effect of rotation on the main-sequence lifetime is modest across the entire grid considered ($\lesssim 8\%$). It is practically nil in low-mass stars (0.5 – $1 M_{\odot}$), in which the mean-molecular-weight gradient (∇_{μ}) that develops between the core and the envelope suppresses rotational mixing — the same stable stratification that distinguishes the Ledoux criterion from the Schwarzschild criterion (Maeder, 2009). In the remaining masses the increase is small, reaching a maximum of $\sim 8\%$ around $8 M_{\odot}$. This non-monotonic dependence on mass reflects the interplay between rotational mixing, which replenishes fuel in the central regions and prolongs hydrogen burning, and the specific internal structure of each star; a detailed mechanistic modeling of each mass regime is beyond the scope of this didactic work.

Underlying Physical Mechanisms Rotation influences stellar evolution through several interconnected mechanisms:

- **Rotational Mixing:** Transport of chemical elements that alters the composition of the nuclear-burning regions, potentially prolonging specific evolutionary phases.
- **Changes in Internal Structure:** The centrifugal force modifies the density and pressure profiles, affecting the conditions for nuclear fusion.
- **Changes in Opacity:** Surface chemical enrichment alters the opacity of the envelope, influencing the efficiency of energy transport.

Implications for Dating Stellar Populations These results have important consequences for astrophysical dating methods:

1. Precision in Different Populations: Clusters rich in low-mass or very massive stars may have their ages significantly underestimated or overestimated if rotation is neglected.
2. Isochrone Calibration: Isochrones must be calibrated separately for different mass ranges, taking into account the specific effects of rotation in each regime.
3. Interpretation of HR Diagrams: The position of stars in the HR diagram as a function of age is affected non-uniformly, requiring mass-specific corrections.

Limitations and Future Perspectives It is also worth noting that the models presented in this work consider a uniform, constant initial rotation ($v/v_{\text{crit}} = 0.4$), but do not include a rich, detailed evolution of the internal rotation profile or the magnetic coupling between layers. In real observed cases, rotation evolves due to angular-momentum transport and mass loss. Furthermore, metallicity (which was not addressed in this analysis) interacts strongly with rotation, affecting both the opacity and the efficiency of stellar winds. Future studies may show us how different evolutions of rotation and chemical composition influence the trajectory in the HR diagram.

The comparative analysis reveals that stellar rotation has modest, mass-dependent effects on main-sequence lifetimes. While stars of $0.5\text{--}1\text{ M}_{\odot}$ show little sensitivity ($\lesssim 0.1\%$), intermediate- and high-mass stars show increases of up to 7.7% (peaking at $8\text{ M}_{\odot}$). This heterogeneity underscores the importance of considering rotation as a parameter in evolutionary models, particularly for dating stellar populations rich in stars of $5\text{--}15\text{ M}_{\odot}$. The results demonstrate that there is no universal correction for the effects of rotation, and that mass-specific treatment is necessary.

11.3 Time Evolution of the Parameters

Stellar evolution is governed by physical processes intimately dependent on the star's initial mass, which determines everything from lifetimes to internal structure and energetic behavior across the different evolutionary phases. To investigate this relation, we analyze the time trajectory of stars with initial masses of $0.5, 1, 5, 10, 15$, and $25\text{ M}_{\odot}$, from the main sequence to the final stages of life. Using the evolutionary tracks provided by the MIST models, we examine the variation of fundamental parameters — such as radius, effective temperature, nuclear energy production (pp chain, CNO cycle, and triple- α process), central temperature, and central density — with the aim of identifying how each mass regime responds to the different stages of nuclear burning. These diagnostics allow us to connect the internal physics (temperature, density, composition, and transport mechanisms) directly to the observable properties of stars and to the drastic differences in evolutionary timescales between low-, intermediate-, and high-mass stars. In the following subsections, we discuss each parameter in detail, highlighting the structural transitions and the physical mechanisms that define the stellar life cycle in each mass regime.

11.3.1 Radius

The time evolution of the stellar radius shows a behavior strongly dependent on the initial mass, reflecting the structural differences and the differences in nuclear evolutionary timescales between low-, intermediate-, and high-mass stars. Figure 11.5 shows the evolutionary trajectories for stars of 0.5, 1, 5, 10, 15, and 25 $M_{\odot}$ from the start of the main sequence to the final stages of life, considering only ages above 1 Myr to eliminate the pre-main sequence.

For low-mass stars, a slow evolution is observed, dominated by long timescales on the main sequence. The 0.5 $M_{\odot}$ star shows modest radius variations throughout practically its entire life, staying close to $R \approx 0.5\text{--}1 R_{\odot}$ over more than 10^{10} years. The solar case (1 $M_{\odot}$), meanwhile, shows an initial contraction period until reaching the main sequence, followed by a stable phase with a small decrease in radius during central hydrogen burning. At ages close to the end of the main sequence ($\sim 10^{10}$ years), an abrupt increase in radius marks entry into the red-giant phase, reaching values above $R \approx 300 R_{\odot}$.

For intermediate masses, the evolution is much faster: their lifetimes vary between 10^7 and 10^8 years. These stars show a shorter initial contraction phase, quickly reach the main sequence, and, after the exhaustion of central hydrogen, undergo significant envelope expansion, reaching maximum radii on the order of hundreds of solar radii. The peak observed in the curves — particularly pronounced at 5 $M_{\odot}$ — corresponds to the envelope expansion during the red-giant or supergiant phase, when the star's main fuel is helium.

In high-mass stars, the evolutionary timescales are even shorter, ranging from a few million to approximately 10^7 years. Already on the main sequence, these stars show larger radii compared to lower masses. The expansion is also more abrupt and occurs earlier, with maxima exceeding $R \approx 10^3 R_{\odot}$, reaching regimes typical of red supergiants. This extreme expansion reflects the strong dependence of stellar structure on luminosity and on the efficiency of convective transport in the envelope. After this peak, the dynamical collapse of the radius (visible at the end of the curves) marks the beginning of the final phases that precede the formation of the compact remnant.

In summary, Figure 11.5 highlights three essential behaviors:

- (i) the strong dependence of the radius on the initial mass throughout stellar life;
- (ii) the drastic acceleration of evolution for larger masses, reducing the total lifetime by several orders of magnitude;
- (iii) the extreme expansion of the envelope as a universal signature of post-main-sequence phases, with amplitudes that increase with stellar mass.

These results are consistent with the numerical values obtained from the *Python* code, which indicate maximum radii ranging from only a few $R_{\odot}$ for 0.5 $M_{\odot}$ to more than 1200 $R_{\odot}$ for stars of 25 $M_{\odot}$. While low-mass stars show a single, late expansion, intermediate-mass stars show multiple phases of expansion and contraction, and high-mass stars exhibit an early, massive expansion followed by terminal collapse.

Radius in Very Massive Stars

For extremely massive stars (40 – 150 $M_{\odot}$), the radius evolution shows striking characteristics that differ significantly from the pattern observed in lower-mass stars. The analysis reveals that the lifetime decreases drastically with increasing mass, ranging from approximately 4.8 million years for 40 $M_{\odot}$ down to only 3 million years for 150 $M_{\odot}$.

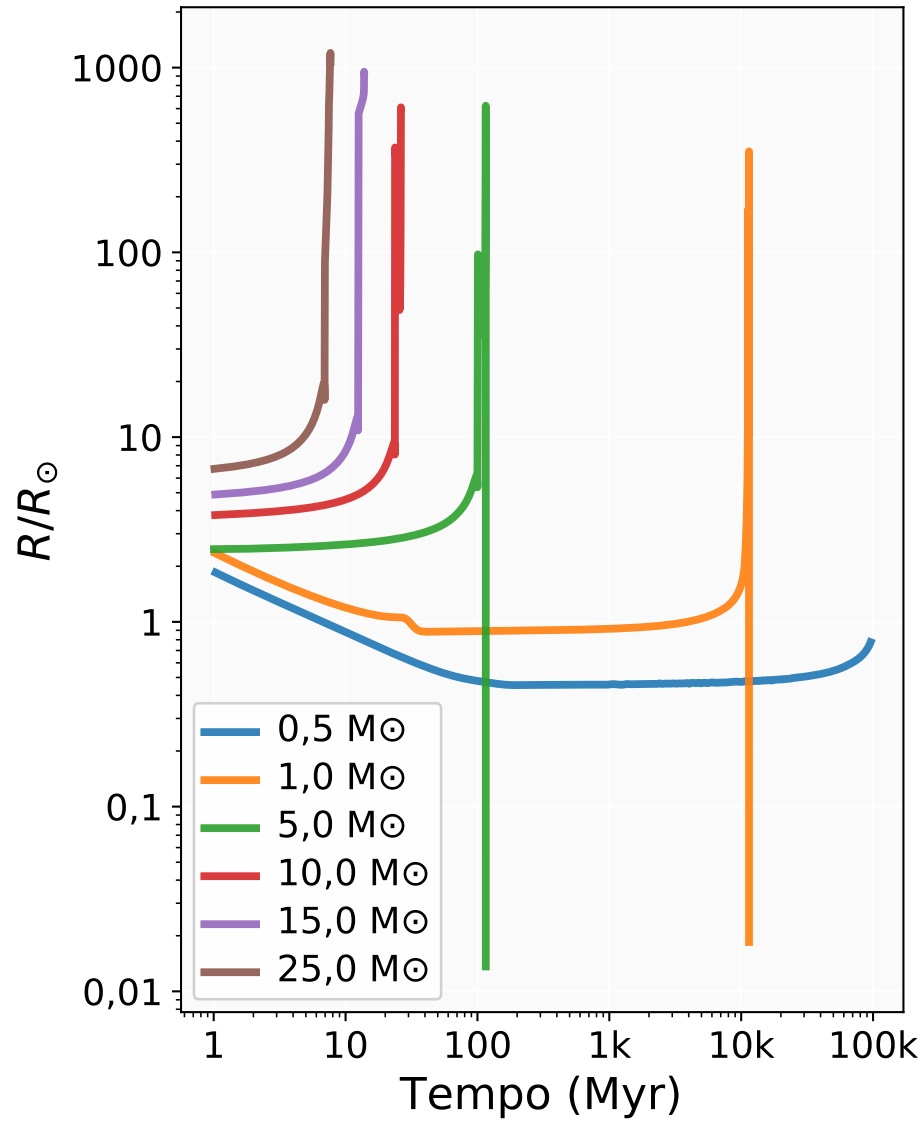


Figure 11.5: Time evolution of the radius for stars between 0.5 and $25 M_{\odot}$, evidencing the strong dependence of the radius on the initial mass and the pronounced expansion in the post-main-sequence phases.

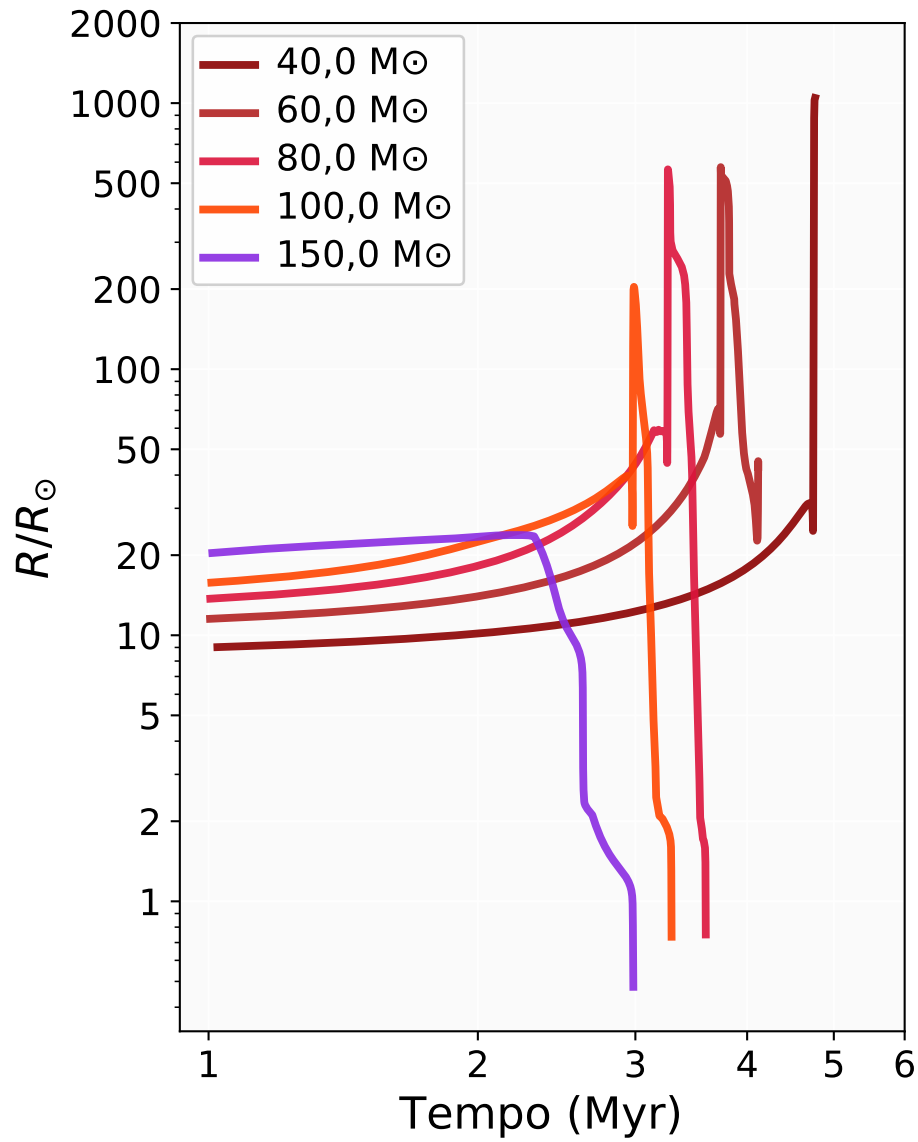


Figure 11.6: Radius evolution for very massive stars (40–150 $M_{\odot}$). The plot highlights the suppression of expansion at high masses due to radiation pressure and intense mass loss through stellar winds.

Notably, a non-monotonic behavior is observed in the maximum expansion as a function of mass. The $40 M_{\odot}$ star reaches the largest absolute expansion, while more massive stars show progressively smaller expansions. The most massive star ($150 M_{\odot}$) shows minimal expansion of only $\approx 24 R_{\odot}$.

This inverse pattern, in which more massive stars expand less, can be attributed to several physical factors: (i) the intense radiation pressure in stars above $60 M_{\odot}$ inhibits envelope expansion; (ii) significant mass losses through strong stellar winds remove material before expansion can occur; (iii) the extremely rapid nuclear burning in hotter cores leads to more abrupt evolutionary transitions. The final collapse of the radius, visible particularly in the $80, 100$, and $150 M_{\odot}$ stars, marks the final stages of nuclear burning before core collapse.

The $150 M_{\odot}$ star shows the most peculiar behavior, practically maintaining its initial radius throughout the evolution, suggesting that stars in this mass range may evolve directly into Wolf-Rayet stars without going through a significant red-supergiant phase, due to the extreme mass-loss rates.

The suppression of expansion in stars above $60 M_{\odot}$ constitutes a striking example of how fundamental physical limits, such as the Eddington limit, impose observable constraints on stellar evolution. This behavior, which contrasts with expectations based on extrapolations from lower masses, highlights the importance of dedicated models for extremely massive stars.

11.3.2 Effective Temperature

Figure 11.7 presents the variation of the logarithm of the effective temperature for stars of $0.5, 1, 5, 10, 15$, and $25 M_{\odot}$ from the start of the main sequence to the final stages.

For low-mass stars, a trajectory marked by stability is observed during the long main-sequence phase, with effective temperatures remaining approximately constant around $\log(T_{\text{eff}}/\text{K}) \approx 3.5\text{--}3.7$. In the case of a $1 M_{\odot}$ star, after the exhaustion of the central hydrogen, the envelope expansion during the red-giant phase causes a marked decline in the effective temperature, which drops below $\log(T_{\text{eff}}/\text{K}) \approx 3.5$, reflecting the surface cooling associated with the increase of the stellar radius.

In addition, a $0.5 M_{\odot}$ star shows temperatures around $T_{\text{eff}} \sim 3000 \text{ K}$, exhibiting a more reddish color and belonging to spectral type M. Its evolutionary process is slow and stable, with a gentle rise in temperature over time due to the low instabilities in the core. A $1 M_{\odot}$ star, on the other hand, shows a typical effective temperature of $\sim 6000 \text{ K}$, with a yellow color. When the central hydrogen is exhausted, the core contracts and the envelope expands, causing the effective temperature to drop abruptly, as observed in the plot.

In intermediate-mass stars, such as those of $5 M_{\odot}$, the evolution is more dynamic. During the main sequence, they show higher temperatures, typically around $T_{\text{eff}} \sim 20000 \text{ K}$, belonging to spectral type B and exhibiting a white color. The exhaustion of hydrogen in the core leads to a rapid collapse, producing a more pronounced and prolonged peak in the effective temperature before the star expands into the red-giant phase. In addition, these stars frequently show characteristic *loops* in the HR diagram during helium burning, associated with thermal instabilities and structural adjustments. At the end of their lives, these stars can show significant thermal fluctuations, with the $5 M_{\odot}$ star showing a final reheating associated with the thermal pulses of the AGB phase (and not with a core collapse, for the same reason discussed above: $5 M_{\odot}$ is below the threshold of $\sim 8 M_{\odot}$ for a core-collapse supernova), while the $10 M_{\odot}$ star maintains lower temperatures throughout the red-supergiant phase until the

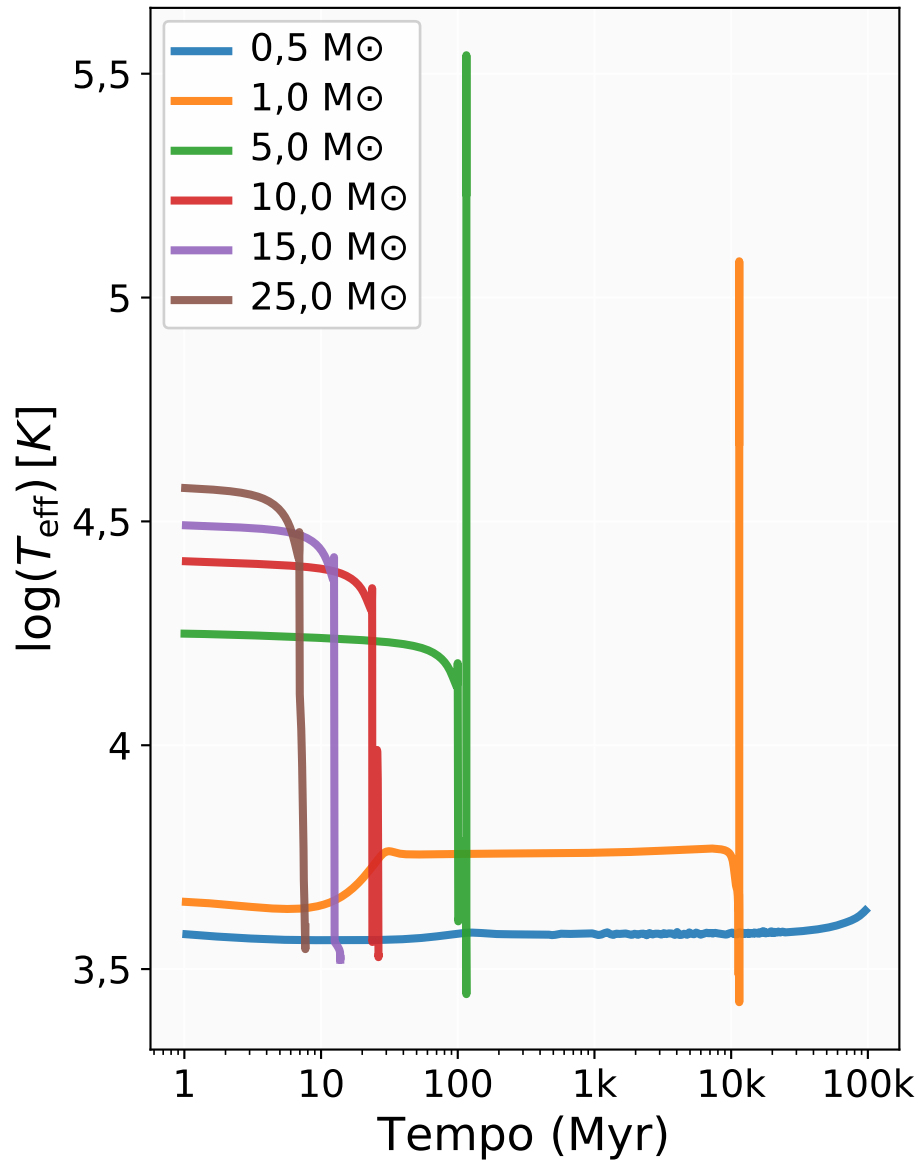


Figure 11.7: Effective temperature as a function of time for stars from 0.5 to 25 $M_{\odot}$, evidencing thermal stability at low masses, pronounced cooling during the giant and supergiant phases, and the abrupt temperature increase in the final stages of evolution.

supernova event.

The effective temperatures start at even higher values for high-mass stars, and remain relatively high during most of their short lives. The cooling during the red-supergiant phase is more pronounced compared with lower-mass stars when related to their initial temperatures. The end of the evolution for these massive stars is marked by an abrupt collapse of the radius with a consequent rapid increase in effective temperature, immediately before the supernova, as a result of the mass loss that exposes hotter internal layers or of the final contraction of the core.

High-mass stars, such as those of 10, 15, and 20 $M_{\odot}$, show the highest effective temperatures, often above 30 000 K, belonging to spectral type O, with a bluish color. They begin their lives with extremely high temperatures and maintain them during most of their short evolutionary timescales. When they leave the main sequence, they expand rapidly and enter the red-supergiant phase, showing more pronounced cooling compared with lower-mass stars, due to the contrast with their extremely high initial temperatures. In the final stages of evolution, the abrupt collapse of the core results in a sudden increase in effective temperature immediately before the supernova, either from the final contraction that heats the outer layers or from the mass loss that exposes hotter internal regions.

11.3.3 Energy-Generation Processes

Nuclear energy production is the fundamental engine of stellar evolution, responsible for determining luminosities, internal structure, and timescales throughout the lives of stars. Different fusion mechanisms become relevant as the central temperature and density evolve, establishing distinct nuclear-burning regimes for each mass range. In low-mass stars, the proton-proton chain dominates practically the entire main sequence, while in more massive stars the CNO cycle quickly takes over energy generation. In more advanced phases, when the central hydrogen is exhausted and temperatures on the order of 10^8 K are reached, the triple- α process becomes the main mechanism responsible for helium fusion. The following subsections discuss each of these processes separately, comparing their efficiency, mass dominance, and time signature in the evolutionary trajectories studied.

All times discussed in this section are measured from the start of the main sequence, allowing direct comparisons between stars of different masses. The pre-main-sequence evolution time is not considered, since it does not contribute significantly to the highlighted nuclear phases.

From the general formalism to the nuclear luminosity curves

The following three subsections are the *observational/time-dependent* counterpart of the entire formalism derived in Chapter 10: the reaction rate $\langle\sigma v\rangle$, the Gamow factor, and the reaction-network master equation, applied there to a single instant and to isolated pairs of nuclei, are exactly what a code such as MESA integrates, nucleus by nucleus and time step by time step, to produce the curves $L_{\text{pp}}(t)$, $L_{\text{CNO}}(t)$, and $L_{3\alpha}(t)$ shown in the following figures. The strong temperature dependence of each process — $\varepsilon_{\text{pp}} \propto T^4$, $\varepsilon_{\text{CNO}} \propto T^{15-20}$, and $\varepsilon_{3\alpha} \propto T^{40}$, already derived and discussed in Chapter 10 — is precisely what quantitatively explains why the transition of dominance among the three processes occurs so abruptly in the evolutionary trajectories discussed below.

Proton-Proton Chain

The proton-proton chain is one of the main nuclear fusion mechanisms through which stars convert hydrogen into helium, and it is particularly dominant in low-mass stars. This process begins with the collision of two protons that, through quantum tunneling across the Coulomb barrier, form a deuterium nucleus:



Subsequently, deuterium fuses with another proton, producing ${}^3\text{He}$, which can then follow different reaction branches. The PP-I branch, dominant at typical temperatures of $\sim 10^7$ K, is completed through the fusion of two ${}^3\text{He}$ nuclei into ${}^4\text{He}$, releasing approximately 26.7 MeV of energy.

Figure 11.8 shows the mass dependence of the PP chain's contribution to nuclear energy generation. For low-mass stars (0.5 and $1 M_\odot$), the plot includes the pre-ZAMS (PMS) phase, since the curves span all ages > 1 Myr without filtering by evolutionary phase. The initial segment with $\log(L_{\text{PP}}/L_\odot) \approx -7$ to -4 corresponds to this pre-ZAMS phase, during which nuclear hydrogen burning is still being established — the PMS of $0.5 M_\odot$ extends to ~ 157 Myr, and that of $1 M_\odot$ to ~ 42 Myr. At the entry into the main sequence (ZAMS), $\log(L_{\text{PP}}/L_\odot) \approx -1.4$ ($0.5 M_\odot$) and ≈ -0.2 ($1 M_\odot$), stabilizing throughout central hydrogen burning. These stars show only a modest decrease in energy production via PP until the end of central hydrogen burning.

For intermediate- and high-mass stars, a distinct behavior is observed: the contribution of the PP chain decays abruptly within the first million years, with $\partial \log(L_{\text{PP}})/\partial \log(t) \propto -M/M_\odot$. This accelerated decline marks the transition to CNO-cycle dominance, which is sensitive to higher-central-temperature environments. The more massive the star, the earlier and more complete this transition is, with the 15 and $25 M_\odot$ stars showing marginal contributions from the PP chain ($\log(L_{\text{PP}}/L_\odot) < -10$) already in the early evolutionary phases.

Although more massive stars exhibit higher initial nuclear luminosities, this does not imply a higher efficiency of the PP chain. This increase reflects only the hotter and denser internal conditions before the complete transition to the CNO cycle, which quickly takes over energy generation.

The decreasing efficiency of the PP chain in more massive stars naturally leads to the second major hydrogen-fusion mechanism: the CNO cycle, whose rates are strongly temperature-dependent and become dominant as the stellar core heats up.

CNO Cycle

The CNO cycle (carbon-nitrogen-oxygen) represents the second main mechanism of hydrogen fusion into helium, becoming dominant in stars with masses above $\sim 1.5 M_\odot$. In the MIST models, the predominant cycle during the main sequence is CNO-I, although CNO-II and CNO-III can contribute marginally at central temperatures above 2.5×10^7 K. These sub-cycles do not alter the overall chemical composition, but they sensitively affect the energy-production rate due to differences in the reaction cross sections. Unlike the proton-proton chain, this process acts as a catalytic cycle in which carbon, nitrogen, and oxygen nuclei participate in the reactions without being consumed, serving as intermediaries for the conversion of four protons into a ${}^4\text{He}$ nucleus.

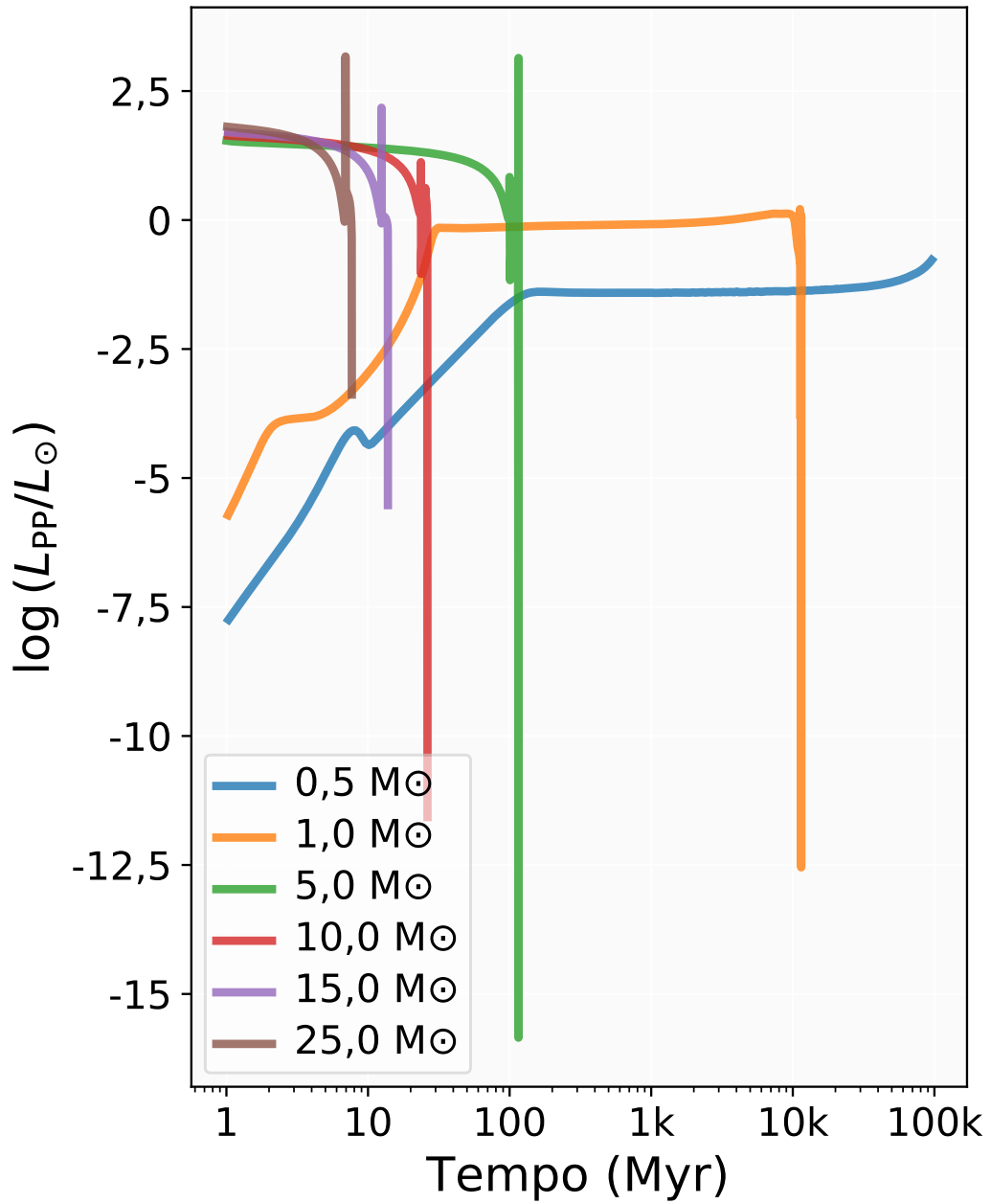


Figure 11.8: Time evolution of the nuclear luminosity from the proton-proton chain for different stellar masses, evidencing its dominance in low-mass stars and its rapid decline in more massive stars due to the transition to the CNO cycle.

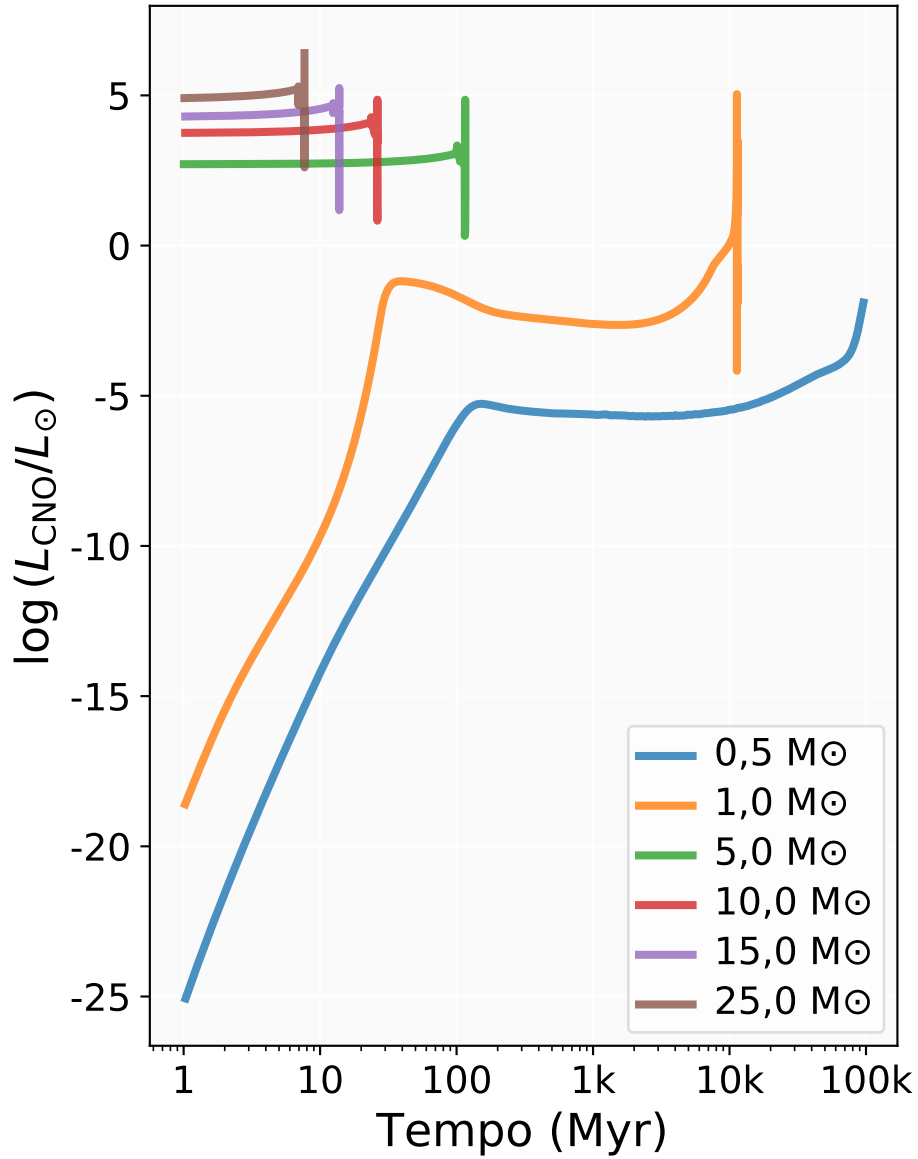


Figure 11.9: Time-dependent contribution of the CNO cycle to nuclear energy generation in stars between 0.5 and 25 $M_{\odot}$, highlighting its extreme sensitivity to temperature and its dominant role on the main sequence of intermediate- and high-mass stars.

Figure 11.9 shows the transition of dominance between the nuclear mechanisms as a function of stellar mass. For the $1 M_{\odot}$ star, the CNO cycle reaches a maximum of $\log(L_{\text{CNO}}/L_{\odot}) = 5.04$, but with a very limited duration (1.6 billion years), being subsequently overtaken by other processes. This behavior reflects the strong temperature dependence of the CNO cycle ($\epsilon_{\text{CNO}} \propto T^{20}$), which makes it extremely sensitive to the thermodynamic conditions of the stellar core.

In intermediate-mass stars, the CNO cycle establishes itself as the dominant mechanism during the main sequence. For $5 M_{\odot}$, it reaches $\log(L_{\text{CNO}}/L_{\odot}) = 4.86$ with a duration of ~ 115 million years, while at $10 M_{\odot}$ it maintains similar values ($\log(L_{\text{CNO}}/L_{\odot}) = 4.86$) for ~ 25 million years. The reduction in the dominance time as mass increases is caused by the acceleration of the evolutionary processes.

For high-mass stars, the CNO cycle reaches its maximum efficiency. The $25 M_{\odot}$ star reaches $\log(L_{\text{CNO}}/L_{\odot}) = 6.41$ – approximately one order of magnitude above intermediate-mass stars – although for only ~ 6.7 million years. This rise in energy production accompanies the increase in central temperatures, which, in massive stars, can exceed 3×10^7 K; ideal conditions for the efficient activation of the cycle.

The abrupt drop in the CNO-cycle contribution at the end of each trajectory marks the exhaustion of the central hydrogen and the transition to more advanced evolutionary phases, in which the burning of helium and heavier elements takes over the dominant role in stellar energy production.

Once hydrogen burning is complete, the contraction of the core raises the temperatures above 10^8 K, inaugurating the helium-burning phase through the triple- α process, which marks the beginning of advanced nuclear evolution.

Triple- α Process

The triple- α process represents a crucial milestone in stellar evolution, marking the transition from hydrogen burning to the fusion of heavier elements. The high rate of this reaction depends critically on the existence of the Hoyle resonant state in ^{12}C , whose energy coincides with the sum of the intermediate states of ^8Be and α . This resonance increases the fusion probability by many orders of magnitude when $T \gtrsim 10^8$ K, allowing carbon to be produced efficiently (Fowler, 1984). This nuclear mechanism, which converts three helium nuclei into carbon through a two-step reaction, becomes significant only when stars reach central temperatures above $\sim 10^8$ K, conditions typical of post-main-sequence phases.

Figure 11.10 reveals distinct temporal patterns for the triple- α process across different mass ranges. For the $1 M_{\odot}$ star, the process manifests late, with $\log(L_{\text{tri_alfa}}/L_{\odot}) = 6.72$ reached only after $\sim 10^{10}$ years of evolution, though with a relatively brief duration of ~ 127 million years. This behavior reflects the slow evolutionary progression of low-mass stars toward the red-giant phases, in which the conditions necessary for helium ignition are finally reached.

In intermediate-mass stars, the triple- α process shows earlier and energetically more significant activation. The $5 M_{\odot}$ star begins helium burning via triple- α in the core-helium-burning phase (CHeB, phase 3, ~ 101 – 115 Myr), with $\log(L_{\text{tri_alfa}}/L_{\odot}) = 6.71$ reached over ~ 15 million years in total (triple- α active from ~ 100.7 to ~ 115.6 Myr). It is important to note that this maximum value *does not* occur during the CHeB, but rather during the thermal pulses of the TPAGB phase (*Thermal Pulse Asymptotic Giant Branch*, phase 5, ~ 115.6 Myr), when $\rho_c \approx 1.25 \times 10^7 \text{ g cm}^{-3}$, consistent with the partial degeneracy of the C/O core at

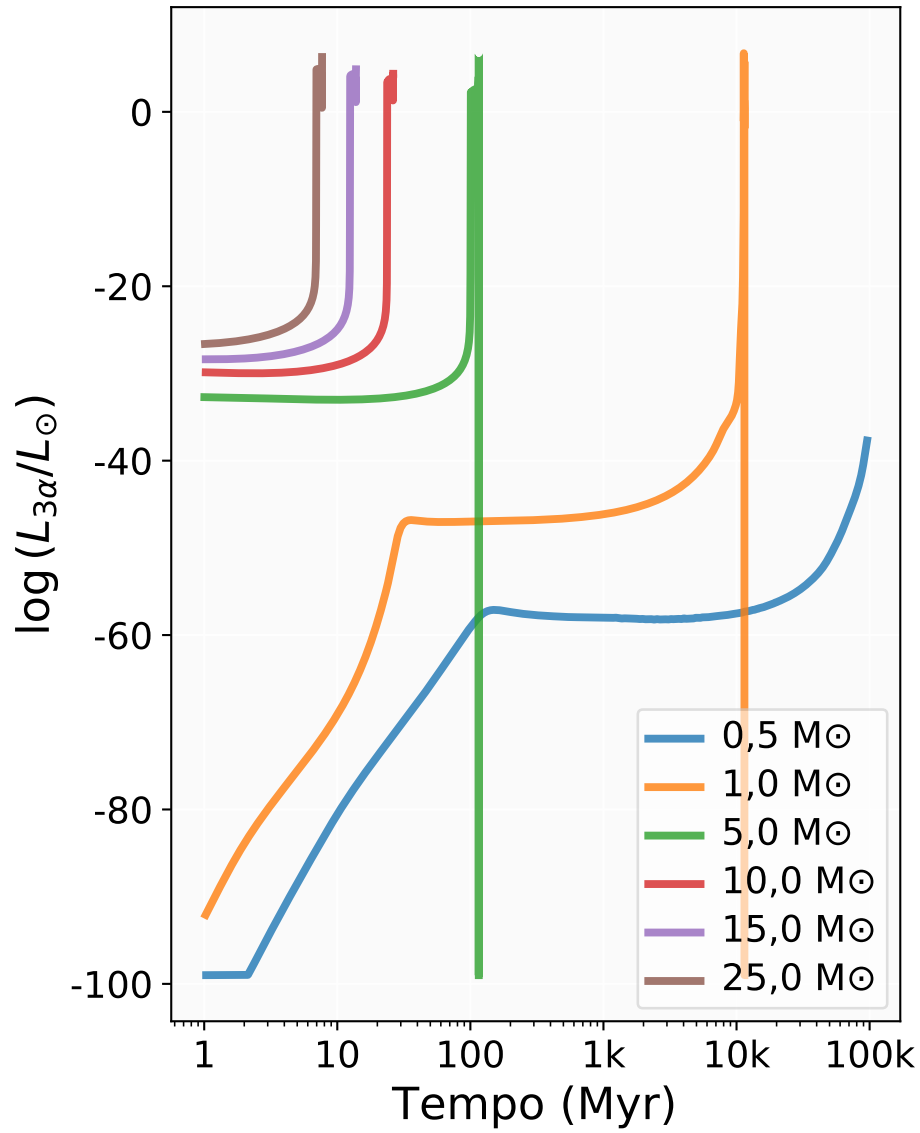


Figure 11.10: Time-dependent variation of the nuclear luminosity associated with the triple- α process, showing its late activation in low-mass stars and its rapid intensification and short duration in intermediate- and high-mass stars during helium burning.

this final stage of the star's life. Notably, in this mass range the triple- α process becomes mechanistically dominant, surpassing both the PP chain and the CNO cycle.

For high-mass stars, the triple- α process shows an even more accelerated evolution. The $25 M_{\odot}$ star activates helium burning in only ~ 6 million years, reaching $\log(L_{\text{tri_alfa}}/L_{\odot}) = 6.29$ with a duration of only ~ 0.7 million years. The drastic reduction in the characteristic timescales accompanies the exponential increase of central temperatures and nuclear consumption rates in massive stars.

It is important to note that, while in $5 M_{\odot}$ stars the triple- α process constitutes the primary mechanism, in stars of $10 M_{\odot}$ and above it is superseded by the CNO cycle during the main sequence, indicating that CNO hydrogen burning remains competitive even in the early helium-burning phases of massive stars.

The abrupt peak followed by rapid decay in the triple- α curves for all masses demonstrates the transient nature of this process, which rapidly consumes the available helium fuel before the ignition of heavier elements or the final collapse of the core, thus marking a critical inflection point in the life of stars.

Together, the evolution of the relative contributions of the PP chain, the CNO cycle, and the triple- α process shows us how central temperature, chemical composition, and mass determine the hierarchy of nuclear mechanisms throughout the lives of stars.

11.3.4 Core Temperature

Low-mass stars (0.5 and $1.0 M_{\odot}$) show a very steady evolution of the central temperature over $\sim 10^{10}$ years. Their highly opaque, partially degenerate cores prevent aggressive contractions, resulting in a smooth increase in T_c until the end of the main sequence. For these stars, the central temperatures remain relatively moderate during most of the main sequence, typically at $T_c \sim (5\text{--}15) \times 10^6$ K. This stability stems from the weak temperature dependence of the proton-proton chain ($\epsilon_{\text{PP}} \propto T^4$), which confers self-regulation on the core. Only in the late phases, when hydrogen is exhausted and gravitational contraction intensifies over an already partially degenerate core, is a rapid temperature increase observed, reaching values close to 10^8 K, necessary for helium ignition via the triple- α process. In stars of $\sim 1 M_{\odot}$, this ignition occurs explosively, in the so-called helium *flash*.

In intermediate-mass stars, the central temperatures start at higher values ($\sim 2\text{--}4 \times 10^7$ K) and increase more sharply throughout the evolution. In these masses, hydrogen burning quickly transitions from a mixed PP-CNO regime to CNO-cycle dominance, whose extreme temperature sensitivity ($\epsilon_{\text{CNO}} \propto T^{15\text{--}20}$) causes a much faster rise of T_c . Since the cores of these stars do not yet reach significant degeneracy levels during hydrogen exhaustion, the subsequent gravitational contraction is efficient: small reductions in the central radius rapidly raise the temperature. Thus, helium ignition occurs in weakly or non-degenerate cores, avoiding the helium *flash* typical of low-mass stars and allowing a smooth transition to helium burning in hydrostatic equilibrium.

The extreme temperature values found in high-mass stars allow the sequential burning of helium, carbon, neon, oxygen, and silicon, since their central temperatures already begin above 5×10^7 K and rapidly reach $T_c > 3 \times 10^8$ K within just a few million years. In these cases, degeneracy plays practically no relevant structural role, since the high temperatures and convective cores keep the gas firmly in the non-degenerate regime. The continuous escalation of T_c reflects both the intense gravitational contraction needed to sustain high luminosities and the activation of nuclear reactions with extremely high power-law temperature dependences.

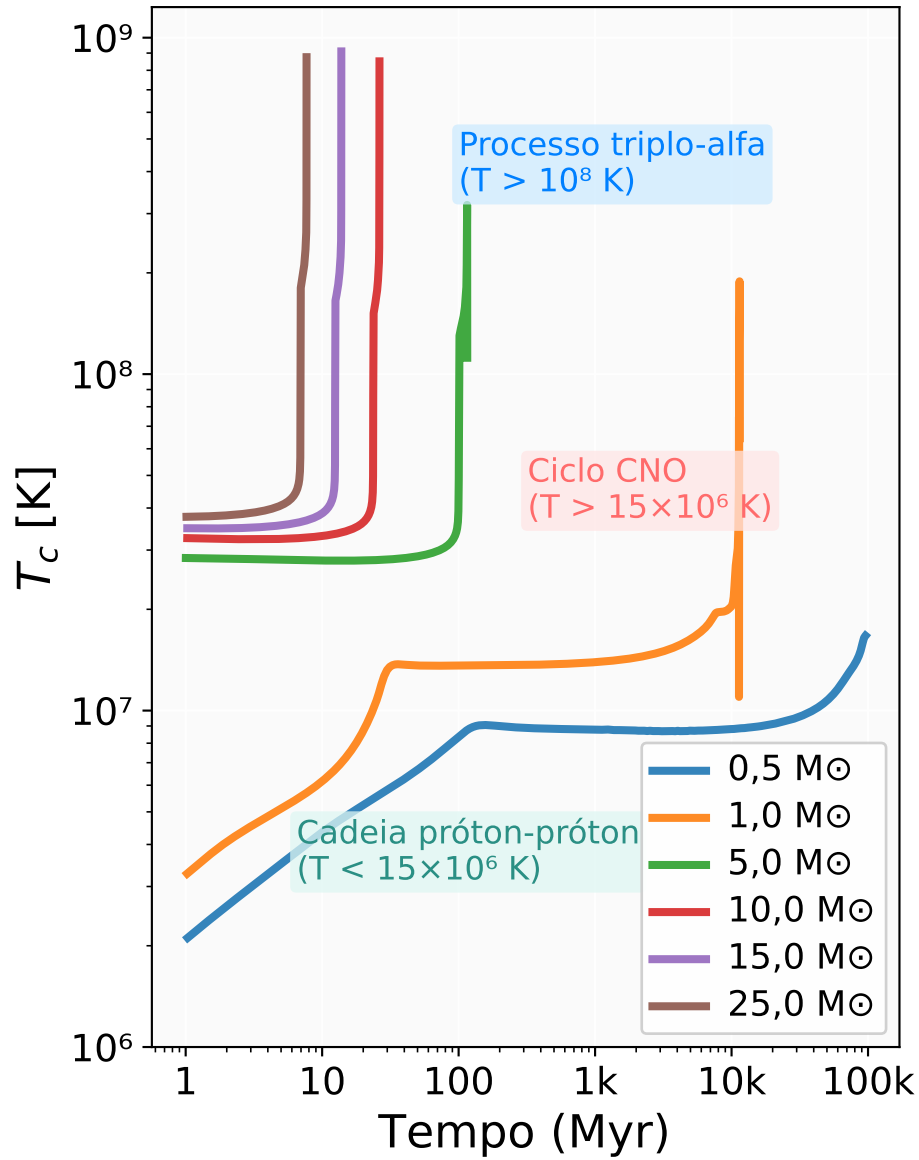


Figure 11.11: Time evolution of the central temperature for stars of different masses, from the main sequence to advanced nuclear-burning phases. The increasing progression of T_c reflects the gravitational contraction of the core as fuels are exhausted, clearly revealing the transitions between the dominant fusion regimes for each mass range.

In general, it is observed that the higher the star's initial mass, the higher the central temperature at all evolutionary stages. The discontinuities and slope changes in the curves highlight critical transitions, such as the end of hydrogen burning, helium ignition, and the onset of advanced fusion cycles. Thus, the evolution of T_c not only clearly distinguishes the degenerate and non-degenerate regimes but also makes central temperature one of the most sensitive parameters for identifying a star's internal evolutionary stage.

11.3.5 Core Density

The central density (ρ_c) is one of the internal parameters most sensitive to the balance between gravitational contraction and the pressure-support mechanisms active in the stellar interior. Figure 11.12 shows non-monotonic behaviors throughout the evolution and among different initial masses, revealing that the relationship between mass and central density is more complex than a direct dependence on gravity. In particular, the presence or absence of electron degeneracy plays a decisive role in the shape of these curves.

For low-mass stars, such as 0.5 and $1 M_\odot$, the plot covers the pre-ZAMS phase (PMS, ages > 1 Myr): at the start of the curves (~ 1 Myr), $\rho_c \approx 0.7 \text{ g cm}^{-3}$, growing to $\rho_c \approx 79\text{--}86 \text{ g cm}^{-3}$ at the entry into the main sequence (ZAMS — ~ 42 Myr for $1 M_\odot$ and ~ 157 Myr for $0.5 M_\odot$), consistent with cores dominated by thermal pressure during the main sequence. In stars of $0.5 M_\odot$, the slow evolution and low luminosity keep the core close to the partially degenerate regime, ensuring great structural stability: hydrogen burning proceeds smoothly and in a regulated manner, without abrupt contractions. In stars of $1 M_\odot$, although the core remains non-degenerate during most of the main sequence, the contraction after hydrogen exhaustion leads to the formation of a degenerate helium core. This growing degeneracy is responsible for the many-orders-of-magnitude increase in density, culminating in the helium *flash* when ρ_c reaches values high enough to allow explosive ignition.

The $5 M_\odot$ star shows the most notable behavior among all the masses analyzed, presenting the highest central-density peak, with $\rho_c \sim 1.25 \times 10^7 \text{ g cm}^{-3}$ during the thermal pulses of the TPAGB phase (~ 115.6 Myr). The absence of full convection and the high opacity amplify the compression of the C/O core, producing the abrupt peak observed. Intermediate-mass stars are particularly sensitive to structural reorganizations that appear as intense variations in central density.

The peak $\rho_c \approx 1.25 \times 10^7 \text{ g cm}^{-3}$ during the TPAGB thermal pulses (~ 115.6 Myr) reflects the electron degeneracy of the C/O core of the $5 M_\odot$ star. It is worth noting that this C/O core forms at the end of *non-degenerate* helium burning in the CHeB phase ($\sim 101\text{--}115$ Myr) and only becomes electrically degenerate in the subsequent contraction, already in the AGB and TPAGB phases — not during core helium burning itself. The density reached, although high, remains far below the conditions for non-degenerate carbon ignition ($\rho_c \gtrsim 2 \times 10^8 \text{ g cm}^{-3}$); and the mass of this core remains below the Chandrasekhar limit ($M_{\text{Ch}} \approx 1.4 M_\odot$), as expected from the empirical initial–final mass relation for progenitors in this mass range (Salaris and Cassisi, 2005), below which the electron-degeneracy pressure is sufficient to sustain the structure without gravitational collapse.

The star ejects the envelope during the TPAGB and leaves this core as a remnant: a C/O white dwarf. The contrast with low-mass stars ($M \lesssim 2 M_\odot$) is instructive: in those stars the *helium* core is already degenerate *before* reaching the ignition temperature ($\sim 10^8$ K), so that burning begins explosively — the *helium flash*. In the $5 M_\odot$ star this does not occur: the helium core ignites non-degenerately, with core helium burning (CHeB) proceeding stably; only the

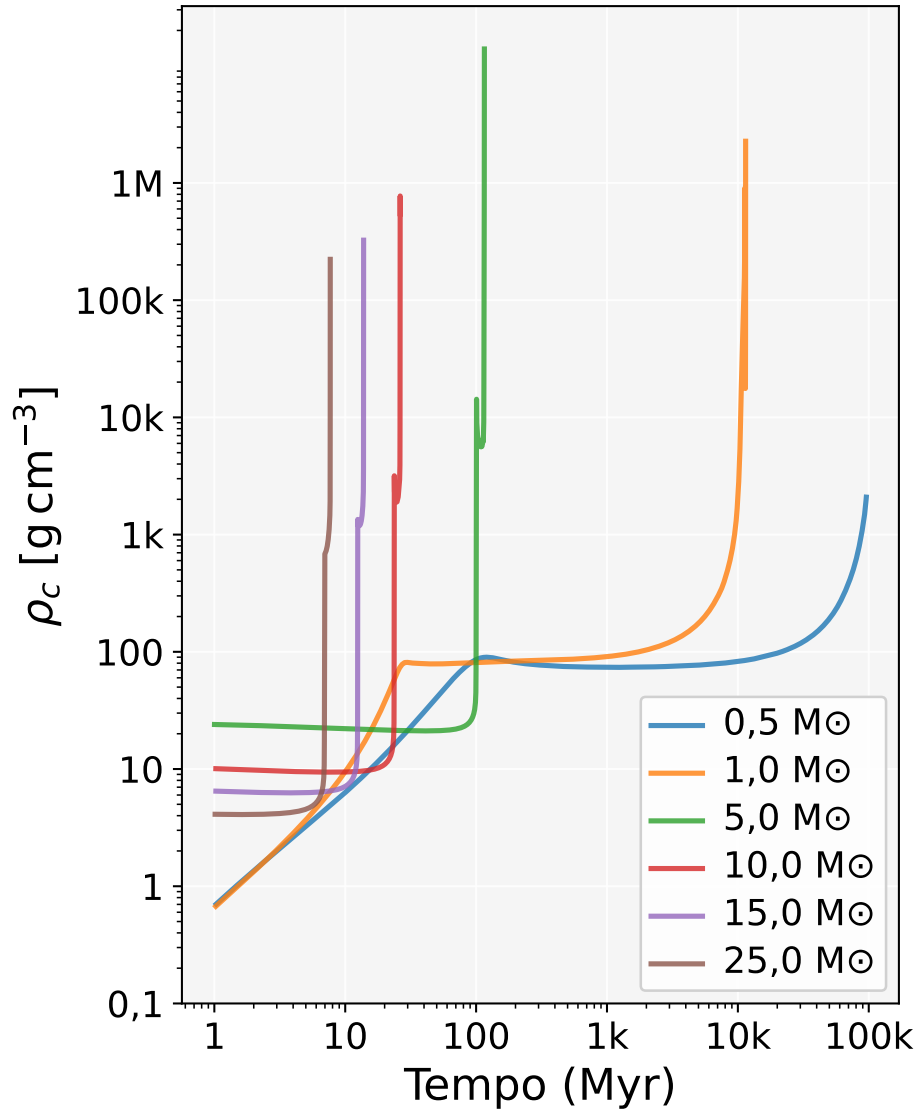


Figure 11.12: Time evolution of the central density for stars of different masses. The growth of ρ_c reflects the gravitational contraction of the core and the transition between support regimes by thermal pressure, electron degeneracy, and, in massive stars, radiation pressure.

resulting C/O residue reaches the degenerate regime during the post-CHeB contraction.

Connecting with Chapters 7 and 8: from degeneracy to the Chandrasekhar mass

This discussion of the degenerate C/O core of the $5 M_{\odot}$ star is a direct application of two central results already derived in this book. The pressure that sustains this core — even at the practically constant temperature of the TPAGB phase, while the density rises by six orders of magnitude — is precisely the **non-relativistic electron-degeneracy pressure**, $P \propto \rho^{5/3}$, derived from scratch from the quantum mechanics of a Fermi gas in Chapter 8. And the mass limit cited above, $M_{\text{Ch}} \approx 1.4 M_{\odot}$, is precisely the **Chandrasekhar mass** derived in Chapter 9 from Chandrasekhar’s differential equation (the relativistic generalization of the Lane–Emden equation). The fact that this star’s residual core ends up *below* this limit is what ensures that it ends its life as a stable white dwarf, and not as a supernova — the same physics, with the same number, appears here as a natural consequence of a complete time-dependent evolutionary calculation.

For massive stars ($10\text{--}25 M_{\odot}$), the trend reverses: their maximum central densities are lower than those of intermediate-mass stars, reaching only $\sim 10^5\text{--}10^6 \text{ g cm}^{-3}$. This behavior results from the dominant role of radiation pressure, which keeps the cores more expanded and hotter, avoiding electron degeneracy during most of the evolution. Thus, even at very high temperatures, their cores do not become compact enough to reach extreme densities until the final phases. As they progress through the sequential burning of helium, carbon, neon, oxygen, and silicon, the density increases gradually, but remains limited by the intense thermal support. Only at the end of life, when the iron core becomes highly compact, does the density grow to the point where degenerate electrons begin to be captured by protons, removing the degeneracy support itself and triggering the free-fall gravitational collapse that precedes the supernova.

In addition, the maximum central densities in these stars are lower than those of intermediate masses, with values of $\sim 7.7 \times 10^5 \text{ g cm}^{-3}$ ($10 M_{\odot}$), $\sim 3.3 \times 10^5 \text{ g cm}^{-3}$ ($15 M_{\odot}$), and $\sim 2.2 \times 10^5 \text{ g cm}^{-3}$ ($25 M_{\odot}$). This inverse trend with mass reflects the dominant role of radiation pressure in these objects, which keeps the cores hotter and less compact. In these extreme regimes, thermal- and radiation-pressure support prevents the core from reaching high degeneracy levels during most of the evolution, resulting in relatively lower central densities.

The comparison between the time-dependent trajectories reveals three patterns seen in Figure 11.12: (i) Low-mass stars show a slow and then accelerated increase in ρ_c , reflecting the progressive approach to electron degeneracy; (ii) Intermediate-mass stars show abrupt density peaks, a direct result of thermonuclear ignitions sensitive to internal conditions and to the degree of partial degeneracy; (iii) Massive stars show smoother evolutions, dominated by radiation pressure, with relatively low densities until the final collapse.

These results demonstrate that central density does not simply grow with stellar mass. Instead, it reflects the competition between thermal pressure, electron degeneracy, and radiation pressure, as well as the impact of the nuclear transitions that restructure the core throughout the evolution.

11.3.6 Convective and Radiative Zones

The internal structure of a star results from the balance between radiative and convective energy transport. In practice, the Schwarzschild criterion is used: a region is convective if the

radiative gradient exceeds the adiabatic gradient, $\nabla_{\text{rad}} > \nabla_{\text{ad}}$, which implies that photon transport cannot carry the energy flux and matter mobilizes to transport heat (convection). Since

$$\nabla_{\text{rad}} \propto \frac{\kappa L}{m T^4}, \quad (11.4)$$

an increase in luminosity L , in opacity κ , or a decrease in the interior mass m favors convection. These physical principles explain the trends observed in Figures 11.13 and 11.14. (Kippenhahn et al., 2012; Hansen et al., 2004; Maeder, 2009; Paxton et al., 2011).

The same Schwarzschild criterion from Chapter 5

∇_{rad} and ∇_{ad} here are exactly the same logarithmic gradients $\nabla \equiv d \ln T / d \ln P$ defined and derived in detail in Chapter 5 — the Schwarzschild criterion $\nabla_{\text{rad}} > \nabla_{\text{ad}}$ cited here is literally the same criterion (with the same notation) used to explain the convective zone of the Sun in that chapter. The mean-molecular-weight gradient ∇_{μ} mentioned earlier in Section 11.3.6 (rotation effects) also connects directly: when the chemical composition varies with depth, the more general **Ledoux** criterion (which adds to ∇_{ad} an extra term proportional to ∇_{μ} , also derived in Chapter 5) replaces the Schwarzschild criterion, which is the physical reason why μ gradients can suppress or modify convective regions.

Low- and intermediate-mass stars (0.5–10 $M_{\odot}$)

Low-mass stars show a pattern dominated by convective envelopes during most of the evolution. For the case of 0.5 $M_{\odot}$, this deep convective region arises from the high opacity and partial ionization of hydrogen, which favor the inefficiency of radiative transport in the surface layers (Chabrier and Baraffe, 2000; Kippenhahn et al., 2012). The core remains essentially radiative, reflecting still-moderate central temperatures and smoother gradients.

As the mass increases (1–5 $M_{\odot}$), the envelope becomes progressively less deep. The increase in central temperature reduces the opacity and makes radiative transport more efficient, allowing convection to be restricted to the outer regions. The 1 $M_{\odot}$ star represents this classical regime: a radiative core during the main sequence and a convective surface envelope, which deepens in the giant phase during the *first dredge-up* (Paxton et al., 2011). At 5–10 $M_{\odot}$, the growing efficiency of radiative transport favors a smaller convective envelope, while internal convective regions can arise associated with composition gradients and helium ignition.

Massive stars (20–50 $M_{\odot}$)

At high masses, the internal structure is typically dominated by extensive convective cores. The figures show growing internal convective fractions (approx.: $\sim 7\%$ at 20 $M_{\odot}$, $\sim 17\%$ at 30 $M_{\odot}$, $\sim 30\%$ at 50 $M_{\odot}$), although non-monotonic variations occur (e.g., a reduction at 40 $M_{\odot}$) due to the interplay between mass loss, composition gradients, and rotation. The physical reason is the extreme sensitivity of the CNO cycle to temperature ($\epsilon_{\text{CNO}} \propto T^{15-20}$): higher central temperatures generate steep radiative gradients that only convection can remove, forming large convective cores. (Maeder, 2009; Hirschi et al., 2005)

In addition, the radiative winds (visible in the total-mass line in the figures) remove outer layers, alter μ gradients, and can expose interior layers, modifying both the extent of the convective zones and the yields (*final nucleosynthetic yields*, i.e., the amounts of each element/isotope produced and ejected by the star). These interactions explain the rapid

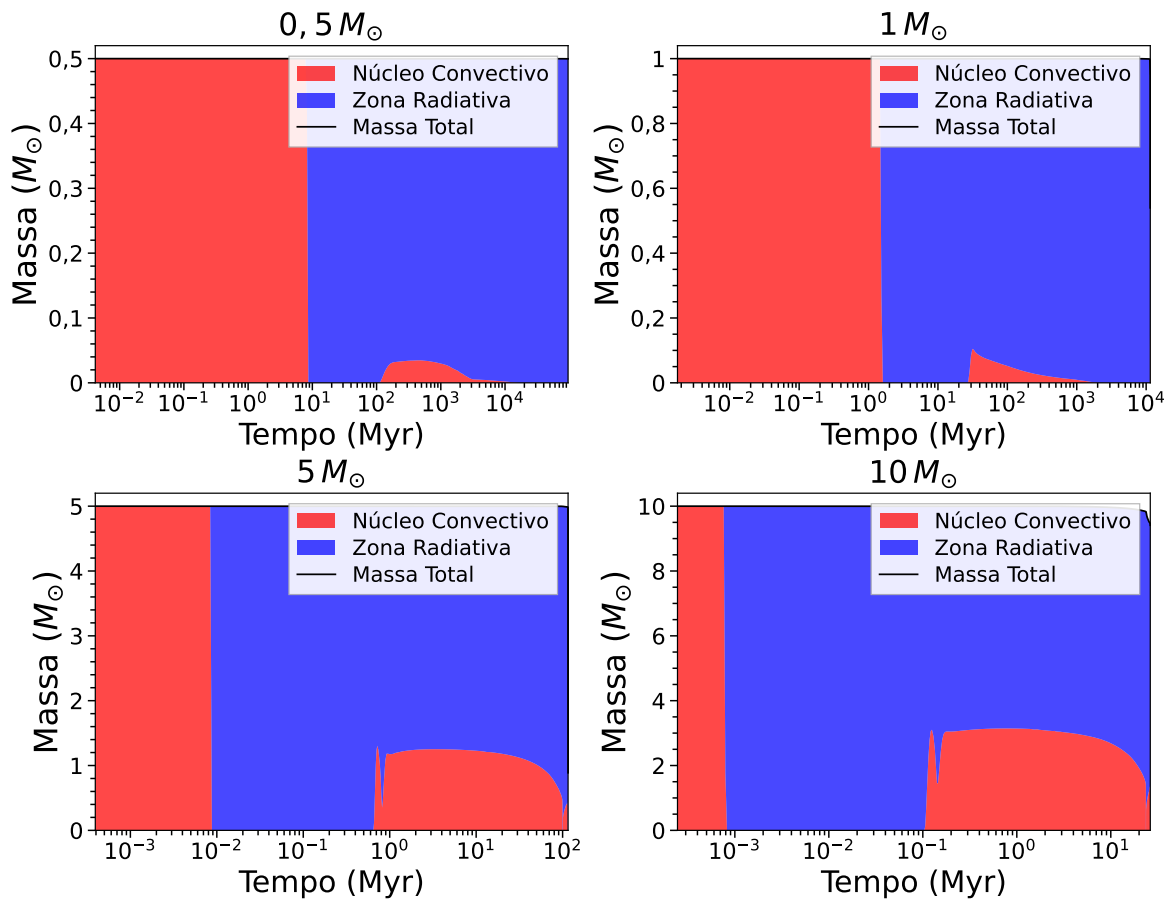


Figure 11.13: Time evolution of the convective (red) and radiative (blue) zones for masses between 0.5 and 10 $M_{\odot}$. The black line represents the total mass, showing wind losses when present.

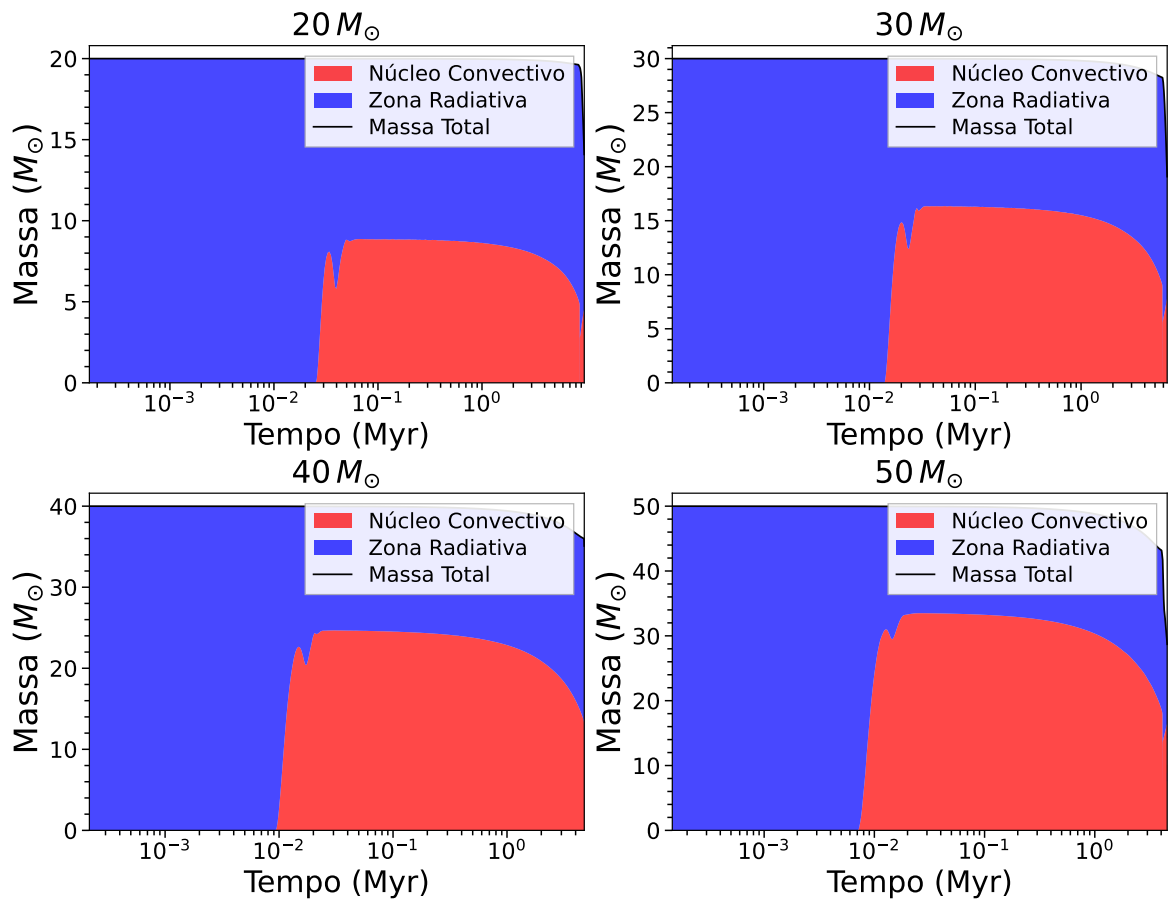


Figure 11.14: Convective zones for massive stars ($20\text{--}50 M_{\odot}$). Note the growing convective core and the mass loss (black line) that modifies the internal structure.

evolution (short lifetimes) and the structural complexity at higher masses. (Vink et al., 2001; Langer, 2012)

Physical and nucleosynthetic consequences

The differences in convective architecture have direct consequences for:

- *Lifetimes*: for a given mass, larger convective cores increase the amount of well-mixed fuel available, which can even prolong the time spent on the main sequence. The shorter lifetime of massive stars stems mainly from the strong dependence of luminosity on mass ($L \propto M^{\sim 3-4}$, already discussed in the mass–luminosity relation), and not simply from the existence of a larger convective core.
- *Chemical mixing*: convection homogenizes internal regions, altering yields and the surface composition (dredge-ups).
- *Nuclear burning*: convective shells favor layers rich in fusion products that participate in the nuclear cycles and affect the yields.
- *Wind and rotation effects*: mass loss and rotation alter thermal and chemical gradients, influencing the extent of the convective zones and the star’s final fate (Langer, 2012; Maeder, 2009).

We can conclude that the comparison between masses 0.5 , 1 , 5 , and $10 M_{\odot}$ confirms the rule of thumb: very light stars show strongly convective envelopes and slow evolution; the solar-type star has a radiative core and a convective envelope; intermediate/massive stars develop significant convective cores and evolve rapidly. Quantitative details (exact extent of convection, presence of shells, non-monotonic variations) depend on mass loss, composition gradients, and rotation — processes included in the MESA models used to generate the figures. (Paxton et al., 2011; Kippenhahn et al., 2012; Maeder, 2009)

11.3.7 Mass Loss

The inclusion of rotation and metallicity effects allows relevant modifications to be assessed in the mass-loss rates. Figures 11.15 and 11.16 present, respectively, the instantaneous mass-loss rates and the main evolutionary episodes responsible for the ejection of material throughout stellar life.

Low-mass regime and the AGB phase In low-mass stars, mass loss is minimal during the main sequence, reflecting their low luminosity and inefficient winds. For $1 M_{\odot}$, the loss rate remains low until the advanced phases, when it reaches values on the order of $10^{-5} M_{\odot} \text{ yr}^{-1}$. The final ejected-mass value ($0.46 M_{\odot}$) is compatible with evolutionary models of solar-type stars.

Intermediate-mass stars show the highest absolute mass-loss values. For $5 M_{\odot}$, a peak of $\sim 10^{-3} M_{\odot} \text{ yr}^{-1}$ is observed throughout the evolution. These values reflect the action of intense winds during the supergiant and AGB⁵ phases, operating on short timescales ($\sim 10^6$ years for the combined EAGB and TPAGB phases).

⁵From the English *Asymptotic Giant Branch*. This is an advanced, luminous evolutionary phase, following core helium burning, in which the star has a degenerate carbon-and-oxygen core surrounded by helium- and hydrogen-burning shells.

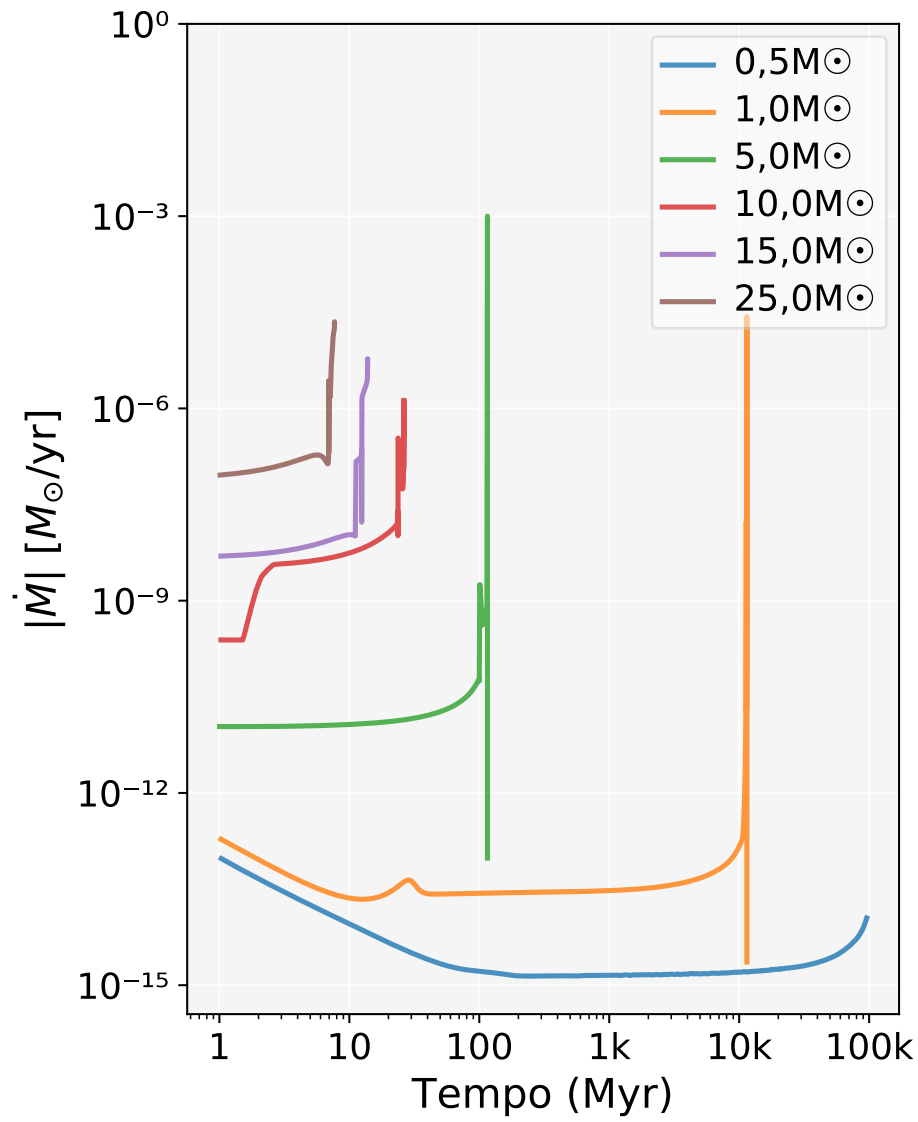


Figure 11.15: Time evolution of the mass-loss rate for stars of different masses, highlighting the episodes associated with stellar winds, thermal pulses, and giant/supergiant phases.

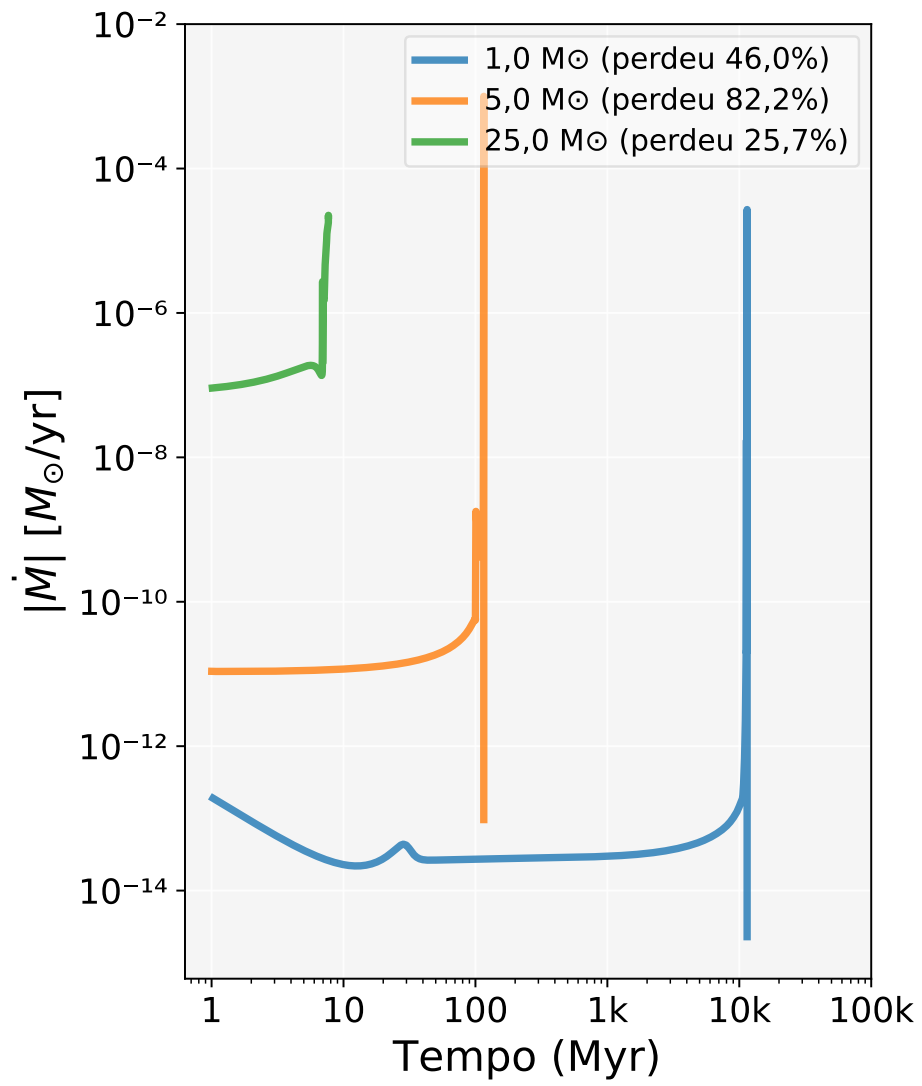


Figure 11.16: Main evolutionary phases responsible for mass loss over time for different stellar masses.

Winds of massive stars (OB and supergiant regime) In massive stars ($10\text{--}25\,M_{\odot}$), mass loss shows a more complex dependence. While the $10\,M_{\odot}$ star loses only about $0.59\,M_{\odot}$, the $15\,M_{\odot}$ and $25\,M_{\odot}$ stars reach total losses of 3.12 and $6.41\,M_{\odot}$, respectively. This difference arises from the combined action of radiative winds and the rapid evolution toward luminous states, in which such winds become highly efficient.

Metallicity Effect The inclusion of rotation and metallicity allows a more realistic analysis of stellar winds, since both factors significantly alter the efficiency of mass loss. Figure 11.17 presents a comparison among three configurations: rotation with $v/v_{\text{crit}} = 0.4$, rotation with near-solar metallicity ($[\text{Fe}/\text{H}] = +0.02$, $Z \approx 0.015$), and rotation with high metallicity ($[\text{Fe}/\text{H}] = +0.20$, $Z \approx 0.022$, $\approx 1.6\times$ the solar metallicity).

For low-mass stars, higher metallicity reduces the maximum mass-loss rate by approximately 18.5%. In this regime, the winds are governed by empirical and pulsational prescriptions that do not explicitly depend on Z : the Reimers (1975) prescription (RGB) and the Blöcker (1995) prescription (TPAGB) in the MIST models, and the de Jager et al. (1988) (*Dutch*) prescription in the MESA *inlist* for cold winds. In contrast to the Vink et al. (2001) law, none of these prescriptions contains Z as a direct wind-strength parameter: the influence of Z on $\dot{M}$ manifests indirectly, through variations in opacity and stellar structure.

The $5\,M_{\odot}$ star, which shows the highest percentage loss among all the cases analyzed (82.54%), shows little sensitivity to metallicity in the maximum value of $\dot{M}$. The most notable difference is temporal: the mass-loss peak occurs about 2 million years earlier for $[\text{Fe}/\text{H}] = +0.20$. This suggests that, for this mass range, the loss mechanisms are dominated by processes related to luminosity and internal structure, being less sensitive to composition variations.

For the most massive star analyzed, high metallicity reduces the maximum loss rate by about 14% and decreases the total loss by 8.5%. This result may seem contradictory to the relation $\dot{M} \propto Z^{0.6-0.9}$ (Vink et al., 2001), which predicts a positive dependence in the line-driven-wind regime (OB and hot-supergiant phases). The apparent contradiction is resolved by recognizing that: (a) the law $\dot{M} \propto Z^{0.6-0.9}$ applies specifically to the hot phases; in cold-supergiant (RSG) and advanced phases, other mechanisms govern the ejection with a different Z dependence; (b) the comparison is made at the same *stellar age* — not at the same equivalent evolutionary point (EEP) — such that the higher-metallicity star may be in a structurally distinct phase at the same instant. Empirical observations and modeling studies confirm that environments with a higher metal abundance favor photon absorption by heavy ions and, therefore, increase the wind strength; on the other hand, in certain regimes (e.g., approach to the Eddington limit, *clumping* effects, and changes in envelope structure) the response is not linear (Vink et al., 2001; Mokiem et al., 2007; Puls et al., 2008).

In summary, the dependence of $\dot{M}$ on metallicity is regime-dependent: positive ($\dot{M} \propto Z^{0.6-0.9}$) in the OB/hot-supergiant regime, and weaker or indirect in low-mass stars and in the AGB/RSG phases, where empirical and pulsational prescriptions dominate without explicit scaling with Z .

Rotation Effect Rotation acts as an enhancer of stellar winds, especially in phases in which the interplay between rotational flattening and opacity becomes relevant. In the MIST models analyzed, the rotation rate is initialized at $v/v_{\text{crit}} = 0.40$ at the ZAMS and evolves throughout the main sequence. A direct audit of the `surf_avg_v_div_v_crit` column in the rotating tracks (folders 1.b2/) confirms this behavior for stars of $8\text{--}25\,M_{\odot}$: the ratio drops slightly to ≈ 0.39 at the start of the MS (surface loss of angular momentum through winds) and

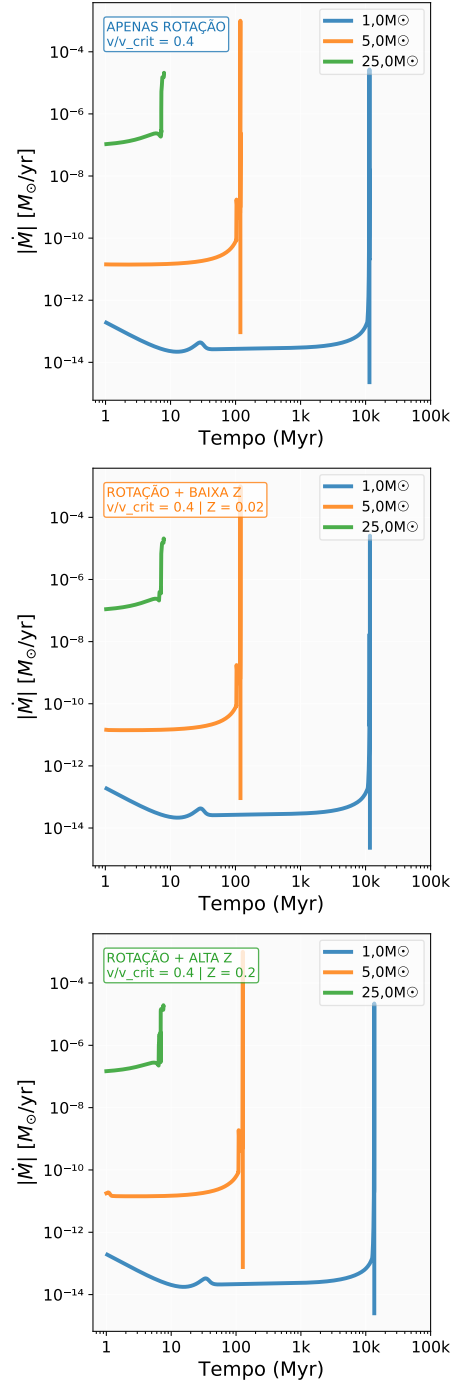


Figure 11.17: Comparison of mass loss over time for different rotation and metallicity configurations, evidencing the combined role of opacity and rotational flattening on the efficiency of stellar winds.

grows to 0.46–0.57 at the TAMS, since the envelope expansion reduces $v_{\text{crit}} = \sqrt{GM/R}$ more rapidly than v_{rot} decreases. The value $v/v_{\text{crit}} = 0.40$ should therefore be understood as a boundary condition at the ZAMS, not as a constant parameter throughout the entire evolution. Although rotation also affects the internal structure and angular-momentum transport, its direct influence on mass loss manifests mainly as a modulation of the rate, superimposed on the primary effects of mass and luminosity.

Instantaneous loss rate and total ejected mass The results reveal three distinct regimes: (i) low-mass stars show late peaks associated with the giant phases; (ii) intermediate-mass stars undergo intense, concentrated episodes of mass loss; (iii) massive stars lose mass throughout most of the evolution, with multiple peaks distributed in time.

Figure 11.16 shows that the total mass-loss efficiency (ejected mass/initial mass) exceeds 80% for $5 M_{\odot}$, while reaching about 25% for $25 M_{\odot}$. This non-monotonicity reflects the evolutionary coverage of the tracks: the $5 M_{\odot}$ track traversed more advanced phases (including the EAGB and TPAGB phases, where mass loss is intense), while the $25 M_{\odot}$ track covers stages prior to core collapse, with relatively less efficient winds in these stages. These results show that mass loss does not scale linearly with initial mass, but depends on the structural properties and the specific evolutionary phases of each star.

11.4 Isochrones

The Kiel Diagram ($\log g \times \log T_{\text{eff}}$) is a complementary tool to the HR Diagram for the study of stellar evolution, especially in observational contexts where absolute luminosity is difficult to determine. The surface gravity, defined by

$$g = \frac{GM}{R^2}, \quad (11.5)$$

allows the stellar radius to be inferred directly, providing essential structural information when combined with the effective temperature (Paxton et al., 2011; Langer and Kudritzki, 2014).

Unlike the HR Diagram, which depends on luminosity and therefore on distance, the Kiel Diagram uses quantities obtained directly from spectroscopy, making it especially useful for studying distant populations or those with parallax uncertainties.

Figure 11.18 presents isochrones for solar metallicity ($Z_{\odot}$) at ages of 0.1, 1, and 10 Gyr, revealing characteristic patterns of stellar evolution. For the 0.1 Gyr isochrone, a well-defined main sequence is observed in the region of high surface gravities and high effective temperatures, corresponding to young stars in the hydrogen-burning phase.

The *turn-off* point provides the main age signature of a cluster. For ages on the order of 1 Gyr, the *turn-off* lies around $\log(T_{\text{eff}}) \approx 3.8$ – 3.9 , indicating typical masses of 1.5 – $2.5 M_{\odot}$. For ages on the order of 10 Gyr, the *turn-off* shifts to cooler regions with lower gravities, showing that only low-mass stars remain on the main sequence. This shift characterizes the consumption of central hydrogen and marks the onset of evolution toward the red-giant branch.

The 10 Gyr isochrone exhibits characteristics of an evolved stellar population, with:

- A main sequence restricted to low-mass stars ($\log T_{\text{eff}} < 3.7$).
- A well-developed red-giant branch in the low-gravity region ($\log g < 3$).

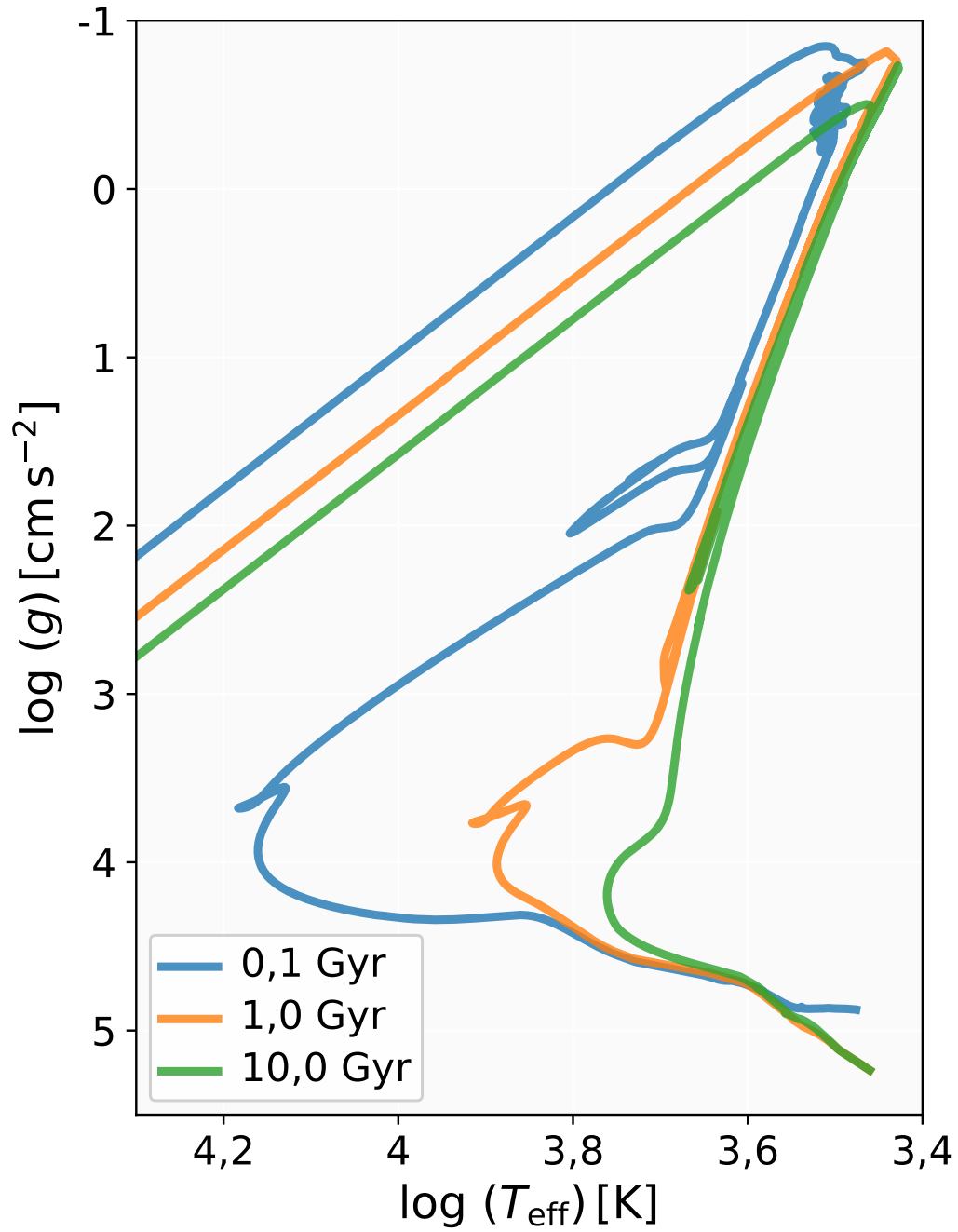


Figure 11.18: Kiel Diagram ($\log g$ versus $\log T_{\text{eff}}$) showing isochrones for solar metallicity at three representative ages: 0.1, 1, and 10 Gyr. The curves highlight the evolution of the main sequence, the shift of the *turn-off* point, and the development of the red-giant branch over time.

- The *turn-off* shifted to lower temperatures ($\log T_{\text{eff}} \approx 3.7$).

The evolution of the isochrone curves in the Kiel Diagram reflects fundamental changes in stellar structure:

- Main-sequence stars occupy the high-surface-gravity region.
- Envelope expansion during the giant phases results in a drastic reduction in $\log g$.
- The horizontal shift in the curves corresponds to surface cooling during the evolution.

The practical application of this diagram includes determining ages of stellar clusters by fitting isochrones to observational data, as well as characterizing stellar atmospheric parameters. The sensitivity of the *turn-off* to age makes the Kiel Diagram particularly useful for studies of stellar evolution and galactic archaeology. This dating method is well established in the literature and constitutes one of the main tools of modern galactic archaeology (Gallart et al., 2005; Deason and Belokurov, 2024).

11.4.1 Effect of Metallicity on Fixed-Age Isochrones

Metallicity is one of the fundamental parameters that modulate the internal structure of stars, altering opacities, nuclear efficiencies, and the morphology of the isochrones themselves. It is frequently expressed by the quantity $[\text{Fe}/\text{H}] = \log_{10}(N_{\text{Fe}}/N_{\text{H}})_{\star} - \log_{10}(N_{\text{Fe}}/N_{\text{H}})_{\odot}$, that is, the logarithm of the ratio between the number of iron and hydrogen atoms in the star, compared with the same value for the Sun. Since this is a relative logarithmic scale, $[\text{Fe}/\text{H}] = 0$ corresponds to solar metallicity, negative values indicate stars poorer in metals than the Sun (for example, $[\text{Fe}/\text{H}] = -1$ corresponds to 1/10 of the solar iron abundance), and positive values indicate stars richer in metals than the Sun (for example, $[\text{Fe}/\text{H}] = +0.5$ corresponds to about $10^{0.5} \approx 3.16$ times the solar abundance). For a fixed age, variations in $[\text{Fe}/\text{H}]$ systematically shift stars in the $\log g$ - $\log T_{\text{eff}}$ plane, allowing us to comparatively study stellar populations belonging to different galactic environments.

The analysis of the metallicity effect on stellar evolution helps to visualize patterns in the structure and development of stars as a function of their chemical composition, which goes beyond hydrogen and helium. Figure 11.19 presents isochrones for a fixed age of 4 Gigayears with four different metallicities: $[\text{Fe}/\text{H}] = -2, -1, 0$, and $+0.5$, allowing us to observe how chemical composition influences the position of stars in the Kiel Diagram.

The results show a systematic shift of the isochrone curves as a function of metallicity. Metal-poor stars ($[\text{Fe}/\text{H}] = -2$) occupy the region of higher effective temperatures, while high-metallicity stars ($[\text{Fe}/\text{H}] = +0.5$) lie at lower effective temperatures. This behavior stems from the increase in opacity with metal content: metals introduce numerous lines and continuum absorptions that hinder radiative transport. As energy accumulates more efficiently in the internal layers, the structure adjusts by slightly expanding the envelope and reducing T_{eff} to maintain hydrostatic equilibrium. Thus, for the same age and mass, metal-rich stars appear cooler in the Kiel Diagram.

Surface gravity ($\log g$) remains within similar ranges for all metallicities, but the distribution along the main sequence varies. At low metallicity, the lower opacity produces more compact stars, resulting in slightly higher values of $\log g$ for the same evolutionary phase. High metallicities, on the other hand, favor more expanded envelopes and slightly lower values of $\log g$, especially near the *turn-off*.

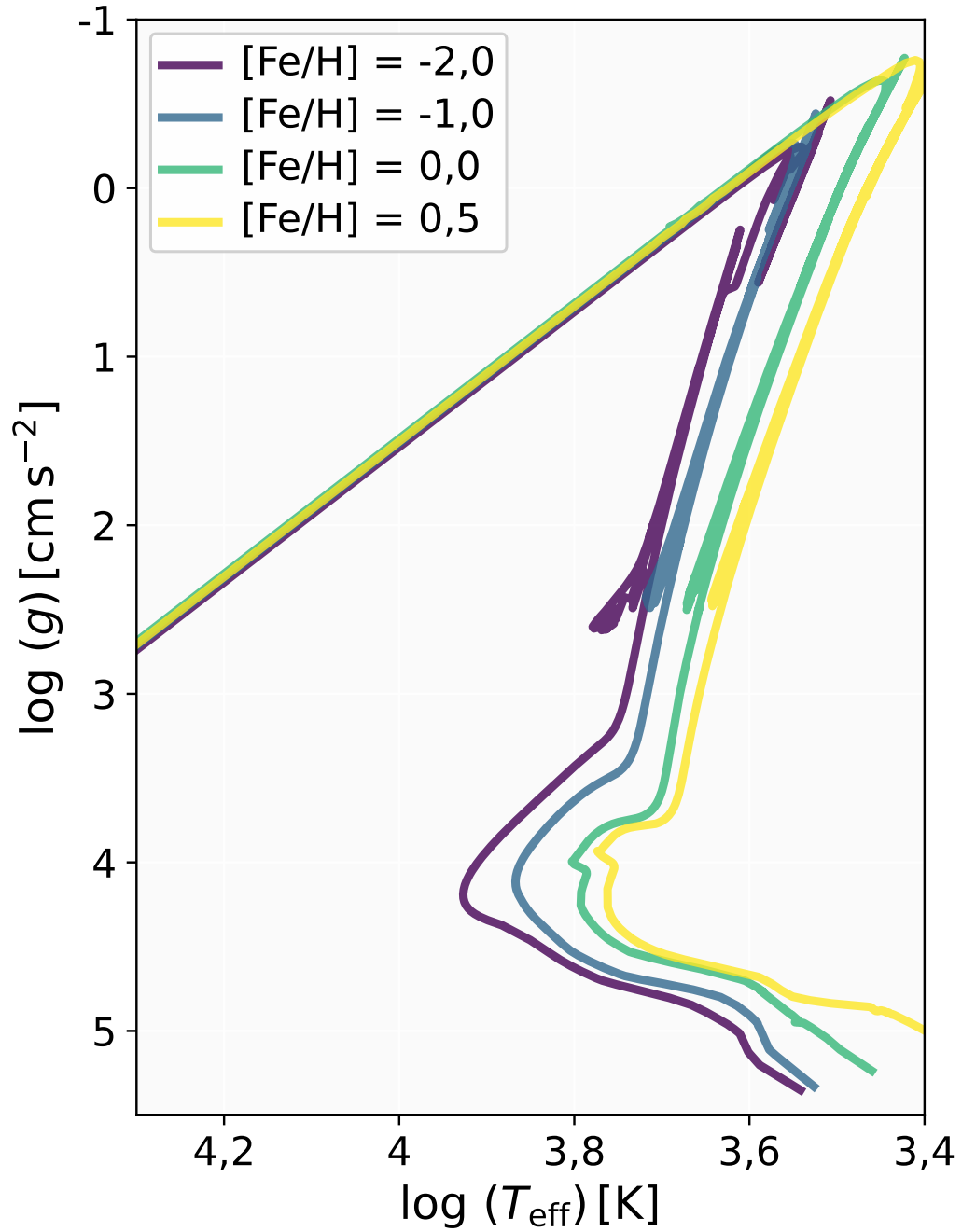


Figure 11.19: Kiel Diagram for 4 Gyr isochrones with different metallicities. The curves demonstrate the shift of the main sequence as a function of metal content.

The astrophysical implications are that metal-poor populations, such as those of globular clusters, show isochrones shifted to higher temperatures, while younger, more chemically enriched populations — typical of the galactic disk, for example — occupy cooler regions of the diagram. Thus, the interpretation of $\log g$ and T_{eff} must always take chemical content into account, since the age–metallicity degeneracy can lead to incorrect estimates if isochrone fitting neglects the effect of composition.

11.4.2 Age Effects on Fixed-Metallicity Isochrones

In this subsection we analyze the effect of age on the morphology of the isochrones for a fixed metallicity. Since age directly controls the position of the *turn-off* point and the extent of the giant branch, comparing isochrones of 1, 4, 7, and 10 Gyr allows us to clearly visualize how stellar populations of the same chemical composition, but at different evolutionary stages, are distributed in the Kiel Diagram.

The progression of the isochrones with age follows a universal pattern: (i) the *turn-off* shifts to lower effective temperatures; (ii) the giant branch becomes more extended; and (iii) the main sequence shortens, reflecting the absence of higher-mass stars that have already evolved to advanced phases. These patterns appear at all metallicities, but their intensity depends strongly on the metal content.

For the extremely low metallicity ($[\text{Fe}/\text{H}] = -2$), the isochrones show a narrow, hot main sequence, characteristic of low opacity. The *turn-off* point shifts to progressively lower temperatures as age increases, and the giant branch becomes more pronounced, although more compact than at higher metallicities due to the smaller envelope expansion.

For $[\text{Fe}/\text{H}] = -1$, the same general pattern is observed, but with an overall cooling compared with the lower metallicity. The isochrones become slightly wider and the giant branch more extended, reflecting higher opacity and greater envelope-expansion efficiency. The 10 Gyr isochrone shows a cooler *turn-off*, shifted vertically, indicating a more advanced structural evolution.

At solar metallicity ($[\text{Fe}/\text{H}] = 0$), the isochrones become more dispersed on the main sequence, and the giant branch is more extended at all ages. The *turn-off* of the oldest isochrones appears considerably cooler relative to the metal-poor cases, a direct consequence of the higher opacity that delays the evolution and favors more expanded envelopes.

For the high metallicity ($[\text{Fe}/\text{H}] = +0.5$), the most pronounced shift toward lower effective temperatures is observed. The main sequence shows strong vertical and horizontal dispersion, especially at advanced ages, due to the combination of high opacity and less compact internal structures. The giant branch becomes significantly expanded and cooler, highlighting the cumulative effect of metallicity on stellar evolution. These effects are widely documented in modern evolutionary models, such as in the PARSEC isochrones (Bressan et al., 2012).

Thus, the evolution of the isochrones with age provides a direct diagnostic of the time evolution of stellar populations, while the comparison among metallicities reveals how differences in opacity modulate evolutionary trajectories. The Kiel Diagram thus becomes an essential tool for characterizing ages, masses, and evolutionary stages of stars in different galactic environments.

Fitting isochrones to stellar clusters is one of the main observational tools for age determination: the theoretical isochrone whose *turn-off* point position in the HR (or Kiel) diagram coincides with that of the observed cluster is chosen. The MIST isochrone files used in this work (for $[\text{Fe}/\text{H}] \in \{-2.0; -1.0; 0.0; +0.5\}$) are the same ones that would be used in such

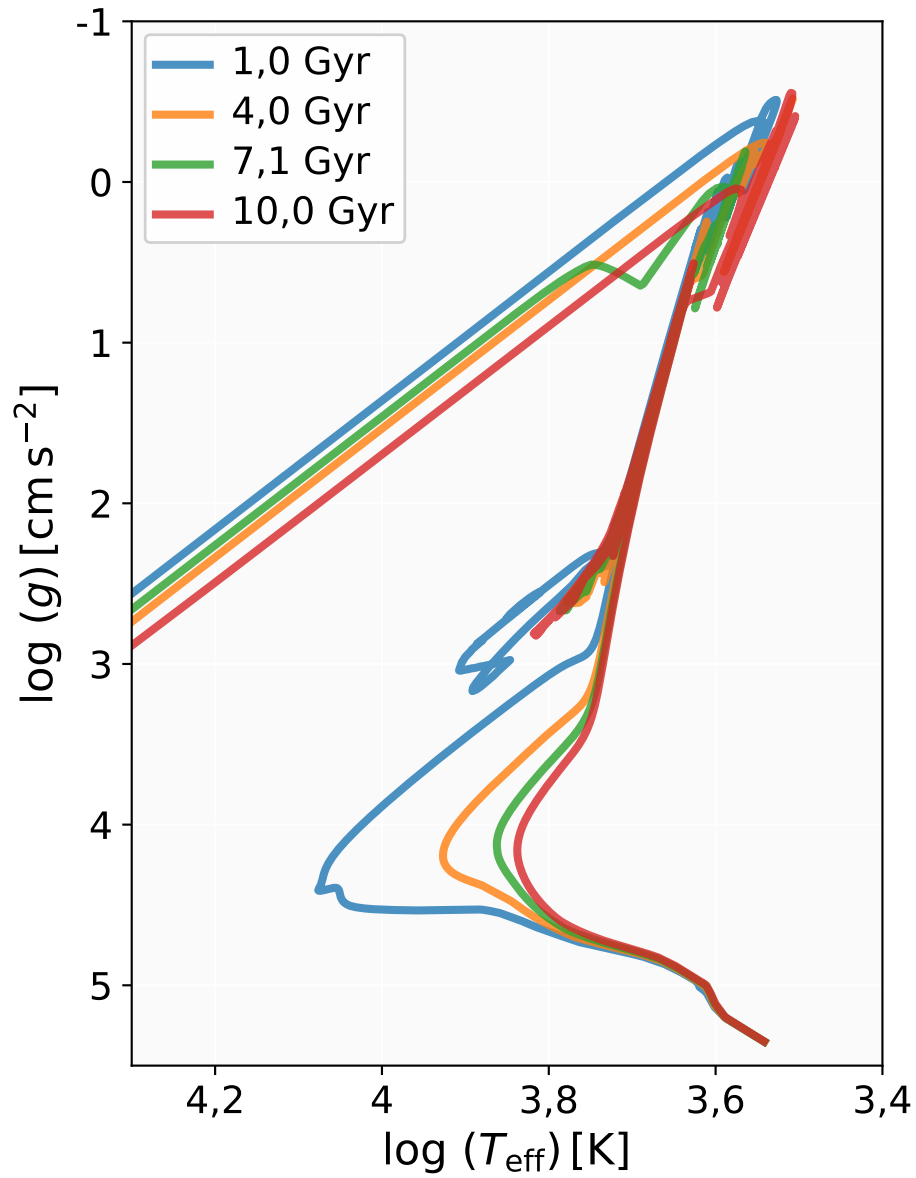


Figure 11.20: Isochrones in the Kiel Diagram for $[Fe/H] = -2$ and ages of 1, 4, 7, and 10 Gyr. The low opacity shifts the main sequence to higher temperatures and compresses the giant branch.

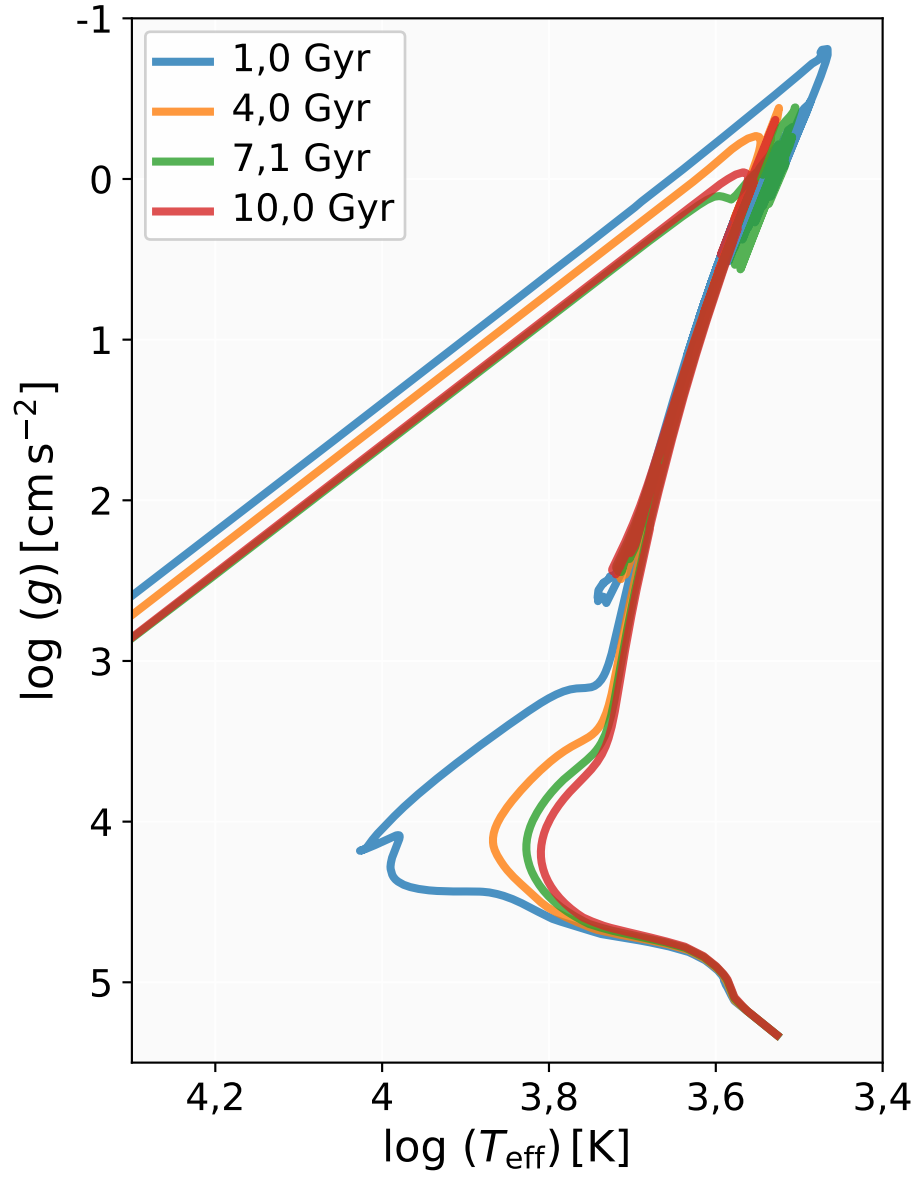


Figure 11.21: Isochrones for $[Fe/H] = -1$. The moderate increase in opacity relative to $[Fe/H] = -2$ shifts the curves to lower temperatures and expands the giant branch.

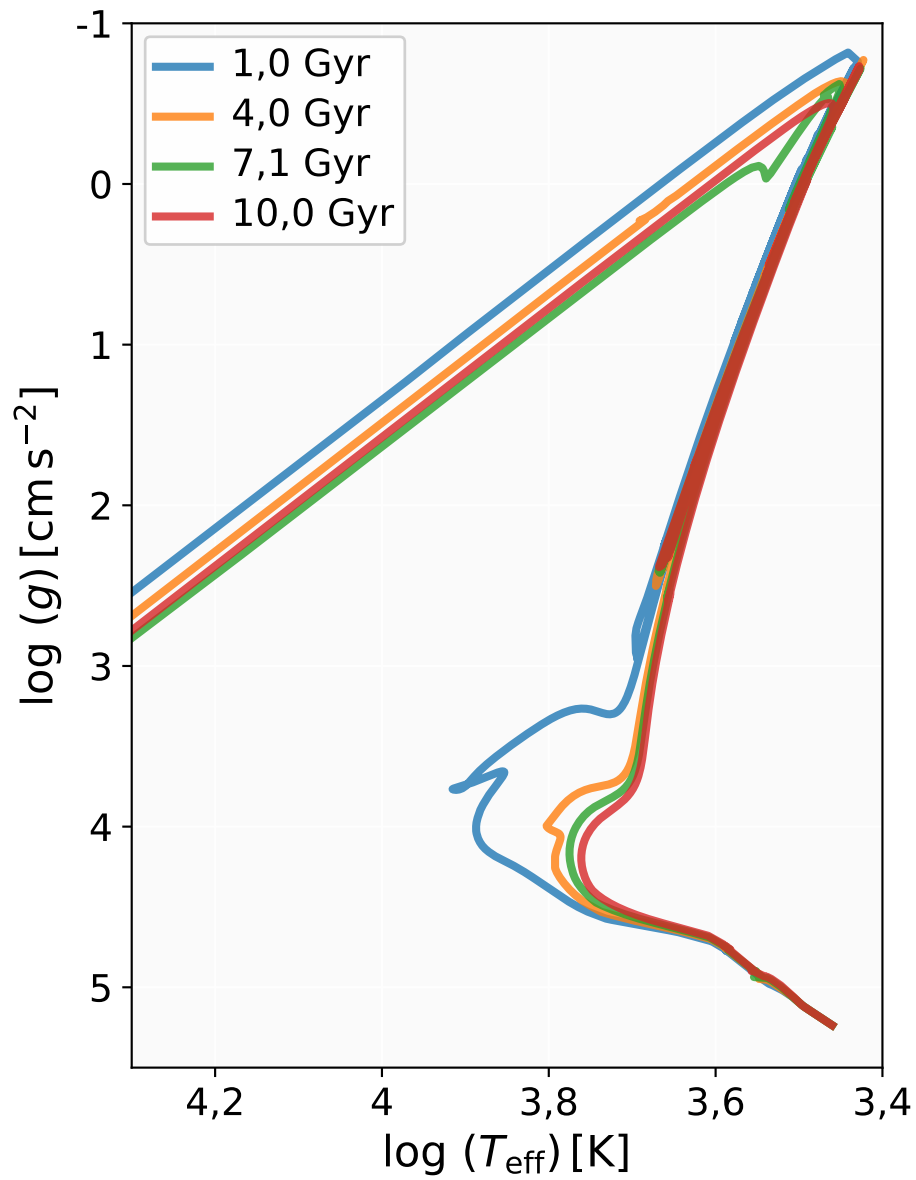


Figure 11.22: Isochrones for solar metallicity. A wider main sequence and a more extended giant branch are observed, reflecting higher opacity.

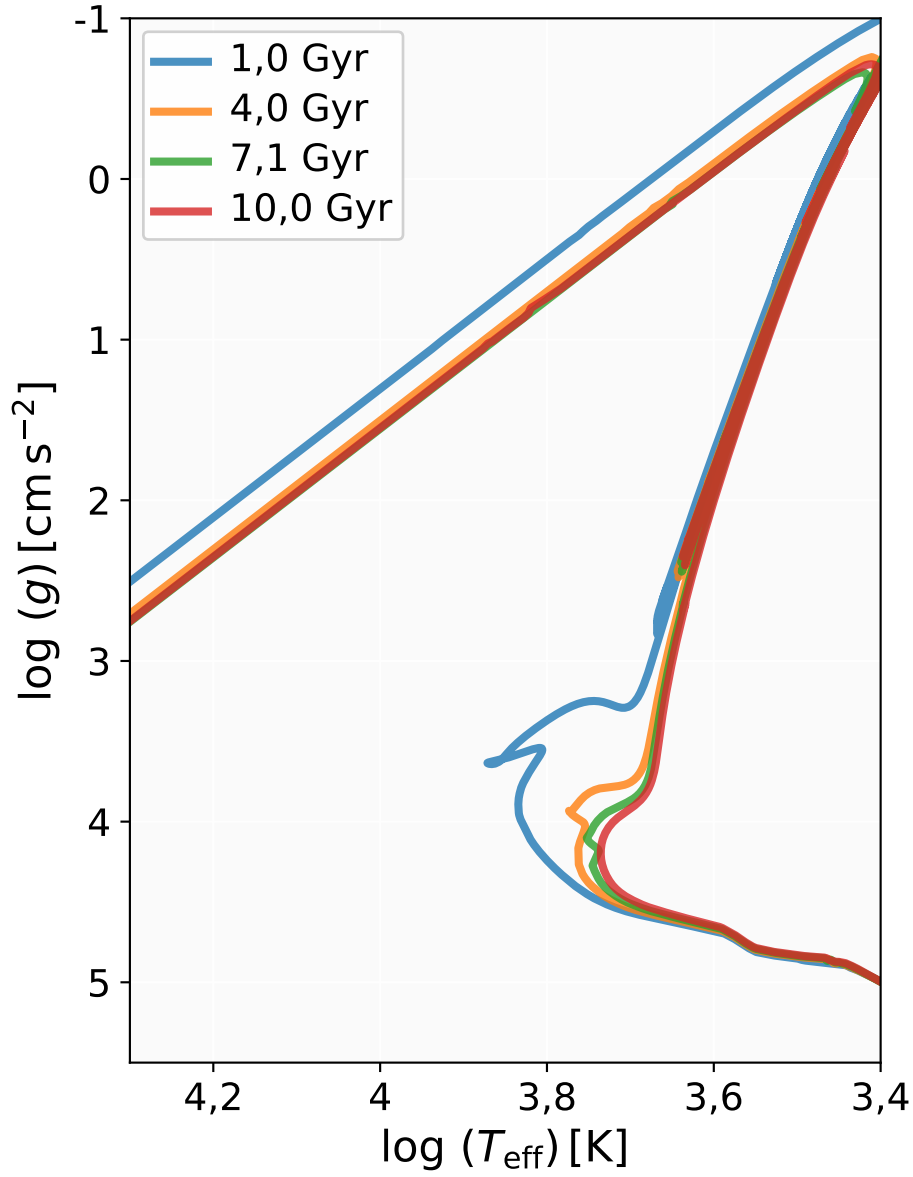


Figure 11.23: Isochrones for $[Fe/H] = +0.5$. The high metallicity produces the largest shift toward low temperatures and the most expanded giant branch among all the cases analyzed.

an analysis, and are available on the project’s portal (Dotter, 2016). From the Kiel diagrams presented in the previous figures, the reader can explore the fitting procedure by combining the MIST isochrones with photometric data of real clusters obtained from public catalogs.

11.5 Stellar Evolution Model

In this section we describe the stellar evolution model used in the simulations. For this, we employed the MESA code, version r23.05.1, adopting as a case study a massive star of $50 M_{\odot}$ with solar metallicity ($Z = 0.02$). The choice of a high-mass star allows us to explore phenomena characteristic of this regime — such as intense radiative winds, short lifetimes, and accelerated structural evolution — in contrast with the slower evolution typical of lower-mass stars.

The evolution of a star is described by a system of coupled differential equations that govern its internal structure and time evolution. These equations express the conditions of hydrostatic equilibrium, mass conservation, energy transport, and energy conservation in the stellar interior (Kippenhahn et al., 2012). In the mass coordinate $m(r)$, the fundamental equations can be written as

$$\frac{dP}{dm} = -\frac{Gm}{4\pi r^4}, \quad (11.6)$$

representing the hydrostatic-equilibrium condition, establishing the balance between the pressure gradient and the gravitational force.

$$\frac{dr}{dm} = \frac{1}{4\pi r^2 \rho}, \quad (11.7)$$

which expresses mass conservation and relates the radius of the stellar shell to its local density.

$$\frac{dL}{dm} = \epsilon_{\text{nuc}} - \epsilon_{\nu}, \quad (11.8)$$

which describes energy conservation, where ϵ_{nuc} represents the energy-generation rate from nuclear reactions and ϵ_{ν} the energy loss from neutrino emission.

Finally, energy transport in the stellar interior can occur by radiation or convection. In the radiative regime, the temperature gradient is given by

$$\frac{dT}{dm} = -\frac{3\kappa L}{64\pi^2 a c r^4 T^3}, \quad (11.9)$$

where κ is the opacity of the stellar plasma, a is the radiation constant, and c is the speed of light.

The numerical integration of these equations, combined with appropriate equations of state, opacity tables, and nuclear-reaction networks, allows the internal structure of the star and its evolution over time to be determined. Modern stellar evolution codes, such as MESA (*Modules for Experiments in Stellar Astrophysics*), solve this system incorporating a detailed description of stellar physics (Paxton et al., 2011).

The same four equations from Chapter 5

The four equations presented above are exactly the **four fundamental stellar structure equations** derived in detail in Chapter 5: hydrostatic equilibrium (dP/dm), mass conservation (dr/dm), energy conservation (dL/dm , with ϵ_{nuc} given by the reaction rates

of Chapter 10), and radiative transport (dT/dm , in the form of photon diffusion already obtained from the Rosseland Approximation in Chapter 4). What MESA does — and what this chapter illustrates in practice — is precisely to integrate this system numerically, coupled to an equation of state (which can include the degenerate regimes of Chapter 8) and to a nuclear-reaction network (Chapter 10), for thousands of time steps throughout the life of a star. It is also worth noting that polytropic models such as those derived in Chapter 6 (the Lane–Emden equation) remain relevant even in the era of numerical codes: MESA uses them as an initial model (*initial guess*) to relax the structure of a pre-main-sequence star before starting the full evolution.

In the present work, the model was evolved from an initial age of 1 Myr to approximately 5.24 Myr, resulting in 333 successive models. This interval covers the late main sequence and the initial phases of envelope expansion toward the supergiant stage, before the onset of advanced burning of heavier elements.

11.5.1 Physical Configuration of the Model

The configuration used in the `inlist` file (made available in Section 11.8) included the following fundamental parameters:

- Mass and initial composition: $M_{\text{ini}} = 50 M_{\odot}$, $Z = 0.02$.
- Initial age: 1 Myr, already starting within the main sequence.
- Wind schemes:
 - *Vink* wind for hot phases ($T_{\text{eff}} \gtrsim 10^4$ K), based on the formalism of Vink et al. (2001), widely used for OB stars due to its sensitivity to luminosity, metallicity, and ionization rate;
 - *Dutch* scheme for cold phases, which uses the empirical prescriptions of de Jager et al. (1988), being the standard recommended by MESA for cold supergiants and advanced phases;
 - Scale factor increased to 2.0 in both regimes, reflecting observational evidence of intensified winds in luminous massive stars.
- Stopping condition: maximum time of 10^7 years. The simulation ended at ~ 5.24 Myr, covering the main sequence and the beginning of envelope expansion, but not the advanced burning phases (He, C, Ne, O, Si) nor core collapse.
- Convective overshoot: enabled with an exponential scheme at the boundaries of the convective core, with parameters `overshoot_f(1) = 0.02` and `overshoot_f0(1) = 0.01` (see *inlist* in Appendix A).
- Rotation: not included in this simulation. The model considers a spherical structure without rotational mixing.
- Opacity and equation of state: OPAL tables for high temperatures combined with low-temperature molecular opacities (Ferguson et al., 2005); equation of state for partially ionized and degenerate matter (Saumon et al., 1995); implementation via MESA `r23.05.1` (Paxton et al., 2011).

- Nuclear network: `basic.net` (MESA default), which includes the nuclear reactions relevant to the main sequence of massive stars (pp, CNO, triple- α , and α capture).
- Semiconvection: not explicitly enabled; this choice prioritizes computational feasibility for didactic purposes.
- Wind scale factor ($Vink_scaling_factor = Dutch_scaling_factor = 2.0$): this factor was adopted in the original configuration of the simulation without a documented physical justification, constituting a non-standard choice that tends to overestimate the mass-loss rate. The reported mass-loss values for this star should, therefore, be interpreted as illustrative estimates, probably higher than those that would be obtained with the standard prescriptions (factor 1.0).
- **Note on convergence:** the parameters above were adopted as an operational starting point and were not subjected to a systematic parameter-convergence study. The reported integrated quantities (mass lost, final radius, final T_{eff}) are illustrative in character and should be interpreted within the didactic context of this work.

The combined use of the Vink and Dutch prescriptions is justified by the strong temperature dependence of the winds of massive stars. Radiative winds dominate the hot phases ($T_{\text{eff}} \gtrsim 2.5 \times 10^4$ K), while empirical supergiant winds become more appropriate as the envelope expands and the star becomes cooler. This hybrid approach is adopted by the community in order to consistently reproduce mass loss throughout the entire evolutionary cycle of high-mass stars.

11.5.2 Simulation Results

The evolution proceeds extremely fast: the star completes all the simulated phases in just 5.24 Myr, a value consistent with what is expected for stars of 40–60 $M_{\odot}$.

The quantitative analysis obtained from the `history.data` file reveals:

- Initial mass: 50.0 $M_{\odot}$
- Final mass: 26.42 $M_{\odot}$
- Mass lost: 23.58 $M_{\odot}$ (47% of the initial mass) — an upper-bound estimate, due to the non-standard wind scale factor adopted in the simulation (see note above).
- Final luminosity: $\log(L/L_{\odot}) = 5.79$
- Final temperature: $\log(T_{\text{eff}}/\text{K}) = 3.86$
- Final radius: $\log(R/R_{\odot}) = 2.696 \Rightarrow R \approx 497 R_{\odot}$
- Maximum loss rate: $8.19 \times 10^{-5} M_{\odot}/\text{yr}$

The initial mass determines the fate of the star. Stars with $M < 8M_{\odot}$ evolve into white dwarfs, exposing He, C/O, or O/Ne cores. For intermediate masses (8–25 $M_{\odot}$), gravitational collapse leads to a Type II supernova explosion and the formation of neutron stars. More massive stars can form black holes via direct collapse or with *fallback*, in agreement with recent core-collapse models (Griffiths et al., 2025).

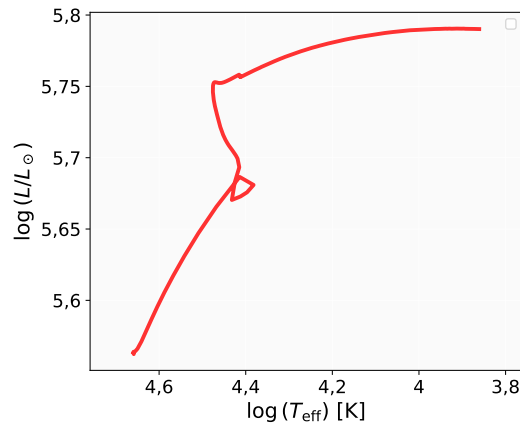


Figure 11.24: Evolutionary trajectory of the $50 M_{\odot}$ star in the HR diagram. The star starts hot and luminous on the main sequence and evolves toward the supergiant region.

11.5.3 HR Diagram

Figure 11.24 shows that the star begins its evolution with $\log(T_{\text{eff}}) \approx 4.66$ and $\log(L/L_{\odot}) \approx 5.56$, evolving rapidly toward lower temperatures while the luminosity increases continuously. This behavior is typical of massive stars that leave the main sequence after exhausting the central hydrogen.

11.5.4 Time Evolution of the Luminosity

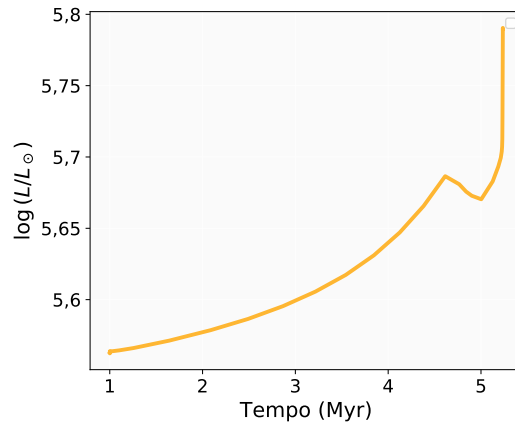


Figure 11.25: Evolution of the luminosity throughout the simulation. The luminosity increases almost monotonically, characteristic of the continuous increase of the nuclear-burning rate.

According to Figure 11.25, the luminosity increases from $\log(L/L_{\odot}) = 5.56$ to 5.79 over 5.2 Myr, a growth of approximately 70%. This behavior is typical of massive stars on the main sequence and at the beginning of the supergiant phase: as the central hydrogen is consumed, the core contracts and heats up, increasing the nuclear-burning rate.

Since the nuclear lifetime of a $50 M_{\odot}$ star is extremely short, even relatively small changes in the central hydrogen abundance result in noticeable increases in luminosity.

In addition, other physical parameters can significantly influence the evolution of massive stars. Stellar rotation, for example, can prolong the main-sequence lifetime through rotational-

mixing processes, which transport fresh hydrogen to the core and bring nuclear-burning products to the surface. This process can lead to surface enrichment in elements such as nitrogen.

Metallicity also plays a fundamental role, since it controls the intensity of the radiatively driven stellar winds. In massive stars, more intense winds can remove a significant fraction of the stellar mass over the evolution, altering the star's internal structure and significantly influencing its final fate.

11.5.5 Evolution of the Effective Temperature

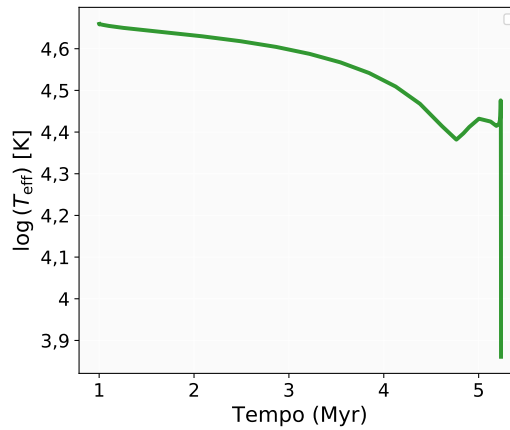


Figure 11.26: Decrease in effective temperature indicating expansion of the stellar envelope.

The effective temperature decreases from $\log(T_{\text{eff}}) = 4.66$ (≈ 46000 K) to 3.86 (≈ 7200 K), representing a surface cooling of almost one order of magnitude.

This behavior directly reflects the envelope expansion in Figure 11.26. Although the core becomes more luminous as it contracts, the stellar surface moves away from the center, greatly increasing the radiating area. Thus, the higher luminosity is compensated by the drop in temperature, in agreement with the Stefan–Boltzmann relation. The trajectory in the HR diagram confirms this trend: the star migrates from the hot region, of spectral type O, to the domain of yellow/red supergiants as the envelope expands.

11.5.6 Evolution of the Radius

The radius increases from $9.7 R_{\odot}$ to $497 R_{\odot}$, an expansion factor of approximately 51 times. This abrupt growth, observed in Figure 11.27, is characteristic of the transition phase between the main sequence and the red-supergiant phase.

The exponential behavior of the growth suggests that the star is approaching advanced evolutionary regimes, in which the structure becomes strongly influenced by steep energy and radiation-pressure gradients.

11.5.7 Evolution of the Mass

In Figure 11.28, we analyze the mass loss, which is dominated by radiative winds described by the formalism of Vink et al. (2001). As a consequence, the final mass ($26.4 M_{\odot}$) places the star in the typical regime of a pre-Wolf–Rayet or evolved supergiant, evidencing that mass loss is one of the key processes that define the star's final fate.

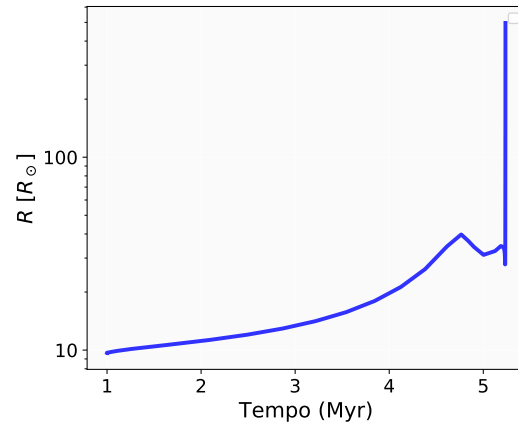


Figure 11.27: Exponential growth of the radius, reaching nearly $500 R_{\odot}$.

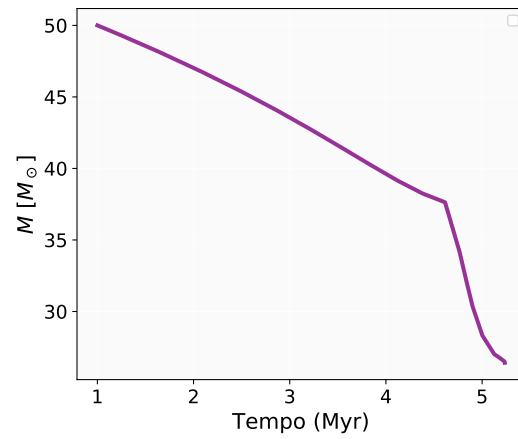


Figure 11.28: Continuous, intense mass loss, reaching 47% of the initial mass.

11.5.8 Evolutionary Timeline

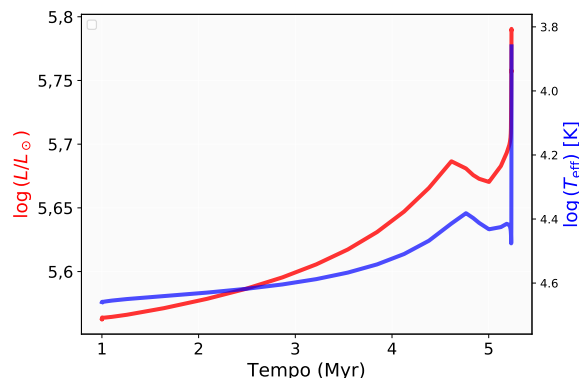


Figure 11.29: Time comparison between luminosity and temperature, showing the temperature decrease accompanied by an increase in luminosity.

The timeline highlights the inverse behavior between luminosity and temperature. While the luminosity grows continuously due to the contraction of the core and the increasing burning rate, the surface temperature decreases markedly. This relation is a classic signature of the evolution of massive stars and is in perfect agreement with the observed and simulated HR diagrams.

11.5.9 Kippenhahn Diagram

The Kippenhahn diagram (**Figure 11.30**) reveals the internal structure of the star throughout the evolution.

Throughout the main sequence, an extensive convective core is maintained, characteristic of massive stars in which hydrogen burning occurs via the CNO cycle, which is extremely sensitive to temperature. As the central hydrogen is exhausted, the star develops shell hydrogen burning, visible as energy-generation zones immediately above the inert He core.

The progressive mass loss pushes the mass-coordinate lines downward, showing that the envelope is being continuously removed by the intense winds. In the later phases, the internal structure becomes more complex, with multiple burning regions and transient convective boundaries, foreshadowing the entry into the supergiant phase.

This diagram is crucial because it confirms, internally, the phenomena inferred from the HR diagrams, the radius curves, and the temperature curves.

11.5.10 Conclusion

The simulations conducted with MESA allow us to consistently characterize the evolution of a massive star of $50 M_{\odot}$ over its first 5.2 million years. The model discussed in this section synthesizes the contemporary view of stellar evolution, in which the initial mass establishes the dominant nuclear regime, while chemical composition, rotation, and mass loss modulate the structural properties and the star's final fate. These conclusions are consistent with results obtained by modern stellar evolution codes described in the literature.

The model adequately captures the main structural and nuclear mechanisms that govern the evolutionary trajectory of this type of object, reproducing trends expected for stars dominated by radiation pressure and intense radiative winds. The results obtained show:

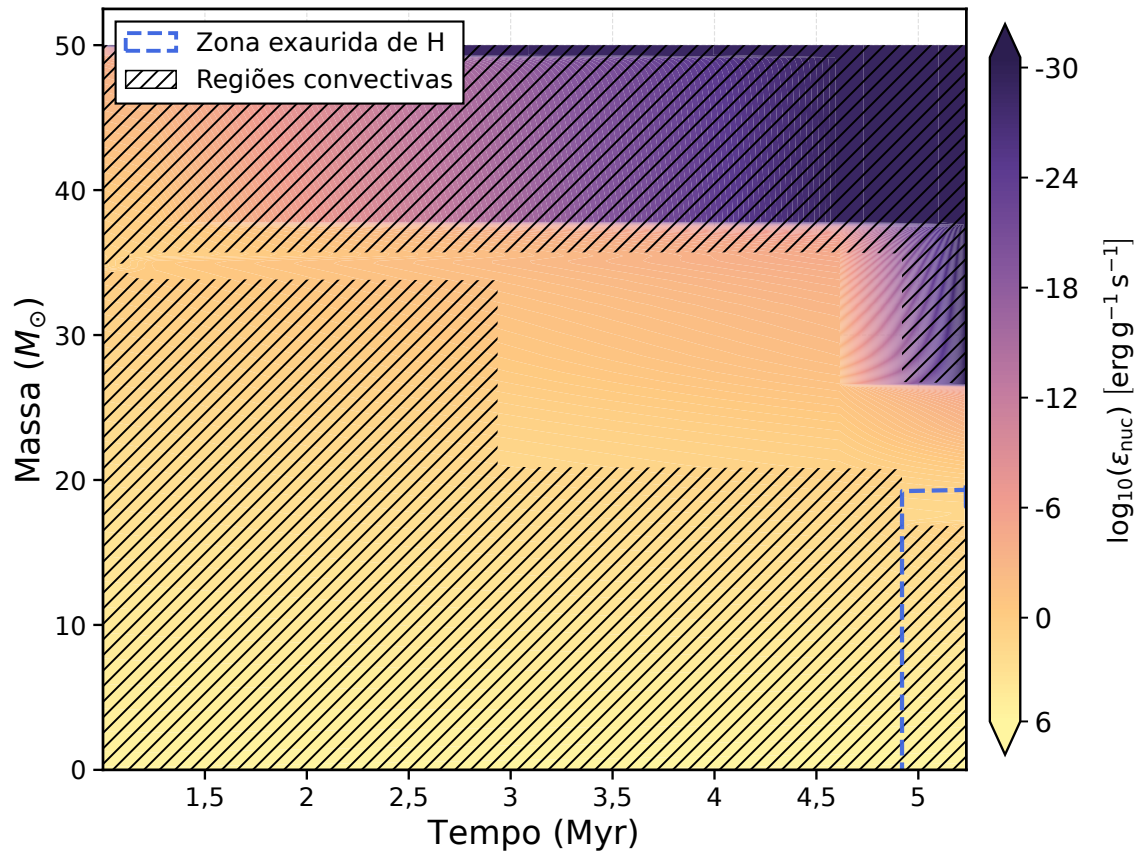


Figure 11.30: Kippenhahn diagram showing convective regions (in blue) and nuclear-burning zones (shades of orange).

- Extreme mass loss: the star loses approximately 47% of its initial mass, a result directly associated with the application of the Vink et al. (2001) winds and of the Dutch formalism in cold and hot regimes. This continuous removal of the envelope profoundly modifies the evolutionary pace and the future of the star.
- Large-amplitude expansion: the radius grows by a factor of ~ 51 , reaching nearly $500 R_{\odot}$. This expansion, accompanied by the reduction of the effective temperature, is an unmistakable signature of the transition to the red-supergiant phase.
- Gradual increase in luminosity: the luminosity grows almost monotonically, reflecting the rise in the nuclear-burning rate and the increase in envelope opacity as the star expands.
- Evolution consistent with theoretical models: the trajectory in the HR diagram, the behavior of the central abundances, and the convective structure observed in the Kippenhahn diagram faithfully reproduce the expectations for massive stars dominated by central hydrogen burning and subsequent shell-burning ignition.

It is important to emphasize that the simulation covers only the first ~ 5.2 Myr of the star's life — the hydrogen main sequence and the beginning of envelope expansion — and *does not* include the advanced burning phases (He, C, Ne, O, Si) nor core collapse. The statements about Wolf–Rayet, supernova type, and remnant formation are based on theoretical extrapolation from the simulated final mass ($\approx 26.4 M_{\odot}$) and on literature models (Griffiths et al., 2025), not on direct results from this simulation. In pedagogical contexts, it is essential to distinguish what was *computed* (internal structure and evolution up to ~ 5.2 Myr) from what is *predicted by theory* (the final fate). Complete simulations of massive stars through core collapse require extensive nuclear networks and numerical stability during the Si- and O-burning phases, demanding significant computational resources (Paxton et al., 2011). From a didactic standpoint, the modeled interval is already sufficient to illustrate the intense mass loss, envelope expansion, and the transition to the supergiant regime — phenomena central to the evolution of massive stars.

11.6 Nucleosynthesis

Nucleosynthesis comprises the processes responsible for the formation of atomic nuclei and encompasses everything from the first minutes after the *Big Bang* to the explosive, violent events that end the lives of stars of different masses. By understanding the processes responsible for the production of the different chemical elements, it becomes possible to directly relate astrophysical phenomena to observed abundance patterns in stars and galaxies.

Below we describe, in summary form, the main mechanisms of nucleus generation, highlighting the elements produced, the physical environments, and the associated uncertainties. For each process, we explicitly list the elements (atomic number Z and name) whose production is significantly associated with that process.

Physical foundations

The viability of each nucleosynthesis process is determined by the competition between the thermal kinetic energy of the nuclei and the *Coulomb barrier* of electrostatic repulsion. For two

nuclei with atomic numbers Z_1 and Z_2 and nuclear radius $R \sim R_0 A^{1/3}$, the barrier height is on the order of

$$E_C = \frac{Z_1 Z_2 e^2}{R} \approx \frac{Z_1 Z_2}{A^{1/3}} \text{ MeV}, \quad (11.10)$$

much higher than the typical thermal energy $k_B T \sim 1 \text{ keV}$ under stellar conditions. Fusion occurs mainly through quantum tunneling across this barrier, with a probability proportional to the Gamow factor $\exp(-2\pi\eta)$, where $\eta = Z_1 Z_2 e^2 / \hbar v$ (Kippenhahn et al., 2012). For this reason, reactions between heavier nuclei (larger product $Z_1 Z_2$) require progressively higher temperatures: H burning occurs at $T \sim 10^7 \text{ K}$, He burning at $\sim 10^8 \text{ K}$, and the burning of C, Ne, O, and Si at $T \sim 10^8\text{--}10^9 \text{ K}$ in massive stars.

The Gamow factor, identical to the one from Chapter 9

This is exactly the same tunneling probability and the same Sommerfeld parameter $\eta = Z_1 Z_2 e^2 / \hbar v$ derived in detail in Chapter 10, including the same discussion of the Gamow peak (the competition between the decreasing Maxwellian tail and the increasing tunneling probability). The dependence of η on $Z_1 Z_2$ quantitatively explains, here, why nucleosynthesis follows an ordered sequence of temperatures — H burning, then He, then progressively heavier elements — the same physical logic used in that chapter to explain why the Sun burns hydrogen so slowly.

Notation collision: Z as metallicity vs. Z as atomic number

Throughout the previous sections of this chapter, Z denoted the **mass fraction in metals** ($Z \equiv 1 - X - Y$, the same convention as Chapter 5). From this section onward, returning to the standard nuclear-physics notation, Z_1, Z_2 (and the Z accompanying each element in the list below) denote the **atomic number** of a nucleus — the number of protons, with no direct relation to the star’s overall metal mass fraction. Both are standard conventions in their respective subfields (stellar evolution and nuclear physics), and the context of each section makes clear which is in use; the original article itself already operates with this dual convention, common in the nucleosynthesis literature.

In the innermost layers of a massive star close to collapse, when $T \gtrsim 4 \times 10^9 \text{ K}$ and $\rho \gtrsim 10^9 \text{ g cm}^{-3}$, the nuclear reactions become fast enough in both directions for the system to reach *nuclear statistical equilibrium* (NSE): the abundance of each species becomes determined only by temperature, density, and the proton/neutron ratio, no longer by the historical sequence of reactions. In this regime, the distribution converges toward the iron peak (^{56}Fe , ^{56}Ni , ^{56}Co), since these isotopes have the highest binding energy per nucleon (Woosley and Weaver, 1995). This is the characteristic that makes iron an “end of the line” for fusion nucleosynthesis: fusing nuclei heavier than iron would consume energy rather than release it.

In addition to hydrostatic fusion reactions, two neutron-capture processes produce the elements heavier than iron. In the s-process (*slow neutron capture*), the neutron number density is low enough ($n_n \lesssim 10^{10} \text{ cm}^{-3}$) that the resulting nuclei have time to decay via β^- before capturing another neutron, following the “valley of stability” of the chart of nuclides (Käppeler et al., 2011; Busso et al., 1999). In the r-process (*rapid neutron capture*), the neutron number density is extremely high ($n_n \gtrsim 10^{20} \text{ cm}^{-3}$), preventing decay before further captures; the resulting highly neutron-rich nuclei decay in cascade after the event (Cowan et al., 2021; Sneden et al., 2008). This fundamental distinction — established in a seminal way by Burbidge et al. (1957) — is the organizing principle behind the abundance patterns of heavy elements observed in stars and in the Galaxy.

- Primordial Nucleosynthesis: occurred within the first minutes after the *Big Bang*, up to about 200 s. It primordially produced:
 - Hydrogen ($Z = 1$, H) — free protons.
 - Helium ($Z = 2$, He) — ^3He and mainly ^4He .
 - Lithium ($Z = 3$, Li) — traces of ^7Li (and much smaller amounts of ^6Li).
 - Related light isotopes: deuterium (^2H), ^3He .
- Cosmic-ray *spallation*: has a galactic origin, through the fragmentation of heavier nuclei by energetic particles. It mainly generates:
 - Lithium ($Z = 3$, Li), Beryllium ($Z = 4$, Be), Boron ($Z = 5$, B).

These elements are little produced in stars and explain the observations in metal-poor stars (Fields, 2011).

- Stellar hydrostatic burning and processes in massive stars / core-collapse SNe: massive stars and core-collapse supernovae produce a large fraction of the light elements up to the iron peak through successive burnings. Elements with significant contributions include:
 - Carbon ($Z = 6$, C), Nitrogen ($Z = 7$, N), Oxygen ($Z = 8$, O) – He- and C-burning; AGB and high masses.
 - Fluorine ($Z = 9$, F) – AGB, ν -process, and some supernova contributions.
 - Neon ($Z = 10$, Ne), Sodium ($Z = 11$, Na), Magnesium ($Z = 12$, Mg), Aluminum ($Z = 13$, Al) – He-, C-, and Ne-burning.
 - Silicon ($Z = 14$, Si), Phosphorus ($Z = 15$, P), Sulfur ($Z = 16$, S), Chlorine ($Z = 17$, Cl), Argon ($Z = 18$, Ar) – advanced stages and explosive burning.
 - Potassium ($Z = 19$, K), Calcium ($Z = 20$, Ca) – explosive burning and pre-explosion layers.
 - Scandium ($Z = 21$, Sc), Titanium ($Z = 22$, Ti), Vanadium ($Z = 23$, V), Chromium ($Z = 24$, Cr), Manganese ($Z = 25$, Mn) – variable production in core-collapse SNe and explosive burning.
 - Iron ($Z = 26$, Fe), Cobalt ($Z = 27$, Co), Nickel ($Z = 28$, Ni) – iron peak; ^{56}Ni produced in the explosion (decaying into ^{56}Fe).
 - Copper ($Z = 29$, Cu), Zinc ($Z = 30$, Zn) – production via explosive burning, the α -rich process, and the *weak-s* component.

These yield patterns depend strongly on the stellar mass, metallicity, rotation, and details of the explosion (Woosley and Weaver, 1995; Nomoto et al., 2013).

- S-process (slow neutron capture) — AGB stars and *weak-s* at high masses: the s-processes build nuclei heavier than iron through slow neutron captures. Elements produced include:
 - First peak ($N \sim 50$): Strontium ($Z = 38$, Sr), Yttrium ($Z = 39$, Y), and Zirconium ($Z = 40$, Zr).

- Second peak ($N \sim 82$): Barium ($Z = 56$, Ba), Lanthanum ($Z = 57$, La), Cerium ($Z = 58$, Ce), and Neodymium ($Z = 60$, Nd).
- Third peak: Lead ($Z = 82$, Pb) is a typical end product of s-processes at low metallicity; Polonium/Bismuth in certain branches (though Bi is *borderline*).
- Lanthanides ($Z = 57\text{--}71$) are partly produced by the s-process (with significant r-contribution for many isotopes).

Environments: low- to intermediate-mass AGB stars (dredge-ups) and the weak-s component in massive stars (Käppeler et al., 2011; Busso et al., 1999).

- R-processes (rapid neutron capture): In extremely neutron-rich environments, rapid captures build nuclei well beyond iron. Strongly affected elements:
 - Lanthanides and actinides: Europium ($Z = 63$, Eu — an r-process indicator), Gadolinium ($Z = 64$, Gd), up to Lutetium ($Z = 71$, Lu).
 - Heavy metals: Platinum ($Z = 78$, Pt), Gold ($Z = 79$, Au), and Silver ($Z = 47$, Ag).
 - Actinides: Thorium ($Z = 90$, Th) and Uranium ($Z = 92$, U) — produced by the r-process and used as cosmochronometers⁶.
 - Others: Osmium ($Z = 76$, Os) and Iridium ($Z = 77$, Ir) have important r-contributions.

Environments: neutron-star mergers are currently the main confirmed sites; rare magnetorotational supernovae may contribute (Thielemann et al., 2017; Metzger, 2019; Sneden et al., 2008).

- Type Ia Supernovae: Mainly produce the iron peak on a large scale in galaxies:
 - Iron ($Z = 26$, Fe), Nickel ($Z = 28$, Ni), Cobalt ($Z = 27$, Co) — large amounts of ^{56}Ni that decays into ^{56}Fe .
 - Intermediate-mass elements (Si, S, Ca) are produced in the outer layers by incomplete burning.

Role: dominate the Fe enrichment in the late stages of galactic history (Nomoto et al., 2013).

- AGB stars and planetary nebulae (non-explosive production): In addition to the s-process listed above, AGB stars and their winds produce:
 - Carbon ($Z = 6$, C) — formation of carbon stars via dredge-up.
 - Nitrogen ($Z = 7$, N) — CNO processing and dredge-up.
 - Rare light elements: ^7Li ($Z = 3$, Li, via the Cameron–Fowler process in some AGB stars), Fluorine ($Z = 9$, F).
- P-processes and γ -processes (photodisintegration): Produce rare isotopes that are not born via neutron capture. Notable examples:

⁶Cosmochronometers are long-half-life radioactive nuclei (as in the case of ^{232}Th and ^{238}U) whose isotopic ratios relative to their decay products or to stable elements make it possible to estimate stellar and galactic ages. The comparison between the observed abundances and those predicted by nucleosynthesis models (especially of the r-process) provides timescales for stellar formation events and for the chemical history of the Galaxy itself (Cowan et al., 2021; Schatz et al., 2002).

- P isotopes: Mo ($Z = 42$, especially $^{92,94}\text{Mo}$) and Ru ($Z = 44$, $^{96,98}\text{Ru}$).
- νp -processes in neutrino-rich environments (e.g., certain regions of SNe) contribute to the nucleosynthesis of some light nuclei.

Environments: interior layers of core-collapse supernovae subjected to intense radiation; regions with intense neutrino flux (Arnould and Goriely, 2003; Rauscher et al., 2002).

- Rare explosive processes and proton capture (rp-process): In thermonuclear explosions at the surface of compact stars (X-ray bursts), rapid proton captures occur that can synthesize proton-rich nuclei up to mass numbers $A \sim 100$. Elements involved (mostly in exotic, often unstable, isotopes):
 - Isotopic production in the Ga–Sn region ($Z \approx 31\text{--}50$) and beyond, depending on conditions.

Examples: rp-processes have an impact on isotopic abundances, but rarely produce abundant, observable stable isotopes on a large scale (Schatz et al., 2001).

- Novae (explosions in cataclysmic systems): Thermonuclear novae (Jose and Hernanz, 1998) on the surface of white dwarfs contribute to some light/intermediate isotopes:
 - ^{13}C , ^{15}N , and ^{17}O (light elements with specific isotopes).
 - ^{22}Na and ^{26}Al (radioactive isotopes observable in γ rays) — combined contributions with massive stars.
 - Under certain conditions, production of ^7Be (decaying into ^7Li).
- Neutrino processes (ν -process): Neutrino interactions during a SN explosion can synthesize some rare nuclei:
 - ^7Li ($Z = 3$, Li), ^{11}B ($Z = 5$, B), ^{19}F ($Z = 9$, F), ^{138}La ($Z = 57$), ^{180}Ta ($Z = 73$) — examples dependent on the neutrino flux.

Highlight: crucial contributions to certain rare abundances that are not explained solely by s- or r-processes (Woosley and Weaver, 1990; Heger and Woosley, 2005).

- Mixed / minor contributions and transition elements: Many elements between Ga ($Z = 31$) and Te ($Z = 52$), and elements of the d- and f-blocks ($Z \approx 30\text{--}80$), have origins distributed among:
 - *weak-s* in massive stars,
 - s-processes in AGB stars,
 - s-processes in violent events,
 - explosive processing in core-collapse SNe/Type Ia SNe,
 - p-processes for proton-rich isotopes.

Examples: Ga (31), Ge (32), As (33), Se (34), Br (35), Kr (36), Rb (37), Sr (38), Y (39), Zr (40), Nb (41), Mo (42), Tc (43 — no stable isotopes), Ru (44), Rh (45), Pd (46), Ag (47), Cd (48), In (49), Sn (50), Sb (51), Te (52), I (53), Xe (54), Cs (55), Ba (56), La (57), Ce (58), Pr (59), Nd (60), Pm (61, radioactive), Sm (62), Eu (63), Gd (64), Tb (65), Dy (66), Ho (67), Er (68), Tm (69), Yb (70), Lu (71), Hf (72), Ta (73), W (74), Re (75), Os (76), Ir (77), Pt (78), Au (79), Hg (80), Tl (81), Pb (82), Bi (83). For each element in this range there is an isotopic “fingerprint” that galactic models seek to reproduce.

- Heaviest natural elements:
 - Protactinium ($Z = 91$, Pa) — also associated with the s-process.
 - Elements with $Z > 92$ (from Neptunium onward):
Neptunium ($Z = 93$), Plutonium ($Z = 94$), Americium ($Z = 95$), etc. — generally unstable with short half-lives; although the r-process can produce very heavy nuclei, these elements decay and do not accumulate in significant cosmic abundances. Only Th and U are abundant and detectable in the spectra of old stars.
- Synthetic / superheavy elements ($Z > 92$ up to $Z = 118$):
 - Most transuranic elements ($Z \geq 93$) were synthesized in terrestrial laboratories (reactors, accelerators) and have very short half-lives. They do not contribute to the observable cosmic abundances.
 - Examples: Np (93), Pu (94), Am (95), Cm (96), . . . , Og (118). They are not relevant as stable sources in the interstellar medium.

11.7 Conclusions

This work proposed a didactic-computational synthesis of stellar evolution, combining two types of resources: (i) public evolutionary tracks and isochrones from the MIST project (Dotter, 2016), used to build HR and Kiel diagrams with and without rotation, as well as to analyze temporally the structural and nuclear parameters of stars between 0.5 and $25 M_{\odot}$; and (ii) our own simulation with the MESA code (version r23.05.1) for a $50 M_{\odot}$ star with winds prescribed by the *Vink* and *Dutch* schemes. The goal was not to produce novel astrophysical results, but rather to integrate, in a single study guided by quantitative data, the stellar life cycle from the main sequence to the advanced evolutionary phases, including nucleosynthesis, for use as support material for teaching Physics and Astronomy.

The results obtained with the MIST tracks confirmed robust physical trends. In the HR diagrams, the strong dependence of luminosity and effective temperature on initial mass was directly visualized: low-mass stars remain close to the main sequence for timescales comparable to or exceeding the age of the Universe, while massive stars evolve rapidly into supergiant phases. The time-dependent analysis of the nuclear processes confirmed the expected regimes: the pp chain dominates in solar-mass stars, the CNO cycle prevails from the ZAMS onward in stars of $\gtrsim 1.5\text{--}2 M_{\odot}$, and the triple- α process becomes active after the exhaustion of central hydrogen — with the maximum peak for $5 M_{\odot}$ occurring during the thermal pulses of the TPAGB phase (~ 115.6 Myr), not during core helium burning. The central density grows from $\sim 80 \text{ g cm}^{-3}$ at the ZAMS of solar-mass stars to $\sim 1.25 \times 10^7 \text{ g cm}^{-3}$ during the thermal pulses of $5 M_{\odot}$. The isochrones showed the shift of the *turn-off* point with age and the effect of metallicity on the position of the main sequence in the Kiel diagram. The MESA simulation captured the onset of envelope expansion and the increase in mass loss as the $50 M_{\odot}$ star leaves the main sequence — the mass dropped by approximately 47% over the simulated period of ~ 5.2 Myr.

Some limitations should be considered when interpreting the results. The evolutionary trajectories analyzed were obtained from the MIST grid and therefore depend on the physical assumptions, mass-loss prescriptions, mixing criteria, and other parameters adopted in these models. In addition, the tracks used were computed without rotation, with $v/v_{\text{crit}} = 0$, so that effects such as rotation-induced mixing, changes in the main-sequence lifetime, and variations

in mass-loss rates were not considered. The temporal filter for ages above 10^6 years removes only the earliest pre-main-sequence phases but does not correspond to a direct selection by evolutionary stage. As a consequence, part of the pre-ZAMS evolution remains represented in the curves of low-mass stars. The temporal coverage of the tracks is also not uniform for all masses, since some sequences do not reach the same final evolutionary phases within the grid considered. The comparisons presented should therefore be interpreted within the range of masses, metallicities, ages, and evolutionary phases actually covered by the models. Finally, the predominantly didactic and comparative character of this work does not include a systematic analysis of the numerical and theoretical uncertainties associated with the different stellar evolution prescriptions.

The main contribution of this work is didactic-computational in nature: by integrating public evolutionary tracks, isochrones, time-dependent analysis of internal parameters, and our own computational simulation into a single narrative and quantitative material, we offer a structured resource for teaching stellar evolution in advanced undergraduate and graduate courses in Physics and Astronomy. The annotated periodic table in the nucleosynthesis section visually synthesizes the connection between astrophysical processes and the origin of the elements, linking stellar evolution to galactic chemical enrichment. The MESA *inlist* made available in the Appendix allows readers to reproduce the simulation presented and also to modify parameters such as initial mass, metallicity, rotation, mixing prescriptions, and mass loss, according to the specific objectives of each study or pedagogical activity. It is hoped that this material will serve as a reference or starting point for computational laboratories in Astrophysics and Computational Physics courses in undergraduate and graduate Physics programs.

Further Reading and Current References

For a very recent extension of the MIST project itself used in this chapter — now covering the white-dwarf cooling sequence, connecting directly with Chapter 9 — see Bauer et al. (2026). On the impact of main-sequence mass loss on the structure and evolution of Wolf-Rayet stars (relevant to the $50 M_{\odot}$ MESA simulation discussed in this chapter), see the recent review by Josiek et al. (2024). For a complete and up-to-date treatment of stellar nuclear physics (the Gamow factor, reaction rates, the pp chain, and the CNO cycle) behind the nuclear luminosity curves presented here, see Iliadis (2015), already cited in Chapter 10.

Acknowledgments

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11.8 Appendix A: Inlist Used

In this section, we make available the contents of the `inlist` file used in the MESA simulation.

&star_job

```
    create_pre_main_sequence_model = .false.
    pgstar_flag = .false.
    set_initial_age = .true.
    initial_age = 1d6
/

&eos
/

&kap
    Zbase = 0.02
/

&controls
    initial_mass = 50
    initial_z = 0.02

    ! WINDS - 50 Msun
    hot_wind_scheme = 'Vink'
    Vink_scaling_factor = 2.0
    cool_wind_RGB_scheme = 'Dutch'
    cool_wind_AGB_scheme = 'Dutch'
    Dutch_scaling_factor = 2.0

    ! Stopping conditions
    max_age = 1.0d7          ! Reduced - 50 Msun lives ~6-8 million years
    max_model_number = 100000

    ! xa_central_lower_limit_species(1) = 'he4'
    ! xa_central_lower_limit(1) = 1d-3
    ! xa_central_lower_limit_species(2) = 'c12'
    ! xa_central_lower_limit(2) = 1d-3

    ! Timestep controls
    varcontrol_target = 1d-2 !
    delta_lgL_H_limit = 0.5  ! Increased
    delta_lgL_He_limit = 0.5 ! Increased
    delta_lgL_z_limit = 0.5  ! Increased

    ! Extra mixing for massive stars
    overshoot_scheme(1) = 'exponential'
    overshoot_zone_type(1) = 'any'
    overshoot_zone_loc(1) = 'core'
    overshoot_bdy_loc(1) = 'top'
    overshoot_f(1) = 0.02
    overshoot_f0(1) = 0.01
/
```

The `inlist` presented in this appendix, together with the methodological description of Section 2, allows the configuration of the MESA simulation used in this work to be reproduced. To enhance the transparency and reproducibility of the study, we make available in the computational repository associated with the article the Python post-processing codes, the analysis notebooks, the selected `inlist` files, the `history.data` file, the generated figures, and detailed instructions for setting up and running the computational environment: <https://github.com/magcosmo/stellar-structure-evolution-rbef>. The complete MIST track and isochrone files are not redistributed in the repository and must be obtained from the official MIST project. Likewise, the `profiles.index` file and the `profile*.data` files needed to build the Kippenhahn diagram are not made available and must be produced or obtained by running the corresponding MESA simulation. The instructions in the repository describe the expected directory structure and the path configuration for these external data. Thus, full reproduction of the analyses requires installing the indicated dependencies and obtaining the necessary MIST and MESA files beforehand.

11.9 Appendix B: Synthesis Mechanisms and Physical Environments of the Chemical Elements

Table 11.4: Main mechanisms and physical environments for the synthesis of the known chemical elements. The references underlying this table are described in Section 11.6.

Progenitor Process	Z	Elements	Comments
Primordial Nucleosynthesis	1	Hydrogen (H)	In the form of free protons (^1H) and deuterium (^2H).
	2	Helium (He)	Mainly ^4He , but also ^3He .
	3	Lithium (Li)	Traces of ^7Li , an even smaller amount of ^6Li .
Cosmic-ray spallation	3	Lithium (Li)	Produced through the fragmentation of nuclei of heavier elements by cosmic rays.
	4	Beryllium (Be)	
	5	Boron (B)	
Stellar hydrostatic burning and supernovae	6	Carbon (C)	Produced during He- and C-burning and dredge-ups in the AGB phase of low- to intermediate-mass stars, or during the MS of massive stars.
	7	Nitrogen (N)	
	8	Oxygen (O)	Produced in AGB stars and core-collapse SNe in massive stars.
	9	Fluorine (F)	Produced in AGB stars and by ν -processes in core-collapse supernovae.
	10	Neon (Ne)	Produced during He-, C-, and Ne-burning.
	11	Sodium (Na)	
	13	Aluminum (Al)	
	12	Magnesium (Mg)	α element, produced in massive stars in the phases before core collapse.

	14	Silicon (Si)	Produced in C- and Ne-burning (static component) and explosive burning (dynamic component); dominant α element in massive stars.
	15	Phosphorus (P)	Produced during advanced stages and explosive burning.
	16	Sulfur (S)	α element; produced in O-burning and explosive burning.
	17	Chlorine (Cl)	Produced during advanced stages and explosive burning.
	18	Argon (Ar)	
	19	Potassium (K)	Originates from explosive burning and pre-explosion layers.
	20	Calcium (Ca)	α element; produced in O- and Si-burning and explosive burning.
	21	Scandium (Sc)	
	22	Titanium (Ti)	Variable production in Type II supernovae and explosive burning.
	23	Vanadium (V)	
	24	Chromium (Cr)	
	25	Manganese (Mn)	
	26	Iron (Fe)	Iron-peak elements, generated during Type Ia supernovae or in massive stars (<i>weak-s</i> component); ^{56}Ni decays, generating ^{56}Fe .
	27	Cobalt (Co)	
	28	Nickel (Ni)	
	29	Copper (Cu)	Produced via explosive burning, the α -rich process, and s-processes, constituting the <i>weak-s</i> component.
	30	Zinc (Zn)	
Mixed contributions (s-, r-, p- and explosive processes)	31–83	—	Elements in this range generally have multiple production pathways. The most relevant cases for galactic enrichment are detailed in the s-, r-, and p-process blocks below.

Progenitor Process	Z	Elements	Comments
S-processes (slow neutron capture) — low- to intermediate-mass AGB stars (<i>main-s</i> component) and massive stars (<i>weak-s</i> component)	38	Strontium (Sr),	Constitute the <i>main-s</i> component.
	39	Yttrium (Y)	
	40	Zirconium (Zr)	
	56	Barium (Ba)	
	57	Lanthanum (La)	
	58	Cerium (Ce)	
	60	Neodymium (Nd)	Typical end product of s-processes at low metallicity.
	82	Lead (Pb)	
	83	Bismuth (Bi)	
	84	Polonium (Po)	
R-processes (rapid neutron capture) – Neutron-star mergers and rare magnetorotational supernovae	63	Europium (Eu)	Lanthanides and actinides produced in part by the s-process, but with significant r-process contribution for many isotopes.
	...		
	71	Lutetium (Lu)	
	47	Silver (Ag)	Heavy metals.
	78	Platinum (Pt)	
	79	Gold (Au)	
	76	Osmium (Os)	R-processes contribute significantly.
	77	Iridium (Ir)	
	90	Thorium (Th)	Used as cosmochronometers.
	92	Uranium (U)	
P-processes and γ -processes	42	Molybdenum (Mo)	Mainly the isotopes ^{92}Mo and ^{94}Mo .
	44	Ruthenium (Ru)	Mainly the isotopes ^{96}Ru and ^{98}Ru .
Rp-processes and ν p-processes	31	Gallium (Ga)	Generate isotopes up to $A \sim 100$.
	...		
	50	Tin (Sn)	

ν -processes	19	Potassium (K)	Occur in core-collapse supernovae (II, Ib, and Ic).
	21	Scandium (Sc)	
	23	Vanadium (V)	
Novae – explosions in cataclysmic systems	4	Beryllium (Be)	Production of specific light and intermediate isotopes: ^{13}C , ^{15}N , ^{17}O , ^{22}Na , ^{26}Al , and ^7Be (which decays into ^7Li).
	6	Carbon (C)	
	7	Nitrogen (N)	
	8	Oxygen (O)	
	11	Sodium (Na)	
Synthetic/superheavy elements	13	Aluminum (Al)	Unstable and with short half-lives, these elements decay and do not accumulate in significant cosmic abundances. They are not relevant as stable sources in the stellar medium, being synthesized in terrestrial laboratories.
	93	Neptunium (Np)	
	94	Plutonium (Pu)	
	95	Americium (Am)	
	...		
	118	Oganesson (Og)	

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